General Aptitude

Part II : Numerical Ability

NUMBERS

Natural Numbers

All positive integers are natural numbers.

e.g. 1, 2, 3, 4, 5,.....

There are infinite natural numbers and number 1 is the *least natural number*.

Types of Natural Numbers : *Based on divisibility there are following two types of natural numbers :*

1. ***Prime Numbers:***

A natural number larger than unity is a prime number if it does not have other divisors except itself and unity.

Note: Unity *i.e* 1 is not a prime number.

To Check A Number is Prime or not:

(*i*) Take square root of the number.

(*ii*) Round of the square root to the next highest integer and call this number as Z.

(*iii*) Check for divisibility of the number N by all prime numbers below Z. If there is no number below the value of Z which divides N then the number will be prime.

e.g. 241 is prime or not?

$\sqrt{241}$ lies between 15 or 16.Hence take the value of Z =16.

Prime numbers less than 16 are 2, 3, 5, 7, 11 and 13.

241 is not divisible by any of these. Hence we can conclude that 241 is a *prime number.*

2. ***Composite Numbers:***

The numbers which are not prime are known as composite numbers.

Note: 1 is neither prime not composite

Co-Primes: Two numbers a and b are said to be co-primes, if their H.C.F is 1.

e.g. (2, 3), (4, 5), (7, 9), (8, 11).....

Tests for Divisibility

A number is divisible by

(1) 2, when its unit's digit is even or 0.

(2) 3, when the sum of its digits is divisible by 3.

(3) 4 when number formed by last two right hand digits is divisible by 4, or if last two digits are 0's.

(4) 5, when its unit's digit is 5 or 0.

(5) 6, when it is divisible by 2 and 3 both.

(6) 8 when number formed by last three right hand digits is divisible by 8, or when last three digit are 0's.

(7) 9, when sum of its digits is divisible by 9.

(8) 10, when its unit's digit is 0.

(9) 11, when difference between sum of the digits in the odd and sum of the digits in the even places is 0 or a multiple of 11.

Note: When any number with an even number of digits is added to its reverse, the sum is always a multiple of 11.

(10) 12, when it is divisible by 3 and 4 both.

HCF AND LCM

Greatest Common Divisor (GCD)

It is the greatest divisor common to two or more given numbers. It is also called HCF (Highest Common Factor).

Method to find HCF of given Numbers.

1. ***By method of factorization:***

Express each given number as product of primes. Now take product of common factors, which is HCF.

e.g., Find HCF of 136, 144, 168.

$136 = 2 \times 2 \times 2 \times 17 = 2^3 \times 17$

$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = 2^4 \times 3^2$

$168 = 2 \times 2 \times 2 \times 3 \times 7 = 2^3 \times 3 \times 7$

$HCF = 2^3 = 8$

2. ***By division method:***

Suppose two numbers are given.

Divide greater number by the lesser one; divide lesser by the remainder; divide first remainder by the new remainder, and so on till there is no remainder. Last divisor is the required H.C.F.

e.g., Find H.C.F. of 12 and 15.

$$\begin{array}{r} 12)15(1 \\ \underline{-12} \\ 3)12(4 \\ \underline{-12} \\ 00 \end{array}$$

$\therefore$ HCF = 3

In case more than two numbers are given, then choose any two of them and find their HCF. HCF of these two and third gives HCF of three numbers and so on.

Note: If 'Z' is HCF of 'X' and 'Y', then Z is also a factor of aX + bY where a, b are integers.

Least Common Multiple (LCM)

LCM of two or more given number is the least number which is exactly divisible by each of the given numbers.

Methods to find LCM of given Numbers.

1. By method of factorization:

Resolve each one of the given numbers into prime factors, then their LCM is product of highest powers of all factors, that occur in these numbers.

e.g., **Find LCM of 136, 144, 168**

$136 = 2^3 \times 17$

$144 = 2^4 \times 3^2$

$168 = 2^3 \times 3 \times 7$

$LCM = 2^4 \times 3^2 \times 17 \times 7 = 17136$

2. Using the formula:

Product of numbers = HCF × LCM

If two numbers are given, find their HCF.

$$LCM = \frac{\text{Product of two numbers}}{HCF}$$

e.g., Find LCM of 136, 144.

$$LCM = \frac{136 \times 144}{8} = 2448$$

If more than two numbers are given, find LCM of any two of them. Then LCM of this LCM and third number gives LCM of these three numbers and so on.

3. Short cut method using factorization:

e.g. Find LCM of 18, 28, 108, 105.

2	18,	28,	108,	105
2	9,	14,	54,	105
3	9,	7,	27,	105
3	3,	7,	9,	35
3	1,	7,	3,	35
5	1,	7,	1,	35
7	1,	7,	1,	7
	1,	1,	1,	1

$LCM = 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 7 = 3780$

Alternatively : We can strike out 18 at once, since any multiple of 108 must contain 18 as a factor.

2	28,	108,	105
2	14,	54,	105
3	7,	27,	105
7	7,	9,	35
	1,	9,	5

$LCM = 2 \times 2 \times 3 \times 7 \times 9 \times 5 = 3780$

Note: For any given numbers, HCF is necessarily a factor of their L.C.M.

HCF and LCM of Fractions.

$$\text{HCF of Fractions} = \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}}$$

$$\text{LCM of Fractions} = \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}$$

e.g. Find HCF and LCM of $\frac{4}{5}, \frac{8}{35}, \frac{2}{15}, \frac{6}{25}$

$$\therefore \quad HCF = \frac{\text{HCF of } 4, 8, 2, 6}{\text{LCM of } 5, 35, 15, 25}$$

$$= \frac{2}{5 \times 7 \times 3 \times 5} = \frac{2}{525}$$

$$LCM = \frac{\text{LCM of } 2, 8, 4, 6}{\text{HCF of } 5, 35, 15, 25} = \frac{24}{5}$$

PROGRESSIONS

Arithmetic Progression (A.P.)

Let 'a' be the first term of an arithmetic progression, 'd' the common difference and 'n' the number of terms in the progression.

$T_n = n^{th}$ term $= a + (n-1)d$

$S_n =$ Sum of n terms $= \frac{n}{2} \times [2a + (n-1)d]$

Progression can be represented as a, a + d, a + 2d, ..., [a + (n − 1) d]. Here, quantity d is to be added to any chosen term to get next term of the progression. When there are n terms in an arithmetic progression, a is first term and [a + (n − 1) d] is last term.

Then $S_n = \frac{n}{2} \times [\text{First Term} + \text{Last Term}]$

Geometric Progression (G.P.)

Geometric progression can be represented as a, ar, ar^2, ... where a is first term and r is common ratio of the geometric progression.

n^{th} term of the geometric progression $= ar^{n-1}$.

$$\text{Sum to n terms} = \frac{a(1-r^n)}{1-r} \text{ or } \frac{a(r^n-1)}{r-1}$$

$$= \frac{r \times \text{last term} - \text{First term}}{r-1}$$

Infinite Geometric Progression.

If $-1 < r < +1$ or $|r| < 1$, then sum of a geometric progression does not increase infinitely; it "converges" to a particular value. Such a G.P. is called infinite geometric progression.

Sum of an infinite geometric progression, $S_\infty = \frac{a}{1-r}$

Remember

Sum of first n natural numbers, $= S_n = \sum n = \frac{n(n+1)}{2}$

Sum of squares of first n natural numbers,

$$\sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

Sum of cubes of first n natural numbers,

$$\sum n^3 = \left[\frac{n(n+1)}{6}\right]^2 \frac{n^2(n+1)^2}{4} = \left[\sum n\right]^2$$

AVERAGES, MEAN, MODE and MEDIAN

Averages

A central value around which a group of values shows a tendency to concentrate is called *average*. Thus an average is a single value that is in some way indicative of a group of values.

Measures of Central Tendency

Following are five measures of central tendency :

1. **Mode :** It is the number that occurs most frequently in a given set of numbers.

   e.g., Find mode of the set of numbers 2, 4, 6, 6, 5, 4, 3, 6, 2, 6, 4, 1, 4, 5, 4.

   Since number 4 occurs maximum number of times, hence mode is 4.

   Note : In a given set of data, if two or more values occur the same number of times, then a unique mode does not exist.

2. **Median:** It is the middle value of a group of numbers arranged in an ascending or descending order.

   If number of values (n) in a given set of data is odd, then Median is the $\left(\frac{n+1}{2}\right)^{th}$ value.

   e.g., Find median of the numbers 40, 41, 33, 48, 41, 39 and 38.

   Arrange values in ascending order, i.e. 33, 38, 39, 40, 41, 41, 48.

   Median is the $\left(\frac{7+1}{2}\right)^{th}$ value, 4th value *i.e.,* 40

   If number of values (n) in a given set of data is even, then there will be two middle values say a and b, and hence median is taken as $\frac{a+b}{2}$.

   e.g., Find median of the numbers 16, 15, 13, 14, 23, 44.

   Arrange values in descending order, i.e. 44, 23, 16, 15, 14, 13. Two middle values are 16 and 15.

   $\therefore$ Median $= \frac{16+15}{2} = \frac{31}{2} = 15.5$

MEAN

1. **Arithmetic Mean (A.M.):** The most commonly used average is the arithmetic mean (AM) or simply the average.

   Arithmetic mean of n numbers $x_1, x_2, x_3 \ldots\ldots x_n$ is denoted by $\bar{x}$ and calculated as

   $$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} \quad \text{or} \quad \bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

   where Greek letter Σ(sigma) indicates *"Sum of the terms of the type"*. It takes value from 1 to n.

   Arithmetic mean of a and b is $\frac{a+b}{2}$.

   e.g., Find arithmetic mean of 10, 15, 17, 20.

   $$\bar{x} = \frac{10+15+17+20}{4} = 15.5$$

   When large numbers are involved, an Arithmetic Mean is generally assumed to facilitate calculation. If a is assumed Arithmetic mean, then true Arithmetic mean $\bar{x}$ of group of n numbers $x_1, x_2, x_3 \ldots\ldots x_n$ is

   $$\bar{x} = a + \frac{\sum_{i=1}^{n} d_i}{n},$$

   where $d_i = x_i - a$ is deviation from the assumed mean.

   e.g., Find the Arithmetic mean of 42, 46, 48, 50, 55, 60, 70

   Let assumed mean be 50

x_i	d_i
42	42 − 50 = − 8
46	46 − 50 = − 4
48	48 − 50 = − 2
50	50 − 50 = 0
55	55 − 50 = 5
60	60 − 50 = 10
70	70 − 50 = 20

   $$\therefore \quad \bar{x} = 50 + \frac{\sum_{i=1}^{7} di}{7} = 50 + \frac{21}{7} = 53.$$

   Weighted Arithmetic Mean: If values $x_1, x_2, \ldots\ldots x_n$ are assigned weights $w_1, w_2, \ldots\ldots w_n$ respectively, then

   $$\text{weighted arithmetic mean }, \bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

   The term *'weight'* stands for relative importance that is attached to the different values.

   'w' can also be replaced by 'f' in a frequency distribution where 'f' denotes frequency or simply the number of times a particular observation occurs.

   e.g.,

x_i	w_i	$w_i x_i$
20	3	60
30	1	30
40	5	200
50	1	50

   $\Sigma w_i = 10$; $\Sigma x_i w_i = 340$

   $$\therefore \quad \bar{x}_w = \frac{\sum_{i=1}^{4} w_i x_i}{\sum_{i=1}^{4} w_i} = \frac{340}{10} = 34$$

   When two or more groups are combined, then combined average of the groups is

   $$\bar{x}_c = \frac{x_1 n_1 + x_2 n_2 + x_3 n_3 + \ldots + x_n n_n}{n_1 + n_2 + n_3 + \ldots + n_n}$$

where $x_1, x_2, x_3, \ldots x_n$ individual averages of the groups and $n_1, n_2, \ldots, n_n$ are number of observations or values in each of the groups.

e.g., Mean marks scored by 50 boys are 70 and those scored by 100 girls are 55. Find average marks scored by this group.

$$x_c = \frac{n_1x_1 + n_2x_2}{n_1 + n_2} = \frac{70 \times 50 + 100 \times 55}{150} = \frac{9000}{150} = 60$$

2. Geometric Mean (G.M.):

G.M. of n numbers $x_1, x_2, \ldots x_n$ is nth root of their products

i.e., $\text{G.M.} = \sqrt[n]{x_1 \times x_2 \times x_3 \times \ldots \times x_n}$

e.g., Geometric mean of 2, 3 and 4 is $\sqrt[3]{2 \times 3 \times 4}$

$= \sqrt[3]{24} = 2.\sqrt[3]{3}$

Geometric mean of a and b is $\sqrt{ab}$

e.g., Find Geometric mean of 9 and 4.

$$\text{G.M.} = \sqrt{9 \times 4} = 6$$

Geometric mean is generally used to calculate rate of growth.

If geometric mean of one group of 'a' numbers is 'x' and that of another group of 'b' numbers is 'y', then Geometric mean of the combined groups

$$= (x_a + y_b)\frac{1}{a+b}$$

3. Harmonic Mean (H.M.):

Harmonic mean of n numbers $x_1, x_2, \ldots x_n$ is calculated as $\frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$

Harmonic mean of a and b is $\frac{2ab}{a+b}$.

e.g., Find Harmonic mean of 1, 0.5, 10.

x_i	$\frac{1}{x_i}$
1	1
0.5	2
10	0.1

$\sum \frac{1}{x_i} = 3.1$

$\text{H.M.} = \frac{3}{3.1} = \frac{30}{31}$

This is an important result and is generally used to find average speed when equal distances are covered at different speeds. (when different distances are covered in the same time, arithmetic mean is used to determine the average speed).

e.g., A person travels from A to B at 6 kmph and returns at 4 kmph, find his average speed.

Average Speed = Harmonic Mean of 6 and 4

$$= \frac{2 \times 6 \times 4}{6+4} = \frac{48}{10} = 4.8 \text{ kmph}$$

Note: This concept will again be covered in time, speed and distance.

Weighted Harmonic Mean: Weighted Harmonic Mean of 3 numbers x_1, x_2 and x_3 with weights w_1, w_2 and w_3 respectively is $\frac{\sum_{i=1}^{3} w_i}{\sum_{i=1}^{3} \frac{w_i}{x_i}}$

This is used to calculate average speed when different distances are covered at different speeds.

e.g., A cyclist covers his first 5 km at an average speed of 10 kmph, another 3 km at 8 kmph and last 2 km at 5 kmph. Find his average speed during the journey.

Average Speed = Weighted HM

$$= \frac{5+3+2}{\frac{5}{10} + \frac{3}{8} + \frac{2}{5}} = 7.84 \text{ kmph}$$

Relation between A.M., G.M. and H.M.

1. Consider two number a and b.

Arithmetic Mean $= \frac{a+b}{2}$; Geometric mean $= \sqrt{ab}$

Harmonic mean $= \frac{2ab}{a+b}$

$\therefore \quad (\text{GM})^2 = \text{AM} \times \text{HM}$

e.g., If Arithmetic mean of two numbers is 6.5 and their Geometric mean is 6, what is their Harmonic Mean?

$$\text{HM} = \frac{(\text{G.M.})^2}{\text{A.M.}} = \frac{6 \times 6}{6.5} = 5.54$$

2. Arithmetic Mean > Geometric Mean > Harmonic Mean

e.g., Arithmetic Mean of 2 and 4 is 3.
Geometric Mean of 2 and 4 is 2.82.
Harmonic Mean of 2 and 4 is 2.66

$\therefore \quad 3 > 2.82 > 2.66$

ALGEBRAIC FORMULAE AND THEIR APPLICATION

An equation is a statement that *two algebraic expressions are equal*. If an equation is satisfied by any value of the variable, then equation is said to be an *identity*.

1. $(a+b)^2 = a^2 + 2ab + b^2 = (a-b)^2 + 4ab$
2. $(a-b)^2 = a^2 - 2ab + b^2 = (a+b)^2 - 4ab$
3. $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
4. $(a+b)^2 - (a-b)^2 = 4ab$
5. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
6. $(a+b+c+d)^2 = a^2 + b^2 + c^2 + d^2 + 2a(b+c+d) + 2b(c+d) + 2cd$
7. $(a+b)(a-b) = a^2 - b^2$
8. $(x+a)(x+b) = x^2 + (a+b)x + ab$
9. $(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$
10. $(a+b)^3 = a^3 + 3ab(a+b) + b^3$
11. $(a-b)^3 = a^3 - 3ab(a-b) - b^3$
12. $a^3 + b^3 = (a+b)^3 - 3ab(a+b) = (a+b)(a^2 - ab + b^2)$

13. $a^3 - b^3 = (a-b)^3 + 3ab(a-b) = (a-b)(a^2 + ab + b^2)$
14. $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - ac - bc)$
    If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$
15. $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \ldots + b^{n-1})$ for all n.
16. $a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \ldots - b^{n-1})$
    if n is even
17. $a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 + \ldots + b^{n-1})$
    if n is odd.
18. $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$

POLYNOMIALS

Factors of polynomials: An expression is said to be resolved into factors when expressions of its product are found.

HCF of polynomials: When two or more polynomials are factorized, then product of all the common factors is HCF of the polynomials.

LCM of polynomials: When two or more polynomials are factorized, then product of all the factors with highest power is LCM of the polynomials.

INEQUATIONS

Properties

1. An inequality will still hold after each side has been increased, diminished, multiplied or divided by the same positive quantity.

i.e., if $a > b$ and $c > 0$

$a + c > b + c$

$ac > bc$

$\frac{a}{c} > \frac{b}{c}$

2. In an inequality, any term may be transposed from one side to the other if its sign is changed.

i.e., if $a - c > b$, then $a > b + c$ or $c < a - b$

3. If sides of an inequality be multiplied by same negative quantity, then sign of inequality must be reversed.

i.e., if $a > b$ and $c < 0$, then $ac < bc$

4. If $a > b$, $a, b \geq 0$ then $a^n > b^n$ and $\frac{1}{a^n} < \frac{1}{b^n}$, or $a^{-n} < b^{-n}$; if n is a positive quantity.

5. Square of every real quantity is positive and therefore must be greater than zero

i.e., for $a \neq b$, $(a-b)^2 > 0$; $a^2 + b^2 > 2ab$;

Similarly, if $x > 0$, $y > 0$, then

$$\frac{x+y}{2} \geq \sqrt{xy}$$

Hence, Arithmetic mean of two positive quantities is greater than or equal to their Geometric mean.

6. If sum of two positive quantities is given, their product is greatest when they are equal; and if product of two positive quantities is given, their sum is least when they are equal.

7. If a, b, c.... k are n unequal quantities, then

$$\left(\frac{a+b+c+\ldots+k}{n}\right)^n > a \times b \times c \times d \ldots \times k$$

i.e., $\frac{a+b+c+\ldots+k}{n} > (a \times b \times c \times \ldots \times k)^{1/n}$

Note: Arithmetic mean of any number of positive quantities is greater than their Geometric mean.

8. If a and b are positive and unequal, then

$\frac{a^m b^m}{2} > \left(\frac{a+b}{2}\right)^m$ except when m is a positiveproper fraction.

If m is a positive integer or any negative quantity, then $\frac{a^m b^m}{2} > \left(\frac{a+b}{2}\right)^m$

If m is positive and less than 1, then $\frac{a^m b^m}{2} < \left(\frac{a+b}{2}\right)^m$

If there are n positive quantities a, b, c ... k, then $\left(\frac{a^m + b^m + c^m + \ldots + k^m}{n}\right) > \left(\frac{a+b+c+\ldots+k}{n}\right)^m$ unless m is a positive proper fraction.

9. If a, b, c are positive and not all equal, then

$(a+b+c)(ab+bc+ca) > 9abc$

and, $(b+c)(c+a)(a+b) > 8abc$.

10. If x is positive and $a < b$, then $\frac{a+x}{b+x} > \frac{a}{b}$

If x is positive and $a > b$, then $\frac{a+x}{b+x} < \frac{a}{b}$

11. If x is positive and $a > b > x$, then $\frac{a-x}{b-x} > \frac{a}{b}$

If x is positive and $x < a < b$, then $\frac{a-x}{b-x} < \frac{a}{b}$

12. $\frac{a+c+e+\ldots}{b+d+f+\ldots}$ is less than greatest and greater than least of the fractions $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}, \ldots$

13. For positive numbers, if $a > x$, $b > y$, $c > z$, then $a + b + c + \ldots > x + y + z + \ldots$ and $abc \ldots > xyz \ldots$

14. $a^2 + b^2 + c^2 \geq bc + ca + ab$

15. $(n!)^2 > n^n$, for $n > 2$.

16. For any positive integer n, $2 \leq \left(1 + \frac{1}{n}\right)^n \leq 3$

17. $a^2b + b^2c + c^2a \geq 3abc$

18. $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$

19. $a^4 + b^4 + c^4 + d^4 \geq 4abcd$

QUADRATIC EQUATION WITH APPLICATION TO INEQUALITIES

In a quadratic equation $ax^2 + bx + c = 0$, if $b^2 - 4ac > 0$, then

$ax^2 + bx + c = a(x-\alpha)(x-\beta)$, where α and β are two roots of the equation with $\alpha < \beta$.

$(x-\alpha)(x-\beta)$ is always positive when $x < \alpha$ or $x > \beta$ i.e., x does not lie between α and β.

$(x-\alpha)(x-\beta)$ is always negative when $\alpha < x$ or $x > \beta$ i.e., x lies between α and β.

EXERCISE – I

MCQ TYPE QUESTIONS

1. If $x = b + c$, $y = c - a$, $z = a - b$, then $x^2 + y^2 + z^2 - 2xy - 2xz + 2yz$ is equal to
(a) $a + b + c$ (b) $4b^2$
(c) abc (d) $a^2 + b^2$

2. HCF of $y^3 - 4y$ and $4y(y^3 + 8)$ is
(a) $y(y + 2)$ (b) $y - 2$
(c) $2(y + 2)$ (d) $(y + 2)(y^2 - 2y + 4)$

3. A tiled floor of a room has dimensions $m \times m$ sq.m. Dimensions of the tile used are $n \times n$ sq.m. All tiles used are green tiles except diagonal tiles which are red. After some years some green tiles are replaced by red tiles to form an alternate red and green tile pattern. How many green tiles are removed?
($m \neq n$ and total number of tiles are odd).
(a) $\frac{m^2 - 4mn + 2n^2}{2n^2}$ (b) $\frac{(m-2n)^2 - n^2}{2n^2}$
(c) $\frac{m^2 - 4mn - n^2}{2n^2}$ (d) $\frac{m^2 - 4mn - 2n^2}{2n^2}$

4. If $\frac{x-4}{-x^2-6} \leq \frac{2}{-x^2-6}$ then
(a) $x \leq 2$ (b) $x \geq 2$
(c) $x \leq 6$ (d) $x \geq 6$

5. Which of the following is true?
(a) $\sqrt{2} + \sqrt{5} = \sqrt{7}$ (b) $\sqrt{2} + \sqrt{5} \leq \sqrt{7}$
(c) $\sqrt{2} + \sqrt{5} < \sqrt{7}$ (d) $\sqrt{2} + \sqrt{5} > \sqrt{7}$

6. If $6 \geq x \geq -2$ and $4 \geq y \geq -4$, then limits for $\frac{y}{x}$, where x and y are non zero integers, is
(a) $\frac{y}{x} \geq 2, \frac{y}{x} \leq \frac{2}{3}$ (b) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq 2$
(c) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq \frac{1}{4}$ (d) $\frac{y}{x} \geq -4, \frac{y}{x} \leq 4$

7. What are the limits of $\frac{2^n(n+1)^n}{n^n}$; where n is a positive integer?
(a) $2 \leq \frac{2^n(n+1)^n}{n^n} \leq 3$
(b) $2^{n+1} \leq \frac{2^n(n+1)^n}{n^n} \leq 3.2^n$
(c) $2^n \leq \frac{2^n(n+1)^n}{n^n} \leq 3.2^n$
(d) None of these

8. If $g = 10^{100}$ and $H = 10^g$, then in which interval does $g! = 1.2.3.... 10^{100}$ lie?
(a) $10H < g! < H$ (b) $H < g! < 10^H$
(c) $10H < g! < 10^H$ (d) $10^H < g!$

9. Which of the following is true?
(a) Sum of four consecutive even numbers is always divisible by 8.
(b) Sum of four consecutive odd numbers is always divisible by 8.
(c) Product of any n consecutive natural numbers may not be divisible by n!.
(d) Product of 4 consecutive odd numbers is always divisible by 15.

10. If a and b are prime numbers, which of the following is true?
I. a^2 has three positive integer factors.
II. ab has four positive integer factors.
III. a^3 has four positive integer factors.
Codes :
(a) I and II only (b) II and III only
(c) All of these (d) None of these

11. If $7x + 6y = 420$, x and y are natural numbers, then what can be said about x?
(a) x is always odd
(b) x is always even.
(c) x is even only if y is odd.
(d) x is odd if y is even.

12. If a, b, c, d, p and q are non-zero, unequal integers and $\frac{a+bi}{c+di} = \frac{p}{q}$, then $\frac{a^2+b^2}{c^2+d^2}$ equals
(a) $\frac{p}{q}$ (b) $\frac{p^2}{q^2}$
(c) 1 (d) None of these

13. An Egyptian fraction has a numerator equal to 1, and its denominator is a positive integer. What is the maximum number of different Egyptian fraction such that their sum is equal to 1, and their denominators are equal to 10 or less?
(a) 3 (b) 5 (c) 7 (d) 9

14. What is the least number which must be subtracted from 1936 so that the remainder when divided by 9, 10, 15 will leave in each case the same remainder 7?
(a) 32 (b) 53
(c) 46 (d) 39

15. What is the greatest number consisting of six digits which on being divided by 6, 7, 8, 9, 10 leaves 4, 5, 6, 7, 8 as remainders respectively?
(a) 997920 (b) 997918
(c) 999999 (d) 997922

16. What is the least number which on being divided by 5, 6, 8, 9, 12 leaves in each case a remainder 1 but when divided by 13 leaves no remainder?
(a) 2987 (b) 3601
(c) 3600 (d) 2986

17. A group of four numbers has only one prime number amongst them. Which of the following must be true about the group?
I. HCF of the four numbers of the group is either 1 or equal to that prime number.
II. LCM of the four numbers of the group is same as product of the prime number and LCM of the remaining three numbers.
III. Product of four numbers is equal to product of the prime number × HCF of the group × LCM of the group.
(a) I only
(b) II only
(c) I and II only
(d) All of these

18. If (a, n)! is defined as product of n consecutive numbers starting from a, where a and n are both natural numbers, and if H is the HCF of (a, n)! and n!, then what can be said about H?
(a) $H = a!$ (b) $H = n!$
(c) $H \geq n!$ (d) $H \geq a \times n$

19. HCF and LCM of two numbers is given. It is possible to find out the two numbers uniquely if
I. either sum or difference between the two numbers is known.
II. HCF of two numbers = LCM of two numbers.
III. $\frac{LCM}{HCF}$ = Prime number.
(a) I and II only
(b) II only
(c) II and III only
(d) I, II and III

20. A ball is dropped from a height of 12 m and it rebounds 1/2 of the distance it falls. If it continues to fall and rebound in this way, how far will it travel before coming to rest?
(a) 36 m (b) 30 m
(c) 48 m (d) 60 m

21. There are N questions in an exam.
For i = 1, 2, ..., N, there are 2^{N-1} students who answered 1 or more questions wrongly. If total number of wrong answers is 8191, then N will be
(a) 12 (b) 11 (c) 10 (d) 13

22. Two new charity organizations C_1 and C_2 were formed, with x members each, on January 1, 2003. On first day of each, subsequent month, in C_1, number of members increases by a certain number a, while in C_2, number of members increases in such a way that ratio of the number of members in a month to the preceding month bear a ratio equal to b. On May 1, 2003, both organizations had the same number of members. If $a = 20x$, then b will be
(a) 2 (b) 3 (c) 2.5 (d) 3.5

23. What is the ratio of common differences d_1 and d_2 of two arithmetic progressions if respective n^{th} terms are in the ratio of $2n + 3 : n - 11$?
(a) 1 : 2 (b) 2 : 3
(c) 2 : 1 (d) 1 : 3

24. What is the sum upto 20 terms of the series $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + ...$?
(a) $\frac{19.3^{20}+1}{3^{20}}$ (b) $\frac{19.3^{20}+1}{2.3^{20}}$
(c) $\frac{19.3^{20}-1}{2.3^{20}}$ (d) $\frac{39.3^{20}+1}{2.3^{20}}$

25. What is the sum to infinity of the series, $3 + 6x^2 + 9x^4 + 12x^6 + ...$ given $|x| < 1$?
(a) $\frac{3}{(1+x^2)}$ (b) $\frac{3}{(1-x^2)^2}$
(c) $\frac{3}{(1+x^2)^2}$ (d) $\frac{3}{(1-x^2)}$

26. A school has 5 divisions in a class IX having 60, 50, 55, 62 and 58 students. Mean marks obtained in a History test were 56, 64, 72, 63 and 50 by each division respectively. What is overall average of the marks per students?
(a) 56.8 (b) 58.2
(c) 62.4 (d) 60.8

27. Average age of a committee of seven trustees is the same as it was five years ago, a younger man having been substituted for one of them. How much younger was he than trustee whose place he took?
(a) 32 years (b) 35 years
(c) 33 years (d) 34 years

28. Six men A, B, C, D, E, F agree with a seventh man G to provide a sum of money among them. A, B, C, D, E, F are to subscribe ₹10 each, and G is to pay ₹3 more than the average of the seven. What is the whole sum to be provided?
(a) ₹73.50 (b) ₹ 74
(c) ₹ 73 (d) ₹ 72.50

29. Which of the following is/are true?

I. When Arithmetic Mean of a set of values and number of values in the set are known, then Geometric Mean of the set can be found out.

II. When Arithmetic Mean of a set of values and number of values in the set are known, sum of all the elements in the set can be found out.

III. When Geometric Mean of a set of values and number of values in the set are known, then Arithmetic Mean of the set can be found out.

Codes :

(a) Only I (b) Only II

(c) Only I and II (d) All except II

30. Which of the following statement/s is/are true?

I. Average of a set of values will always lie between lowest and the largest of these values.

II. If each of the values in a set is increased by a constant k, then new average of the set is increased by k.

III. If each of the values in a set is multiplied by a constant m, then new average will also be 'm' times the old average.

Codes :

(a) Only I and II (b) Only II and III

(c) Only I and III (d) I, II and III

31. Three people A, B and C weigh themselves in a particular order. First A, B, C weigh themselves individually and then AB, BC CA and ABC together respectively. Recorded weight for the last measured is 180 kgs. Then average of the 7 measures is

(a) 320 kgs (b) $\frac{360}{7}$ kgs

(c) $\frac{720}{7}$ kgs (d) Cannot be determined

32. Average age of a team of 15 employees is 36. Youngest of them is 20 years old and eldest is 36 years old. Two of them with average age 28 leave the team. If one of the two comes back on the condition that he will be made the team leader, then which of the following can possibly be average age of the new team so formed?

(a) 35 (b) 36 (c) 38 (d) 39

NUMERICAL TYPE QUESTIONS

1. If $x - y = 1$, then $x^3 - y^3 - 3xy$ equals to _____

2. When $x + y + z = 9$ and $xy + yz + zx = 11$, then $x^3 - y^3 - z^3 - 3xyz$ equals to _____

3. If $x + y + z = 0$, then $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$ equals to _____

4. In an election for the President, if 261 valid votes are cast, for the 5 contestants then least number of votes a candidate requires to receive to win the election are is _____

5. A number when divided by sum of 555 and 445 gives two times their difference as quotient and 30 as remainder. The number is _____

6. Sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is _____

7. Sum of all odd numbers up to 100 is _____

8. A = 0. $a_1a_1a_1$ ... and B = 0.$a_2a_2a_2$, where a_1 and a_2 are multiples of 3 and also, a_1 and a_2 are distinct integers from 0 to 8. Then value of A + B is _____

9. Anil wants to divide ₹100 into a number of bags so that one can ask for any amount between ₹1 and ₹100, he can give the proper amount by giving certain number of these bags without taking out the amount from them. The minimum number of bags he will require if each bag has whole number of rupees is _____

10. The remainder when $2050 \times 2071 \times 2095$ is divided by 23 is _____

11. The smallest number which when increased by 3 is divisible by 27, 35, 25 and 21 is _____

12. _____is the least number which on being divided by 12, 21 and 35 will leave in each case the same remainder 6

13. A gardener had a number of shrubs to plant in horizontal rows. At first he tried to plant 5 shrubs in each row, then 6, then 8 and then 12, but had always 1 left. On trying 13, in one row he had none left. The smallest number of shrubs that he could have had is _____

14. Sum of an infinite geometric series is 3 and sum of an infinite geometric series formed from squares of the terms of the original series is 6. First term of the first series will be____

15. The maximum sum of the series 60, 58, 56, 54, 52, ... is_____

16. On an average my income for 15 days ₹ 7, average for the first 5 days was ₹ 6 and average for the last 9 days was ₹ 8. The income on the sixth day was _____

17. The harmonic mean of two numbers whose geometric mean and arithmetic mean is 8 and 5 respectively is _____

18. Average marks of 15 students in a class is 145, maximum marks being 150. If two lowest scores are removed, the average increases by 5. Also, two lowest scores are consecutive multiples of 9. The lowest score in the class is _____

EXERCISE – II
(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Read the following table giving sales data of five types of batteries for years 2006 to 2012

Year	Type I	Type I	Type I	Type I	Type I
2006	75	144	114	102	108
2007	90	126	102	84	126
2008	96	114	75	105	135
2009	105	90	150	90	75
2010	90	75	135	75	90
2011	105	60	165	45	120
2012	115	85	160	100	145

Out of the following, which type of battery achieved highest growth between the years 2006 and 2012?

(a) Type V (b) Type III

(c) Type II (d) Type I

2. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval

(a) $\left[1, \frac{2}{3}\right]$ (b) $\left[\frac{-1}{2}, 1\right]$

(c) $\left[-1, \frac{1}{2}\right]$ (d) [2,–4]

3. If the list of letters, P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. 2P, 2R, 2S, 2T, 2U

II. P–3, R–3, S–3, T–3, U–3

III. P^2, R^2, S^2, T^2, U^2

(a) I only (b) I and II

(c) II and III (d) I and III

4. The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q. No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

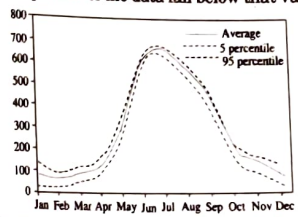
What is the average of the marks obtained by the class in the examination?

(a) 2.290 (b) 2.970

(c) 6.795 (d) 8.795

2014

5. The monthly rainfall chart on 50 years of rainfall in Agra is shown in the following figure. Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)



(i) On average, it rains more in July than in December

(ii) Every year, the amount of rainfall in August is more than that in January

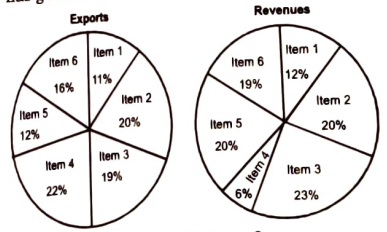
(iii) July rainfall can be estimated with better confidence than February rainfall

(iv) In August, there is at least 500 mm of rainfall

(a) (i) and (ii) (b) (i) and (iii)

(c) (ii) and (iii) (d) (iii) and (iv)

6. The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?



(a) Item 2 (b) Item 3

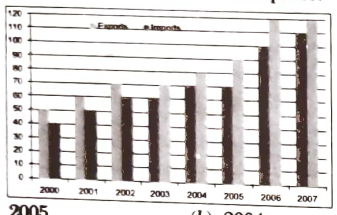
(c) Item 6 (d) Item 5

batsmen are provided in the following table. Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

(a) K (b) L
(c) M (d) N

8. The exports and imports (in crores of ₹) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit 1/5th of the exports?



(a) 2005 (b) 2004
(c) 2007 (d) 2006

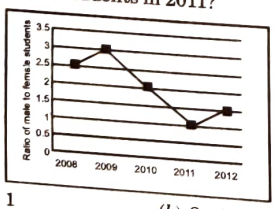
9. The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q.No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

(a) 1.34 (b) 1.74
(c) 3.02 (d) 3.91

10. The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?



(a) 1 : 1 (b) 2 : 1
(c) 1.5 : 1 (d) 2.5 : 1

2 to 198?

(a) 90 (b) 100
(c) 110 (d) 120

12. The value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$ is

(a) 3.464 (b) 3.932
(c) 4.000 (d) 4.444

13. If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

(a) 2, 4 (b) 2, 14
(c) 4, 52 (d) 14, 52

2013

14. A number is as much greater than 75 as it is smaller than 117. The number is

(a) 91 (b) 93
(c) 89 (d) 96

15. Following table provides figures (in rupees) on annual expenditure of a firm for two years - 2010 and 2011.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the following two categories have registered increase by same percentage?

(a) Raw material and Salary & wages
(b) Salary & wages and Advertising
(c) Power & fuel and Advertising
(d) Raw material and Research & Development

16. What will be the maximum sum of 44, 42, 40,... ?

(a) 502 (b) 504
(c) 506 (d) 500

17. The current erection cost of a structure is ₹ 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in ₹ is

(a) 16,500 (b) 15,180
(c) 11,000 (d) 10,120

18. Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots\ldots\ldots + \frac{1}{\sqrt{80}+\sqrt{81}}$$

(a) 7 (b) 8
(c) 9 (d) 10

19. In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in °C on Thursday was

(a) 40 (b) 43
(c) 46 (d) 49

20. The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is

(a) $(-\infty, 0)$ (b) $(0, 1)$
(c) $(1, \infty)$ (d) $(0, \infty)$

21. Find the sum to n terms of the series $10 + 84 + 734 + \ldots$

(a) $\frac{9(9^n+1)}{10}+1$ (b) $\frac{9(9^n-1)}{8}+1$

(c) $\frac{9(9^n-1)}{8}+n$ (d) $\frac{9(9^n-1)}{8}+n^2$

22. If $3 \le X \le 5$ and $8 \le Y \le 11$ then which of the following options is TRUE?

(a) $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{5}$

(b) $\frac{3}{11} \le \frac{X}{Y} \le \frac{5}{8}$

(c) $\frac{3}{11} \le \frac{X}{Y} \le \frac{8}{5}$

(d) $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{11}$

23. Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

(a) USA and Japan
(b) USA and Australia
(c) England and France
(d) Japan and Australia

2012

24. If $(1.001)^{1259} = 3.52$ and $(1.001)^{2052} = 7.85$, then $(1.001)^{793} =$

(a) 2.23 (b) 4.33
(c) 11.37 (d) 27.64

25. Raju has 14 currency notes in his pocket consisting of only ₹ 20 notes and ₹ 10 notes. The total money value of the notes is ₹ 230. The number of ₹ 10 notes that Raju has is

(a) 5 (b) 6
(c) 9 (d) 10

26. The data given in the following table summarizes the monthly budget of an average household.

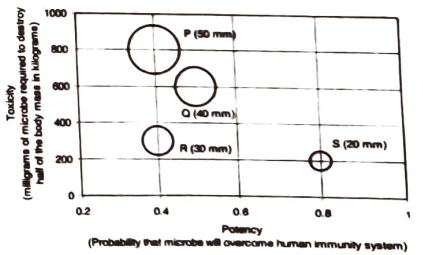
Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

The approximate percentage of the monthly budget **NOT** spent on savings is

(a) 10% (b) 14%
(c) 81% (d) 86%

2011

27. P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below :



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

(a) P (b) Q
(c) R (d) S

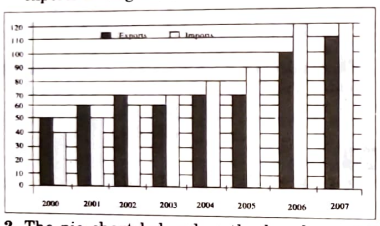
28. The sum of n terms of the series $4 + 44 + 444 + \ldots$ is
(a) $(4/81)[10^{n+1} - 9n - 1]$
(b) $(4/81)[10^{n-1} - 9n - 1]$
(c) $(4/81)[10^{n+1} - 9n - 10]$
(d) $(4/81)[10^{n} - 9n - 10]$

29. Given that $f(y) = |y|/y$, and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is
(a) 0 (b) −1
(c) 1 (d) 2

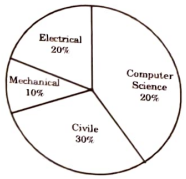
NUMERICAL TYPE QUESTIONS

2015

1. The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest ?



2. The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5 : 4. There are 40 males in Electrical Engineering. What is the difference between numbers of female students in the Civil department and the female students in the Mechanical department?



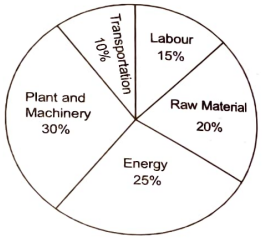
2014

3. A foundry has a fixed daily cost of ₹ 50,000 whenever it operates and variable cost of ₹ 800Q, where Q is the daily production in tonnes. What is the cost of production in ₹ per tonne for a daily production of 100 tonnes?

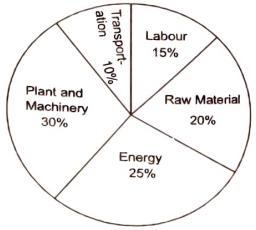
4. The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3 : 4 : 5 : 6. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

5. In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

6. A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is ₹ 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013?

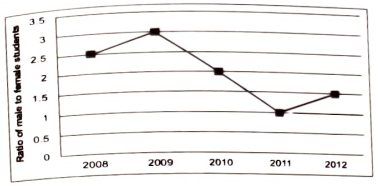


7. A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is ₹ 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. If the company registered a profit of ₹ 10 lakhs in 2012, at what price (in ₹) was each air purifier sold?



8. The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number?

9. The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?



10. If $(z + 1/z)^2 = 98$, compute $(z^3 + 1/z^3)$

11. In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

12. Round–trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is ₹ 100, a group of 5 tourists purchasing round–trip tickets will be charged ₹ _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (b)	2. (a)	3. (b)	4. (d)	5. (d)	6. (d)	7. (b)	8. (c)	9. (b)	10. (c)
11. (b)	12. (b)	13. (a)	14. (d)	15. (b)	16. (b)	17. (a)	18. (b)	19. (d)	20. (a)
21. (d)	22. (b)	23. (c)	24. (d)	25. (b)	26. (d)	27. (b)	28. (a)	29. (b)	30. (d)
31. (c)	32. (b)								

Numerical Type Questions

1. 1	2. 432	3. 3	4. 53	5. 220030	6. 67	7. 2500	8. 1	9. 7	10. 6
11. 4722	12. 426	13. 481	14. 12/5	15. 930	16. 3	17. 12.8	18. 108		

EXERCISE – II

MCQ Type Questions

1. (d)	2. (b)	3. (b)	4. (c)	5. (b)	6. (d)	7. (a)	8. (d)	9. (c)	10. (c)
11. (b)	12. (c)	13. (d)	14. (d)	15. (d)	16. (c)	17. (a)	18. (b)	19. (c)	20. (b)
21. (d)	22. (b)	23. (c)	24. (d)	25. (a)	26. (d)	27. (d)	28. (c)	29. (d)	

Numerical Type Questions

1. (2006)	2. (32)	3. (1300 to 1300)	4. (180 to 180)	5. (495)
6. (22)	7. (20,000)	8. (163)	9. (140 to 140)	10. (96)
11. (48)	12. (850)			

EXERCISE – I

MCQ TYPE QUESTIONS

1. $x^2 + y^2 + z^2 - 2xy - 2xz + 2yz = (x - y - z)^2$
$= (b + c - c + a - a + b)^2 = 4b^2$

2. $y^3 - 4y = y(y^2 - 4)$
$= y(y - 2)(y + 2)$
$4y(y^3 + 8) = 4y(y + 2)(y^2 - 2y + 4)$
$\therefore$ HCF $= y(y + 2)$.

3. Total red tiles initially $= 2\left(\frac{m}{n}\right) - 1$

Total number of tiles $= \frac{m^2}{n^2}$

As number of red tiles > number of green tiles (since room has odd number of tiles and corner tiles are red)

Hence total red tiles required to form an alternate pattern

$$= \frac{\left(\frac{m}{n}\right)^2 + 1}{2} = \frac{m^2 + n^2}{2n^2}$$

$\therefore$ Number of green tiles removed

$$= \frac{m^2 + n^2}{2n^2} - 2\left(\frac{m}{n}\right) + 1$$

$$= \frac{m^2 + n^2 - 4mn + 2n^2}{2n^2}$$

$$= \frac{(m - 2n)^2 - n^2}{2n^2}$$

4. $\frac{x - 4}{-x^2 - 6} \le \frac{2}{-x^2 - 6}$

i.e. $\frac{x - 4}{-(x^2 + 6)} \le \frac{2}{-(x^2 + 6)}$

i.e. $x - 4 \ge 2$ ...(cancelling a negative term reverses sign of the inequality)

i.e. $x \ge 6$.

5. $(\sqrt{2} + \sqrt{5})^2 = 2 + 2\sqrt{10} + 5$

$= 7 + 2\sqrt{10}$

$\therefore$ $\sqrt{2} + \sqrt{5} > \sqrt{7}$

7. $\frac{2^n(n + 1)^n}{n^n} = 2^n\left(\frac{n + 1}{n}\right)^n = 2^n\left(1 + \frac{1}{n}\right)^n$

Now, $2 \le \left(1 + \frac{1}{n}\right)^n \le 3$

$\therefore$ $2.2^n \le 2^n\left(1 + \frac{1}{n}\right)^n \le 3.2^n$

i.e., $2^{n+1} \le 2^n\left(1 + \frac{1}{n}\right)^n \le 3.2^n$

8. Choose k, so that

$10^H = g^{\lambda} = 10^{100k}$,

$k = \frac{H}{100}$

$\therefore$ $10^H = g^{\frac{H}{100}} > g^g > g!$

On the other hand,

$10H = 10 \times 10^g = 10^{g+1} < g!$

(most factors are much bigger than 10)

$10H < g! < 10^H$

9. (*a*) Four consecutive even numbers can be written as 2n, 2n + 2, 2n + 4 and 2n + 6, where n is any natural number.

$\therefore$ Sum $= 2n + 2n + 2 + 2n + 4 + 2n + 6$
$= 8n + 12$
$= 4(2n + 3)$ not always divisible by 8.

Thus, (*a*) is not true.

(*b*) Four consecutive odd numbers can be written as 2n + 1, 2n + 3, 2n + 5, 2n +7

where n is a natural number.

Sum $= 2n + 1 + 2n + 3 + 2n + 5 + 2n + 7 = 8n + 16$

divisible by 8

Thus, (*b*) is true

(*c*) In product of n consecutive natural numbers atleast one is divisible by n, atleast one by n – 1 ... till 1.

Thus product is atleast divisible by

$n \times (n - 1) \times (n - 2) \ldots . 1 = n!$.

Thus, (*c*) is not true.

(*d*) Take four consecutive odd numbers as 7 × 9 × 11 × 13 which is not divisible by 15.

Thus, (*d*) is not true.

10. Factors of a^2 are 1, a and a^2.
Factors of ab are 1, a, b and ab.
Factors of a^3 are 1, a, a^2 and a^3.

11. $7x + 6y = 420$

Equation is of the form:

7x + even number = even number.

$\therefore$ 7x has to be even

Hence x has to be even.

12. $\frac{a + bi}{c + di} = \frac{p}{q}$

$\therefore$ $qa + qbi = pc + pdi$

Equating real and imaginary parts.

$qa = pc$ and $qb = pd$.

$\Rightarrow$ $a = \frac{p}{q}c$ and $b = \frac{p}{q}d$

$\therefore$ $\frac{a^2 + b^2}{c^2 + d^2} = \frac{\frac{p^2}{q^2}c^2 + \frac{p^2}{q^2}d^2}{c^2 + d^2} = \frac{p^2}{q^2}$

13. We ignore $\frac{1}{7}$ and $\frac{1}{9}$ because no sum of other numbers is going to give 7th or 9th in the denominator.

Also, $\frac{1}{5}$ and $\frac{1}{10}$ are not enough to add up to anything $\left(\frac{1}{10}, \frac{2}{10}, \text{and } \frac{3}{10}\right.$ are going to leave tenths left over no matter what else you add$\left.\right)$.

What's left is $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}$.

Sum total of these is $\frac{11}{8}$. So we need all of them except $\frac{1}{8}$, which means $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$. Which is the only way to do this with egyptian fractions whose denominators are 10 or less. Hence maximum number of Egyptian fractions needed is 3.

14. L.C.M. of 9, 10 and 15 = 90

$\therefore$ $1936 = 90 \times 21 + 46$ (Remainder).

But a part of this remainder is 7.

so $46 - 7 = 39$ should be subtracted from 1936.

15. $6 - 4 = 2, 7 - 5 = 2, 8 - 6$
$= 2, 9 - 7 = 2, 10 - 8 = 2$

L.C.M. of 6, 7, 8, 9, 10 = 2520;

Greatest number of 6 digits = 999999

$2520 \times 396 + 2079 = 999999$

$\therefore$ Reminder = 2079.

Subtract 2079 from 999999, then we get

$999999 - 2079 = 997920$.

Subtract 2 from this number to get required number, which is 997918 and which will give the remainders 4, 5, 6, 7, 8 when divided by 5, 6, 7, 8, 9 respectively.

16. L.C.M. of 5, 6, 8, 9 and 12 is 360

Required number = 360 K + 1
$= (13 \times 27 + 9)$ K + 1
$= (13 \times 27)$ K + (9K + 1)

Now this number must be divisible by 13

$\therefore$ K = 10 and required number
= 3601.

17. I. In case all the remaining three numbers are multiples of the prime number, HCF of the group is equal to that prime number. otherwise HCF of the group is 1. Thus, I is true.

II. **Case 1:** Let numbers be 2, 4, 6 and 8, where 2 is only prime number, then LCM = 24.

But LCM of 4, 6 and 8 = 24.

LCM = 24 & 2 = 24

Case 2 : Let numbers be 2, 9, 81 and 27 Then LCM = 162 and LCM of 9, 81 and 27 = 81.

Thus, LCM = 81 × 2 = 162

Hence II is not always true.

III. Let numbers be 2, 4, 6 and 8

$\therefore$ LCM = 24 and HCF = 2

Product of four numbers
$= 2 \times 4 \times 6 \times 8 = 384$

Also, $384 \ne 24 \times 2 \times 2$.

Thus, III is not true.

18. (a. n)! = product of n consecutive natural number starting from 'a' which is atleast divisible by n!.
(n)! = product of n consecutive natural numbers.

For n = 2 :

(a. n)! = a(a + 1) and n! = 2

a(a + 1) is divisible by 2!.

For n = 3 :

(a n)! = a(a + 1)(a + 2) and n! = 6.

One of the factors of a(a + 1)(a + 2) is divisible by 3 and other by 2.

Thus, proceeding in this manner, (a. n)! and n! have HCF = n!

$\therefore$ H = n!.

19. Let HCF be h and LCM be *l*.

I. Let numbers be ah and bh.

Then abh = l and (a + b)h = m

$\Rightarrow$ (a – b)h = n

Using these ah and bh can be uniquely determined. Thus, I is true

II. If HCF = LCM, then two numbers are equal and same as HCF or LCM.

Thus, II is true.

III. $\frac{LCM}{HCF}$ = a prime i.e., $\frac{\ell}{h} = P$.

Then one of the numbers is equal to h and other is equal to ℓ. Thus, III is true.

20. Total distance travelled by the ball before coming to rest

$$= 12 + 2\,(6 + 3 + 1.5 + \ldots \text{ to } ?)$$

$$= 12 + 2.\left(\frac{6}{1-1/2}\right)$$

$$= 12 + 2(12) = 36\text{m}$$

21. Number of students who answered 1 or more questions wrongly = 2^{N-1}.

Number of students who answered 2 or more questions wrongly = 2^{N-2}

Hence number of students who answered 1 question wrongly

$$= 2^{N-1} - 2^{N-2} = 2^{N-2}$$

Similarly it can be shown that number of students who answered 2 questions wrongly

$$= 2^{N-2} - 2^{N-3} = 2^{N-3}$$

Similarly we can find number of students who answered K questions wrongly where $K \geq 3$

Hence total number of questions attempted wrongly

$$s = 2^{N-2} + 2(2^{N-3}) + 3(2^{N-4}) + \ldots + (N-1)(2^0) + N(1) \ldots (i)$$

$$\therefore \quad \frac{s}{2} = 2^{N-3} + 2(2^{N-4}) + \ldots + (N-2)(2^0) + \frac{N-1}{2} + \frac{N}{2} \ldots (ii)$$

Substracting equations (i) from (ii)

$$\frac{s}{2} = 2^{N-2} + 2^{N-3} + \ldots + 2^0 + \frac{1}{2}$$

$$\Rightarrow \quad s = 2^{N-1} + 2^{N-2} + 2^{N-3} + \ldots + 1$$

$$= 2^N - 1 = 8191$$

$$\Rightarrow \quad N = 13$$

22. Number of members in C_1 on May 1, 2003 = $x + 4a$

Number of members in C_2 on May 1, 2003 = xb^4

$$x + 4a = xb^4 \text{ and } a = 20x$$

$$\therefore \quad x(b^4 - 81) = 0$$

As $x \neq 0$, $b^4 - 81 = 0$

$\therefore$ $b = 3$

23. n^{th} terms are in the ratio of $\frac{2n+3}{n-11}$.

If $n = 1$ in $2n + 3$ series, then first term is

$T_1 = 2(1) + 3 = 5$

$n = 2$, then second term is $2(2) + 3 = 7$

$T_2 = 7$

$d = T_2 - T_1 = 7 - 5 = 2$

$\therefore$ $d_1 = 2$

If $n = 1$ in $n - 11$ series, then first term is $1 - 11 = -10$

If $n = 2$, then second term is $2 - 11 = -9$

$\therefore$ $d_2 = T_2 - T_1 = -9 - (-10) = 1$

$\therefore$ $\frac{d_1}{d_2} = \frac{2}{1}$ or 2 : 1

24. $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \ldots$ up to 20 terms

$$= \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{9}\right) + \left(1 - \frac{1}{27}\right) + \left(1 - \frac{1}{81}\right) + \ldots \text{ up to 20 terms}$$

$$= 20 - \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \ldots + \frac{1}{3^{20}}\right)$$

$$= 20 - \frac{1}{3}\left(\frac{1 - \left(\frac{1}{3}\right)^{20}}{1 - \frac{1}{3}}\right)$$

$$= 20 - \frac{3^{20} - 1}{2.3^{20}}$$

$$= \frac{40.3^{20} - 3^{20} + 1}{2.3^{20}} = \frac{39.3^{20} + 1}{2.3^{20}}$$

25. Let $s = 3 + 6x^2 + 9x^4 + 12x^6 + \ldots$

$\therefore$ $x^2 s = 3x^2 + 6x^4 + 9x^6 + 12x^8 + \ldots$

$\Rightarrow$ $s - x^2 s = 3 + 3x^2 + 3x^4 + 3x^6 + \ldots$

$\Rightarrow$ $s(1 - x^2) = 3\,[1 + x^2 + x^4 + x^6 + \ldots]$

$\Rightarrow$ $s(1 - x^2) = 3\left[\frac{1}{1 - x^2}\right] \left[\because |x| < 1\right]$

$\therefore$ $s = \frac{3}{(1 - x^2)^2}$

26. Average

$$= \frac{60\times56 + 50\times64 + 55\times72 + 62\times63 + 58\times50}{60 + 50 + 55 + 62 + 58}$$

$$= \frac{3360 + 3200 + 3960 + 3906 + 2900}{285} = 60.8$$

27. Increase after 5 years should have been

$7 \times 5 = 35$ years.

But since average remains same, difference in their age is 35 years.

28. Let average of the seven men be x.

$$\therefore \quad x = \frac{10 \times 6 + (x + 3)}{7}$$

$$\Rightarrow \quad 60 + x + 3 = 7x$$

$$\Rightarrow \quad x = \frac{63}{6} = 10.5$$

$\therefore$ Total amount = $10.5 \times 7 = 73.5$

29. Consider a set of three numbers a, b and c.

I. $AM = \frac{a + b + c}{3}$

Thus, $a + b + c = 3AM$.

But $GM = \sqrt[3]{abc}$ which cannot be found out.

Thus, I is false,

II. $AM = \frac{a + b + c}{3}$

$\therefore$ Sum of the numbers = $a + b + c = 3AM$.

Thus, II is true.

III. $GM = \sqrt[3]{abc}$.

But $AM = \frac{a + b + c}{3}$, which cannot be found out.

Thus, III is false.

30. I. Average of a set of numbers is greater than smallest and smaller than the greatest number of the set. Thus, I is true.

II. Consider five numbers a, b, c, d and e whose average is $\frac{a + b + c + d + e}{5}$

Now if each of them is increased by k, then we have average

$$= \frac{a + k + b + k + c + k + d + k + e + k}{5}$$

$$= \frac{a + b + c + d + e}{5} + \frac{5k}{5} = \text{old average} + k.$$

Thus, II is true.

III. Consider three numbers x, y and z whose average is $\frac{x + y + z}{3}$

Now if each of them is multiplied by m then new average = $\frac{xm + ym + zm}{3}$

$$= m\left(\frac{x + y + z}{3}\right) = m \times \text{odd average}.$$

Thus, III is true.

31. Order of measures is

A, B, C, A + B, B + C, C + A, A + B + C.

Given: $A + B + C = 180$

Hence, average of the 7 measures

$$= \left[\frac{(A) + (B) + (C) + (A+B) + (B-C) + (C+A) + (A-B-C)}{7}\right]$$

$$= \frac{4}{7}(A + B + C) = \frac{4}{7} \times 180 = \frac{720}{7} \text{ kgs.}$$

32. Total age = 15×36

After two left, to tal age = $15 \times 36 - 2 \times 28$

$= 540 - 56$

$= 484$ years.

Let the age of the person who returns be x years.

Then new average = $\frac{484 + x}{14}$

Now x lies between 20 and 36 (both inclusive)

$\therefore$ New average min = $\frac{484 + 20}{14} = 36$ years

And New average max = $\frac{484 + 36}{14}$

$= 37.14$ years.

NUMERICAL TYPE QUESTIONS

1. $x^3 - y^3 - 3xy$

$= x^3 - y^3 - 3xy(x - y)$ ..... as $(x - y) = 1$

$= (x - y)^3 = 1$.

2. $x^3 - y^3 - z^3 - 3xyz$

$= (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$= 9[(x + y + z)^2 - 3(xy + yz + zx)]$

$= 9(81 - 33) = 432$.

3. $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy}$

... (since, $x + y + z = 0$; $x^3 + y^3 + z^3 = 3xyz$)

$\frac{x^3 + y^3 + z^3}{xyz} = \frac{3xyz}{xyz} = 3$.

4. Worst scenario is when other four get equal number of votes.

Let the winning candidate get x votes.

$\therefore$ $x > \frac{261}{5}$

$\Rightarrow$ $x > 52.2$

$\therefore$ $x = 53$

5. Number $= (555+445) \times (555-445) \times 2 + 30$
$= (555+445) \times 2 \times 110 + 30$
$= 220000 + 30 = 220030.$

6. $x + (x + 36) + y = 100$
$\Rightarrow \quad 2x + y = 64$
Therefore y must be even prime which is 2
$2x + 2 = 64 \quad \Rightarrow \quad x = 31.$
$\therefore$ Third prime number
$= x + 36 = 31 + 36 = 67.$

7. Given numbers are 1, 3, 5.....99.
This is an A.P with a = 1, d = 2.
Let it contain n terms
$\therefore \quad 1 + (n-1)2 = 99 \quad \Rightarrow \quad n = 50$
Hence required sum $= \frac{n}{2}$ (first term + last term)
$= \frac{50}{2}(1 + 99) = 2500.$

8. $A = 0.a_1a_1a_1 \cdots$
$\therefore \quad 10A = a_1.a_1a_1a_1 \cdots$
$\Rightarrow \quad 9A = a_1$
$\Rightarrow \quad A = \frac{a_1}{9}$
Similarly, $B = \frac{a_2}{9}.$
Now, a_1 and a_2 are multiples of 3 and are distinct.
Also, these values are less than 8.
either $a_1 = 3$ and $a_2 = 6$ or $a_1 = 6$ and $a_2 = 3$
$\therefore \quad a_1 + a_2 = 9.$
$\therefore \quad A + B = \frac{a_1}{9} + \frac{a_2}{9} = \frac{a_1 + a_2}{9} = \frac{9}{9} = 1.$

9. If Anil has to give 1 rupee he needs a bag with ₹1. For 2 rupees he had two bags with ₹1 each or ₹2. bag. To have minimum bags, he has a bag with ₹ 2. Now with the two bags he can give ₹3. So next he will require a bag with ₹4. With these three he can give ₹5. ₹6 and ₹7 and next bag will be one containing ₹8 and so on. Thus he would have bags with ₹1. 2. 4. 8. 16. 32. Sum of which is 63 and remaining 37 can be put in the last bag. So total number of bags is 7.

10. $2050 \times 2071 \times 2095$
$(23 \times 89 + 3) \times (23 \times 90 + 1) \times (23 \times 91 + 2)$
$= (23[89 \times 90 + 89 + 90 \times 3] + 3) \times (23 \times 91 + 2)$
$= (23 \times K + 3) \times (23 \times 91 + 2)$
$= (23 \times [K \times 91 + K \times 2 + 91 \times 3] + 6)$
$= 23 \times L + 6$ (where K and L are constants).
Hence remainder is 6.

11. L.C.M. of 27, 35, 25 and 21 = 4725
$\therefore$ Number = 4725 – 3 = 4722.

12. L.C.M. of 12, 21 and 35 = 420
$\therefore$ Number required = 420 + 6 = 426.

13. Number is $120K + 1$
$= ((13 \times 9 + 3)K + 1)$
$= (13 \times 9)K + 3K + 1,$
which is divisible by 13.
3K + 1 is divisible by 13.
$\therefore \quad K = 4.$ Number = 481.

14. $\frac{a}{1-r} = 3$
$\Rightarrow \quad 3 - 3r = a$
Also, $\frac{a^2}{1-r^2} = 6$
$\Rightarrow \quad \frac{a}{1-r} \cdot \frac{a}{1+r} = 6$
$\Rightarrow \quad \frac{a}{1+r} = 2$
$\Rightarrow \quad 2 + 2r = a$
$\therefore \quad 3 - 3r = 2 + 2r$
$\Rightarrow \quad r = \frac{1}{5}$
$\therefore \quad a = 3 - \frac{3}{5} = \frac{12}{5}$

15. Since d = –2, maximum sum of the series will be sum of all the positive terms.
Least positive term of the series is 2 or 0 (both giving equal sum)
Let, 2 be the n^{th} term
$2 = 60 + (n-1)(-2)$
$\Rightarrow \quad n = 30$
$\therefore$ Required sum = 30/2 (60 + 2) = 930

16. Total income for 15 days = 15 × 7 = ₹105
Total income for first 5 days = 5 × 6 = ₹30
Total income for last 9 days = 9 × 8 = ₹72
$\therefore$ Income on sixth day
$= 105 - (30 + 72)$
$= 105 - 102 = ₹3$

17. $(GM)^2 = HM \times AM$
$HM = \frac{8 \times 8}{5} = \frac{64}{5} = 12.8$

18. Total marks of 15 students = 15 × 145 = 2175.
Average marks of 15 students (excluding two lowest scores) = 13 × (145 + 5) = 1950.
$\therefore$ Total of two lowest scores = 2175 – 1950 = 225
Given that two scores are consecutive multiples of 9
$\therefore \quad 9x + 9x + 9 = 225$
$\Rightarrow \quad 18x = 216$
$\Rightarrow \quad x = \frac{216}{18}$
$\therefore$ Lowest score $= 9x = \frac{216}{18} \times 9 = 108.$

EXERCISE – II

MCQ TYPE QUESTIONS

1. Type-I achieved a growth of 53% in the period which is higher than any other type of battery

4. Total number of students in class = 44
Average of marks,
$= \frac{(2\times21) + (3\times15) + (1\times11) + (2\times23) + (5\times31)}{42}$
$= \frac{42 + 45 + 11 + 46 + 155}{44} = \frac{299}{44} = 6.795$
$\therefore$ Option (c) is correct.

5. In the question the monthly average rainfall chart for 50 years has been given.
Let us check the options.
(i) On average, it rains more in July than in December $\Rightarrow$ correct.
(ii) Every year, the amount of rainfall in August is more than that in January.
$\Rightarrow$ may not be correct because average rainfall is given in the question.
(iii) July rainfall can be estimated with better confidence than February rainfall.
$\Rightarrow$ From chart it is clear the gap between 5 percentile and 95 percentile from average is higher in February than that in July $\Rightarrow$ correct.
(iv) In August at least 500 mm rainfall
$\Rightarrow$ May not be correct, because its 50 year average.
So correct option (b) (i) and (iii).

6. Item : 2
$\frac{\frac{20}{100} \times 250 \times 10^7}{\frac{20}{100} \times 500 \times 10^3}$
$0.5 \times 10^4 = 5 \times 10^3 \boxed{1} =$ Item 2
Item : 3
$\frac{23 \times 250 \times 10^7}{19 \times 500 \times 10^3}$
1.2 = Item 3
Item : 6
$\frac{19}{16} = 1.18 =$ Item 6
Item : 5
$\frac{20}{12} = \frac{5}{3} = 1.6 \Rightarrow \boxed{1.6 = \text{Item } 5}$

8. For 2005,
trade defict = (90 – 70) crores = 20 crores
Now $\frac{1}{5}^{th}$ of export $= \frac{1}{5}(70)$ crores
$= 14$ crores $\neq 20$ crores.
Hence option (a) is wrong.
For 2004,
Trade defict = (80 – 70) crores = 10 crores
Now $\frac{1}{5}^{th}$ of export $= \frac{1}{5}(70)$ crores
$= 14$ crores $\neq 20$ crores.
Hence option (b) is wrong.
For 2007
Trade defict = (120 – 110) crores
= 10 crores
Now $\frac{1}{5}^{th}$ of export $= \frac{1}{5}(110)$ crores
$= 22$ crores $\neq 10$ crores.
Hence option (c) is wrong.
For 2006,
Trade defict = (120 – 100) crores
= 20 crores
Now $\frac{1}{5}^{th}$ of export $= \frac{1}{5}(100)$ crores
= 20 crores = Trade deficts
Hence option (d) is correct.

9. Total question
44 × 2 = 88
44 × 3 = 132
$\frac{144}{132} = \frac{88}{308}$
Total marks obtained = (21×2) + (15×3) + (23×2)
= 133
Total Number of students = 44
Average $= \frac{133}{44} = 3.02$

10. Take number of female students in 2011 = 100
$\therefore$ Number of male in 2011 = 100
No. of female in 2012 = 100
No. of male in 2012 = 150
Ratio $= \frac{150}{100}$

11. All multiples of 10 from 2 to 198

$= 10, 20, \ldots\ldots\ldots 190$

Now $a_n = a + (n-1)d$

$190 = 10 + (n-1) \times 10$

$180 = (n-1) \times 10$

$18 = n - 1$

$\therefore \quad n = 19$

Hence total number of multiples are 19.

Now $S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{19}{2}[20 + (18)(10)]$

$= \frac{19}{2}[20 + 180] = 1900$

Now average of all multiples of 10 from 2 to198

$= \frac{10 + 20 + \ldots + 190}{19}$

$\therefore \quad = \frac{1900}{19} = 100$

12. $\sqrt{12 + \sqrt{12 + \sqrt{12 + \ldots}}}$

For a particular type of question

i.e. $\sqrt{a + \sqrt{a + \sqrt{a + \ldots}}}$ we find a two consecutive number whose product is a. The greater number among the two consecutive number is the answer.

Here $a = 12$

i.e. $12 = 3 \times 4$

↓

greater number

Hence $\sqrt{12 + \sqrt{12 + \sqrt{12 + \ldots}}} = 4$

13. Here it is given that x is real

Now $|x^2 - 2x + 3| = 11$

$\Rightarrow \quad x^2 - 2x + 3 = 11$ or $x^2 - 2x + 3 = -11$

For $x^2 - 2x + 3 = 11 \Rightarrow x^2 - 2x - 8 = 0$

$b^2 - 4ac = 4 - 4(1)(-8) = 36 > 0$

Hence roots are real.

and for $x^2 - 2x + 14 = 0$

$b^2 - 4ac = 4 - 4(1)(14) < 0$

Hence roots are imaginary

$\therefore$ The value of x is the root of $x^2 - 2x - 8 = 0$

Now $x = \frac{+2 \pm \sqrt{4 - 4(1)(-8)}}{2}$

$= \frac{2 \pm \sqrt{36}}{2} = \frac{2 \pm 6}{2}$

$\therefore \quad x = 4, -2.$

Now for $x = 4$,

$|-x^3 + x^2 - x| = |-64 + 16 - 4| = 52.$

and for $x = -2$

$|-x^3 + x^2 - x| = |+8 + 4 + 2| = 14$

Hence possible values of $|-x^3 + x^2 - x| = 14, 52$

16. $0 = 44 + (n-1)(-2)$

$n - 1 = 22$

$n = 23$

Sum $= \frac{n}{2}(a + l) = \frac{23}{2}(44 + 0)$

$= 23 \times 22 = 506$

Hence (c) is the answer

17. Let 'd' be the # of days required

Let 'h' be the daily working hours.

$\therefore dh$ can be taken as a measure of the total work be done in order to erect the structure.

Now working hours per day $h' = \frac{23}{24}h$

$\therefore$ # of days $d' = \frac{dh}{h'} = \frac{dh}{\left(\frac{23}{24}h\right)} = \frac{24}{23}d$

Let c = daily wages earlier

$\therefore \quad c = \frac{13,200}{d}$

Now daily wages $c' = \frac{6}{5}c$

$\therefore$ Total new cost $= c' \times d'$

$= \frac{6}{5} \times \frac{13200}{d} \times \frac{24}{23}d \approx 16,500$

18. Given: The expression

$S = \frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \ldots\ldots + \frac{1}{\sqrt{80} + \sqrt{81}}$

To find: Sum of the expression.

Analysis: Each term of the expression is having irrational denominator. One can also see, that both the elements in the denominator of a term differ by only 1.

This gives an idea that on rationalising the denominator, each denominator will become 1, so the sum will be easier than earlier.

Rationalising the denominator,

$S = \frac{\sqrt{2} - \sqrt{1}}{(\sqrt{2})^2 - (\sqrt{1})^2} + \frac{\sqrt{3} - \sqrt{2}}{(\sqrt{3})^2 - (\sqrt{2})^2} + \ldots + \frac{\sqrt{81} - \sqrt{80}}{(\sqrt{81})^2 - (\sqrt{80})^2}$

$= \frac{\sqrt{2} - \sqrt{1}}{1} + \frac{\sqrt{3} - \sqrt{2}}{3 - 2} + \ldots + \frac{\sqrt{81} - \sqrt{80}}{81 - 80}$

$= \sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \ldots + 81 - \sqrt{80}$

Now, one can see that the terms will cancel out.

Each term except $\sqrt{1}$ & $\sqrt{81}$

Occurs twice, once as positive and once a negative.

So, after cancelling out, we are left with the following:

$S = \sqrt{81} - \sqrt{1} = 9 - 1 = 8$

So, the answer is (b).

19. $\frac{\text{Mon} + \text{Tues} + \text{Wed.}}{3} = 41$

Mon + Tues + Wed. = 123 ...(1)

$\frac{\text{Tues} + \text{Wed} + \text{Thurs.}}{3} = 43°$

Tue + Wed + Thu. = 129° ...(2)

(2) − (1)

Tues + Wed + Thu − (Mon + Tues. + Wed.)

$= 129 - 123 = 6°$

Thu. − Mon. $= 6° \Rightarrow \frac{115x}{100} - x = 6°$

Thus. = Mon $\times \frac{115}{100} = \frac{15x}{100} = 6°$

Mon $= x \quad x = 40°$

Thurs $= \frac{115x}{100}$

$\therefore \quad$ Thurs $= \frac{115 \times 40}{100} = 46°$

24. Let $1.001 = x$

$x^{1259} = 3.52$, and $x^{2062} = 7.85$

$\therefore \quad x^{3321} = x^{1259} \cdot x^{2062}$

$= 3.52 \times 7.85$

$= 27.64$

25. Let number of ₹ 20 notes be x and ₹ 10 notes be y

$\therefore \quad 20x + 10y = 230$ ...(1)

and $x + y = 14$ ...(2)

Solving equations (1) and (2), we have

$x = 9$ and $y = 5$

Hence numbers of 10 rupee notes are 5.

26. Total budget = 10,500

Expenditure other than savings = 9000

Hence approximate percentage of monthly budget

$= \frac{9000}{10500} = 86\%$

27. By observation of the table, we can say S

	P	Q	R	S
Requirement	800	600	300	200
Potency	0.4	0.5	0.4	0.8

28. Let $S = 4(1 + 11 + 111 + \ldots)$

$= \frac{4}{9}(9 + 99 + 999 + \ldots)$

$= \frac{4}{9}\{(10 - 1) + (10^2 - 1) + (10^3 - 1) + \ldots\ldots\ldots\}$

$= \frac{4}{9}\{(10 + 10^2 + \ldots + 10^n) - n\}$

$= \frac{4}{9}\left\{10\frac{(10^n - 1)}{9} - n\right\}$

$= \frac{4}{81}\{10^{n+1} - 9n - 10\}$

29. Given : $f(y) = \frac{|y|}{y}$

$\Rightarrow \quad f(q) = \frac{|q|}{q}$

$f(-q) = \frac{|-q|}{-q} = \frac{-|q|}{q}$

$|f(q) - f(q)| = \frac{|q|}{q} + \frac{|q|}{q} = \frac{2|q|}{q} = 2$

NUMERICAL TYPE QUESTIONS

1. Increase in exports in 2006 $= \frac{100 - 70}{70} = 42.8\%$

Increase in imports in 2006 $= \frac{120 - 90}{90} = 33.3\%$

which is more than any other year.

2. According to question,

There are 40 males in Electrical Engg.

$\therefore$ Total number of females in Electrical Engg.

$= 72 - 40 = 32$

$\therefore$ Total number of students in Electrical Engg. = 72

$\Rightarrow 20\% = 72$

$\therefore 30\%$ = Total number of students in Civil Engg.

$\therefore$ Total number of students in Civil Engg.

$= \frac{30 \times 72}{20} = 36 \times 3 = 108$

$\therefore$ Number of female students in Civil Engg.

$$= \frac{4 \times 108}{9} = 4 \times 12 = 48$$

Now, total number of students in Mechanical Engg.

$$= \frac{10 \times 72}{20} = 36$$

$\therefore$ Number of female students in Mechanical Engg.

$$= \frac{4 \times 36}{9} = 16$$

$\therefore$ Difference between female students in Civil and Mechanical departments = 48 – 16 = 32

3. Fixed cost = ₹ 50,000

Variable cost = ₹ 800Q

Q = daily production in tones

For 100 tonnes of production daily, total cost of production = 50,000 + 800 × 100 = 130,000

So, cost of production per tonne of daily production

$$= \frac{130,000}{100} = ₹\ 1300.$$

4. Let the angles of quadrilateral are $3x$, $4x$, $5x$, $6x$

So, $3x + 4x + 5x + 6x = 360$

$$x = 20$$

Smallest angle of quadrilateral = 3 × 20 = 60°

Smallest angle of triangle = $\frac{2}{3} \times 60° = 40°$

Largest angle of triangle = 2 × 40° = 60°

Three angles of triangle are 40°, 60°, 80°

Largest angle of quadrilateral is 120°

Sum (2nd largest angle of triangle + largest angle of quadrilateral) = 60° + 120° =180°.

5. 8th observation is 7 × 2 = 14 more than 1st observation

9th observation is 14 more than 2nd observation

10th observation is 14 more than 3rd observation

11th observation is 14 more than 4th observation

12th observation is 14 more than 5th observation

Total 14 × 5 =70

Sum of the first five numbers = 425

Sum of last five numbers = 495

6.

	2012	2013
Transport (10%)	300,000	360,000
Labour (15%)	450,000	540,000
Raw material (20%)	750,000	780,000
Energy (25%)	750,000	900,000
Plant and Machinery (30%)	900,000	1,080,000
Total	**3,000,000**	**3,660,000**

Percentage increase in total cost = 22%

7. Total expenditure = $\frac{15}{100} x = 4,50,000$

$$x = 3 \times 10^6$$

Profit = 10 lakhs

So, total selling price = 40,00,000

Total purifies = 200 ...(1)

S.P. of each purifier = (1)/(2) = 20,000 ...(2)

8. Eight consecutive odd number = 656

$a - 6, a - 1, a - 2, a, a + 2, a + 4, a + 6$

$$a + 8 = 656$$

$$a = 81$$

Smallest $m = 75$

Average consecutive even numbers ...(1)

$$\Rightarrow \frac{a-2+a+a+2+a+4}{4} = 87$$

$$\Rightarrow a = 86$$

Second largest number = 88 ...(2)

Adding equation (1) and (2),

$$75 + 88 = 163$$

9. Ratio of male to female student in 2008 = 5 : 2

In 2009, let the male student be $5x$ and female student be $2x$

Now if number of female students double in 2009

$\therefore$ Number of female student = $4x$

$\therefore$ Number of male student = $12x$

Gender	2008	2009
Male	$5x$	$12x$
Female	$2x$	$4x$

% of male student increase in 2009

$$= \left[\left(\frac{12x - 5x}{5x}\right) \times 100\right]\%$$

$$= \left[\frac{7}{5} \times 100\right]\%$$

$$= 140\%$$

10. Expanding

$$\left(z + \frac{1}{z}\right)^2 = 98$$

$$\therefore \quad z^2 + \frac{1}{z^2} + 2(z)\left(\frac{1}{z}\right) = 98$$

$$\therefore \quad z^2 + \frac{1}{z^2} = 96$$

11. Total respondents = 300

Now men respondents who do not have scooter

$$= 40 + 20 = 60$$

And women respondents who do not have scooter

$$= 34 + 50 = 84$$

$\therefore$ Total respondents who do not have scooter

$$= 60 + 84 = 144$$

$\therefore$ Required percent $= \left(\frac{144}{300} \times 100\right)\% = 48\%$

12. One way single person fare = ₹ 100

$\therefore$ Two way fare for single person = ₹ 200

$\therefore$ For 5 persons two way fare = ₹ 1000

Now, total discount = (10 + 5)% = 15%

Discount amount = $₹ \left[\frac{15}{100} \times 1000\right] = ₹\ 150$

Amount to be paid = ₹(1000 – 150) = ₹ 850

■■

2 CHAPTER

Percentage and Its Applications

The term percentage is quite frequently used in our day to day life especially to avoid the fractions less than 1. Instead of treating the complete entity as 1, we treat as 100, and take the ratios accordingly.

PERCENTAGE

Per Cent: This term means for every hundred. A fraction whose denominator is 100 is called *percentage* and numerator of the fraction is called *rate per cent*. It is denoted by the symbol %.

To find % equivalent of a fraction.

Express fraction with the denominator 100 and numerator is the required answer.

e.g., $\frac{11}{24} = \frac{11}{24} \times 100 = 45\frac{5}{6}\%$

To find fraction equivalent of 'a %': Divide 'a' by 100.

e.g., $21\frac{7}{8}\% = \frac{21\frac{7}{8}}{100} = \frac{175}{800} = \frac{7}{32}$

To increase a number by a given %:

Multiply the number by the factor $\frac{100 + \text{rate}}{100}$

e.g., increase 20 by 15%,

i.e $20 \times \left(\frac{100+15}{100}\right) = 20 \times \frac{115}{100} = 23$

or $20 \times 1.15 = 23$

To decrease a number by a given %:

Multiply the number by the factor $\frac{100 - \text{rate}}{100}$

e.g., decrease 30 by 20%, i.e $30 \times \left(\frac{100-20}{100}\right)$

$= 30 \times \frac{80}{100} = 24$

or $30 \times 0.8 = 24$

To find the % increase of a number:

$$\%\text{increase} = \frac{\text{total increase}}{\text{initial value}} \times 100$$

$$= \frac{\text{Final value} - \text{Initial value}}{\text{Initial value}} \times 100$$

e.g., Population of a village in 1980 was 3000 and in 1990 was 3200. Find the % increase.

$$\% \text{ increase} = \frac{3200 - 3000}{3000} \times 100$$

To find % decrease of a number:

$$\% \text{ decrease} = \frac{\text{total decrease}}{\text{Initial value}} \times 100$$

$$= \frac{\text{Initial value} - \text{Final value}}{\text{Initial value}} \times 100$$

e.g., Cost of a bike last year was ₹ 19000. Its cost this year is ₹ 17000. Find the % decrease in its cost.

$$\% \text{ decrease} = \frac{19000 - 17000}{19000} \times 100$$

$$= \frac{2000}{19000} \times 100 = 10.5\%$$

Note:

(1) If price of a commodity increases by r%, then reduction in consumption, so as not to increase expenditure is $\left(\frac{r}{100+r} \times 100\right)\%$

(2) If price of a commodity decrease by r%, then the increase in consumption so as not to decrease expenditure is $\left(\frac{r}{100-r} \times 100\right)\%$

(3) If A's income is r% more than that of B, then B's income is less than that of A by $\left(\frac{r}{100+r} \times 100\right)\%$

(4) If A's income is r% less than that of B, then B's income is more than that of A by $\left(\frac{r}{100-r} \times 100\right)\%$.

SIMPLE INTEREST AND COMPOUND INTEREST

When a sum of money is lent by A to B, A is called *lender* (*creditor*), B the *borrower* (*debtor*).

The sum lent is called *principal* (P).

Interest (I) is the extra money paid by the borrowed to the lender for the use of the money for a specified time.

The time for which the money is borrowed is called *period* (N).

The extra amount paid per 100 rupees in a year is called *rate per cent per annum* (R).

The sum of interest and principal is called *Amount* (A).

$$A = P + I$$

Simple Interest (S.I.)

When interest is paid as it falls due, it is called simple interest i.e., throughout the load period, interest is charged on the original sum (principal) borrowed.

$$\text{S.I.} = \frac{PNR}{100}$$

e.g., Find interest to be paid on ₹ 1500 at 9% per annum for a period of 9 years.

Here P = 1500, R = 9%, N = 3 years

$$\therefore \quad \text{S.I.} = \frac{PNR}{100} = \frac{1500 \times 3 \times 9}{100} = ₹\ 405$$

Compound Interest (C.I.)

Money is said to be lent at compound interest when at the end of a year or other fixed period, the interest that has become due is not paid to the lender, but is added to the sum lent, and the amount thus obtained becomes the principal for the next year or period. The process is repeated until the amount for the last period has been found. The difference between final amount and the original principal is the *compound interest* (C.I.).

In compound interest, interest is calculated on the accrued interest also.

$$\text{Amount} = P\left[1+\frac{R}{100}\right]^N$$

Compound Interest = Amount – Principal

Note:

(1) The time period after which the interest is added each time to form a new principal is called *conversion period*. It may be annually, semi annually or quarterly.

(2) In case, interest is paid semi annually (half yearly), N is number of half years and R is rate percent per half year *i.e.*,

Number of years × 2 = Number of half years.

Rate percent per annum ÷ 2 = Rate percent per half year.

(3) In case interest is paid quarterly, N is number of quarters and R is rate percent per quarter.

Number of years ÷ 4 = Number of quarters

Rate percent per quarter + 4 = Rate percent per quarter.

(4) When rates are different for different years, say R_1, R_2, R_3 percent of 1st, 2nd and 3rd years respectively then

$$\text{Amount} = P\left(1+\frac{R_1}{100}\right)\left(1+\frac{R_2}{100}\right)\left(1+\frac{R_3}{100}\right)$$

(5) In case the time is a fraction of a year, say $y\frac{x}{z}$ years, then

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^y \times \left(1+\frac{\frac{x}{z}\times R}{100}\right)$$

Population Formula: The original population of a town is P and annual increase is r%, then

population in n years is $\left(1+\frac{R}{100}\right)^n$ and if annual decrease is r%, then population in n years is given by a change of sign in the formula

i.e., $\left(1-\frac{R}{100}\right)^n$

e.g., If annual increase in the population of a town is 4% and present population is 15625, what will be the population in 3 years?

Required population = $15625\,(1.04)^3 = 17576$

Hire Purchase: In a hire purchase plan, a customer can make use of the goods while paying for them. The amount paid at the time of purchase is called *down payment*. The remainder is paid in equal installments and each is the monthly installment. The difference between total amount to be paid and the cash price is called *installment charge*.

Monthly Installment

$$= \frac{\text{Amount to be paid} - \text{down payment}}{\text{Number of instalments}}$$

e.g., If a transistor is available at ₹ 400 each or ₹ 100 down payment and ₹ 70 per month for 5 months, find total amount paid for it, and the installment charge

Amount paid = 100 + 70 × 5 = ₹ 450

Installment charge = 450 – 400 = ₹ 50

PROFIT AND LOSS

Cost Price (CP): The price for which an article is bought is called its *cost price*.

Selling Price (SP): The price at which an article is sold is called its *selling price*.

Profit (Gain): The difference between selling price and cost price is called *profit*. For profit, selling price should be greater than cost price.

Loss: The difference between cost price and the selling price is called *loss*. When cost price is greater than the selling price, there is a loss.

Profit and loss is generally represented as a percent of the cost price, unless otherwise stated.

Overhead charges: If an individual has to spend some money on transportation etc., then this extra expenditure is called *overhead charges*.

Marked price (MP): The price on the label is called *marked price* or *list price*.

Discount: The reduction made on the '*marked price*' of an article is called *discount*. When no discount is given, '*selling price*' is the same as '*marked price*'.

List of Formulae

(1) Profit = SP – CP ... (SP > CP)

(2) Loss = CP – SP ... (CP > SP)

(3) $\%\ \text{Profit} = \frac{\text{Profit}}{CP}\times 100 = \frac{SP-CP}{CP}\times 100$

(4) $\%\ \text{Loss} = \frac{\text{Loss}}{CP}\times 100 = \frac{CP-SP}{CP}\times 100$

(5) $\text{Profit} = \frac{\text{Profit}\,\% \times CP}{100}$

(6) $\text{Loss} = \frac{\text{Loss}\,\% \times CP}{100}$

(7) $SP = \frac{100+\text{Profit}\,\%}{100}\times CP$

(8) $SP = \frac{100-\text{Loss}\,\%}{100}\times CP$

(9) SP = Marked Price – Discount

(10) $CP = SP \times \left(\frac{100}{100+\text{Profit}\,\%}\right)$

(11) $CP = SP \times \left(\frac{100}{100-\text{Loss}\,\%}\right)$

(12) $\text{Discount}\,\% = \frac{\text{Discount}}{\text{Marked Price}}\times 100$

Some important point:

(1) If two items are sold, each at ₹ X, one at a gain of p% and the other at a loss of p%, there is an overall loss given by $\frac{p^2}{100}\%$. The absolute value of the loss is given by $\frac{2p^2x}{100^2-p^2}$

(2) If CP of two items is the same and % Loss and % Gain on the two items are equal, then net loss or net profit is zero.

(3) Buy x get y free *i.e.*, if x + y articles are sold at cost price of x articles, then

$$\text{percentage discount} = \frac{x}{x+y}\times 100.$$

(4) By using false weight, if a substance is sold at cost price, then

$$\text{overall gain}\,\% = \frac{100+\text{Gain}\%}{100} = \frac{\text{True Scale or Weight}}{\text{False Scale or Weight}}$$

(5) In case of successive discount a% and b%,

$$\text{effective discount} = \left(a+b-\frac{ab}{100}\right)\%.$$

PARTNERSHIP

It is an association of two or more persons who invest their money in order to carry on a certain business. A partner who manages the business is called *working partner* and the one who simply invests the money is called *sleeping partner*.

If capitals of the partners are invested for the same time, then partnership is called *simple*, and if invested for different periods, the partner is called *compound*.

If period of investment is the same for each partner, then profit or loss is divided in the ratio of their investments.

e.g., (*i*) If A and B are partners in a business, then

$$\frac{\text{Investment of A}}{\text{Investment of B}} = \frac{\text{Profit of A}}{\text{Profit of B}}$$

or $\frac{\text{Investment of A}}{\text{Investment of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$

(*ii*) If A, B and C are partners in a business, then

Investment of A : Investment of B : Investment of C

= Profit of A : Profit of B : Profit of C

OR Loss of A : Loss of B : Loss of C

Monthly Equivalent Investment: It is the product of the capital invested and the period for which it is invested.

If period of investment is different, then profit or loss is divided in the ratio of their Monthly Equivalent Investment.

e.g., If A and B are partners in a business, then,

$$\frac{\text{Monthly Equivalent Investment of A}}{\text{Monthly Equivalent Investment of B}} = \frac{\text{Profit of A}}{\text{Profit of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$$

i.e., $\frac{\text{Investment of A} \times \text{Period of Investment of A}}{\text{Investment of B} \times \text{Periof of Investment B}} = \frac{\text{Profit of A}}{\text{Profit of B}} = \frac{\text{Loss of A}}{\text{Loss of B}}$

If A, B and C are partners in a business, then

Monthly Equivalent Investment of A : Monthly Equivalent Investment of B : Monthly Equivalent Investment of C = Profit of A : Profit of B : Profit of C

or

Monthly Equivalent Investment of A : Monthly Equivalent Investment of B : Monthly Equivalent Investment of C = Loss of A : Loss of B : Loss of C

STOCKS AND SHARES

Face value: The price of a share is printed on the share certificate is called face value of the share. (Nominal value).

Dividend: Every shareholder of a company is entitled to a proportionate share of the profits of the company. The amount of profits on each share is called *dividend*. Dividend is expressed as a percentage of the face value.

Market Value: In the market, selling price of the share may be different from its face value. Thus, market value of a share is *fluctuable price*.

Shares: *In the market, three types of shares are available* :

(*i*) **Premium share:** If market price of a share is more than nominal value, then it is called *premium share*.

(*ii*) **Par share:** If market price of a share is equal to the nominal value, then it is called *par share*.

(*iii*) **Discount share:** If market price of a share is less than the nominal value, then it is called a *discount share*.

Types of shares: *There are two types of shares:*

(*i*) **Preference shares:** On these shares a fixed rate of dividend is paid to share holders irrespective of whether the company makes any profits or whether it runs into loss. In case the company is not able to pay the agreed dividend to preference shareholders, the dividend on cumulative preference shares goes on accumulating and is paid as and when the company can do so.

(ii) **Equity shares:** Profit of the company that is left over after its distribution to the preference shareholders is distributed among the equity share holders. This dividend on equity share may sometimes be more or sometimes less than the dividend on the preference share.

Stock

In order to meet expenses of a certain plan, the Government of India sometimes raises a load from the public at a certain fixed rate of interest. Bonds or promissory notes called *stocks of a fixed value* are used for sale to the public.

Note:

(1) Income per share = Rate of dividend × Nominal value of 1 share.

(2) Annual income = Income per share × Number of shares.

(3) Number of shares

$$= \frac{\text{Nominal value of all the shares}}{\text{Nominal value of one shares}} = \frac{\text{Market value of the shares (Investment)}}{\text{Market value of one share}} = \frac{\text{Total Dividend}}{\text{Dividend per share}}$$

(4) Rate of interest on the Invest (yield)

$$= \frac{\text{Total income} \times 100}{\text{Total investment}}$$

(5) When stoke/share is purchased, brokerage is added to cost price.

(6) When stock/share is sold, brokerage is subtracted from selling price.

(7) In most problems, face value of a share is taken to be ₹, 100 unless specified.

EXERCISE – I

MCQ TYPE QUESTIONS

1. The Manager (M) of the company along with his Assistant Manager (AM) and a Business Analyst (BA) invest ₹ 55000, ₹ 45000 and ₹ 60000 respectively in a partnership firm for six months. The Manager being the highest authority among the three gets one-tenth of the total profit and Assistant Manager being second highest authority gets $\frac{1}{20}$ th of the remaining and the remaining profit is divided among the three in the ratio of their investments. If the total profit was ₹ 40000, who gets the minimum share and how much?

(a) BA, ₹ 11400 (b) AM, ₹ 11419
(c) AM, ₹ 12500 (d) None of these

2. In a business partnership of three people A, B and C, B's share in total profit is ₹ 250 per ₹ 2000. At this profit his return on investment is 15%. If B earns total of ₹ 900. what is the total investment made by A and C in business? Assume that each of A, B and C have invested their money for the entire year and profits are shared in the ratio of their investments.

(a) ₹ 48000 (b) ₹ 42000
(c) ₹ 6000 (d) Data insufficient

3. Abhay and Bimal started a business in partnership. Abhay invested a capital of ₹ 50000. Bimal agrees to pay Abhay an interest on half the capital at 8% per annum. Also Bimal receives a salary of ₹ 500 per month for managing the business. After six months Abhay withdraws ₹ 20000 and Bimal pitches in an equal amount. Bimal continues to pay Abhay interest at 8% per annum for the balance amount. At the end of the year they divide the profit equally and found that total earning of Bimal in the year was exactly thrice that of Abhay. The total profit ?

(a) ₹ 7200 (b) ₹ 6800
(c) ₹ 5400 (d) ₹ 4800

4. How much stock must be bought in 3 percent $89\frac{1}{8}$ in order that by selling it at $91\frac{3}{8}$ a profit of ₹ 45 may be made?

(a) 20 (b) 30 (c) 40 (d) 50

5. Which is the better investment (the security being equal), 3 percents perpetually at par, or 5 percents at 124, the latter stock being redeemable in 20 years at par? (Simple interest to be reckoned).

(a) 3% at par
(b) 5% at 124
(c) Both are same
(d) Cannot be determined

6. What rate percent per annum does a man get by buying ₹100 share (₹ 70 paid-up) at ₹ 60 when dividend is 5% per annum?

(a) 4.28% (b) 7.14%
(c) $8\frac{1}{3}$ % (d) $5\frac{5}{6}$ %

7. A rejects 0.08% of the meters as defective .How many will he examine to reject 2?

(a) 2500 (b) 1250
(c) 625 (d) 5000

8. (a% of b) + (b% of a) is equal to

(a) a% of b (b) 2% of ab
(c) 20% of ab (d) 2% of 100 ab

9. Ryan's stock of goods is worth $9462, which is 15% more than Harland's, and 15% less than Lambert's. What is the value of stock carried by Harland and Lambert respectively?

(a) $ 8218, $ 10032 (b) $ 1419, $ 10881
(c) $ 8042, $ 10881 (d) $ 8228, $ 11132

10. Numerator of a fraction increases by 25%. What can be said about the resultant fraction?

(a) It is 67% less than original.
(b) It is equal to the original fraction.
(c) It is 67% more than the original fraction.
(d) The change cannot be determine.

11. Which of the following statement/s is/are true?

I. If two numbers are such that one is 25% more than the second, then sum of the two numbers is nine times that of the difference of the two numbers.

II. If 56% of a number A is 24% of another number B then 49% of A is 21% of B.

Codes:

(a) Only I (b) Only II
(c) Only I and II (d) None of these

12. In the Annual General Meeting of the Rotary club a chairman is to be selected for which A and B is the only two candidates. After the first round, each of them gave a speech on their policy and work plan for the coming year after which another round of election was held. The number of votes not cast for A increased by 25% in the second round over those not cast for him in the first round and A lost second round to B by twice as many votes as that by which he had won in the first round. If total 260 people voted each time what was the percent decrease in votes received by A in the second round assuming that no vote was void in any of the rounds?

(a) 18% (b) 25% (c) 21.43% (d) 30%

13. A watermelon weighs 5000 gm. 99% of its weight is water. It is kept in a drying room and after some time it turns out that it is only 98% water by weight. What is its weight now?

(a) 2500 gm
(b) 4500 gm
(c) 4950 gm
(d) None of these

14. A sum of ₹ 5000 is divided into two parts A and B such that simple interests at the rate of 20% p.a. on A and B after 2 and 3 years respectively are equal. Which of the following is false about A and B?

(a) Ratio of A to B is 2 : 3
(b) Ratio of A to B is 3 : 2
(c) A exceeds B by ₹ 1000
(d) Total interest earned on A and B is ₹ 2400

15. Which amongst the following two offers is the better one? Consider with respect to interest accrued.

I. Investing an amount compounded annually at 1% per annum for 100 years.

II. Investing the amount compounded annually at 100% per annum for 1 year.

Codes:

(a) First offer
(b) Second offer
(c) Both are same
(d) Cannot be determined

16. The population of rate in a locality x increases by 20% in one year. Observing this, the pest control committee decided to use a special kind of pesticide 'xyz' which effectively kills 160 rats in 3 months. Just after 2 years, what is the net increase or decrease in the population of rats if, initially the population of rats is 3200 and pesticide is used effectively?

(a) Increase of 128 rats.
(b) Decrease of 128 rats.
(c) Neither an increase nor a decrease in the population.
(d) None of these.

17. A man travels from A to B to buy goods which he can get 10% cheaper in B than in A. If expenses of the journey are ₹ 15 and he makes a clear saving of ₹ 10, how much does he pay for the goods?

(a) ₹ 225
(b) ₹ 200
(c) ₹ 150
(d) Cannot be determined

18. By selling 5 dozen mangoes for ₹ 156, it was found that $\frac{3}{10}$ th of the outlay was gained. What should the retail price per mango be in order to gain 60%?

(a) ₹ 4 (b) ₹ 2 (c) ₹ 3.2 (d) ₹ 4.2

19. A man sells sugar to a tradesman at a profit of 20% but the tradesman becoming bankrupt pays only 80 paise in the rupee. How much percentage does the man gain or lose by his sale?

(a) 2.5% (b) 3% (c) 4% (d) 5.2%

20. A trader allows a discount of 5 percent to his customers. What price should he mark on a article the cost price of which is ₹ 800 so as to make clear profit of 25 percent on his outlay?

(a) ₹ 1000 (b) ₹ 1053
(c) ₹ 1200 (d) ₹ 1123

21. A camera costing ₹550 is marked to be sold at a price, which gives a profit of 30%. What will be its selling price in a sale when 20% is taken off the marked price?

(a) ₹ 600 (b) ₹ 572
(c) ₹ 635 (d) ₹ 605

22. Buy two and get one free. Which of the following is true?

(a) The person is selling at a loss.
(b) The person is selling at a profit.
(c) The loss or gain depends on the profit from sale of two articles.
(d) The person is selling at the cost price.

Directions (Q.23–25): *Refer to the data below and answer the questions that follow.*

A profit-maximizing producer has the option of discriminating between markets A and B (i.e., charging different prices in markets A and B for the same product) for a product X. Demand for the product in each of the markets A and B is given by

$$Q_A = 42 - 0.2\,P_A$$
$$Q_B = 100 - 0.8\,P_B$$

Total cost in any market is given by

$C = 1000 + 5Q$, *where Q is quantity sold in that market.*

Consider two cases:

***Case 1** in which she discriminates between the markets and*

***Case 2** in which she does not discriminate.*

23. What is the approximate ratio of the price in market A to price in market B in Case 1?

(a) 1.65 (b) 2.00
(c) 2.25 (d) 2.50

24. What is the ratio of the total quantity of product X sold in Case 1 to the total quantity sold in Case 2?

(a) 0.50 (b) 1
(c) 1.10 (d) 1.05

25. What is the difference in total profit in Case 1 and the total profit in Case 2?

(a) ₹ 289 (b) ₹ 829 (c) ₹ – 289 (d) ₹ – 829

NUMERICAL TYPE QUESTIONS

1. A and B enter into a partnership. A puts in the whole capital of ₹ 45000 on the condition that the profits will be equally divided after which B will pay A interest on half the capital at 10% p.a. and receive ₹ 60 per month from A for carrying on the concern. The yearly profit in rupees, if B's income is half of A's income is _____

2. A, B and C started a business by investing ₹ 72000, ₹ 27000 and ₹ 81000. It was decided that servant's salary ₹ 2000 per month should be given from the profit. If profit is ₹ 25000 in the first month and ₹ 20000 in the second month and average profit of remaining 10 months is ₹ 21000, then share of C is _____

3. Three friends started a business of renting out air conditioners by investing ₹ 20000, ₹ 24000 and ₹ 16000, respectively. C gets 20% of total profit for repair and maintenance of the air conditioner. If in a particular year, C gets ₹ 487.50 less than total earnings of the other two, then total profit for the year in ₹ is _____

4. A person holds forty ₹ 500 shares in a concern, which pays dividend at the rate of 6 percent per annum. When the shares are at ₹ 675, he sells out and invests half the proceeds in 4 percent stock at 90. With the other half he buys a house. For which he receives an annual rent of ₹ 1440 subject to a deduction of 25P per rupees for repairs and taxes. Alteration in rupees in his annual income will be _____.

5. A person invests ₹ 34539 in three percent at 87. After receiving one year's dividend, he sells out at 89. He then invests the whole in railway stock paying 5 percent at 115.The difference in his income will be_____

6. Difference of two numbers is 1660. If 7.5 % of one number is 12.5% of the other number, then two numbers are_____

7. A school has only three classes which contain 40,50,60 students respectively. The pass percent of these classes are 10, 20 and 10 respectively. Then pass percent in the school will be_____

8. For a sphere of radius 10cm, the numerical value of surface area is _____ percent of the numerical value of its volume.

9. At an election, there are two candidates only, a candidate who gets 43 per cent of the votes is rejected by a majority of 420 votes. Then total number of votes recorded assuming that there was no void vote are_____

10. If price of wheat be raised 30%, then by _____ percent a householder must reduce his consumption of the article so as not to increase his expenditure?

11. A fraction in reduced form is such that when it is squared and then its numerator is reduced by $33\frac{1}{3}$% and denominator is reduced to 20%, its result is twice the original fraction. Sum of the numerator and the denominator is _____

12. A person borrows two equal sums at the same time at 5 and 4 percent respectively and finds that if he repays the former sum with simple interest on certain date 6 months before the latter, he will have to pay in each case the same amount, viz., ₹ 1100. The amount borrowed in rupees is _____

13. Effective annual rate corres-ponding to a nominal rate of 6 percent per annum, payable half-yearly is _____

14. A tradesman puts two prices on his goods, one for ready money and the other for 6 month's credit interest being calculated at $12\frac{1}{2}$% per annum.If credit price on an article be ₹ 26.56, then its cash price in rupees is _____

15. An article costing ₹9000 is sold at a discount which is equal to the simple interest on ₹ 3000 for N months. If rate of discount is same as the rate of interest, the N will be_____months

16. A lent B ₹ 900 for a certain time at a certain rate percent per annum, which is equal to the square root of the number of the months of the time. After the time B wanted to return the money, but A, instead of taking the interest which amounted to ₹8 asked for a certain sum of money at the same rate for a certain number of years which is equal to the square root of the rate percent. Sum of money in rupees so that none of them should be the loser will be_____ (Take simple interest)

17. A shopkeeper sells a microwave at a discount on its marked price of ₹ 2500. But in a clearance sale he gives an additional discount of 20% thus selling it at ₹ 1500. Then first discount was_____%

18. A sports goods shop owner has 6 packets of balls each containing the same number of balls. If he sells each ball at ₹ 8 he gains ₹ 96. If there were 1 ball less in each packet and he sells each ball at ₹ 9 then he gains ₹ 90 in the transaction. The cost price of each packet of balls in rupees is _____

19. One merchant correctly calculates his percentage profit on the cost price; another wrongly calculates it on the selling price. The difference in actual profits in rupees if both claim to make $17\frac{1}{2}$% profit on goods sold at ₹ 3760 is_____

20. A manufacturer marks his goods in advance at 80 percent more than the cost price, but he allows 15 articles to the dozen and also 10 percent discount for cash. The profit on his outlay does he obtain from a customer who pays cash is _____%

21. A trader buys a certain amount of goods worth ₹ 22520. He decides to make a profit of 5.36% on the sale of goods worth ₹ 5000 and increase the profit percent by 3.14% for sales upto ₹15000 and then increase the profit percent for the sale of remaining lot such that he is able to make a profit of 25% on the sale of the full lot. Then profit in rupees that he makes on the third lot of goods is_____

22. Three shepherds A, B and C rented a pasture for a year. A grazed 22 sheeps for 4 months, B grazed 16 sheeps for 8 months and C grazed 32 sheeps for 6 months. If C's share of rent is ₹600, then total rent in rupees for the year was_____

23. Three bachelors, Amar, Akbar and Anthony rented a house for a year. But, Amar left after 4 months, Akbar stayed for 8 months and only Anthony stayed for the entire year. If the annual rent was ₹ 6000, then share of Akbar is_____

24. Ram, Shyam and Ghanshyam invest ₹2000, ₹ 10000 and ₹ 5000 to set up a stall. Ram gets 15% of the total profit for running the stall. Ram gets ₹ 500 less than Shyam and Ghanshym together, then total profit of that year is_____

25. x, y and z start a joint venture wherein they make an annual profit of ₹ 3600. x invested one-third of the capital for one-fourth of the time, y invested one-fourth of the capital for one-half of the time, while z invested the remainder of the capital for the entire year. Then share of x in the profit is_____.

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2014

1. The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

(a) 3 – 4 years (b) 4 – 5 years

(c) 5 – 6 years (d) 6 – 7 years

2. One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft?

(a) 3.0 (b) 2.5

(c) 1.5 (d) 1.25

3. Industrial consumption of power doubled from 2000-2001 to 2010-2011. Find the annual rate of increase in percent assuming it to be uniform over the years.

(a) 5.6 (b) 7.2

(c) 10.0 (d) 12.2

4. The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from ₹ 50/ USD to ₹ 60/ USD. India's GDP in USD during the period 2012-2013

(a) increased by 5% (b) decreased by 13%

(c) decreased by 20% (d) decreased by 11%

2013

5. A firm is selling its product at ₹ 60 per unit. The total cost of production is ₹ 100 and firm is earning total profit of ₹ 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

(a) 5 (b) 10

(c) 15 (d) 30

2012

6. An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

(a) 0.288

(b) 0.334

(c) 0.667

(d) 0.720

2011

7. There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

(a) 100

(b) 110

(c) 90

(d) 95

NUMERICAL TYPE QUESTION

2015

1. From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (b)	2. (b)	3. (a)	4. (a)	5. (b)	6. (d)	7. (a)	8. (b)	9. (d)	10. (c)
11. (c)	12. (c)	13. (a)	14. (a)	15. (a)	16. (c)	17. (a)	18. (c)	19. (c)	20. (b)
21. (b)	22. (c)	23. (a)	24. (b)	25. (a)					

Numerical Type Questions

1. 9180	2. 103950	3. 2812.50	4. 480	5. 345.2	6. (4150, 2490)
7. $13\frac{1}{3}$	8. 30	9. 3000	10. $23\frac{1}{13}$	11. 8	12. 1000
13. 6.09	14. 25	15. 4	16. 600	17. 25	18. 48
19. 98	20. 29.6	21. 4512	22. 1275	23. 2000	24. 1000
25. 480					

EXERCISE – II

MCQ Type Questions

1. (a)	2. (d)	3. (b)	4. (d)	5. (a)	6. (b)	7. (a)

Numerical Type Question

1. (2.06)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. The ratio of their investments is 11 : 9 : 12.

So, Manager gets $\frac{1}{10}(40000) = 4000 +$ his share

Remaining amount = 36000.

Asstt.Manager gets $\frac{1}{20}(36000) = 1800 +$ his share

Remaining amount = 36000 – 1800 = 34200

$\therefore$ M's share $= \frac{11}{32} \times 34200 + 4000$

$= 11756.25 + 4000$

$=$ ₹ 15756.25

$\equiv$ ₹ 15756

AM's share $= \frac{9}{32} \times 34200 + 1800$

$= 9618.75 + 1800$

$=$ ₹ 11418.75

$\approx$ ₹ 11419

BA's share $= \frac{12}{32} \times 34200 =$ ₹ 12825.

2. B's share in the profit = 250 per 2000

If B gets ₹ 900, then

total profit $= 2000 \times \frac{18}{5} = 7200$

B's return on investment = 15%

$\therefore \frac{900}{\text{B's investment}} \times 100 = 15$

$\therefore$ B's investment = 6000

Hence profits are shared in the ratio of their investments.

Total investments $= 6000 \times \frac{7200}{900} = 48000$

$\therefore$ Investment made by A and C = 48000 – 6000

= 42000.

3. Interest paid by Bimal to Abhay for first six months

$= \frac{50000}{2} \times \frac{8}{100} \times \frac{1}{2} =$ ₹ 1000

For last six months

$= \frac{30000 - 20000}{2} \times \frac{8}{100} \times \frac{1}{2}$

= ₹ 200

$\therefore$ Total salary paid to Bimal = $12 \times 500 = ₹\ 6000$

Let total profit be ₹ x

$\therefore$ Abhay's earing = $1200 + \frac{x - 6000}{2}$

Bimal's earning = $6000 + \frac{x - 6000}{2} - 1200$

$\therefore \quad 3\left(1200 + \frac{x - 6000}{2}\right) = 4800 + \frac{x - 6000}{2}$

$\Rightarrow \quad 3600 + \frac{3x}{2} - 9000 = 4800 + \frac{x}{2} - 3000$

$\Rightarrow \quad \frac{3x}{2} - \frac{x}{2} = 7200;$

$\therefore \quad x = ₹\ 7200$

4. Profit per stocks = $91\frac{3}{8} - 89\frac{1}{8} = ₹\ 2\frac{2}{8}$

$\therefore \quad \frac{45}{2\frac{1}{4}} = 20$ stocks

5. Income from a stock at 3% in 20 years at par

$$= \frac{100}{100} \times 3 \times 20 = 60$$

Income from a stock at 5% at 124 in 20 years

$$= \frac{100}{124} \times 5 \times 20 = 80.64.$$

6. He invests ₹ 60 and gets ₹ 100 share

Income = ₹ 5

Required % = $\frac{5}{60} \times 100(0.70) = 5.83\% = 5\frac{5}{6}\ \%$

7. Let number of meters to be examined be x.

Then 0.08% of $x = 2$.

$\Rightarrow \quad \frac{0.08}{100} \times x = 2$

$$x = \frac{2 \times 100}{0.08} = \frac{2 \times 100 \times 100}{8} = 2500$$

8. $\left(\frac{a}{100} \times b\right) + \left(\frac{b}{100} \times a\right) = \frac{ab}{100} + \frac{ab}{100} = \frac{2ab}{100}$

$$= \frac{2}{100} \times ab = 2\% \text{ of } ab.$$

9. Ryan's stock = $1.15 \times$ Harland's stock

$\therefore$ Harland's stock = $\frac{9462}{1.15} = \$8228$

Ryan's stock = $0.85 \times$ Lambert's stock

$\therefore$ Lambert's stock = $\frac{9462}{0.85} = \$11132.$

10. Let the fraction be $\frac{x}{y}$.

Let $x_1 = x + 0.25x = 1.25x$

$y_1 = y - 0.25x = 0.75x$

$\therefore \quad \frac{x_1}{y_1} = \frac{1.25x}{0.75x} = \frac{5}{3}\frac{x}{y}$

$\therefore$ Increase = $\frac{\frac{5}{3}\frac{x}{y} - \frac{x}{y}}{\frac{x}{y}} \times 100 = \frac{2}{3} \times 100 = 66.67\%$

Thus, resultant fraction is more than the original fraction by 67%. Thus III is true.

11. I. Let first number be x.

$\therefore$ Second number = 1.25x

Sum = x + 1.25x = 2.25x

Difference = 1.25x – x = 0.25x

Also, $9 \times 0.25x = 2.25x$

Thus I is true.

II. $\frac{56}{100} \times A = \frac{24}{100} \times B$

Multiplying by $\frac{49}{56}$ on both the sides.

$\therefore \quad \frac{49}{56} \times \frac{56}{100} \times A = \frac{49}{56} \times \frac{24}{100} \times B$

$\Rightarrow \quad \frac{49}{100}A = \frac{21}{100}B$

$\Rightarrow$ 49% of A = 21% of B.

Thus, II is true.

12. In the first round, let A get 'a' votes and B get 'b' votes.

Then, $a + b = 260$

i.e., $a = 260 - b \quad \ldots (i)$

In second round, B gets

$$b \times \frac{125}{100} = \frac{5b}{4}$$

and A gets $260 - \frac{5b}{4}$

Also, $\frac{5b}{4} - \left(260 - \frac{5b}{4}\right) = 2(a - b)$

$\frac{10b}{4} - 260 = 2(260 - b - b)$

$\frac{10b}{4} - 260 = 520 - 4b$

$4b + \frac{10b}{4} = 520 + 260$

$b = 120$

Hence first round, A gets 140 and B gets 120.

In second round A gets 110 and B gets 150.

$\therefore$ % decrease in votes received by A

$$= \frac{140 - 110}{140} \times 100 = 21.43\%$$

13. Non-water matter is 1% of 5000 = 50 gm.

After drying, this matter becomes 2% of the weight of the water-melon. Hence weight of the watermelon now is 2500 gm.

14. Let A = ₹ x and B = ₹ (5000 – x)

Then, $\frac{x \times 2 \times 20}{100} = \frac{(5000 - x) \times 3 \times 20}{100}$

$\Rightarrow \quad 2x = 15000 - 3x$

$\Rightarrow \quad 5x = 15000$

$\Rightarrow \quad x = 3000$

$A = x = ₹\ 3000$

and $B = 5000 - x$

$= 5000 - 3000$

$= ₹\ 2000$

$\therefore \quad \frac{A}{B} = \frac{3000}{2000} = \frac{3}{2}$

Thus, (a) is not true and (b) is true.

A – B = 3000 – 2000 = ₹ 1000.

Thus, (c) is true.

(d) The total interest

$$= \frac{3000 \times 2 \times 20}{100} + \frac{2000 \times 3 \times 20}{100}$$

$$= ₹\ 2400.$$

Thus, (d) is true.

15. First investment gives a return

$$P\left(1 + \frac{1}{100}\right)^{100} - P$$

First term is $P\left(1 + \frac{1}{100}\right)^{100} = X,$

such that $2P < X \leq 3P$

Hence return X – P lies between P and 2P

i.e., $2P - P < X - P \leq 3P - P.$

i.e., $P < X - P \leq 2P.$

Whereas second offer.

$$= \text{Return} = P\left(1 + \frac{100}{100}\right)^{1} - P$$

$$= 2P - P = P.$$

Alternatively

Option of investing at 1% for 100 years is obviously better than investing at 100% for 1 year, because total compound interest for 100 years will be atleast 100% of the total amount (1% for each year), plus whatever the compounding effect will have.

16. Growth rate of rat population in 3 months

$$= 20 \times \frac{3}{12} = 5\%.$$

Increase in first 3 months = $3200 \times 1.05 = 3360$

Also, net decrease in 3 months = 160

$\therefore$ Rat population = 3360 – 160 = 3200

In the same way, after every 3 months, rat population remains the same.

Hence, even after 3×8 months i.e., 2 years, the population is maintained

17. Let CP in A be ₹ x

$\therefore \quad$ CP in B = ₹ 0.9x

$\therefore \quad 0.9x + 15 + 10 = x$

$\Rightarrow \quad 0.1x = 25$

$\Rightarrow \quad x = 250$

$\therefore \quad$ Price in B = $250 \times 0.9 = ₹\ 225$

18. Let CP of 5 dozen mangoes be ₹ x.

SP = ₹ 156 and Gain = 0.3x

$\therefore \quad 156 - x = 0.3x$

$\Rightarrow \quad x = 120$

$\therefore$ SP of 60 mangoes = $120 \times 1.6 = 192$

$\therefore$ SP per mango = ₹ 3.2

19. Let CP be ₹ x ; SP = 1.2x

But he gets only $1.2x \times 0.8 = 0.96x$

$\therefore \quad$ Loss = 0.04x

$\therefore \quad$ Loss = 4 %

20. Selling price = $800 \times 1.25 = ₹\ 1000$

$\therefore$ Marked price = $₹\ \frac{1000}{0.95} = 1053$

21. CP = ₹ 550

$\therefore$ Marked price = 1.3×550 = ₹ 715

$\therefore$ Sales price = 0.8×715 = ₹ 572.

22. Profit or loss of the sale depends upon how much profit the person is earning on the sale of two articles.

If SP > CP of 1 article, there is a profit.

If SP < CP of 1 article, there is a loss.

If SP = CP of 1 article, there is not profit no loss.

Thus, only (c) is true.

23. *Following two key points need to be noted* :

(A) Producer is profit maximizing which implies she will set prices that maximize the profit function.

(B) Quantity demanded in market A is independent of quantity demanded in market B.

CASE 1:

In this case, producer charges different prices in markets A and B. This implies that she will maximize profits individually.

Equations can be re-written as

$$P_A = 210 - 5Q_A$$

$$P_B = 125 - 1.25\,Q_B$$

Profit in Market A = π_A

= Revenue – Cost

$= Q_A \times P_A - C_A$

$= Q_A \times (210 - 5Q_A) - (1000 + 5Q_A)$

At the maximum profit level, differential π_A with respect to Q_A is 0

or

$\therefore \quad 210 - 10Q_A - 5 = 0$

$\Rightarrow \quad Q_A = 20.5,\ P_A = 107.5$

and total revenue in market A

$= 20.5 \times 107.5$ = ₹ 2203.75

Similarly

Profit in Market B = π_B

= Revenue Cost

$= Q_B \times P_B - C_B$

$= Q_B \times (125 - 1.25\,Q_B) - (1000 + 5Q_B)$

At maximum profit level, differ-ential of π_B with respect to Q_B is 0

or $\quad 125 - 2.5Q_B - 5 = 0$

$\Rightarrow \quad Q_B = 48,\ P_B = 65$

and total revenue in market B

$= 48 \times 65$ = ₹ 3120

Using the above, we get

$$\frac{P_A}{P_B} = \frac{107.5}{65} \sim 1.65.$$

24. Data for 1 is available above. Consider case 2. In this case, prices across the two markets are the same. Hence, quantity equations can be added to give

$$Q = 142 - P$$

where Q is total quantity and P is price in both markets

$\Rightarrow \quad P = 142 - Q$

$\therefore \quad \pi = \text{Profit} = Q \times (142 - Q) - (1000 + 5Q)$

At the maximum profit level, differential of π with respect to Q is 0

$\therefore \quad Q = 68.5, \quad P = 73.5$

and Total Revenue = 5034.75

Given this, we note that quantities sold in both Case 1 and Case 2 are the same. Hence, the ratio is exactly 1.

25. Profit Difference between case 1 and case 2

= Revenue Difference (since quantity sold and hence, cost is the same)

$\therefore$ Profit difference = 2203.75 + 3120 – 5034.75

= 289

NUMERICAL TYPE QUESTIONS

1. Interest on ₹ 22500 = 0.1×22500 = ₹2250

Charges for managing the concern

$= 60 \times 12$ = ₹ 720

If yearly profit is ₹ x, then

$$\text{B's share} = \frac{\text{A's Share}}{2}$$

$$\therefore \quad \frac{x}{2} - 2250 + 720 = \frac{1}{2}\left(\frac{x}{2} - 720 + 2250\right)$$

$\Rightarrow \quad x$ = ₹ 9180.

2. Total annual profit

$= 25000 + 20000 + 21000 \times 10$

= ₹ 255000

Actual profit

$= 255000 - (2000 \times 12)$

= ₹ 231000

Total investment

= 72000 + 27000 + 81000

= ₹180000

$$\text{A's share} = \frac{72000}{180000} \times 231000$$

= ₹ 92400

$$\text{B's share} = \frac{27000}{180000} \times 231000$$

= ₹ 34650

$$\text{C's share} = \frac{81000}{180000} \times 231000$$

= ₹ 103950.

3. 80% of the total profit is divided in the ratio

20000 : 24000 : 16000 = 5 : 6 : 4

80% of total profit = 5x + 6x + 4x = 15x

$$\text{Total profit} = \frac{15x}{80\%} = 18.75x$$

$\therefore$ Share of C in profit = 4x + 20% of 18.75

= 4x + 3.75x = 7.75x

Share of A in profit = 5x

Share of B in profit = 6x

$\therefore \quad (6x + 5x) - 7.75x = 487.50$

$\Rightarrow \quad 3.25x = 487.50$

$\Rightarrow \quad x = 150$

$\therefore$ Total profit = 18.75×150 = ₹ 2812.50.

4. Selling price of 40 shares

= 675×40 = ₹ 27000

Income from half the proceeds

$$= \frac{13500}{90} \times 4 = ₹\ 600$$

Income from house = $1440 \times 0.75 = 1080$

Present total income = 1080 + 600 = ₹ 1680

$$\text{Earlier Income} = 40 \times 500 \times \frac{6}{100} = ₹\ 1200$$

Increase in income = ₹480

5. One years dividend = $\frac{34539}{87} \times 3$ = ₹ 1191

$$\text{Selling Price} = \frac{34539}{87} \times 89 = ₹\ 35333$$

Number of railway stocks bought

$$= \frac{35333}{115} = 307.24$$

Income from railway stocks

= 307.24×5 = ₹ 1536.2

Difference in income = 345.2

6. Let two numbers be x and y.

7.5% of x = 12.5% of y

$\therefore \quad 75x = 125\,y$

$\Rightarrow \quad 3x = 5y$

$\Rightarrow \quad x = \left(\frac{5}{3}\right)y.$

Now $x - y = 1660$

$\Rightarrow \quad \left(\frac{5}{3}\right)y - y = 1660$

$\Rightarrow \quad \left(\frac{2}{3}\right)y = 1660$

$\Rightarrow \quad y = 2490$

$\therefore \quad x = 2490 + 1660 = 4150.$

Hence numbers are 4150, 2490.

7. Number of passed candidates

$$= \frac{10}{100} \times 40 + \frac{20}{100} \times 500 + \frac{10}{100} \times 60$$

$= 4 + 10 + 6 = 20$

Total students in school

$= 40 + 50 + 60 = 150$

$\therefore$ Required percentage

$$= \frac{20}{150} \times 100 = \frac{40}{3} = 13\frac{1}{3}\,\%$$

8. Surface area = $4 \times \frac{22}{7} \times r^2$

$$= \frac{3}{r}\left(\frac{4}{3} \times \frac{22}{7} \times r^3\right) = \frac{3}{r} \times \text{Volume}$$

where r = 10 cm

So we have

$$S = \frac{3}{10}\,V = \frac{3}{10} \times 100\,\%\ \text{of}\ V = 30\,\%\ \text{of}\ V$$

So surface area is 30 % of volume.

9. Difference in % of votes = 57 – 43 = 14%

$\therefore$ 14% is represented by 420

$$\therefore \text{ Total number of votes} = 420 \times \frac{100}{14} = 3000.$$

10. % reduction in consumption = $\left(\frac{30}{100 + 30} \times 100\right)\%$

$$= \left(\frac{30}{130} \times 100\right)\% = 23\frac{1}{13}\,\%$$

11. Let the fraction be $\frac{a}{b}$

$$\therefore \quad \frac{a^2}{b^2} \times \frac{\frac{2}{3}}{\frac{1}{5}} = 2\frac{a}{b}\ \frac{a}{b} \neq 0 \text{ if } a \neq 0,$$

then $\quad \frac{a}{b} \times \frac{2}{3/5} = 2$

$$\Rightarrow \quad \frac{a}{b} = \frac{3}{5}$$

$\therefore \quad a + b = 8.$

12. $1100 = P + \dfrac{P \times 5 \times \left(T - \frac{1}{2}\right)}{100}$

$1100 = P + \dfrac{P \times 4 \times T}{100}$

$\therefore \dfrac{P \times 5 \times \left(T - \frac{1}{2}\right)}{100} = \dfrac{P \times 4 \times T}{100}$

$\Rightarrow 5T - \dfrac{5}{2} = 4T$

$\therefore T = \dfrac{5}{2}$ years

$\therefore 1100 = P\left(1 + \dfrac{4 \times \frac{5}{2}}{100}\right) = P\left(\dfrac{110}{100}\right)$

$\therefore P = ₹\ 1000$

13. $\left(1 + \dfrac{3}{100}\right)^2 = 1.0609$

$\therefore$ Effective annual rate = 6.09%

14. Let cash price be ₹P

$\therefore 26.56 = P + \dfrac{P \times 12.5 \times 0.5}{100}$

$\Rightarrow P \approx ₹\ 25.$

15. Let R be the rate of interest.

$\text{Discount} = \dfrac{9000 \times R}{100}$

$\text{Interest} = \dfrac{3000 \times R \times T}{100}$

By the given information:

$\dfrac{9000 \times R}{100} = \dfrac{3000 \times R \times T}{100}$

$\Rightarrow T = \dfrac{1}{3}$ years = 4 months.

16. Let N be the number of years

$\therefore R\% = \sqrt{N \times 12}$

$\Rightarrow 48 = \dfrac{900 \times \sqrt{12N} \times N}{100}$

Squaring both sides

$N^2 = \dfrac{48 \times 48}{9 \times 9 \times 12} = \dfrac{16 \times 4}{9 \times 3} = \dfrac{64}{27}$

$\therefore R = \sqrt{12 \times \dfrac{4}{3}} = 4\%$

$\therefore 48 = \dfrac{P \times 4 \times 2}{100}$

$\Rightarrow P = ₹\ 600$

17. Let first discount be x%

$\therefore \dfrac{(100-20)}{100} \times \dfrac{100-x}{100} \times 2500 = 1500$

$\Rightarrow \dfrac{80}{100} \times \dfrac{100-x}{100} \times 2500 = 1500$

$\Rightarrow 100 - x = \dfrac{1500 \times 100 \times 100}{80 \times 2500}$

$\Rightarrow 100 - x = 75;$

$\Rightarrow x = 25$

Alternatively,

Successive discount $= \left(a + b - \dfrac{ab}{100}\right)\%$

$\therefore \left(\dfrac{2500-1500}{2500}\right) \times 100 = a + 20 - \dfrac{20a}{100}$

$\Rightarrow \dfrac{1000}{2500} \times 100 = a + 20 - \dfrac{a}{5}$

$\Rightarrow 40 - 20 = a - 0.2a$

$\Rightarrow a = \dfrac{20}{0.8} = 25\%$

18. Let there be n number of balls in each packet.

$\therefore 6n \times 8 = CP + 96$

$\Rightarrow 48n = CP + 96$ ...(i)

Also, $6(n-1) \times 9 = CP + 90$

$\Rightarrow 54n - 54 = CP + 90$

$\Rightarrow 54n = CP + 144$ ...(ii)

Solving equations (i) and (ii), we get,

$n = 8$

$\therefore CP = 48 \times 8 - 96 = 384 - 96 = ₹\ 288$

Each packet cost = ₹48.

19. CP of the merchant who calculates his % profit on

$CP = \dfrac{3760}{1.175} = ₹\ 3200$

$\therefore$ His profit $= 0.175 \times 3760 = ₹\ 560$

Profit of the merchant who calculates his % profit on SP

$= 0.175 \times 3760 = ₹\ 658$

$\therefore$ Difference in profit = 658 − 560 = ₹ 98.

20. Let CP be ₹ x

$\therefore$ MP = 1.8x

SP of 15 articles = $12 \times 1.8x = 21.6x$

$\therefore$ Discount = $0.9 \times 21.6x = 19.44x$

$\therefore$ Profit on 15 articles = $(19.44 - 15)x = 4.44x$

$\therefore$ Profit % = $\dfrac{4.44x}{15x} \times 100 = 29.6\%.$

21. C.P. of goods for the trader = ₹ 22520

$\text{S.P.} = \dfrac{125 \times 22520}{100} = ₹\ 28150$

$\therefore$ Profit = ₹ 5630

Now, $5630 = 0.0536 \times 5000 + 0.085 \times 10000 + \dfrac{x}{100} \times 7520$

$\Rightarrow 5630 = 268 + 850 + 75.2x$

$\Rightarrow 75.2x = ₹\ 4512,$

Which is nothing but profit from sale of third lot.

22. A's Monthly Equivalent Rent = 22 × 4

B's Monthly Equivalent Rent = 16 × 8

C's Monthly Equivalent Rent = 32 × 6

$\dfrac{\text{B's Monthly Equivalent Rent}}{\text{C's Monthly Equivalent Rent}} = \dfrac{\text{Rent paid by B}}{\text{Rent paid by C}}$

Rent paid by $B = \dfrac{16 \times 8}{32 \times 6} \times 600 = ₹\ 400$

Similarly, rent paid by A

$= \dfrac{22 \times 4 \times 600}{32 \times 6} = ₹\ 275$

$\therefore$ Total rent = 400 + 600 + 275 = ₹1275

23. Rate in which the rent is to be divided = 4 : 8 : 12

$\therefore$ Akbar's share of rent

$= \dfrac{8}{24} \times 6000$

$= ₹\ 2000.$

24. 85% of the total profit is divided in the ratio 2 : 10 : 5.

85% of total profit = 2x + 10x + 5x = 17x

$\text{Total profit} = \dfrac{17x}{85\%} = 20x$

$\therefore$ Share of Ram in the profit

= 2x + 15% of 20x

= 2x + 3x = 5x

Share of Shyam in the profit = 10x

Share of Ghanshyam in the profit = 5x

Now, $(10x + 5x) - 5x = 500$

$\Rightarrow 10x = 500$

$\Rightarrow x = 50$

$\therefore$ Total profit = 20 × 50 = ₹ 1000.

25. Let capital invested be C.

Monthly Equivalent Investment of

$x = \dfrac{C}{3} \times \dfrac{T}{4}$

Monthly Equivalent Investment of

$y = \dfrac{C}{4} \times \dfrac{T}{2}$

Monthly Equivalent Investment of

$z = \dfrac{5C}{12} \times T$

$\therefore$ Profit is to be shared in the ratio

$\dfrac{CT}{12} : \dfrac{CT}{8} : \dfrac{5CT}{12}$

i.e., 2 : 3 : 10

$\therefore$ Share of x in the profit

$= \dfrac{2}{15} \times 3600 = ₹\ 480$

EXERCISE – II

MCQ TYPE QUESTIONS

1. $A = P\left[1 + \dfrac{r}{100}\right]^t$

10 million = 5 million $\left[1 + \dfrac{20}{100}\right]^t$

$2 = \left[1 + \dfrac{1}{5}\right]^t$

$2 = \left(\dfrac{6}{5}\right)^t = \log 2 = t \log\left(\dfrac{6}{5}\right)$

$0.301 = t\ 0.0791$

$t = \dfrac{0.301}{0.0791} = 3.8$ years

$\therefore$ Hence around 3–4 years would it take to double the growth rate

2. Let number of people in country y = 100

So, number of people in country x = 300

Total number of people taller than 6ft in both the countries $= 300 \times \dfrac{1}{100} + 100 \times \dfrac{2}{100} = 5$

% of people taller than 6ft in both the countries

$= \dfrac{5}{400} \times 100 = 1.25\%.$

4. Per ₹ 100 final value ₹ 107

$\Rightarrow$ per $\frac{100}{50}$ Dollars final value $\frac{107}{60}$

for 100 dollars ____?

$$= \frac{100 \times 50}{100} \times \frac{107}{60} = 89.16$$

Descreased by 11%

6.

	X	Y
Supply	60%	40%
Reliable	96%	72%
Overall	0.576	0.288

$$\therefore \quad P(x) = \frac{0.288}{0.576 + 0.288} = 0.334$$

7.

P	Q
40%	60%
–6%	+6%
+15%	–15%
49%	51%

$\therefore$ 2% = 2 $\Rightarrow$ 100% = 100

NUMERICAL TYPE QUESTION

1. 90% of area of sheet = Cross sectional area of cone

$\Rightarrow 0.9 \times \pi \times 30 \times 30 = \pi \times r_1 \times 30$

($\because$ Slant height of cone (l) = 30)

$\Rightarrow$ 27 cm = r_1

$\therefore$ Height of the cone = $\sqrt{30^2 - 27^2}$ = 13.08 cm

Hence ratio of radius and height is 2.06.

■■

CHAPTER 3

Time and Work

TIME, SPEED AND DISTANCE

Speed.

Speed of a body is defined as the distance covered by it in unit time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Average Speed: If a body travels $d_1, d_2, d_3 \ldots d_n$ distances, with speeds $s_1, s_2, s_3 \ldots s_n$ in time $t_1, t_2, t_3 \ldots t_n$ respectively, then average speed of the body through the total distance is given by

$$\text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{d_1 + d_2 + d_3 + \ldots d_n}{t_1 + t_2 + t_3 + \ldots t_n}$$

$$= \frac{s_1t_1 + s_2t_2 + s_3t_3 + \ldots + s_nt_n}{t_1 + t_2 + t_3 + \ldots + t_n}$$

$$= \frac{d_1 + d_2 + d_3 + \ldots + d_n}{\frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + \ldots + \frac{d_n}{s_n}}$$

Some Important Points:

(1) If a body covers the distance d_1 and d_2 at a speed of s_1 and s_2 km/hr, respectively, in time t_1 and d_2, then

Total time taken, $T = t_1 + t_2 = \frac{d_1}{s_1} + \frac{d_2}{s_2}$.

Total distance covered, $d = d_1 + d_2 = s_1t_1 + s_2t_2$.

(2) While travelling a certain distance d, if a man changes his speed in the ratio m: n, then ratio of time taken becomes n : m.

(3) If a certain distance (d), say from A to B, is covered at 'a' km/hr and same distance is covered again say from B to A in 'b' km/hr, then average speed during the whole journey is given by

$$\text{Average speed} = \left(\frac{2ab}{a+b}\right) \frac{\text{km}}{\text{hr}} \ldots.$$

(which is harmonic mean of a and b)

Also, if t_1 and t_2 is taken to travel from A to B and B to A respectively, then distance 'd' from A to B is given by

$$d = (t_1 + t_2)\left(\frac{ab}{a+b}\right); \quad d = (t_1 - t_2)\left(\frac{ab}{a-b}\right);$$

$$d = (a - b)\left(\frac{t_1t_2}{t_1 - t_2}\right)$$

(4) If a body travels a distance 'd' from A to B with speed 'a' in time t_1 and travels back from B to A i.e., same distance with $\frac{m}{n}$ of the usual speed 'a', then change in time taken to cover the same distance is given by

$$\text{Change in time} = \left(\frac{n}{m} - 1\right) \times t_1; \text{ for } n > m$$

$$= \left(1 - \frac{n}{m}\right) \times t_1; \text{ for } m > n$$

(5) If two persons A and B start at the same time in opposite directions from two points and meet at any point in 'a' and 'b' hr, respectively after having met, then,

$$\frac{\text{A's Speed}}{\text{B's Speed}} = \frac{\sqrt{b}}{\sqrt{a}}$$

Relative speed.

(1) Time taken by a moving object 'x' meters long in passing a stationary object of negligible length from the time they meet is same as the time taken by the moving object to cover 'x' meters with its own speed.

(2) Time taken by a moving object 'x' meters long in passing a stationary object 'y' meters long from the time they meet, is same as the time taken by the moving object to cover 'x + y' meters with its own speed.

(3) If two objects of length 'x' and 'y' meters move in the same direction at 'a' and 'b' m/s, then time taken to cross each other from the time they meet

$$= \frac{\text{Sum of their length}}{\text{Relative speed}}$$

i.e., $\frac{x+y}{a-b}$ if a > b or else, $\frac{x+y}{b-a}$

(4) If two objects of length 'x' and 'y' meters, move in the opposite direction at 'a' and 'b' m/s, then time taken to cross each other from the time they meet

$$= \frac{\text{Sum of their length}}{\text{Relative speed}} = \frac{x+y}{a+b}$$

(5) If speed of a boat in still water is x km/hr and speed of the stream is y km/hr, then
Relative speed while travelling with the stream i.e., speed downstream = (x + y) km/hr.
Relative speed while travelling against the stream i.e., speed upstream = (x – y) km/hr.

(6) Also speed of the boat in still water = $\frac{1}{2}$ (Speed with stream + Speed against stream)

Speed of the river = $\frac{1}{2}$ (Speed with stream – Speed against stream)

RACES

Some Important Points:

(1) 'A gives B a start of x meters', implies that, if distance between starting point and finishing point is L meters, then A covers L meters while B covers L – x metres.
e.g., In a 100 metre race. A gives B a start of 10 metres means, while A runs 100 metres, B runs 100 –10 = 90 metres.

(2) *'A beats B by x metres'*, implies that, if distance between starting point and finishing point is L meters, then A wins the race by covering L metres, while B covers L – x metres only.

(3) 'A gives B a start of t seconds', implies that A starts the race t seconds after B starts from the starting point.

(4) *'A beats B by t seconds'*, implies that, A and B start together from the starting point, but A reaches the finishing point t seconds before B finishes.
Note: (*iii*) and (*iv*) both imply that B takes t seconds more than A to finish the distance.

(5) A beats B by 'x' metres or 't' seconds means, B runs 'x' metres in 't' seconds.

(6) Winner's distance = Length of the race.

(7) Distance covered by loser = Winner's distance – (Beat distance + Start distance)

(8) Time taken by winner = Time taken by loser – (Beat time + Start time)

(9) $\frac{\text{Winner's Time}}{\text{Loser's Distance}} = \frac{\text{Loser's time}}{\text{Winner's distance}}$

$= \frac{\text{Beat time + Start time}}{\text{Beat distance + Start distance}}$

(10) If a race ends in a dead heat, then beat time = 0 and beat distance = 0.

(11) Two persons staring at the same time and from the same point along a circular path will be together again for the first time. When faster gains one complete round over the other, then time taken by faster person to complete one round over the other

$= \frac{\text{Length of race course}}{\text{Relative Speed}}$

(12) Two persons, starting at the same time from the same point along a circular path, will be together again for the first time at the same starting point, at a time which is LCM of the time taken by each to complete a round.

(13) Three persons, staring at the same time and from the same point along a circular path, will be together for the first time after the start at a time which is equal to LCM of the time taken by the fastest to gain a complete round over each of the other two.

(14) A overtakes B $\frac{1}{n}$ th of x^{th} round means, when A has completed $\left(x - \frac{1}{n}\right)$ rounds, B has completed $\left[(x-1) - \frac{1}{n}\right]$ round.

Also, $\frac{\text{A's Speed}}{\text{B's Speed}}$

$= \frac{\text{Rounds completed by A in a given time}}{\text{Rounds completed by B in the same time}}$

e.g., A overtakes B in the middle of the 4^{th} round implies, when A has completed $3\frac{1}{2}$ rounds, B has completed $2\frac{1}{2}$ rounds. $2\ \frac{\text{A's Speed}}{\text{B's Speed}} = \frac{3\frac{1}{2}}{2\frac{1}{2}}$

WORK

Some Important Points:

(1) If a can do a work in 'a' number of days, then in one day $\frac{1^{th}}{a}$ work is done.
Conversely, if a man does $\frac{1^{th}}{a}$ of a work in 1 day, then he can complete the work in $1 \div \frac{1}{a}$ = a days.

(2) If A is 'x' time as good a workman as B, then he will take $\frac{1^{th}}{x}$ of the time taken by B to do the same work.

(3) If A and B can do a piece of work in 'x' and 'y' days respectively. then working together, they will take $\frac{xy}{x+y}$ days to finish the work and in one day. They finish $\left(\frac{x+y}{xy}\right)^{th}$ part of the work.

(4) To compare the work done by different people, first find the amount of work each can do in the same time.

(5) If number of men to do a job is changed in the ratio a : b, then time required to do the work will be in the ratio b : a, assuming amount of work done by each of them in the given time is the same, or they are identical.

(6) If two men A and B together can finish a job in 'x' days and if A working alone takes 'a' days more than A and B working together and B working alone takes 'b' days more than A and B working together, then $x = \sqrt{ab}$.

(7) To do a piece of work, the number of men employed and the number of days required to do the work are in inverse proportion. Also, the number of men employed and the hours worked per day are in inverse proportion.

PIPES AND CISTERNS

A pipe connected with a cistern is called an *inlet*, if it fills the cistern.
A pipe connected with a cistern is called an *outlet*, if it empties the cistern.
filling or emptying a cistern can be considered as work done.

Some Important Points:

(1) If an inlet pipe fills a cistern in 'a' hours, then $\frac{1^{th}}{a}$ part is emptied in 1 hour.
Similary, if an outlet pipe empties a cistern in 'a' hours, then $\frac{1^{th}}{a}$ part is emptied in 1 hour.

(2) If pipe A is 'x' times bigger than pipe B, then pipe A will take $\frac{1^{th}}{x}$ of the time taken by pipe B to fill the cistern.

(3) If A and B fill a cistern in 'm' and 'n' hours, respectively, then together they will take $\frac{mn}{m+n}$ hours to fill the cistern and in one hour $\frac{m+n^{th}}{mn}$ part of the cistern will be filled.
Similarly, 'A' and 'B' empty a cistern in 'm' and 'n' hours respectively, then, together they will take $\frac{mn}{m+n}^{th}$ hours to empty the cistern and in one hour $\frac{m+n^{th}}{mn}$ part of the cistern will be empty.

(4) If an inlet pipe fills a cistern in 'm' hours and an outlet pipe empties the cistern in 'n' hours, then the net part filled in 1 hour when both the pipes are opened is $\left(\frac{1}{m} - \frac{1}{n}\right)$ i.e., $\frac{n-m}{mn}$ and the cistern will get filled in $\left(\frac{mn}{n-m}\right)$ hours.
For the cistern to get filled, m < n. If m > n, the cistern will never get filled.
In general,
Net part filled of a cistern = (Sum of work done by inlets) – (Sum of work done by outlets)

(5) If an inlet pipe fills a cistern in 'a' minutes, takes 'x' minutes longer to fill the cistern due to a leak in the cistern, then the time in which the leak will empty the cistern is given by $a \times \left(1 + \frac{a}{x}\right)$.

(6) If two pipes A and B can fill a cistern in 'x' minutes and if A alone can fill it in 'a' minutes more than 'x' minutes and B alone can fill it in 'b' minutes more than 'x' minutes, then $x = \sqrt{ab}$.

EXERCISE – I

MCQ TYPE QUESTIONS

1. A barrel full of beer has 2 taps, one midway which draws a litre in 6 minutes and the other at the bottom which draws a litre in 4 minutes. The lower tap is normally used after the level of beer in the barrel is lower than midway. The capacity of the barrel is 36 litres. A new assistant opens the lower tap when the barrel is full and draws out some beer. As a result the lower tap has been used 24 minutes before the usual time. For how long was the beer drawn out by the new assistant?
(*a*) 10 mins. (*b*) 8 mins.
(*c*) 16 mins. (*d*) 32 mins.

2. Pipe A can fill a cistern in 36 minutes and B in 48 minutes. If both the pipes are opened together, when should pipe B be closed so that the cistern may be just full in 24 minutes?
(*a*) 8 mins. (*b*) 9 mins.
(*c*) 12 mins. (*d*) 16 mins.

3. Pipes P, Q and R are attached to a tank and each can act as either an inlet or outlet pipe. Pipes P, Q and R respectively take 8, 10 and 12 hours to fill the empty tank or empty the full tank. In the first hour, pipes P and R work as inlet and Q work as outlet. In the second hour, pipes P and Q work as inlet and pipe R as outlet. In the third hour pipes Q and R work as inlet and pipe P as outlet and the process goes on like this. When will the cistern be filled?
(*a*) In the 8th hour. (*b*) In the 9th hour.
(*c*) In the 10th hour. (*d*) In the 11th hour.

4. There are 2 inlet pipes and an outlet pipe. The efficiency of one of the inlet pipes is double than that of the other. Also, the efficiency of the outlet pipe is half that of the lesser efficient inlet pipe. The empty tank gets filled in 16 hours when all the pipes are opened. How many hours will be taken to fill the empty tank when the 'lesser' efficiency inlet pipe is plugged and the rest kept opened?
(*a*) $33\frac{1}{3}$ hours (*b*) $26\frac{2}{3}$ hours
(*c*) 100 hours (*d*) 80 hours

5. A train overtakes 2 persons walking at 2 km/hr and 4 km/hr respectively in the same direction and completely passes them in 9 seconds and 10 seconds. Length of the train and its speed in km/hr respectively are
(a) 75 m, 18 km/hr (b) 80 m, 21 km/hr
(c) 60 m, 20 km/hr (d) 50 m, 22 km/hr

6. A policeman goes after a thief who has a 100 m start. The policeman runs 1 km. in 6 mins. and the thief 1 km in 10 mins. How far did the thief go before he was overtaken?
(a) 100 m (b) 125 m
(c) 150 m (d) 250 m

7. Everyday Sachin exercises for $2\frac{1}{2}$ hours. He exercises by walking, jogging and running. His speeds while walking, jogging and running in the ratio 1 : 2 : 4. Which of the following statements are true?
I. If he spends equal time on the three activities, then he runs twice the distance that he jogs.
II. If he runs at 8 km per hour and distance covered by running and jogging together is 12 km, then he walks a distance of 1 km in 1 hour.
III. If the time for which he walks, jogs and runs is in the ratio 2 : 2 : 1 and he covers a total distance of 10 km, then his speeds while walking, jogging and running are 2 km/hr, 4 km/hr and 8 km/hr res-pectively.
Codes:
(a) I and II (b) II and III
(c) I and III (d) I, II and III

8. A monkey climbing up a greased pole ascends 10 metres and slips down 2 metres in alternate minutes. If the pole is 64 metres high, how long will it take him to reach the top?
(a) 16 min
(b) 14 min and 48 sec
(c) 12 min
(d) 14 min

9. A train travelling at 10 m/sec from A to B at 7 a.m. meets a train leaving B at 7:20 a.m. and coming to A at a speed $\frac{1}{3}$ times faster than the first train. If the distance from A to B is 68 km. then, at what distance from A will the two trains meet?
(a) 72 km (b) 36 km
(c) 60 km (d) 50 km

10. Two trains A and B start from stations X and Y towards each other. B leaves station Y half an hour after train A leaves station X. Two hours after train A has started, the distance between trains A and B is $\frac{19}{30}$ th of the distance between stations X and Y. How much time would it take each train (A and B) to cover the distance X to Y, if train A reaches half an hour later to its destination as compared to B?
(a) 8 hrs, 6 hrs (b) 5 hrs, 4 hrs
(c) 10 hrs, 9 hrs (d) 9 hrs, 8 hrs

11. Two men, A and B, run a 4 km race on a circular course of $\frac{1}{4}$ km. If their speeds are in the ratio of 5 : 4, how often does the winner pass the other?
(a) Once (b) Twice
(c) Thrice (d) Four times

12. A and B start from the same point to run in opposite directions round a circular path 550 yards in length, A giving B a start of 100 yards. They pass each other when A has run 250 yards. Who will come first to the starting point and at what distance will they be apart?
(a) A, 20 yards (b) B, 10 yards
(c) B, 20 yards (d) A, 10 yards

13. Two runners run in the same direction along a circular track 2 km long. The faster runner overtakes the slower one every 30 minutes. What are their speeds, if faster one completes a round one minute sooner than the other?
(a) 24 km/hr and 20 km/hr
(b) 44 km/hr and 28 km/hr
(c) 20 km/hr and 18 km/hr
(d) 20 km/hr and 16 km/hr

14. To do certain work alone 'A' takes 4 hrs, 'B' takes 5 hrs and 'C' takes 6 hrs. How long would A and B together take to do a work which C can do in $4\frac{1}{2}$ hrs.?
(a) 2 hrs 10 mins. (b) 2 hrs 30 mins.
(c) 1 hr 20 mins. (d) 1 hr 40 mins.

15. A can do piece of work in 8 days which B and C can do in 12 and 16 days respectively. If they work on alternate days, then which of the following is true?
(a) If A started the work on the first day followed by B and c, then the work will be finished on the 10th day.
(b) If B started the work on the first day followed by C and A, then the work will be finished on the 11th day.
(c) If C started the work on the first day followed by A and B, then the work will be finished on the 11th day.
(d) None of these

NUMERICAL TYPE QUESTIONS

1. Meera paints walls for a living. She sometimes has Ashok to work with her. Meera takes 2 days to paint a 10 ft by 10 ft wall. Ashok takes 3 days to do the same job. Meera has to pay Ashok for his work on a per day basis, so she'd rather do without him. Meera has recently got a contract to paint the walls and ceiling of a room of dimensions 10 ft by 15 ft and a height of 10 ft. She had promised to do the job within 9 days. The _____ number of days does she need to employ Ashok?

2. 3 pipes can fill a reservoir in 10, 15 and 20 hours respectively. If the three taps are opened one after another in the given order, with a certain fixed time gap between them, the reservoir fills in 5 hours. The time gap will be _____ hr.

3. Two pipes A and B can fill a tank in 8 hours. If only pipe A is open, then it would take 4 hours longer to fill the tank. The _____ hours it would take if only pipe B is open.

Directions (Q.4 – 5): *Refer to the data below and answer the questions that follow.*
The boiler tank in a chemical factory holds 10^5 litres. 5 tanks each having one-fifth the capacity of the boiler tank fill in 'hard water' at same rates in the boiler tank in 2 hours. The outlet of the two of smaller tanks work as inlet pipes and other two work as outlet and the fifth tank fill in the main 'boiler' at half its efficiency.

4. The number of hours taken to fill in the main boiler when all the pipes are opened at once is _____

5. Only three pipes are working, two at their full efficiency and the third one with half of its efficiency and all three are acting as an inlet. If all the three are opened alternately with the latter pipe being first to be opened to fill the boiler such that each smaller tank fills 'hard water' in the main boiler for equal time, then number of hours taken to fill the boiler to half its capacity is_____

6. A contract is to be completed in 46 days and 117 men were set to work, each working 8 hours a day. After 33 days, 4/7 of the work is completed. The _____ number of additional men may be employed so that the work may be completed in time, each man now working 9 hours a day?

7. A train travelling at 42 km/hr. passes a cyclist going in the same direction in 9 secs.; if cyclist had been going in the opposite direction, the train would have passed him in 5 secs. Length of the train is_____ metres

8. A man rows upstream 13 km. and downstream 28 km. taking 5 hrs. each time. Velocity of the current in km/hr_____ is

Directions (Q. 9 – 11): *Refer to the data below and answer the questions that follow.*
The variation in the speed of a car on a particular day at the respective times is shown in the table below:

s(km/hr)	t(hr)
0	11.00 am
40	11.30 am
50	1.00 pm
85	1.30 pm
10	3.30 pm
10	4.30 pm

9. The distance travelled by the car from 11 a.m. to 1 p.m. is _____km

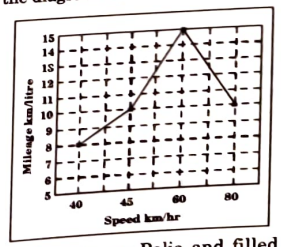
10. If car maintains the speed it has at 4.30 p.m. then at _____ time will the car cover the same distance as it had covered from 11.00 a.m. to 1.00 p.m.?

11. Average speed of the car from 11.00 a.m. to 4.30 p.m. is_____ kmph

12. A cheetah chase a deer which is 100 m ahead. The time in which the deer takes 10 leaps the cheetah takes only 6 leaps. In one leap, the deer covers 1 m while the cheetah covers 2 m. In _____ leaps would the cheetah catch up the deer.

13. A hare pursued by a hound who is 50 of her own leaps before him. When the hare takes 4 leaps, the hound takes 3. In one leap, the hare goes $1\frac{3}{4}$ metres and the hound $2\frac{3}{4}$ metres. In how _____ leaps will the hound overtake the hare.

Directions (Q. 14 – 15): *Refer to the graph below and answer the questions that follow.*
The engine of the new car Palio brought by Fiat in the market follows certain mileage characteristics as shown in the diagram below:



Vijay bought the new car Palio and filled exactly 10 litres in his car. He goes at a speed of 40 km/hr for first 20 km and then at a speed of 60 km/hr for next 60 km and driving at a constant speed reaches his destination which was 115 km away from his starting point.

14. The time in hours taken by Vijay if he reaches his destination in minimum possible time and finishes all the fuel.

15. In the above question, if Vijay had travelled at a constant average speed rather than changing his speed, taking the same minimum time, then the approximate saving in the fuel consumed is _____ litres

16. In a 2 km race on a circular course of $\frac{1}{4}$ of a km. A overlaps B in the middle of his 6th round. Then_____ distance (in km) will A win at the same rate of running

17. A runs $1\frac{3}{8}$ times faster than B. A gives B a start of 120 metres. _____ metre must the winning post be so that it may be a dead heat.

18. The work done by a man, a woman and a child is in the ratio of 3 : 2 : 1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount of ₹ 780, which is divided in the ratio of work done by the men, women and children. The wages of 15 men, 21 women and 30 children for 2 weeks is _____

19. A can do a work in 6 days and B in 8 days. With the help of a boy, the three complete the work in 3 days and get ₹ 200. The boy's share in rupees, if the money is distributed in the ratio of the work done is _____

20. A can finish a job in 12 days and B when working at twice his efficiency finishes a job in 9 days. The _____ number of days will they take if they work for two days alternately working at their standard rate given that A starts first.

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O and P, in that order. The cumulative distances travelled and the corresponding electricity consumption are shown in the table below.

Stretch	Comulative distance(km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

(a) M (b) N
(c) O (d) P

2014

2. It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes?

(a) 4 times the draining rate
(b) 3 times the draining rate
(c) 2.5 times the draining rate
(d) 2 times the draining rate

2013

3. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

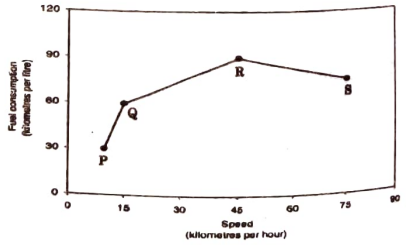
(a) 36
(b) 30
(c) 24
(d) 18

4. A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

(a) 30 (b) 36
(c) 40 (d) 24

2011

5. The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometers)	Average speed (kilometers per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

(a) P (b) Q
(c) R (d) S

2010

6. 5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(a) 20 days (b) 18 days
(c) 16 days (d) 15 days

NUMERICAL TYPE QUESTIONS

2015

1. A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5 metre per leap respectively, what distance in meters will the tiger have a run before it catches the deer?

2014

2. A man can row at 8 km per hour in still water. If it takes him thrice as long to row upstream, as to row downstream, then find the stream velocity in km per hour.

3. A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (c) 2. (d) 3. (c) 4. (b) 5. (d) 6. (c) 7. (c) 8. (b) 9. (b) 10. (c)
11. (c) 12. (d) 13. (a) 14. (d) 15. (c)

Numerical Type Questions

1. 6 2. $\frac{1}{2}$ 3. 16 4. 20 5. 6 6. 81 7. 75 8. 1.5 9. 77.5
10. 12.15 a.m. 11. 39.3 12. 300 13. 210 14. 2 15. $2\frac{1}{3}$ 16. $\frac{4}{11}$ 17. 440
18. 1170 19. 25 20. 14

EXERCISE – II

MCQ Type Questions

1. (b) 2. (a) 3. (c) 4. (c) 5. (a) 6. (d)

Numerical Type Questions

1. (800) 2. (4) 3. (560 to 560)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. Top tap is operational till 18 litres is drawn out.

$\therefore$ Time after which the lower tap is usually open
$= 18 \times 6 = 108$ minutes.

$\therefore$ Time after which it is open now
$= 108 - 24 = 84$ minutes.

$\therefore$ Litres drawn $= \frac{84}{6} = 14$ litres

$\therefore$ $18 - 14 = 4$ litres

were drawn by the new assistant.

$\therefore$ Time $= 4 \times 4 = 16$ minutes.

Alternatively

Upper tap was not used for 24 minutes.

$\therefore$ Litres drawn $= \frac{24}{6} = 4$ litres.

$\therefore$ Beer was drawn for $4 \times 4 = 16$ minutes.

2. Let pipe B be turned-off after x minutes.

Part filled in x minutes $= x\left(\frac{1}{36} + \frac{1}{48}\right) = \frac{7x}{144}$.

Part filled in $24 - x$ minutes $= \frac{24 - x}{36}$

$\therefore$ $\frac{7x}{144} + \frac{24 - x}{36} = 1$

$\Rightarrow$ $\frac{7x + 96 - 4x}{144} = 1$

$\Rightarrow$ $x = 16$ minutes

3. In a cycle of 3 hours : pipes P, Q and R are working as inlet pipes for 2 hours each and they are working as outlet pipes for an hour each. So part of tank filled in 3 hours.

$$= 2x\left(\frac{1}{8} + \frac{1}{10} + \frac{1}{12}\right) - \left(\frac{1}{8} + \frac{1}{10} + \frac{1}{12}\right)$$

$$= \frac{1}{8} + \frac{1}{10} + \frac{1}{12} = \frac{15 + 12 + 10}{120} = \frac{37}{120}$$

Hence they will take $\frac{120 \times 3}{37} = 9\frac{27}{37}$ hours.

Thus, tank will be filled in 10th hour.

4. Let number of hours taken by the outlet and the 2 inlet pipes be 4x, 2x and x respectively. So in 1 hour the part of empty tank filled is

$$\frac{1}{x} + \frac{1}{2x} - \frac{1}{4x} = \frac{1}{16}$$

$$\frac{5}{4x} = \frac{1}{16}$$

$\Rightarrow$ $x = 20$

Hence, inlet pipe with higher efficiency fills the tank in 20 hours.

So in one hour, the desired pipes fill

$$\frac{1}{x} - \frac{1}{4x} = \frac{3}{4x} \text{ th of the tank.}$$

Tank gets filled in

$$\frac{4x}{3} = \frac{80}{3} \text{ hours} = 26\frac{2}{3} \text{ hours.}$$

5. In each case, train has to travel its own length, to pass each man.

If x km/hr is speed of the train, then

Length = Relative speed × Time

$$= (x - 2) \times \frac{9}{60 \times 60}$$

$$= (x - 4) \times \frac{10}{60 \times 60}$$

$\therefore$ $9x - 18 = 10x - 40$

$\Rightarrow$ $x = 22$ km/hr.

$\therefore$ Length $= (22 - 2) \times \frac{9}{3600}$

$$= \frac{20 \times 9}{3600} \text{ km} = 50 \text{ m.}$$

6. Policeman gains $= \frac{1}{6} - \frac{1}{10} = \frac{4}{60}$ km/min

$$= 66\frac{2}{3} \text{ m/min.}$$

Hence to gain100 m, time required

$$= \frac{100}{66\frac{2}{3}} = 1\frac{1}{2} \text{ min.}$$

Hence thief has gone ahead by

$$1\frac{1}{2} \times \frac{1}{10} \times 1000 \text{ metres} = 150 \text{ m.}$$

7. I. Ratio of speed = 1 : 2 : 4

If equal time is spent on each of these, then ratio of distance = 1 : 2 : 4

So, Sachin runs twice the distance that he jogs.

Thus, I is true.

II. He runs at 8 km/hr and therefore walks at 2 km/hr.

Hence II is not true.

III. Ratio of speed= 1 : 2 : 4

Ratio of time = 2 : 2 : 1

Ratio of distance
$= 2 \times 1 : 2 \times 2 : 4 \times 1 = 2 : 4 : 4$

If Sachin covers a total distance of 10 km, then he

Walks for $\frac{2}{2+4+4} \times 10 = 2$ km,

Joggs for $\frac{4}{2+4+4} \times 10 = 4$ km.

Runs for $\frac{4}{2+4+4} \times 10 = 4$ km.

Total time taken is $2\frac{1}{2}$ hours $= \frac{5}{2}$ hours.

Thus, he walks for $\frac{2}{2+2+1} \times \frac{5}{2} = 1$ hour.

jogs for $\frac{2}{2+2+1} \times \frac{5}{2} = 1$ hour.

runs for $\frac{1}{2+2+1} \times \frac{5}{2} = \frac{1}{2}$ hour.

So speed while,

Walking $= \frac{2}{1} = 2$ km/hr

Jogging $= \frac{4}{1} = 4$ km/hr

Running $= \frac{4}{\frac{1}{2}} = 8$ km/hr.

Thus, III is true.

8. In 1 minute, monkey ascends 10 metres but he takes 1 minute to slip down 2 metres. Thus, at the end of 2 minutes, net ascending of the monkey is $=10 - 2 = 8$ metres.

Thus, to have a net ascending of 8 metres, process of ascending and then slipping happens once. So, to cover 64 metres, above process is repeated $\frac{64}{8}$ or 8 times.

It is clear that in 8 such happenings, the monkey will slip 7 times, because 8th time, he will ascend to the top. Thus, in climbing 7 times and slipping 7 times, he covers (7×8) or 56 metres.

Time taken to cover 56 metres

$$= \frac{56 \times 2}{8} = 14 \text{ minutes}$$

Remaining distance $= 64 - 56 = 8$ metres

Time taken to ascend 8 metres

$$= \frac{8}{10} \text{ min} = \frac{4}{5} \text{ min.}$$

Total time taken = 14 minutes $+ \frac{4}{5}$ min

$= 14$ mins. 48 sec.

9. If they meet t hrs after 7 a.m,

$$10 \times \frac{18}{5} \times t + 10 \times \frac{18}{5} \times \frac{4}{3}\left(t - \frac{1}{3}\right) = 68$$

$\Rightarrow$ $36t + 48t - 16 = 68$

$\Rightarrow$ $t = 1$ hr.

They meet at a distance of

$10 \times 60 \times 60 = 36000$ m

i.e., 36 km from A.

10. Two equations are

$$T_1 - T_2 = \frac{1}{2} + \frac{1}{2} = 1 \qquad ...(i)$$

where T_1 and T_2 are time taken by trains A and B to cover the whole distance

and $\frac{2}{T_1} + \frac{1.5}{T_2} = \frac{11}{30}$ ...(ii)

Solving equations (i) and (ii), we get

$T_1 = 10$ hrs. and $T_2 = 9$ hrs.

11. When A runs 5 rounds, B runs 4 rounds (ratio of speeds)

$\therefore$ A passes B each time A has run 5 rounds or

$5 \times \frac{1}{4} = \frac{5}{4}$ km. $= 1\frac{1}{4}$ km.

$1\frac{1}{4}$ km is contained in 4 km.

3 times.

Hence A passes B thrice.

12. When A has run 250 yards, B should have run

$550 - 250 = 300$ yards.

$\therefore$ B runs $300 - 100 = 200$ yards in the same time taken by A to run 250 yards.

$\therefore$ Ratio of the speeds of A and B
$= 250 : 200 = 5 : 4$

If speed of A is 5x yards per minute, then speed at B = 4x yards/min.

Time taken by B to reach the starting point

$$= \frac{450}{4x} = \frac{112.5}{x} \text{ min.}$$

Time taken by A to reach the starting point

$$= \frac{550}{5x} = \frac{110}{x} \text{ min.}$$

$\therefore$ A wins by $\frac{2.5}{x}$ min

$\therefore$ Distance travelled by B in $\frac{2.5}{x}$ min.

$$= \left(\frac{2.5}{x}\right) \times (4x) = 10 \text{ yards}$$

Hence B is 10 yards behind A.

13. Let n_1 and n_2 be the speeds in km/hr and let n_1 be the greater speed i.e., $n_1 > n_2$.

$\therefore$ $\frac{2}{n_2} = \frac{2}{n_1} + \frac{1}{60}$.

$\Rightarrow$ $n_1 n_2 = 120(n_1 - n_2)$

Faster runner covers $(n_1 - n_2)$ km. more than the slower runner in 1 hour.
Hence to cover one round of 2 km, he will take

$\frac{2}{n_1 - n_2}$ or $\frac{1}{2}$ hour.

$\therefore \quad \frac{2}{n_1 - n_2} = \frac{1}{2}$

$\Rightarrow \quad n_1 - n_2 = 4$

$\Rightarrow \quad n_1 = 4 + n_2$

$\Rightarrow \quad n_2(4 + n_2) = 120 \times 4$

$\Rightarrow \quad n_2^2 + 4n_2 - 480 = 0$

$\Rightarrow \quad n_2 = 20 \text{ or } -24 \text{ (rejected)}$

$\therefore \quad n_1 = 4 + n_2 = 24 \text{ km/hr}$

Hence speeds are 24 km/hr and 20 km/hr

Alternatively

Options may also be substituted to find the result.

14. Work done by A and B in 1 hr

$= \frac{1}{4} + \frac{1}{5} = \frac{9}{20}^{\text{th}}$ part.

$\therefore$ Ratio of work done by (A + B)

and $C = \frac{9}{20} : \frac{1}{6}$

$\therefore$ Ratio of time taken

$= \frac{1}{6} : \frac{9}{20} = 10 : 27$

If C takes 27 hrs. (A + B) will take 10 hrs.

If C takes $4\frac{1}{2}$ hrs. (A + B) will take

$\frac{10 \times 9}{2 \times 27} \cong 1$ hr. 40 mins.

15. I. Work done by A, B and C together in 3 days

$\frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}$ th of the work.

Thus, in 9 days, total work done will be

$3 \times \frac{13}{48} = \frac{39}{48}$ th of the work

$\therefore$ Work remaining $= 1 - \frac{39}{48} = \frac{9}{48}$ th of the work is left of which A will finish $\frac{6}{48}$ th of work on 10th day and B will finish remaining work on 11th day.

Thus, I is false

II. Work done by A, B, C together in 9 days $= \frac{39}{48}$

Work left $= 1 - \frac{39}{48}$

$= \frac{9}{48}$ th of the whole.

On the 10th day B will finish $\frac{1}{12}$ th of the whole.

$\therefore$ Work left $= \frac{9}{48} - \frac{1}{12}$

$= \frac{5}{48}$ th of the whole.

On 11th day C will finish $\frac{1}{16}$ th of the work

$\therefore$ Work left $= \frac{5}{48} - \frac{1}{16}$

$= \frac{2}{48}$ th of the whole

which will be done by A on the 12th day. Thus, work will be finished on the 12th day. Thus, II is false.

III. Work done by A, B and C in 9 days $= \frac{39}{48}$

Work left after 9 days

$= 1 - \frac{39}{48} = \frac{9}{48}$ th of the whole.

On the tenth day $\frac{1}{16}$ th will be finished by C.

$\therefore$ Work left $= \frac{9}{48} - \frac{1}{16}$

$= \frac{6}{48}$ th of the whole,

which will be finished by A on the eleventh day. Thus, III is true.

NUMERICAL TYPE QUESTIONS

1. Total area to be painted

$= 10 \times 15$ (ceiling) $+ 2 \times 10 \times 10$ (wall) $+ 2 \times 10 \times 15$ (wall)

$= 650$ sq.ft

In one day Meera can paint

$10 \times \frac{10}{2} = 50$ sq.ft

Ashok can paint

$10 \times \frac{10}{3} = \frac{100}{3}$ sq.ft,

So, together they can paint $\frac{250}{3}$ sq.ft per day.

Let number of days Ashok works be x.

Work done in 9 days $= 50(9 - x) + \frac{250}{3}x = 650$

$\therefore \quad 450 - 50x + \frac{250x}{3} = 650$

$\Rightarrow \quad \frac{100}{3}x = 200$

$\Rightarrow \quad x = 6$ days.

2. Let fixed interval be x

$\therefore \quad \frac{5}{10} + \frac{5-x}{15} + \frac{5-2x}{20} = 1$

$\Rightarrow \quad (5)6 + (5 - x)4 + (5 - 2x)3 = 60$

$\Rightarrow \quad 30 + 20 - 4x + 15 - 6x = 60$

$\Rightarrow \quad x = \frac{1}{2}$ hour.

3. Let pipe B alone take b hours longer than if both A and B were open.

$8 = \sqrt{4 \times x}$

$64 = 4 \times x$

$\Rightarrow \quad x = 16$ hours.

Alternatively

Part filled if both A and B are open $= \frac{1}{8}$

Part filled if only A is open $= \frac{1}{12}$

$\therefore$ Part filled if only B is open $= \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$

$\therefore$ B alone would take 24 hours, i.e., B alone would take 24 – 8 = 16 hours longer to fill the tank.

4. Since 5 tanks of same efficiency fill the boiler in 2 hours each tank individually takes

$2 \times 5 = 10$ hours to fill it.

Tank with half the efficiency will take

$10 \times 2 = 20$ hours.

In one hour, fraction of boiler getting filled when 4 fully efficient pipes with two as inlet and two as outlet and 5th acts at 50% efficiency as an inlet is

$\frac{1}{10} + \frac{1}{10} + \frac{1}{20} - \frac{1}{10} - \frac{1}{10} = \frac{1}{20}$

Hence it will take 20 hours to fill the boiler tank.

5. Here fraction of tank filled in 1 hour

$\frac{1}{10} + \frac{1}{10} + \frac{1}{20} = \frac{5}{20} = \frac{1}{4}$ th

of the tank.If they are opened alternately, then tank gets filled in $4 \times 3 = 12$ hours

Hence half of the tank gets filled in $\frac{12}{2} = 6$ hours.

6. 4/7 of work is completed.

Remaining work $= 1 - \frac{4}{7} = \frac{3}{7}$

Remaining period $= 46 - 33 = 13$ days

Less work, less men(direct proportion)

Less days, more men(Indirect proportion)

More hours/day, less men(Indirect proportion)

work	4/7 : 3/7
Days	13 : 33 :: 117 : x
hrs/day	9 : 8

$\frac{4}{7} \times 13 \times 9 \times x = \left(\frac{3}{7}\right) 33 \times 8 \times 117$

$x = (3 \times 33 \times 8 \times 117) / (4 \times 13 \times 9)$

$= 198$ men

So, additional men to be employed

$= 198 - 117 = 81$

7. Let L is length of train in km and x is speed of the cyclist, Then

(i) $\frac{3600}{9} L = 42 - x = 400L$

(ii) $\frac{3600}{5} L = 42 + x = 720 L$

On adding, we get

$1120 L = 84$

$\Rightarrow \quad L = \frac{84 \times 1000}{1120} = 75$m

8. $\frac{13}{\text{Speed upstream}} = 5$

and $\frac{28}{\text{Speed downstream}} = 5$

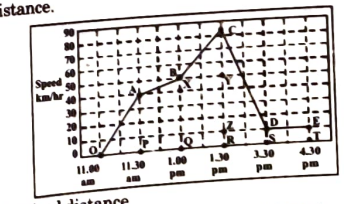
$\therefore$ Speed upstream $= \frac{13}{5}$ km/hr.

and Speed downstream $= \frac{28}{5}$ km/hr.

$\therefore$ Speed of river $= \frac{1}{2}\left(\frac{28}{5} - \frac{13}{5}\right)$

$= \frac{1}{2} \times \frac{15}{5} = 1.5$ km/hr

9. Graph of speed vs time is plotted as shown. Area under the graph and time axis gives the distance.



Required distance

$= A(\Delta OAP) + A(\Delta BXA) + A(\square APQX)$

$= \frac{1}{2} \times \left(\frac{1}{2}\right) \times 40 + \frac{1}{2} \times 10 \times \left(\frac{3}{2}\right) + \left(\frac{3}{2}\right) \times 40$

$= 10 + \frac{30}{4} + 60 = 77.5$ km.

10. Distance covered are the same.
Speed in the latter is maintained at 10 kmph.
$\therefore \quad 77.5 = 10 \times x$
$\Rightarrow \quad x = 7.75$ hours.
Hence, clock time will be 4.30 + 7 hours 45 mins
= 12.15 a.m.

11. Average speed = $\frac{\text{total distance travelled}}{\text{total time taken}}$
Distance travelled from 1.00 pm to 4.30 p.m.
= A(□BQRY) + A(ΔBYC) + A(ΔCZD) + A(□ZRTE)

$$= 50 \times \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) \times 35 + \frac{1}{2} \times (2) \times 75 + 3 \times 10$$

$$= \frac{50}{2} + \frac{35}{4} + 75 + 30$$
$$= 130 + 8.75$$
$$= 138.75 \text{ km.}$$

Total distance travelled = 77.5 + 138.75
= 216.25 km.

$\therefore$ Average speed = $\frac{216.25}{5.5} = 39.3$ kmph.

12. While deer makes 10 leaps, cheetah makes 6 leaps.
$\therefore$ Distance covered in 10 leaps by deer
$= 10 \times 1 = 10$m
Distance covered in 6 leaps by cheetah
$= 6 \times 2 = 12$ m
Hence in six leaps, cheetah gains
$12 - 10 = 2$ m over the deer.
Cheetah has to gain 100 m over the deer.
So, total leaps required by cheetah

$$= \frac{6}{2} \times 100 = 300.$$

13. 50 leaps of the hare $50 \times 1\frac{3}{4} = \frac{175}{2} = 87.5$ m.

So, hound should gain $\frac{175}{2}$ m over the hare.

When hound travels $3 \times 2\frac{3}{4}$ m, then

hare travels $4 \times 1\frac{3}{4}$ m.

Hence in 3 leaps of the hound the hound gains

$$\frac{33}{4} - \frac{28}{4} = \frac{5}{4} \text{ metres}$$

$\therefore$ Number of leaps required = $\frac{\frac{175}{2}}{\frac{5}{4}} \times 3$

$$= \frac{175}{2} \times \frac{4}{5} \times 3 = 210 \text{ leaps.}$$

14. Fuel required for first 20 km = $\frac{20}{8} = 2.5$ litres.

Fuel required for next 60 km = $\frac{60}{15} = 4$ litres.

$\therefore$ Petrol left = 10 – 6.5 = 3.5 litres.
Now, he has to travel = 115 – (60 + 20)
= 35 km in 3.5 litres.
So he can go at a speed of either 45 km/hr or 80 km/hr as mileage is 10 km/litre in both cases. But since he reaches in minimum time, he travels at 80 km/hr

$\therefore$ Time taken = $\frac{20}{40} + \frac{60}{60} + \frac{35}{80} = \frac{155}{80}$

$$= 1 + \frac{75}{80} \text{ hours}$$

= 1 hours 56 minutes and 25 seconds.

15. Total distance travelled = 115 km

Total time taken = $\frac{155}{80}$ hours

$\therefore$ Average speed = $\frac{115}{\frac{155}{80}} = 60$ km/hr

$\therefore$ Fuel consumption = $\frac{115}{15} = 7\frac{2}{3}$ litres.

So, fuel saved = $10 - 7\frac{2}{3} = 2\frac{1}{3}$ litres.

16. A overlaps B in the middle of the 6th round
So, when A runs $5\frac{1}{2}$ rounds, B runs only $4\frac{1}{2}$ rounds.

$\therefore \quad \frac{\text{A's Speed}}{\text{B's Speed}} = \frac{5\frac{1}{2}}{4\frac{1}{2}} = \frac{11}{9}$

So when A runs 11 km B runs 9 km.
Thus, when A runs 2 km, B runs $2 \times \frac{9}{11} = \frac{18}{11}$ km.

$\therefore$ A wins by $\left(2 - \frac{18}{11}\right) = \frac{4}{11}$ km.

17. Speeds of A and B are as

$$1\frac{3}{8} : 1 = \frac{11}{8} : 1 = 11 : 8$$

In a race of 11 metres, A gain
$11 - 8 = 3$ metres over B.
To gain 120 metres, race must be of

$$120 \times \frac{11}{3} = 440 \text{ metres.}$$

18.

	Work	Number
Men	3	20
Women	2	30
Children	1	36

Ratio of wages = (3 × 20) : (2 × 30) : (1 × 36)
= 5 : 5 : 3

Total wages of 20 men = $\frac{5}{13} \times 780 = ₹ 300$

$\therefore$ Wages of a man = ₹ 15.
Similarly, wages of 30 women

$$\frac{5}{13} \times 780 = ₹ 300$$

Wages of a woman = ₹ 10

Wages of 36 children $\frac{3}{13} \times 780 = ₹ 180$

Wages of a child = ₹ 5
$\therefore$ Total wages of 15 men, 21 women and 30 children
= 15 × 15 + 21 × 10 + 30 × 5
= ₹ 585

$\therefore$ Total wages for 2 weeks = ₹ 1170.

19. Work done by A in 3 days
$= 3 \times \frac{1}{6} = \frac{1}{2}$ of the work.
Work done by B in 3 days
$= 3 \times \frac{1}{8} = \frac{3}{8}$ of the work.
$\therefore$ Work done by the boy in 3 days

$$= 1 - \left(\frac{1}{2} + \frac{3}{8}\right)$$

$$= \frac{1}{8} \text{ of the work.}$$

$\therefore$ Ratio of work done

$$= \frac{1}{2} : \frac{3}{8} : \frac{1}{8} = 4 : 3 : 1$$

$\therefore$ Boy's share = $\frac{1}{8} \times ₹ 200 = ₹ 25$.

20. A finishes the job at standard rate in 12 days,
So in 1 day, the part of job done by A = $\frac{1}{12}$
Similarly the part of job done by B in 1 day

$$= \frac{1}{2 \times 9} = \frac{1}{18}$$

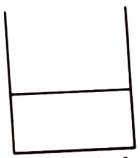
If a works for 2 days followed by 2 days of B at standard rate.
Part of work done = $\frac{2}{12} + \frac{2}{18} = \frac{5}{18}$.
So, in (2 alternate days work of 'A' and 'B')
3 × 4 = 12 days
They will finish $3 \times \frac{5}{18} = \frac{15}{18}$ th of the work.
Work remaining = $1 - \frac{15}{18} = \frac{3}{18} = \frac{1}{6}$
This is finished by A alone in $\frac{\frac{1}{6}}{12} = 2$ days.
Hence, total number days = 12 + 2 = 14 days.

EXERCISE – II

MCQ TYPE QUESTIONS

1. For M $\Rightarrow \frac{12}{20} = 0.6$
N $\Rightarrow \frac{25}{45} = 0.55$
O $\Rightarrow \frac{45}{75} = 0.6$
P $\Rightarrow \frac{57}{100} = 0.57$

2. $V_{half} = 30(s)$ drawing rate = s
Total volume = 60 S tank
$(s^1)(10) - (s)10 = 30s$
$s^1(s) - s = 3s$
$s1 = 4s$
$s^1 = 4$ drawing rate



3. **Given:** A tourist covers half the journey at 60 km/h, one fourth at 30 km/h and the remaining one fourth at 10 km/h
To find: Avearge speed of the whole journey.
Analysis: One approach to this is to find individual time consumed in each of the three parts and use that value to get total time, and find average speed.
So,
Lets assume total length of journey = x km
$\therefore$ First phase at 60 km/h

$$\text{Time } t_1 = \frac{\text{Distance}}{\text{Speed}} = \frac{x}{2} \frac{1}{60} = \frac{x}{120} h$$

Second phase at 30 km/h
Distance = $\frac{x}{4}$ km

$$\text{Time } t_2 = \frac{\frac{x}{4}}{30} = \frac{x}{120}$$

Third phase at 10 km/h
Distance = $\frac{x}{4}$ km

$$\text{Time } t_3 = \frac{\frac{x}{4}}{10} = \frac{x}{40} h$$

$\therefore$ Total time $= \frac{x}{120}+\frac{x}{120}+\frac{x}{40} = \frac{x}{40}\left(\frac{1}{3}+\frac{1}{3}+1\right)$

$= \frac{x}{40}\frac{(1+1+3)}{3} = \frac{5x}{120}$

$\therefore$ Average speed $= \frac{\text{Total Distance}}{\text{Total time}}$

$= \frac{x \cdot 120}{5x} = 24$ km/h

Hence, the answer is (c)

4. Average Speed $= \frac{\text{Total distance}}{\text{total time taken}}$

Total Distance $= \frac{8+6+16}{15+15+15} = \frac{30\times 60}{45}$

$= 40$km/h.

5.

	Fuel consumption	Actual
P	60 km/l	$\frac{15}{60}=\frac{1}{4}l$
Q	90 km/l	$\frac{75}{90}=\frac{5}{6}l$
R	75 km/l	$\frac{40}{75}=\frac{8}{15}l$
S	30 km/l	$\frac{10}{30}=\frac{1}{3}l$

6. 5 skilled workers build wall in 20 days

1 skilled worker build wall in 20×5 days

Hence in 1 day, part of work done by skilled work

$= \frac{1}{100}$

Similarly in 1 day part of work done by semi-skilled workers

$= \frac{1}{25\times 8}$

and in 1 day part of work done by un-skilled worker

$= \frac{1}{30\times 10}$

So part of work done in 1 day by 2 skilled, 6 semi-skilled and 5 unskilled

$= \frac{2}{100}+\frac{6}{200}+\frac{5}{300} = \frac{1}{15}$

So work done by given workers in days = 15

NUMERICAL TYPE QUESTIONS

1. Tiger – 1 leap $\Rightarrow$ 8 meter

Speed = 5 leap/hr = 40m/min

Deer $\rightarrow$ 1 leap = 5 meter

speed = 4hr = 20m/min

Let at time 't' the tiger catches the deer.

$\therefore$ Distance travelled by deer + initial distance between them

$50 \times 8 \Rightarrow 400$m = distance covered by tiger.

$\Rightarrow 40 \times t = 400 + 20t$

$\Rightarrow t = \frac{400}{200} = 20$ min

$\Rightarrow$ total distance $\Rightarrow 400 + 20 \times t = 800$ m

2. Speed of man = 8;

Left distance = d

Time taken = $\frac{d}{8}$

Upstream: Speed of stream = s

$\Rightarrow$ speed upstream = S' = $(8 - s)$

$$t' = \left(\frac{d}{8-s}\right)$$

Downstream:

Given speed downstream = $t'' = \frac{d}{8+s}$

$\Rightarrow 3t' = t'' \Rightarrow \frac{3d}{8-s} = \frac{d}{8+s}$

$\Rightarrow \frac{3d}{8-s} = \frac{d}{8+s}$

$\Rightarrow s = 4$ km/hr

3. Let the length of platform = x.m

and train length = 280 m (given)

According to Question

$$\frac{x+280}{60} = \frac{280}{20}$$

$\therefore x = 560$ m.

■■

4 CHAPTER

Ratio, Proportion and Mixtures

RATIO

Ratio of two terms 'a' and 'b' is denoted by a : b (read as 'a is to b') and is measured by $\frac{a}{b}$.

Numerator 'a' is called *'antecedent'* and denominator 'b' is called 'consequent'.

While comparing two quantities in terms of ratio :

(*i*) Two quantities must be of the same kind.

(*ii*) Units of measurement of the two quantities must be the same.

e.g., if a = ₹ 2 and b = 50 paise,

then a : b = 200 : 50 = 4 : 1.

(*iii*) Ratio is a pure number, i.e., without any unit of measurement as it denotes how many times a given quantity is in comparison to another.

(*iv*) Ratio would stay unaltered even if both antecedent and the consequent are multiplied or divided by the same number.

e.g., $a : b = \frac{a}{b} = \frac{a\times m}{b\times m} = \frac{\frac{a}{m}}{\frac{b}{m}}$.

Properties of Ratios

1. **Invertendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$ i.e., inverse ratios of two equal ratios are equal.

2. **Alternendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$ i.e., ratios of antecedents and consequents of two equal ratios are equal.

3. **Componendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$ i.e., adding 1 to both sides.

4. **Dividendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-b}{b} = \frac{c-d}{d}$ i.e., subtracting 1 from both sides.

5. **Componendo-Dividendo :** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ i.e., dividing results of componendo by dividend.

Equal Ratio : If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$; then each ratio

$= \frac{\text{Sum of Numerators}}{\text{Sum of Denominators}} = \frac{a+c+e\ldots}{b+d+f\ldots}$

The principle can also be applied after multiplying Numerator and Denominator of any fraction by the same number.

If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=\ldots$, then each ratio is equal to

$\left(\frac{pa^n+qc^n+re^n+\ldots}{pb^n+qd^n+rf^n+\ldots}\right)^{1/n}$, where p, q, r, n ... may have any values except that they must not all be zeros.

PROPORTION

When ratio of two terms is equal to the ratio of two other terms, then these four terms are said to be in **proportion,** i.e., if a : b = c : d, then **a, b, c and d are** in proportion.

a, b, c and d are called first, second, third and fourth proportionals respectively. The terms a and d are called **extremes,** while b and c are called **means.**

If a : b = c : d, then ad = bc i.e., product of extremes is equal to product of means.

Continued Proportion

When $\frac{a}{b} = \frac{b}{c}$, then a, b and c are said to be in continued proportion and b is called *geometric mean* or *mean proportional* between a and c.

Also, $b^2 = a \times c$, $\Rightarrow$ $b = \sqrt{ac}$

If $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}=\ldots\ldots$, then a, b, c, d, e ... are said to be in continued proportion.

Concept of continued proportion is useful in the following situation.

(*i*) If four quantities a, b, c and d and a : b, b : c and c : d, are know, then we can find a : d as

$\frac{a}{d} = \frac{a}{b}\times\frac{b}{c}\times\frac{c}{d}$

(*ii*) If a, b, c and d are four quantities and a : b, b : c, c : d are known, then we can find a : b : c : d.

e.g., If a : b = 2 : 3; b : c = 4 : 5, c : d = 7 : 9, find a : b : c : d.

a : b = 8 : 12, b : c = 12 : 15 (LCM of 3 and 4 is 12)

$\therefore$ a : b : c = 8 : 12 : 15

a : b : c = 56 : 84 : 105, c : d = 105 : 135 (LCM of 7 and 15 is 105)

$\therefore$ a : b : c : d = 56 : 84 : 105 : 135

MIXTURES AND ALLIGATIONS

Simple Mixture : When two different ingredients are mixed together, it is called *simple mixture.*

Compound Mixture : When two or more simple mixtures are mixed together to form another mixture, it is called compound mixture.

Alligation Rule

Alligation rule states that "When different quantities of the same or different ingredients, of different costs (one cheap and other dear) are mixed together to produce a mixture of a mean cost, then ratio of their quantities is inversely proportional to the difference in their cost from the mean cost."

$$\frac{\text{Quantity of Cheap}}{\text{Quantity of Dear}} = \frac{\text{Price of Dear} - \text{Mean Price}}{\text{Mean Price} - \text{Price of Cheap}}$$

Application of Alligation Rule :

(1) To find mean value of a mixture when prices of two or more ingredients, which are mixed together and the proportion in which they are mixed are given.

(2) To find proportion in which ingredients at given prices must be mixed to produce a mixture at a given price.

Some Important Results :

(1) When x_1 quantity of ingredient A of cost C_1 and x_2 quantity of ingredient B of cost C_2 are mixed, the, cost of the mixture,

$$C_m = \frac{C_1x_1 + C_2x_2}{x_1 + x_2}$$

(2) Similarly, when more than two ingredients are mixed, then $C_m = \frac{C_1x_1 + C_2x_2 + C_3x_3 + \ldots C_nx_n}{x_1 + x_2 + x_3 + \ldots x_n}$

(3) When two mixtures M_1 and M_2 each containing ingredient A and B in the ratio a : b and x : y, respectively, are mixed, then proportion of the ingredients A and B i.e., $q_A : q_B$, in the compound mixture is given by.

$$\frac{q_A}{q_B} = \frac{M_1 \times \left(\frac{a}{a+b}\right) + M_2 \times \left(\frac{x}{x+y}\right)}{M_1 \times \left(\frac{b}{a+b}\right) + M_2 \times \left(\frac{y}{x+y}\right)}$$

and, quantity in which M_1 and M_2 is to be mixed when quantity of A and B i.e., q_A and q_B in the compound mixture is known is given by

$$\frac{\text{Quantity of } M_1}{\text{Quantity of } M_2} = \frac{\left(\frac{x}{x+y}\right) - \left(\frac{q_A}{q_A+q_B}\right)}{\left(\frac{q_A}{q_A+q_B}\right) - \left(\frac{a}{a+b}\right)}$$

(4) When a mixture of three ingredients A, B and C is given, take any two ingredients such that cost of the mixture is between costs of the two chosen ones and find the ratio. Once again, take two more ingredients and find their ratio. Then find combined ratio. This will give infinite number of solutions.

(5) If a vessel contains 'a' litres of liquid A, and 'b' litres be withdrawn and replaced by liquid B, then if 'b' litres of mixture be withdrawn and replaced by liquid B, and operation repeated 'n' times in all, then

$$\frac{\text{Liquid A left after } n^{th} \text{ operation}}{\text{Initial quantity of liquid A in vessel}} = \frac{\left(\frac{a-b}{a}\right)^n}{1-\left(\frac{a-b}{a}\right)^n}$$

$$\frac{\text{Liquid A left after } n^{th} \text{ operation}}{\text{Liquid B left after } n^{th} \text{ operation}} = \frac{\left(\frac{a-b}{a}\right)^n}{1-\left(\frac{a-b}{a}\right)^n}$$

EXERCISE - I

MCQ TYPE QUESTIONS

1. A cask of wine when fully filled holds 10 litres. 2 litres of wine is removed and filled with water. Then 4 litres in the solution is replaced with water. Then, 6 and 8 litres respectively. At the end of the 4th operation, the ratio of wine to water is

(*a*) $\frac{4!}{(5)^4}$ (*b*) $\frac{8!}{(10)^4}$

(*c*) $\frac{4!}{5^4 - 4!}$ (*d*) $\frac{8!}{(10)^4 - 8!}$

2. A 20 litre vessel is filled with alcohol. Some of the alcohol is poured out into another vessel of an equal capacity, which is then completely filled by adding water. Then mixture obtained is poured into the first vessel to capacity. Then $6\frac{2}{3}$ litres is poured from the first vessel into second. Both vessels now contain an equal amount of alcohol. How much alcohol was originally poured from the first vessel into the second?

(*a*) 9 litres

(*b*) 10 litres

(*c*) 12 litres

(*d*) 12.5 litres

3. What is the ratio compounded of x – y : x + y and $y^2 + xy : x^2 - xy$?

(*a*) $y^2 : x^2$ (*b*) xy : 1

(*c*) y : x (*d*) $\sqrt{x} : \sqrt{y}$

4. If ratio of two natural numbers x and y is 'a' and that of y and x is 'b', then value of 'a + b' is

(*a*) greater than 2

(*b*) lesser than 1

(*c*) greater than 1

(*d*) data insufficient.

5. The charges of a goldsmith is partly fixed and partly variable with the amount in grams of gold. The charge is ₹ 300 for 20 grams of gold and ₹ 550 for 45 grams of gold. Find the charge of work for 100 grams of gold.

(*a*) ₹ 900

(*b*) ₹ 1000

(*c*) ₹ 1100

(*d*) ₹ 9000

Directions (Q.6–7) : *Refer to the data below and answer the questions that follow.*

The ratio of white collar to blue collar employees in a firm is 8 : 3. Also ratio of male employees to female employees is 7 : 4. It is observed that 60% of the white collar employees are males.

6. What is the ratio of female white collar employees to male blue collar employees?

(*a*) 16 : 11 (*b*) 24 : 11

(*c*) 3 : 1 (*d*) Data insufficient

7. If there are 48 female blue collar workers, what is the difference between male white collar employees and female blue collar employees?

(*a*) 288 (*b*) 240

(*c*) 220 (*d*) Data insufficient

8. In a chemical experiment, two NaOH solution bottles are used. Bottle A contains salt and water in the ratio 7 : 3 and bottle B contains salt and water in the ratio 4 : 3. In what proportion should the quantities be taken from A and B to give the 2 : 1 NaOH solution?

(*a*) 2 : 1 (*b*) 10 : 7

(*c*) 20 : 7 (*d*) 1 : 2

9. A triswitch mixer grinder operates in three stages of speed in revolutions per min i.e., rpm. The speed of the motor with some load in the second stage varies as a square of the speed in stage 1 and that in stage 3 varies as a cube of that in stage 2. If mixer motor makes 50 revolutions in 3 minutes in stage 1, its speed in stage 2 and 3 with the same load is 2500 and 12500 rpm respectively. What will be the speed of motor (in rpm) in stage 3, if at the same load motors speed in rpm is 25 revolutions in 6 minutes in stage 1?

(*a*) $\frac{3125}{16}$ rpm (*b*) $\frac{5^5}{2^{10}}$ rpm

(*c*) $\frac{5^{10}}{2^5}$ rpm (*d*) None of these

10. Two bags have certain number of mangoes. If half the mangoes from bag 1 are transferred to bag 2, the ratio of mangoes in bag 1 to bag 2 gets reversed. Now one-third and two-third of the mangoes from the first and the second bag are transferred to the second and first bag respectively. The ratio of mangoes in bag 1 to bag 2 now is

(*a*) 2 : 1 (*b*) 2 : 2

(*c*) 4 : 5 (*d*) 5 : 4

NUMERICAL TYPE QUESTIONS

1. Five litres are drawn from a cask full of wine and it is then filled with water. Five litres of the mixture are drawn and the cask is again filled with water. The quantity of wine now left in the cask to that of the water in it is in the ratio 361 : 39. The cask hold _____ litres.
2. In a mixture of wheat and barley, wheat is 60%. To 400 kg of the mixture a quantity of barley is added and then the wheat is $53\frac{1}{3}\%$. The_____kg of barley are added?
3. Two jars of capacity 3 and 5 litres are filled with mixtures of alcohol and water. In the smaller jar 25% of the mixture is alcohol and in the larger 25% of the mixture is water. The jars are emptied into a 9 litre cask and remaining volume is filled up with water. Then percentage of alcohol in the cask will be _____
4. A man buys milk at ₹ 5 a litre and after adding water, sells it at ₹ 6 a litre, thereby making a profit of $33\frac{1}{3}\%$. The proportion of water to milk in the mixture is_____
5. The percentage of alcohol in two wine tanks is 20 and 40 respectively and rest is grape juice. If a wine glass is filled to its half by the tank 1 and 2 in the ratio 2 : 3 and the rest is filled again by pure alcohol, then the ratio of the grape juice to alcohol in the wine glass is _____
6. In _____ ratio must a grocer mix two varieties of tea worth ₹ 60 a kg and ₹ 65 a kg so that by selling the mixture at ₹ 68.20 a kg he may gain 10%?
7. A container contains 40 lit of milk. From this container 4. lit of milk was taken out and replaced by water. This process was repeated further two times. The _____litres of milk is now contained by the container?
8. A merchant has 1000kg of sugar,part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is_____ kg
9. Volumes in the proportion 3 : 4 : 7 of three different substances are mixed together. The densities of equal volume of the substances are in the ratio 5 : 2 : 6 respectively. The weight of the third substance contained in 130 lbs. of the mixture is _____ lbs
10. In a bag, there are 50 paise coins, 25 paise coins and rupee coins, proportional to the numbers 3, 4 and $2\frac{1}{2}$. If they amount to ₹ 210, then number of rupee coins is_____

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible.
   (*a*) 1 : 4
   (*b*) 1 : 3
   (*c*) 1 : 2
   (*d*) 2 : 3

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2. A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water, Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?
   (*a*) 7.58 litres
   (*b*) 7.84 litres
   (*c*) 7 litres
   (*d*) 7.29 litres

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (*c*)	2. (*b*)	3. (*c*)	4. (*c*)	5. (*c*)	6. (*a*)	7. (*b*)	8. (*c*)	9. (*b*)	10. (*a*)

Numerical Type Questions

1. 100	2. 50	3. 50	4. 1 : 9	5. 17 : 33	6. 3 : 2	7. 29.16	8. 600	9. 84	10. 105

EXERCISE – II

MCQ Type Questions

1. (*c*)	2. (*d*)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. Stage $\frac{\text{Wine left}}{\text{Original quantity}}$

0 $\frac{10}{10}$

1 $\frac{8}{10}$

2 $\frac{8}{10}\times\frac{6}{10}$

3 $\frac{8}{10}\times\frac{6}{10}\times\frac{4}{10}$

4 $\frac{8}{10}\times\frac{6}{10}\times\frac{4}{10}\times\frac{2}{10} \Rightarrow \frac{4\times3\times2\times1}{5^4} = \frac{4!}{5^4}$

$\therefore \quad \frac{\text{Wine}}{\text{Water}} = \frac{\frac{4!}{5^4}}{1-\frac{4!}{5^4}} = \frac{4!}{5^4-4!}$

2.

	C_1	C_2
Operation1 Operation 2	20 litres alcohol (20 – x) alcohol	Empty x litres alcohol (20 – x) litres water
Operation 3	(20 – x) litres alcohol + xlitres mixture i.e., $\left(\frac{x^2}{20}\text{litres alcohol}\right)$	(20 – x) litres mixture

Now $\frac{1}{3}$ rd of C_1 poured into C_2

Quantity of alcohol in $C_1 = C_2 = 10$ litres

In C_1 : $\frac{2}{3}\left[20 - x + (x)\left(\frac{x}{20}\right)\right] = 10$

Solving we get, $x = 10$ litres

Hence10 litres of alcohol was initially poured from C_1 to C_2.

3. Required ratio $= \frac{x-y}{x+y}\times\frac{y^2+xy}{x^2-xy} = \frac{y}{x} = y : x$

4. *Given* : $\frac{x}{y} = a$ and $\frac{y}{x} = b$

If $\frac{x}{y} > 1$, then $\frac{y}{x} < 1$,

$\Rightarrow \quad \frac{x}{y} + \frac{y}{x} > 1.$

Hence option (*c*) is correct.

5. Total charge = fixed + variable

$T_C = m + kx$

where m and k are constants.

$\therefore \quad 300 = m + k \times 20$

Also, $\quad 550 = m + k \times 45$

Solving for m and k; we get

$m = 100; k = 10.$

Hence, for 100 grams,

total charge = 100 + 10 × 100 = ₹ 1100.

Q. 6 and 7.
Using given data, we can draw following table

	Male	Female	
White Collar	4.8x	3.2x	8x
Blue Collar	2.2x	0.8x	3x
	7x	4x	11x

Now, all the questions can be answered.

6. $\dfrac{\text{Female white collar employess}}{\text{Male blue collar employees}} = \dfrac{3.2x}{2.2x} = \dfrac{16}{11}$

7. Since $0.8x = 48$

$\Rightarrow \quad x = 60$

$\therefore$ Required difference $= 4.8x - 0.8x$

$= 4.8 \times 60 - 0.8x = 240$

8. Let 10x be taken from bottle A and 7y be taken from bottle B.

$$\text{Total salt taken} = \left(\frac{7}{10} \times 10x\right) + \left(\frac{4}{7} \times 7y\right)$$

$$= 7x + 4y$$

Total water taken $= 3x + 3y$.

$\therefore \quad \dfrac{7x+4y}{3x+3y} = \dfrac{2}{1}$

$\Rightarrow \quad 7x + 4y = 6x + 6y$

$\Rightarrow \quad x = 2y$

$\therefore \quad \dfrac{x}{y} = \dfrac{2}{1}$

$\therefore$ Ratio of quantities $= \dfrac{10x}{7y}$

$$= \frac{10}{7} \times \frac{2}{1} = \frac{20}{7}$$

9. Let speed of the motor in stage 1 be x rpm. Then its speed in stage 2 be $(x^2) = ax^2$ (say) and in stage 3 be $(ax^2)^3 = b.a^3x^6$ (say)

Given that $x = \dfrac{50}{3}$ ...(i)

$ax^2 = 2500$ ...(ii)

$ba^3x^6 = 12500$ ...(iii)

Substituting value of x in (ii), we get

$$a = \frac{2500}{(50)^2} \times 3^2 = 9$$

Again, substituting values of a and x, we get

$$b = \frac{12500}{(2500)^3} = \frac{1}{125} \times 10^{-4}$$

Given : $x = \dfrac{25}{6}$ rpm

$\therefore$ Speed in stage 3 $= \dfrac{10^{-4}}{125} \times (9)^3 \times \left(\dfrac{25}{6}\right)^6$

$$= \frac{10^{-4}}{5 \times 25} \times (3)^6 \times \frac{(25 \times 5) \times (5 \times 25^4)}{(2)^6 \cdot (3)^6}$$

$$= \frac{5 \times 25^4}{2^6 \times 10^4} = \frac{5^5}{2^{10}} \text{ rpm}$$

10. Let bag 1 and 2 have

	x mangoes	y mangoes
Step 1	$\dfrac{x}{2}$	$y + \dfrac{x}{2}$

But $\dfrac{x}{2}$: $y + \dfrac{x}{2} = y : x$

Step 2	$\dfrac{2}{3}y + \dfrac{2}{3}x$	$\dfrac{1}{3}y + \dfrac{1}{3}x$

$\therefore$ Required ratio $= \dfrac{\frac{2}{3}y + \frac{2}{3}x}{\frac{1}{3}y + \frac{1}{3}x} = \dfrac{2\left(\frac{x+y}{3}\right)}{1\left(\frac{x+y}{3}\right)}$

$= 2 : 1$

NUMERICAL TYPE QUESTIONS

1. Let quantity of wine be a, i.e., capacity of the cask be 'a'

i.e., $\dfrac{\text{Quantity of wine}}{\text{Quantity of water}} = \dfrac{\left(1 - \frac{5}{a}\right)^2}{1 - \left(1 - \frac{5}{a}\right)^2}$

$\therefore \quad \dfrac{361}{39} = \dfrac{\left(1 - \frac{5}{a}\right)^2}{1 - \left(1 - \frac{5}{a}\right)^2}$

$\Rightarrow \quad \dfrac{361}{39} - \dfrac{361}{39}\left(1 - \frac{5}{a}\right)^2 = \left(1 - \frac{5}{a}\right)^2$

$\Rightarrow \quad \left(1 - \dfrac{5}{a}\right) = \sqrt{\dfrac{361}{400}} = \dfrac{19}{20}$

$\Rightarrow \quad \dfrac{5}{a} = \dfrac{1}{20}$

$\therefore \quad a = 100$ litres

2. Quantity of wheat in 400 kg of mixture

$= 0.6 \times 400 = 240$ kg

Quantity of barley $= 400 - 240 = 160$ kg

Let x kg of barley be added to 400 kg of the mixture

$\therefore \quad \dfrac{240}{160 + x} = \dfrac{53\frac{1}{3}}{46\frac{2}{3}} = \dfrac{160}{140} = \dfrac{8}{7}$

$\Rightarrow \quad 240 \times 7 = 8 \times 160 + 8x$

$\Rightarrow \quad x = 50$ kg

3. Alcohol content in 1st jar $= 0.25 \times 3 = 0.75$ litre

Alcohol content in 2nd jar $= 0.75 \times 5 = 3.75$ litres

$\therefore$ Total alcohol in 9 litre cask $= 0.75 + 3.75$

$= 4.5$ litres

$\therefore$ Percentage of alcohol in the cask

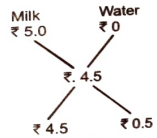
$= \dfrac{4.5}{9} \times 100 = 50\%$

4. Cost price of milk = ₹ 5/litre

Cost price of mixture $= \dfrac{100}{133\frac{1}{3}} \times 6 =$ ₹ 4.5/litre

$\therefore \quad \dfrac{\text{Quantity of water}}{\text{Quantity of milk}} = \dfrac{5 - 4.5}{4.5 - 0} = \dfrac{1}{9}$

i.e.,



Milk : Water = 9 : 1

5.

	Alcohol	Grape juice	Alcohol : Grape
Tank 1	20%	80%	1 : 4
Tank 2	40%	60%	2 : 3

Ratio of grape juice to alcohol in half filled wine glass

$$= \frac{2 \times \frac{4}{5} + 3 \times \frac{3}{5}}{2 \times \frac{1}{5} + 3 \times \frac{2}{5}} = \frac{17}{8}$$

So, half capacity of glass $= 17 + 8 = 25$

So, full capacity $= 25 \times 2 = 50$

Now rest 25 parts has to be filled with alcohol.

So, ratio of the grape juice to alcohol

$= 17 : (8 + 25) = 17 : 33.$

Alternatively,

As the wine glass is filled to half by taking two parts from tank 1 and three parts from tank 2 and second half is equivalent to five parts which is pure alcohol,

percentage concentration of grape juice

$$= \frac{2 \times 80 + 3 \times 60 + 5 \times 10}{2 + 3 + 5} = 34\%$$

Hence ratio of grape juice to alcohol

$= 34 : 66 = 17 : 33.$

6. S.P of 1 kg of mixture = ₹ 68.20, Gain = 10%

S.P of 1 kg of mixture $=$ ₹ $\left(\dfrac{100}{110} \times 68.2\right) =$ ₹ 62.

Cost of 1 kg tea of 1st kind		Cost of 1 kg tea of 2nd kind
60		65
	Mean Price ₹ 62	
3		2

$\therefore$ Required ratio = 3 : 2.

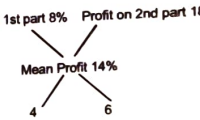
7. Amount of milk left after 3 operations

$$= 40\left(1 - \frac{4}{40}\right)^3$$

$$= 40 \times \frac{9}{10} \times \frac{9}{10} \times \frac{9}{10}$$

$= 29.16$ lit

8.



Ratio of 1st and 2nd parts = 4 : 6 = 2 : 3.

Quantity of 2nd part $= \dfrac{3}{5} \times 1000$ kg $= 600$ kg.

9. Ratio of volumes $= \dfrac{3}{14} : \dfrac{4}{14} : \dfrac{7}{14}$

Ratio of densities $= \dfrac{5}{13} : \dfrac{2}{13} : \dfrac{6}{13}$

$\therefore$ Ratio of weights $= \dfrac{3}{14} \times \dfrac{5}{13} : \dfrac{4}{14} \times \dfrac{2}{13} : \dfrac{7}{14} \times \dfrac{6}{13}$

$= 15 : 8 : 42$

... (Mass = Volume × Density)

Weight of the third substance $= \dfrac{42}{65} \times 130 = 84$ ℓbs.

10. Let number of 50p coins be 3x, number of 25p coins be 4x and number of rupee coins be $2\frac{1}{2}x$.

$\therefore \quad 2\frac{1}{2}x + \frac{1}{2} \times 3x + \frac{1}{4} \times 4x = 210$

$\Rightarrow \quad \frac{5}{2}x + \frac{3}{2}x + x = 210$

$\Rightarrow \quad 5x = 210$

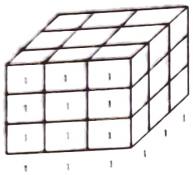
$\Rightarrow \quad x = 42$

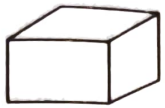
$\therefore$ Number of rupee coins $= \frac{5}{2} \times 42 = 105.$

EXERCISE – II

MCQ TYPE QUESTIONS

1.





Number of faces per cube = 6

Total number of cubes = 9 × 3 = 27

$\therefore$ Total number of faces = 27 × 6 = 162

$\therefore$ Total number of non visible faces = 162 – 54 = 108

$\therefore \quad \frac{\text{Number of visible faces}}{\text{Number of non visible faces}} = \frac{54}{108} = \frac{1}{2}$

2. $10\left(\frac{10-1}{10}\right)^3 = 10\left(\frac{9}{10}\right)^3 = \frac{729}{1000}$

$\therefore \quad \frac{729}{1000} \times 1 = 7.29$ litres

■■

5 CHAPTER

Permutations and Combinations & Probability

PERMUTATIONS AND COMBINATION

PERMUTATIONS

Each of the arrangements which can be made by taking some or all of a number of items is called a *Permutation*. Permutation implies "*arrangement*" or that "*order of the items*" is important.

Permutations of three items a, b and c taken two at a time are ab, ba, ac, ca, cb and bc. Since the order in which items are taken is important, ab and ba are counted as two different permutations. The words "permutation" and "arrangement" are synonymous and can be used interchangeably."

Number of permutations of n things taking r at a time is denoted by ${}^nP_r$ (and read as "nPr")

COMBINATIONS

Each of the groups or selections which can be made by taking some or all of a number of items is called *Combination*. In combinations, order in which the items are taken is not considered as long as the specific things are included.

Combinations of three items a, b and c taken two at a time are ab, bc and ca. Here, ab and ba are not considered separately because the order in which a and b are taken is not important but it is only required that a combination including a and b is what is to be counted. Words "*combination*" and "*selection*" are synonymous.

Number of combinations of n things taking r at a time is denoted by ${}^nC_r$ (and read as "nCr")

$${}^nP_r = \frac{n!}{(n-r)!}, \quad {}^nC_r = \frac{n!}{r!(n-r)!}$$

Number of ways in which n things may be arranged taking them all at a time, when p of the things are exactly alike of one kind, q of them exactly alike of another kind, r of them exactly alike of a third kind, and the rest all distinct is $\frac{n!}{p!q!r!}$

Number of ways of selecting one or more items from n given items $= 2^n - 1$

Number of ways of dividing (p + q) items into two groups of p and q items respectively is $\frac{(p+q)!}{p!q!}$.

Number of ways of dividing 2p items into two equal groups of p each is $\frac{(2p)!}{(p!)^2}$ where two groups have distinct identity.

Number of ways of dividing 2p items into two equal groups of p each is $\frac{(2p)!}{2!(p!)^2}$ where two groups do not have distinct identity.

Number of ways in which (p + q + r) things can be divided into three groups containing p, q and r things respectively is $\frac{(p+q+r)!}{p!q!r!}$

Number of circular arrangements of n distinct items is (n – 1)! If there is a DIFFERENCE between clockwise and anticlockwise arrangements and (n – 1)!/2 if there is NO DIFFERENCE between clockwise and anticlockwise arrangements.

For $x_1 + x_2 + x_3 + \dots\dots\dots + x_n = s$ where $s \geq 0$, number of positive integral solutions (when $s \geq n$) is ${}^{s-1}C_{n-1}$ and number of non-negative integral solution is ${}^{s+n-1}C_{n-1}$.

PROBABILITY

If in an experiment there are 'n' occurrences out of which 'm' occurrences are favourable to a particular event E, then probability of the event E [written as P(E)] is defined as P(E) = m/n.

For the event E, non-occurrence of the event is denoted by $\bar{E}$. Out of a total n occurrences, if m occurrences are favourable to the event E, it means that (n – m) occurrences are not favourable to the event $\bar{E}$, i.e. (n - m) occurrences are favourable to the event $\bar{E}$.

Hence, probability of the event $\bar{E}$ will be $P(\bar{E}) = \frac{n-m}{n}$ (complementary event).

We can also see that $P(\bar{E}) + P(E) = 1$.

In some problems when we are asked to find out the probability of an event $\bar{E}$, it becomes easier to calculate probability of E and subtract that probability from 1 to get $P(\bar{E})$.

Note:

(*i*) Probability [P(E)] of the happening of an event E is called *probability* of success and probability $\left[P(\bar{E})\right]$ of non-happening of the event is called *probability of failure*.

(*ii*) If P(E) = 1, then event is called a *certain event* and if P(E) = 0, then event is called *impossible event*.

(*iii*) Instead of saying that the chance of happening of an event is m/n, we can also say that odds in favour of the event are m (to n – m). Similarly, if m/n is probability of the occurrences of an event, since (n – m)/n is probability of non-occurrence of the event, we can also say that the odds against the event are (n – m) to m.

Addition Theorem on Probability

If there are two sets A and B, we know that the number of elements in $A \cup B$ is given by

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

A similar relationship exists in probability theory (called Addition Theorem of Probability) as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where A and B are two events.

If A and B are mutually exclusive events (i.e, A and B are disjoint sets), then $P(A \cap B) = 0$

If there are two mutually exclusive events A and B, then $P(A \cup B) = P(A) + P(B)$.

Independent Events

Two events, E_1 and E_2 are said to be independent, if occurrence of the event E_2 is not affected by the occurrence or non-occurrence of the event E_1.

If A and B are two independent events, then

$$P(A \cap B) = P(A) \times P(B)$$

EXERCISE – I

MCQ TYPE QUESTIONS

Directions (Q. 1 – 2) :
If A and B are two possible events of an experiment such that P (A ∩ B) = 0.6 and P(A) = 0.3 then find P (B) given that

1. A and B are mutually exclusive events

(a) $\frac{7}{10}$ (b) $\frac{4}{10}$

(c) $\frac{3}{10}$ (d) $\frac{1}{2}$

2. A and B are independent events

(a) $\frac{4}{7}$ (b) $\frac{9}{10}$ (c) $\frac{3}{10}$ (d) $\frac{3}{7}$

3. Raju throws a fair dice. He is promised an amount (in rupees), which is twice the number showing up if that number is odd and an amount thrice the number showing up, if it is even. What is the maximum amount Raju would be willing to pay each time to throw the dice, if in the long run he wants to make an average profit of ₹ 7 per throw?

(a) ₹ 6 (b) ₹ 7 (c) ₹ 3 (d) ₹ 2

Directions (Q. 4 – 5) :
7 letters are to be placed in seven addressed envelops. If the letters are placed at random into the envelopes, the probability that

4. All of them are placed in the corresponding envelops is

(a) 1

(b) $\frac{1}{6!}$

(c) $\frac{1}{7!}$

(d) None of these

5. Exactly six letters are placed in their corresponding envelopes is

(a) $1-\frac{1}{7!}$ (b) $\frac{1}{7!}$ (c) 1 (d) 0

Directions (Q. 6 – 8) :
If three cards are drawn at random, from a well shuffled pack of cards, then what is the probability that

6. All of them are from the same suit?

(a) $\frac{4\left({}^{13}C_3\right)}{{}^{52}C_3}$ (b) $\frac{\left({}^{13}C_3\right)^3}{{}^{52}C_3}$

(c) $\frac{4\left({}^{13}C_3\right)^2}{{}^{52}C_3}$ (d) $\frac{{}^{13}C_3}{{}^{52}C_3}$

7. All of them are from different suits?

(a) $\frac{13}{{}^{52}C_3}$ (b) $\frac{4(13)^3}{{}^{52}C_3}$

(c) $\frac{{}^{39}C_3}{{}^{52}C_3}$ (d) $\frac{(13)^4}{{}^{52}C_3}$

8. Two of them are number cards from the same suit and the remaining number card is from a different suit?

(a) $\frac{{}^{9}C_2\,{}^{9}C_1}{{}^{52}C_3}$ (b) $\frac{{}^{9}C_2\,{}^{27}C_1}{{}^{52}C_3}$

(c) $\frac{4\left({}^{9}C_2\,{}^{27}C_1\right)}{{}^{52}C_3}$ (d) $\frac{4\left({}^{9}C_2\,{}^{9}C_1\right)}{{}^{52}C_3}$

9. A bag contains 5 white balls and 7 red balls. If 5 balls are drawn at random, then probability that 3 balls are red and the rest are white is

(a) $\frac{175}{198}$ (b) $\frac{175}{396}$

(c) $\frac{38}{132}$ (d) $\frac{221}{396}$

10. A bag contains 6 five rupee coins, 5 two rupee coins and 4 one rupee coins. If 5 coins are selected at random from the bag, then odds in favour of the draw yielding minimum possible amount are

(a) 1:3002 (b) 5:2998

(c) 2:3003 (d) 5:3003

11. In how many ways can 6 girls and 6 boys sit around a circular table so that no two boys sit together?

(a) $(5!)^2$ (b) $(6!)^2$

(c) 5! 6! (d) 11!

12. How many ten-digit numbers can be formed using all the digits of 1324642119 such that the even digits appear only in odd places?

(a) $(5!)^2$ (b) $\frac{(5!)^2}{3!}$

(c) 5! 3! 2! (d) $\frac{(5!)^2}{(2!)^2\,3!}$

Directions (Q. 13 – 14) :
A man has 12 friends whom he wants to invite for lunch. In how many ways can he invite

13. At least one of them?

(a) 4096 (b) 4095

(c) 2047 (d) 2048

14. At least 10 of them?

(a) 66 (b) 140

(c) 153 (d) None of these

15. In how many ways can 6 prizes be distributed among 3 boys, if each boy is eligible to receive one or more prizes?

(a) 3^6 (b) 6^3

(c) ${}^{6}C_3$ (d) ${}^{6}P_3$

Directions (Q. 16 – 17) :
Find the number of ways of dividing 16 different books equally

16. Among 4 boys

(a) $\frac{16!}{(4!)^4}$ (b) $\frac{16!}{(4!)^3}$

(c) $\frac{16!}{(4!)^5}$ (d) $(4!)^4$

17. Into 4 parcels (Parcels are not distinct)

(a) $\frac{16!}{(4!)^4}$ (b) $\frac{16!}{(4!)^3}$

(c) $\frac{16!}{(4!)^5}$ (d) $(4!)^4$

18. How many linear arrangements can be made using all the letters of the word ILLUSTRATE which begin with S and end with a vowel?

(a) $\frac{8!}{(2!)^2}$ (b) 8!

(c) $\frac{8!}{2!}$ (d) $\frac{8!}{(2!)^3}$

19. From a group of 10 professors and 6 assistant professors, a management institute desires to send to delegation of 8 persons consisting of 5 professors and 3 assistant professors to the IIMs annual meet. If Prof. Balamurali, a science professor refuses to be in the delegation if assistant Prof. Sheshdari, an arts professor is included in the delegation, then in how many ways can the delegation be formed?

(a) ${}^{9}C_4\,{}^{6}C_3$

(b) ${}^{9}C_5\,{}^{4}C_2$

(c) ${}^{10}C_5\,{}^{6}C_3 - {}^{9}C_4\,{}^{5}C_2$

(d) ${}^{9}C_4\,{}^{4}C_3 + {}^{9}C_5\,{}^{4}C_2$

20. Two red pencils, three black pencils and two white pencils are to be arranged in a row such that

1. No two adjacent pencils are of the same colour and
2. Pencils at the two ends of the row are of same colour.

(a) 12 (b) 8 (c) 9 (d) 10

NUMERICAL TYPE QUESTIONS

Directions (Q. 1 – 2) :
The odds against an event are 4 to 5 and the odds in favour of another independent event are 3 to 7. The probability that

1. Exactly one of them will occur is _____

2. Neither of them will occur is _____

Directions (Q. 3 – 5) :
Three persons Shiva, Jagan and Rohit aim at a target. Their respective probabilities of hitting the target are 2/3, 5/7 and 3/8. The probability that

3. None of them hits the target is _____

4. At least two of them hit the target is _____

5. Exactly one of them hits the target si _____

6. If three consecutive letters are selected at random from the English alphabet, then the probability that at least one letter is a vowel is _____

7. Eight unbiased coins are tossed together. The probability that the number of heads is equal to the number of tails is _____

8. In a biased coin, Head occurs three times as frequently as tail occurs. If the coin is tossed 3 times, then the probability of getting two heads is _____

Directions (Q. 9 – 10) :

Two cards are drawn at random from a well shuffled pack of cards. Given that both are black, the probability that the cards have

9. Same honour on them is_____
10. Different honours and belong to different suits is _____

Directions (Q. 11 – 13) :

Three mountaineers Akil, Nikil, and Sunil are climbing up a mountain with their respective probability of reaching the summit being $\frac{2}{3}, \frac{5}{8}$ and $\frac{4}{7}$, *respectively. The probability that*

11. None of them reach the summit is _____
12. Exactly two of them reaches the summit is _____
13. Atleast two of them reach the summit is _____

Directions (Q. 14 – 15) :

There are 12 points in a plane of which 4 are on a straight line and no three of the other points lie on a straight line.

14. The _____number straight lines can be formed by joining these points
15. The _____number of triangles can be formed by joining these points
16. The _____ number of non-negative integral solutions does the equation $x_1 + x_2 + x_3 + x_4 = 10$ have?
17. A certain number of students of a school participated in the chess tournament of their annual sports meet. Each player played 1 game against each of the other player. It was found that in 66 games both the players were girls, and in 240 games one was a girl and the other was a boy. The number of games in which both the players were boys is _____
18. Nine points are marked on a straight line and 10 points are marked on another line which is parallel to the first line. The_____ number of triangles can be formed with these points as vertices?
19. The _____ number of positive integers less than 100,000 and divisible by 125 can be formed using the digits 0, 1, 2, 5 and 8, if repetition is allowed?
20. There is an unlimited supply of identical red, blue, and green coloured balls. In _____ number of ways can 12 balls be selected from the supply.
21. Number of positive integral solutions of the equation $a + b + c = 15$ is _____
22. If six unbiased coins are tossed together, then probability that the number of heads exceeds the number of tails is_____

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king?

(a) 4/52 (b) 2/50

(c) 1/52 × (1/52) (d) 1/52 × (1/52) × (1/50)

2. Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have tc be held complete the league round of matches?

(a) 20 (b) 10 (c) 8 (d) 5

3. Right triangle PQIt is to be constructed in the xy – plane so that the right angle is at P and line PR is parallel to the-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequalities: $-4 \le x \le 5$ and $6 \le y \le 16$. How many different triangles could be constructed with these properties?

(a) 110 (b) 1,100

(c) 9,900 (d) 10,000

4. A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occurs in three tosses. Based on the above information, which one of the following statements is TRUE?

(a) X and Y are not independent

(b) Y and Z are dependent

(c) Y and Z are independent

(d) X and Z independent

5. Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is 1/6 and that of Ramesh is 1/8. What is the probability that only one of them will be selected?

(a) $\frac{47}{48}$ (b) $\frac{1}{4}$ (c) $\frac{13}{48}$ (d) $\frac{35}{48}$

6. Given set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is probability that the sum of the two numbers equals 16 ?

(a) 0.20 (b) 0.25

(c) 0.30 (d) 0.33

7. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p.c:

I. $p + m + c = \frac{27}{20}$

II. $p + m + c = \frac{13}{20}$

III. $(p) \times (m) \times (c) = \frac{1}{10}$

(a) Only relation I is true.

(b) Only relation II is true.

(c) Relations II and III are true.

(d) Relations I and III are true.

2014

8. A five digit number is formed using the digits 1,3,5,7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

(a) 6666660

(b) 6666600

(c) 6666666

(d) 6666606

9. You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

(a) 1/4

(b) 1/3

(c) 1/2

(d) 2/3

2013

10. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(a) 13/90

(b) 12/90

(c) 78/90

(d) 77/90

11. What is the chance that a leap year, selected at random, will contain 53 Saturdays?

(a) 2/7 (b) 3/7

(c) 1/7 (d) 5/7

12. In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

(a) 0.35 (b) 0.45

(c) 0.5 (d) 0.4

2012

13. A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

(a) $\frac{1}{4}$ (b) $\frac{1}{16}$

(c) $\frac{7}{16}$ (d) $\frac{9}{16}$

2010

14. Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

(a) 50 (b) 51

(c) 52 (d) 54

NUMERICAL TYPE QUESTIONS

2015

1. How many four digit numbers can be formed with the 10 digits 0, 1, 2, ... 9 if no number can start with 0 and if repetitions are not allowed?

2014

2. In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is _____ years.
3. 10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^- individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____
4. A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch is accepted is _____

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (c)	2. (d)	3. (d)	4. (c)	5. (d)	6. (a)	7. (b)	8. (c)	9. (b)	10. (b)
11. (c)	12. (d)	13. (b)	14. (d)	15. (a)	16. (a)	17. (c)	18. (b)	19. (c)	20. (d)

Numerical Type Questions

1. $\frac{47}{90}$	2. $\frac{14}{45}$	3. $\frac{5}{84}$	4. $\frac{107}{168}$	5. $\frac{17}{56}$	6. $\frac{13}{24}$	7. $\frac{35}{128}$	8. $\frac{27}{64}$	9. $\frac{4}{325}$	10. $\frac{12}{325}$
11. $\frac{3}{56}$	12. $\frac{37}{84}$	13. $\frac{19}{28}$	14. 61	15. 216	16. 286	17. 190	18. 765	19. 99	20. 91
21. 91	22. $\frac{57}{64}$								

EXERCISE – II

MCQ Type Questions

1. (b)	2. (b)	3. (c)	4. (d)	5. (b)	6. (a)	7. (d)	8. (b)	9. (b)	10. (d)
11. (a)	12. (c)	13. (c)	14. (b)						

Numerical Type Questions

1. 4536 2. (25 to 25) 3. (0.47 to 0.48) 4. (0.8145)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1 and 2

Given : $P(A \cup B) = 0.6$

and $P(A) = 0.3$

1. If A and B are mutually exclusive events, then

$P(A \cap B) = 0$

$\therefore$ $P(A \cup B) = P(A) + P(B)$

$0.6 = 0.3 + P(B)$

$\Rightarrow$ $0.3 = P(B)$

Hence $P(B) = 0.3$

2. If A and B are independent events, then

$P(A \cap B) = P(A).P(B)$

$\therefore$ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= P(A) + P(B) - P(A).P(B)$

$P(A \cap B) = P(A) + P(B)[1 - P(A)]$

$\Rightarrow$ $0.6 = 0.3 + P(B)(1 - 0.3)$

$0.3 = P(B)(0.7)$

$\Rightarrow$ $\frac{0.3}{0.7} = P(B)$

$\therefore$ $P(B) = \frac{3}{7}$

3. When a die is rolled, for each number the probability of getting it is $\frac{1}{6}$.

When number on the dice is 1, 3 or 5 Raju receives ₹ 2, ₹ 6 and ₹10 respectively.

When the number on the die is 2, 4 or 6, then Raju receives ₹ 6, ₹ 12 and ₹ 18 respectively.

$\therefore$ Expected value $= \frac{1}{6}[2 + 6 + 10 + 6 + 12 + 18]$

$= \frac{54}{6} = ₹ 9$

Hence to make an average profit of ₹ 7 per throw, Raju must pay ₹ 2 (9 – 7) for each time to throw the die.

4. 7 letters can be arranged in 7 addressed envelopes in 7! ways.

$\therefore$ $n(S) = 7!$

We can arrange 7 letters into corresponding 7 addressed envelopes in only one way.

$\therefore$ Required probability $= \frac{1}{7!}$.

5. If six letters can be placed in their corresponding addressed envelopes, then seventh letter is also in the correct envelope.

Hence required probability is 0.

6, 7 and 8.

Three cards can be drawn from 52 cards in $^{52}C_3$ ways

$n(S) = 52c_3$

6. Since we have 4 suits i.e., Diamonds, Spades, Clubs and Hearts and each suit contains 13 cards.

Three cards can be selected from 13 cards in $^{13}C_3$ ways.

$\therefore$ Number of favorable outcomes $= 4.\ ^{13}C_3$

Required probability $= 4.\frac{^{13}C_3}{^{52}C_3}$

7. From four suits we can select 3 suits in $^4C_3$ ways.

One card can be selected from each suit in $^{13}C_1.\ ^{13}C_1.\ ^{13}C_1$

$\therefore$ Number of favorable outcomes

$= {}^4C_3.\ {}^{13}C_1.\ {}^{13}C_1.\ {}^{13}C_1.\ {}^{13}C1$

$\therefore$ Required probability $= \frac{4(13)^3}{^{52}C_3}$

8. Each suit contains 9 number cards. 2 number cards can be drawn from 9 cards in $^9C_2$ ways..

Third card can be selected from the remaining 27 number cards in $^{27}C_1$ ways.

Hence total number of favorable outcomes is $4(^9C_2)(^{27}C_1)$

Required probability $= \frac{4\left(^9C_2\right)\left(^{27}C_1\right)}{^{52}C_3}$

9. Total number of balls in the bag = 5 + 7 = 12.

5 balls can be drawn from 12 balls in $^{12}C_5$ ways

Number of ways of drawing 3 red balls and 2 white balls from 5 white balls and 7 red balls is $^5C_2.\ ^7C_3$

$\therefore$ Required probability $= \frac{^5C_2.\ ^7C_3}{^{12}C_5}$

$= \frac{10 \times 35}{792} = \frac{175}{396}$

10. Total number of coins in bag = 6 + 5 + 4 = 15.

5 coins can be drawn from 15 coins in $^{15}C_5$ ways.

When 4 coins are one rupee coins and fifth coin is a two rupee coin the amount will be minimum.

$\therefore$ Number of favourable cases $= {}^6C_4.\ {}^5C_1 = 5$

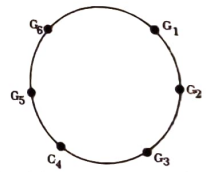
Probability that the amount will be minimum is

$P(E) = \frac{5}{^{15}C_5} = \frac{5}{3003}$

$\therefore$ $P(\bar{E}) = 1 - P(E) = 1 - \frac{5}{3003} = \frac{2998}{3003}$

$\therefore$ Odds in favour of the required event = 5: 2998

11.



Six girls can be arranged around a circle in 5! ways.

Now, the six boys can be arranged in the six places in between the girls in 6! ways.

Hence, required number of ways = 5 ! 6!

12. Of the 10 places available, there are 5 even places.

Digits 2, 4, 6, 4 and 2 can be placed in these 5 even places in $\frac{5!}{2!2!}$ ways.

Remaining digits 1, 3, 1, 1 and 9 can be placed in the remaining 5 places in $\frac{5!}{3!}$ ways.

Hence, required number of ways

$= \frac{5!}{2!2!}\frac{5!}{3!} = \frac{(5!)^2}{(2!)^2\,3!}$

13. Number of ways of inviting at least one friend is
$2^{12} - 1 = 4095$

14. Number of ways of inviting at least 10 friends
$= {}^{12}C_{10} + {}^{12}C_{11} + {}^{12}C_{12}$
$= 66 + 12 + 1 = 79$

15. Each prize can be distributed in 3 ways, so, 6 prizes can be distributed in 3^6 ways.

16. Number of ways of dividing 16 books equally among 4 boys
$$= \frac{16!}{4!4!4!4!} = \frac{16!}{(4!)^4}$$

17. Number of ways of dividing 16 books equally into 4 parcels
$$= \frac{16!}{4!(4!)^4} = \frac{16!}{(4!)^5}$$

18. There are four vowels viz., U, A, I, E in the word "ILLUSTRATE".
First letter must be S and the last letter can be any one of the vowels U, A, I and E.
Remaining 8 places can be filled with the remaining 8 letters in $\frac{8!}{2!2!}$ ways.
Hence, required number of arrangements is
$$= 4 \times \frac{8!}{2!2!} = 8!$$

19. Prof. Balamurali and Asst. B of Sheshadri cannot be included together in the delegation.
Hence, number of ways of forming delegation
$= {}^{10}C_5\,{}^6C_3 - {}^9C_4\,{}^5C_2$

20. Possible ways of arranging the pencils are:
RBWBWBR, WBRBRBW, BWBRWRB, BRBWRWB, BWRBWRB, BWRRBRWB, BRWBRWB, BRWBWRB, BWRWBRB and BRWBBWB
i.e. a total of 10 possible arrangements.

NUMERICAL TYPE QUESTIONS

1 and 2 :
Let A and B be the two events.

Given: $P(\bar{A}) : P(A) = 4 : 5$

$\Rightarrow \quad P(\bar{A}) = \frac{4}{9}; \quad P(A) = \frac{5}{9}$

Also, $P(B) : P(\bar{B}) = 3 : 7$

$\Rightarrow \quad P(B) = \frac{3}{10}$ and $P(\bar{B}) = \frac{7}{10}$

1. $A \cap \bar{B}$ and $\bar{A} \cap B$ denote events that exactly one of them will occur.
$$P(A \cap \bar{B}) = P(A)\,P(\bar{B}) = \frac{5}{9}\cdot\frac{7}{10} = \frac{7}{18}$$

$$= P(\bar{A} \cap B) = P(\bar{A}).P(B)$$
$$= \frac{4}{9}\cdot\frac{3}{10} = \frac{2}{15}$$

Required probability is
$$= \frac{7}{18} + \frac{2}{15} = \frac{35+12}{90} = \frac{47}{90}$$

2. P (neither of them occurs)
$$= P(\bar{A} \cap \bar{B}) = P(\bar{A}).P(\bar{B}) = \frac{7}{10}\left(\frac{4}{9}\right) = \frac{14}{45}$$

3, 4 and 5.
Let A, B and C be the events that Shiva, Jagan and Rohit hit the target respectively.

$\therefore \quad P(A) = \frac{2}{3}; \quad P(\bar{A}) = \frac{1}{3}$

$P(B) = \frac{5}{7}; \quad P(\bar{B}) = \frac{2}{7}$

$P(C) = \frac{3}{8}; \quad P(\bar{C}) = \frac{5}{8}$

3. P (None of them hit the target)
$$= P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}).P(\bar{B}).P(\bar{C})$$
$$= \left(\frac{1}{3}\right).\left(\frac{2}{7}\right).\left(\frac{5}{8}\right) = \frac{5}{84}$$

4. Event $(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)$ denotes that exactly two persons hit the target, and $A \cap B \cap C$ is the event that all the three persons hit the target.

$$P(A \cap B \cap \bar{C}) = P(A)\,P(B)\,P(\bar{C})$$
$$= \frac{2}{3}.\left(\frac{5}{7}\right).\left(\frac{5}{8}\right) = \frac{25}{84}$$

$$P(A \cap \bar{B} \cap C) = P(A)\,P(\bar{B})\,P(C)$$
$$= \frac{2}{3}.\left(\frac{2}{7}\right).\left(\frac{3}{8}\right) = \frac{1}{14}$$

$$P(\bar{A} \cap B \cap C) = P(\bar{A})\,P(B)\,P(C)$$
$$= \frac{1}{3}.\left(\frac{5}{7}\right).\left(\frac{3}{8}\right) = \frac{5}{56}$$

$$P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$$
$$= \frac{2}{3}.\left(\frac{5}{7}\right).\left(\frac{3}{8}\right) = \frac{5}{28}$$

Hence required probability
$$= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) + P(A \cap B \cap C)$$
$$= \frac{25}{84} + \frac{1}{14} + \frac{5}{56} + \frac{5}{28} = \frac{50+12+15+30}{168} = \frac{107}{168}$$

5. Exactly one of them hits the target
$(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) + (\bar{A} \cap \bar{B} \cap C)$
denotes that exactly one of them hits the target.
$$P(A \cap \bar{B} \cap \bar{C}) = P(A)\,.\,P(\bar{B})\,.\,P(\bar{C})$$
$$= \frac{2}{3}\times\frac{2}{7}\times\left(\frac{5}{8}\right) = \frac{5}{42}$$

$$P(\bar{A} \cap B \cap \bar{C}) = P(\bar{A}).P(B).P(\bar{C}) = \frac{1}{3}.\frac{2}{7}.\frac{3}{8} = \frac{1}{28}$$

$$P(\bar{A} \cap \bar{B} \cap C) = P(\bar{A}).P(\bar{B})P(C) = \frac{1}{3}.\frac{5}{7}.\frac{5}{8} = \frac{25}{168}$$

Hence required probability is
$$= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap B \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C)$$
$$= \frac{5}{42} + \frac{25}{168} + \frac{1}{28} = \frac{20+25+26}{168} = \frac{51}{168} = \frac{17}{56}$$

6. Three consecutive letters can be selected from English alphabet in 24 ways.
$\therefore \quad n(S) = 24$
Since, no two consecutive letters are vowels, the number of ways that among three letters one letter is vowel is 13.

$\therefore$ Required probability $= \frac{13}{24}$

7. Eight coins are tossed, so $n(S) = 2^8$
Let E be the favorable event. It is required to get 4 heads and 4 tails which is possible in the favorable out come HHHHTTTT.

This can be arranged in $\frac{8!}{4!4!}$ ways.
$$n(E) = \frac{8\times7\times6\times5\times4!}{4\times3\times2\times1\times4!} = 70$$

$\therefore$ Required probability $= \frac{n(E)}{n(S)} = \frac{70}{2^8} = \frac{35}{128}$

8. *Given :* $P(H) = 3P(T)$
We know
$P(H) + P(T) = 1$
$3P(T) + P(T) = 1$

$$P(T) = \frac{1}{4}$$
$$P(H) = \frac{3}{4}$$

Since the coin is tossed 3 times, two heads may occur in 1st and 2nd trial, 2nd and 3rd trial or 1st and 3rd trial.
$\therefore$ Required probability
$$= \frac{3}{4}\times\frac{3}{4}\times\frac{1}{4} + \frac{1}{4}\times\frac{3}{4}\times\frac{3}{4} + \frac{3}{4}\times\frac{1}{4}\times\frac{3}{4}$$
$$= \frac{9}{64} + \frac{9}{64} + \frac{9}{64} = \frac{27}{64}$$

9. Two black cards can be selected in $26c_2$ ways.
A pack contain 8 black honours.
Hence number ways of selecting two same honours is 4.

$\therefore$ Required probability $= \frac{4}{26c_2} = \frac{4}{325}$

10. Let one card is king, then second card can be any of the other 3 cards from the other suit and it can be selected in $4 \times 3 = 12$ ways. Since there are four honour cards, hence

$\therefore$ Required probability $= \frac{12}{325}$

11, 12 and 13
Probability that Akil reach the summit
$= P(A) = 2/3$

$\therefore \quad P(\bar{A}) = 1/3$

Probability that Nikil reach the summit is
$P(N) = 5/8$

$\therefore \quad P(\bar{N}) = 3/8$

Probability that Sunil reach the summit
$$P(S) = \frac{4}{7}$$

$\therefore \quad P(\bar{S}) = \frac{3}{7}$

11. Probability that none of them reaches the summit
$$= P(\bar{A} \cap \bar{N} \cap \bar{S}) = P(\bar{A})P(\bar{N})P(\bar{S})$$
$$= \frac{1}{3}\times\frac{3}{8}\times\frac{3}{7} = \frac{3}{56}$$

12. Probability that exactly two of them reaches the summit
$$= P(A \cap S \cap \bar{N}) + P(A \cap N \cap \bar{S}) + P(\bar{A} \cap N \cap S)$$
$$= P(A)P(S)P(\bar{N}) + P(A)P(N)P(\bar{S}) + P(N)P(S)P(\bar{A})$$
$$= \frac{2}{3}.\frac{4}{7}.\frac{3}{8} + \frac{2}{3}.\frac{5}{8}.\frac{3}{7} + \frac{1}{3}.\frac{5}{8}.\frac{4}{7} = \frac{37}{84}$$

13. Probability that atleast two of them reaches the summit

= Exactly two of them reach summit + three of them reach the summit

$$= \frac{37}{84} + \frac{5}{21} = \frac{57}{84} = \frac{19}{28}$$

14. 2 points determine a line.

Hence12 points can form $^{12}C_2$ straight lines.

But 4 points are collinear, i.e they give only one line.

Number of straight lines that can be formed

$$= {}^{12}C_2 - {}^4C_2 + 1 = 66 - 6 + 1 = 61$$

15. Three points determine a triangle.

$\therefore$ 12 non-collinear points can give $^{12}C_3$ triangles.

But 4 points are collinear, so, these 4 points do not give any triangles.

Hence, number of triangles = $^{12}C_3 - {}^4C_3 = 216$

16. Number of non-negative integral solutions of the equation

$x_1 + x_2 + ... + x_k = n$ is $(^{n+k-1})C_{k-1}$.

Hence number of non-negative integral solutions of

$x_1 + x_2 + x_3 + x_4 = 10$ is $^{13}C_3 = 286$.

17. Let m, n be the number of girls and the number of boys respectively. Then

$$^mC_2 = 66$$

$$\Rightarrow \quad \frac{m(m-1)}{2!} = 66 \quad \Rightarrow \quad m = 12$$

Also, $^mC_1 \times {}^nC_1 = 240 \Rightarrow n = 20$

Hence, number of games in which both the players were boys = $^{20}C_2 = \frac{20(19)}{2} = 190$

18. Points lying on a straight line do not form triangles. Hence, number of triangles that can be formed by using the given points

$$= {}^{19}C_3 - ({}^9C_3 + {}^{10}C_3) = 765$$

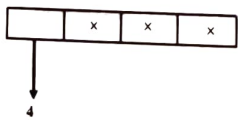
19. A number is divisible by 125 if and only if number formed by its last three digits is divisible by 125.

So, possible combinations for last three digits are: 000, 125, 250 and 500.

Now, number of three-digit number divisible by 125 is 3.

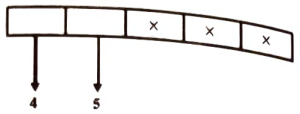
Number of four-digit numbers divisible by 125

= 4 (4) = 16.



Number of five-digit numbers divisible by 125

= 4 (4) (5) = 80.



Hence, required number of numbers

= 3 + 16 + 80 = 99.

20. Let the number of red, blue and green balls selected from the supply be x_1, x_2 and x_3 respectively.

Then, $x_1 + x_2 + x_3 = 12$.

Number of non-negative integral solutions of the above equation

$= (^{n+k-1})C_{k-1} = {}^{14}C_2 = 91$.

21. Number of positive integral solutions of equation $x_1 + x_2 + ... + x_k = n$ is $^{n-1}C_{k-1}$.

Number of positive integral solutions of the equation

Hence $a + b + c = 15$

$$^{15-1}C_{3-1} = {}^{14}C_2 = 91$$

22. When 'n' coins are tossed together, then

probability of getting exactly 'r' tails = $\frac{^nC_r}{2^n}$

Hence probability of getting atleast two tails

= 1 – (probability of getting no tail + probability of getting exactly one tail)

$$= 1 - \left\{\frac{^6C_0}{64} + \frac{^6C_1}{64}\right\} = 1 - \frac{7}{64} = \frac{57}{64}$$

EXERCISE – II

MCQ TYPE QUESTIONS

1. There are 4 kings in a pack of 52 cards.

If 2 cards are selected and both are kings, remaining cards will be 50 out of which 2 will be kings.

2. For a match to be played, we need 2 teams

No. of matches = No. of ways of selections 2 teams out of 5

$$= {}^5C_2 = 10$$

4. $x = \{HHT, HHH\}$

y depends on x

$z = \{TTH, TTT\}$

$\therefore$ 'd' is the correct choice.

5. P (Ram) = 1/6; p(Ramesh) = 1/8

p(only at) = p (Ram) × p (not ramesh) + p(Ramesh) × p($n_o \times R_{am}$)

$$= \frac{1}{6} + \frac{7}{8} + \frac{1}{8} \times \frac{5}{6}$$

$$\Rightarrow \frac{12}{40} = \frac{1}{4}$$

6. Here the given sets are

$A = \{2, 3, 4, 5\}$

and $B = \{11, 12, 13, 14, 15\}$

desired outcome = $\{(2, 14), (3, 13), (4, 12), (5, 11)\}$

$\therefore \quad n(E) = 4$

and total outcome = 4 × 5 = 20

$\therefore \quad n(s) = 20$

$\therefore$ Probability of an event

$$= \frac{n(E)}{n(s)} = \frac{4}{20} = \frac{1}{5} = 0.2$$

8. The digit in unit place is selected in 4! Ways

The digit in tens place is selected in 4! Ways

The digit in hundreds place is selected in 4! Ways

The digit in thousands place is selected in 4! Ways

The digit in ten thousands place is selected in 4! Ways

Sum of all values for 1

$4! \times 1 \times (10^0 + 10^1 + 10^2 + 10^3 + 10^4)$

$= 4! \times 11111 \times 1$

Similarly for '3' $4! \times (11111) \times 3$

Similarly for '5' $4! \times (11111) \times 5$

Similarly for '7' $4! \times (11111) \times 7$

Similarly for '9' $4! \times (11111) \times 9$

$\therefore$ sum of all such numbers

$4! \times (11111) \times (1 + 3 + 5 + 7 + 9)$

$= 24 \times (11111) \times 25$

$= 6666600$

9. The probability that the other face is tail is $\frac{1}{3}$.

10. **Given:** All the 2-digit numbers

To find: Probability of selecting a number at random not divisible by 7

Analysis: Total number of 2-digit numbers

10, 11, ..., 99 → 90 numbers

Unfavourable numbers *i.e.* numbers divisible by

7 : 14, 21, ..., 98

Number of unfavourable numbers = 13

This can be hand counted or we can use the following properties:

Last term = First time + (no. – 1) Diff

$\Rightarrow \quad 98 = 14 + (\text{no.} - 1)\,7$

$\Rightarrow \quad \text{no.} = 13$

$\Rightarrow$ Probability = $\frac{\text{Total} - \text{unfavourable}}{\text{Total}}$

$$= \frac{90 - 13}{90} = 77/90$$

Hence, the solution is (*d*)

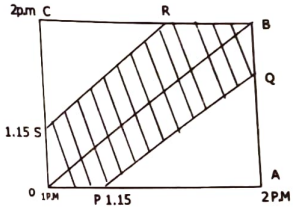
11. at a leap year

52 weeks, and 2 extra day they are (Mon, Tues) (Tues, Wed) (Wed Thu.) (Thus. Fri) (Fri. Sat) (Sat. Sun) (Sun Mond.)

$n(s) = 7$

$n(E) = 2 \quad P(E) = \frac{2}{7}$

13.



OB is the line when both A and B arrive at same time.

Total sample sapce = 60 × 60 = 3600

Favourable cases = Area of OABC – 2(Area of SRC)

$$= 3600 - 2 \times \left(\frac{1}{2} \times 45 \times 45\right)$$

$$= 1575$$

$\therefore$ Required probability = $\frac{1575}{3600} = \frac{7}{16}$

14. 2, 2, 3, 3, 4, 4, 4, 4

322 — 3, 4

323 — 2, 3, 4

324 — 2, 3, 4

332 — 2, 3, 4

333 — 2, 4

334 — 2, 3, 4

342 — 2, 3, 4

343 — 2, 3, 4

$x > 3000$

344 — 2, 3, 4

422 — 3, 4

423 — 2, 3, 4

424 — 2, 3, 4

432 — 2, 3, 4

433 — 2, 3, 4

434 — 2, 3, 4

442 — 2, 3, 4

443 — 2, 3, 4

444 — 2, 3, 4

NUMERICAL TYPE QUESTIONS

1. In thousands place, 9 digits except 0 can be placed

In hundreds place, 9 digits can be placed (including 0, excluding the one used in thousands place)

In tens place, 8 digits can be placed (excluding the ones used in thousands and hundreds place)

In ones place, 7 digits can be placed (excluding the one used in thousands, hundreds and tens place)

Total number of combinations

$= 9 \times 9 \times 8 \times 7 = 4536$

2. Given, in one year, probability of an earthquake > 6 magnitude $= 0.04$

So, average time between successive earthquakes $= 1/0.04 = 25$ years

3. Given, 10% of the population is HIV^+, so probability $= 0.1$ for HIV^+ and 0.9 for HIV^-

The diagnostic kit identifies $\rightarrow 0.95$ HIV^+ and 0.89 HIV^- correctly

So, the required probability that the person is actually $HIV^+ = 0.95/2 = 0.475$

4. Probability for one bulb to be non-defective is

$$\frac{95}{100}$$

$\therefore$ Probabilities that none of the bulbs is defectives $\left(\frac{95}{100}\right)^4 = 0.8145$

■■

6 CHAPTER Miscellaneous

EXERCISE – I

MCQ TYPE QUESTIONS

1. If one root of a quadratic equation is $3 + \sqrt{2}$, then quadratic equation is

(a) $x^2 - 6x - 7 = 0$ (b) $x^2 + 6x + 7 = 0$

(c) $x^2 + 6x - 7 = 0$ (d) $x^2 - 6x + 7 = 0$

2. Three consecutive positive integers such that square of their sum exceeds the sum of their squares by 214, are

(a) 5, 6, 7 (b) 10, 11, 12

(c) 9, 10, 11 (d) 7, 8, 9

3. A and B attempt to solve a quadratic equation of the form $ax^2 + bx + c = 0$. A starts with a wrong value of b and gets the roots as –3 and –5. B starts with a wrong value of C and gets the roots as 6 and 2. The correct roots are

(a) 3, 4 (b) 3, 5

(c) 4, 6 (d) 2, 3

4. What is the value of

$$\frac{1}{5+\frac{1}{5+\frac{1}{5+\ldots.}}} ?$$

(a) $\frac{5+\sqrt{29}}{2}$ (b) $\frac{5-\sqrt{29}}{2}$

(c) $\frac{5\pm\sqrt{29}}{2}$ (d) 7

5. A man buys two horses for ₹ 86. By selling one for three – quarters of its cost price, and the other for four-thirds of its cost price, he makes a profit of ₹ 3 on the whole transaction. The cost price (in rupees) of each horse is

(a) ₹ 39, ₹ 47 (b) ₹ 34, ₹ 52

(c) ₹ 48, ₹ 38 (d) ₹ 44, ₹ 42

6. What is the number of triplets (a, b, c) where a, b, c are positive integers, which satisfy the simultaneous equations ?

(i) $ab + bc = 44$

(ii) $ac + bc = 23$ is/are

(a) 1

(b) 2

(c) 4

(d) Indefinitely many

7. A man sitting beside me at a hotel ate idlis one after the other by ordering plate by plate. He said to me after drinking some water the last one I ate was my hundredth idli in last five days. Each day I ate 6 more than the previous day. How many he ate yesterday?

(a) 26 (b) 24 (c) 22 (d) 20

8. A man, who went out between five and six o'clock and returned between six and seven o'clock, found that the hands of the watch had exchanged places. When did he go out?

(a) $32\frac{4}{13}$ minutes past 5 o'clock.

(b) $27\frac{2}{13}$ minutes past 5 o'clock.

(c) $37\frac{3}{13}$ minutes past 5 o'clock.

(d) $31\frac{5}{13}$ minutes past 5 o'clock.

9. What is the time between 7 a.m. and 8 a.m. when hands of a clock will be in the same straight line but not coincide?

(a) 7 hours $5\frac{5}{11}$ min

(b) 7 hours $3\frac{5}{11}$ min

(c) 7 hours $7\frac{3}{11}$ min

(d) 7 hours $7\frac{7}{11}$ min

NUMERICAL TYPE QUESTIONS

1. When $x^5 + x^4 + 5x^2 - 3$ is divided by $(x + 2)$, the remainder will be_____

2. Pressure of a gas is inversely proportional to the volume. If pressure of oxygen is 84 atm when the volume is 1000 cm^3, then volume when the pressure is 28 atm will be_____ cm^3

3. The value of 'a' will make $x^2 - ax + 25$ a perfect square is _____

6.2

4. The value of $\alpha^3 + \beta^3$ if α and β are roots of the equation : $2x^2 - 7x - 4 = 0$ is _____

5. Sum of Anita's and Sunita's age is 43 years. 11 years hence, Anita's age will be $\frac{7}{6}$ times Sunita's age then. Sunita's present age is___ years

6. In a school trip, the students were accommodated in two rooms A and B. If 5 students are shifted from room A to room B then there will be equal number of students in both the rooms. But if 5 students from room B is shifted to room A, then the resulting number of students in room A is double the number of students left in room B. Total number of students are _____

7. A band of workers unloaded a certain number of boxes from their ship. The number of workers were between 50 and 60 (both inclusive). Each worker handled exactly 11 boxes, and each box was handled by exactly 7 workers. Exactly_____number of boxes were there?

8. If ax = b,by = c ,cz = a, then value of xyz is _____

9. The value of

$$\frac{1}{\left(1+\frac{xb}{xa}+\frac{xc}{xa}\right)}+\frac{1}{\left(1+\frac{xa}{xb}+\frac{xc}{xb}\right)}+\frac{1}{\left(1+\frac{xb}{xc}+\frac{xa}{xc}\right)}$$

is _____

10. A palm tree was 90 cm high, when it was planted. It grows by an equal number of cm each year, and at the end of the seventh year it was one ninth taller than at the end of the sixth year. The height of tree in cm at the end of the twelfth year is _____

11. A bottle and its cork together cost ₹ 1.10, and the bottle costs ₹ 1.00 more than its cork. The price of the bottle is _____

12. The value of is $(100-1)(100-2)(100-3)\ldots\ldots$ $(100+1)(100+2)(100+3)$ is _____

13. The value of $\cfrac{2+\cfrac{1}{3+\cfrac{4}{5}}}{2+\cfrac{1}{3+\cfrac{1}{1+\cfrac{1}{4}}}}$ is _____

14. Three integers, a, b and c, are such that $\frac{b}{c} = 3$. The fraction $\frac{b}{a}$ is in its lowest form. The_____ is the possible value of a.

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. If $\log_x\left(\frac{5}{7}\right) = -\frac{1}{3}$, then the value of x is

(a) $\frac{343}{125}$ (b) $\frac{125}{343}$

(c) $-\frac{25}{49}$ (d) $-\frac{49}{25}$

2. A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

(a) 59 (b) 45

(c) 43 (d) 35

3. If $x > y > 1$, which of the following must be true?

(i) $\ln x > \ln y$ (ii) $e^x > e^y$

(iii) $y^x > x^y$ (iv) $\cos x > \cos y$

(a) *i* and *ii* (b) *i* and *iii*

(c) *iii* and *iv* (d) *ii* and *iv*

4. log tan1° + log tan2° +------+ log tan89° is ____.

(a) 1 (b) $1/\sqrt{2}$

(c) 0 (d) –1

5. Based on the given statements, select the most appropriate option to solve the given question.

What will be the total weight of 10 poles each of same weight?

Statements:

(I) One fourth of the weight of a pole is 5 kg

(II) The total weight of these poles is 160 kg more than the total weight of two poles.

(a) Statement II alone is not sufficient

(b) Statement II alone is not sufficient

(c) Either I or II alone is sufficient

(d) Both statements I and II together are not sufficient.

6. Consider a function $f(x) = 1 - |x|$ on $-1 \le x \le 1$. The value of x at which the function attains a maximum, and the maximum value of function are:

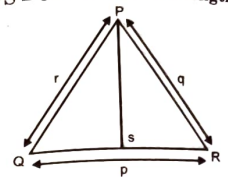
(a) 0, –1

(b) –1, 0

(c) 0, 1

(d) –1, 2

7. In a triangle PQR, PS is the angle bisector of ∠QPR and ∠QPS = 60°. What is the length of PS?



(a) $\frac{(q+r)}{qr}$ (b) $\frac{qr}{(q+r)}$

(c) $\sqrt{(q^2+r^2)}$ (d) $\frac{(q+r)^2}{qr}$

8. Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

I. Each step is ¾ foot high.

II. Each step is 1 foot wide.

(a) Statement I alone is sufficient, but statement II alone is not sufficient.

(b) Statement II alone is sufficient, but statementI alone is not sufficient.

(c) Both statements together are sufficient, but neither statement alone is sufficient.

(d) Statement I and II together are not sufficient.

2014

9. If $y = 5x^2 + 3$, then the tangent at $x = 0, y = 3$

(a) passes through $x = 0, y = 0$

(b) has a slope of +1

(c) is parallel to the x-axis

(d) has a slope of –1

10. Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

(a) 2^{n-m} P (b) 2^{m-n} P

(c) $2(n-m)$ P (d) $2(m-n)$ P

11. A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: $1 \to 0.167$; $2 \to 0.167$; $3 \to 0.152$; $4 \to 0.166$; $5 \to 0.168$; $6 \to 0.180$. We call this die

(a) irregular (b) biased

(c) Gaussian (d) insufficient

12. Consider the equation : $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

(a) 1634 (b) 1737

(c) 3142 (d) 3162

13. At what time between 6 *a.m.* and 7 *a.m.* will the minute hand and hour hand of a clock make an angle closest to 60°?

(a) 6 : 22 *a.m.* (b) 6 : 27 *a.m.*

(c) 6 : 38 *a.m.* (d) 6 : 45 *a.m.*

14. The roots of $ax^2 + bx + c = 0$ are real and positive a, b and c are real. Then $ax^2 + b|x| + c = 0$ has

(a) No roots (b) 2 real roots

(c) 3 real roots (d) 4 real roots

2012

15. The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹ 50 per unit. The number of units to be produced by the firm such that the profit is maximized is

(a) 5 (b) 10

(c) 15 (d) 25

16. Which of the following assertions are **CORRECT ?**

P : Adding 7 to each entry in a list adds 7 to the mean of the list

Q : Adding 7 to each entry in a list adds 7 to the standard deviation of the list

R : Doubling each entry in a list doubles the mean of the list

S : Doubling each entry in a list leaves the standard deviation of the list unchanged

(a) P, Q (b) Q, R

(c) P, R (d) R, S

17. A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

(a) 8 meters (b) 10 meters

(c) 12 meters (d) 14 meters

18. There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

(a) 2 (b) 3

(c) 4 (d) 8

2011

19. If log (P) = (1/2) Log (Q) = (1/3) Log (R), then which of the following options is TRUE?

(a) $P^2 = Q^3 R^2$ (b) $Q^2 = PR$

(c) $Q^2 = R^3 P$ (d) $R = P^2 Q^2$

20. The variable cost (V) of manufacturing a product varies according to the equation V = 4q, where q is the quanity produced. The fixed cost (F) of production of same product reduces with q according to the equation F = 100/q. How many units should be produced to minimize the total cost (V + F)?

(a) 5 (b) 4

(c) 7 (d) 6

21. Three friends R, S and T shared toffee from a bowl. R took 1/3rd of the toffees, but returned four to the bowl. S took 1/4th of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

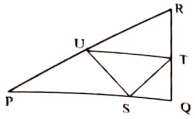
(a) 38 (b) 31

(c) 48 (d) 41

NUMERICAL TYPE QUESTIONS

2015

1. In the given figure angle Q is a right angle, PS : QS = 3:1, RT : QT = 5:2 and PU : UR = 1:1. If area of triangle QTS is 20 cm², then the area of triangle PQR in cm² is _____.



2. If p, q, r, s are distinct integers such that:

$f(p, q, r, s)$ = max (p, q, r, s)

$g(p, q, r, s)$ = min (p, q, r, s)

$h(p, q, r, s)$ = remainder of $(p \times q)/(r \times s)$ if $(p \times q) > (r \times s)$ or remainder of $(r \times s)/(p \times q)$ if $(r \times s) > (p \times q)$

Also a function $fgh(p, q, r, s) = f(p, q, r, s) \times g(p, q, r, s) \times h(p, q, r, s)$

Also the same operations are valid with two variable function of the form $f(p, q)$.

What is the value of $fg(h(2, 5, 7, 3), 4, 6, 3$

2014

3. When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines? _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (d) **2.** (a) **3.** (b) **4.** (a) **5.** (d) **6.** (b) **7.** (a) **8.** (a) **9.** (a)

Numerical Type Questions

1. 1 **2.** 3000 **3.** 10 **4.** $\frac{511}{8}$ **5.** 19 **6.** 60 **7.** 88 **8.** 1 **9.** 1

10. 450 **11.** 1.05 **12.** 0 **13.** 1 **14.** 104

EXERCISE – II

MCQ Type Questions

1. (a) **2.** (c) **3.** (a) **4.** (c) **5.** (c) **6.** (c) **7.** (b) **8.** (a) **9.** (c) **10.** (a)

11. (b) **12.** (c) **13.** (a) **14.** (d) **15.** (a) **16.** (c) **17.** (b) **18.** (a) **19.** (b) **20.** (a)

21. (c)

Numerical Type Questions

1. 280 **2.** (8) **3.** (6)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. If one root = $3 + \sqrt{2}$, then other root = $3 - \sqrt{2}$

∴ Sum of roots = $3 + \sqrt{2} + 3 - \sqrt{2} = 6$

∴ Product of roots

$= (3 + \sqrt{2})(3 - \sqrt{2}) = 9 - 2 = 7$

Quadratic equation is

x^2 – (Sum of roots) x + Product of roots = 0

i.e., $x^2 - 6x + 7 = 0$

2. Let consecutive positive integers be x – 1, x and x + 1

$\therefore [x - 1 + x + x + 1]^2$

$= (x - 1)^2 + x^2 + (x + 1)^2 + 214$

$(3x)^2 = x^2 + 1 - 2x + x^2 + x^2 + 2x + 1 + 214$

$9x^2 = 3x^2 + 216$

$6x^2 = 216$

$x^2 = 36$; $x = \pm 6$

But x is a positive integer, hence x = 6

So, numbers are 5, 6, and 7.

3. Equation formed by A with the roots –3 and –5 is

$x^2 - (-8x) + 15 = 0$

i.e., $x^2 + 8x + 15 = 0$

But value of B is wrong.

Hence coefficient of x is wrong.

Equation formed by B with roots 6 and 2 is

$x^2 - 8x + 12 = 0$

But the value of C is wrong.

hence constant term is wrong.

So, correct equation is

$x^2 - 8x + 15 = 0$.

Hence, real roots are 3 and 5.

4. Let

$$x = 5 + \cfrac{1}{\left[5 + \cfrac{1}{5 + \cfrac{1}{5 + ...}}\right.}$$

i.e., $x = 5 + \frac{1}{x} = \frac{5x + 1}{x}$

$\Rightarrow x^2 = 5x + 1$

$\Rightarrow x^2 - 5x - 1 = 0$

$\Rightarrow x = \frac{5 \pm \sqrt{(-5)^2 - 4(-1)(1)}}{2}$

$= \frac{5 \pm \sqrt{25 + 4}}{2} = \frac{5 \pm \sqrt{29}}{2}$

Since, all the terms are positive. The answer has to be $\frac{5 + \sqrt{29}}{2}$.

5. Let cost price of horses by ₹ x and ₹ y.

$\therefore x + y = 86$ ...(1)

and $\frac{3x}{4} + \frac{4y}{3} = 89$

$\Rightarrow 9x + 16y = 1068$ ...(2)

Multiplying (1) by 9, $9x + 9y = 774$ ...(3)

Subtracting (3) from (2),

$7y = 294$

$\Rightarrow y = 42$

Substituting value of 'y' in equation (1),

$x + 42 = 86$

$\Rightarrow x = 44$.

6. Equation (ii) is c(a + b) = 23, a prime number.

So, 2 factors must be 1, 23.

Since a, b are positive, therefore a + b > 1

$\Rightarrow$ c = 1; a + b = 23

Put c = 1; b = 23 – a in (i)

$\Rightarrow a^2 - 22a + 21 = 0$

$\Rightarrow$ a = 1 or 21

$\Rightarrow$ b = 22 or 2

∴ Solution sets are (1, 22, 1); (21, 2, 1).

7. First day the number of idlis he ate be x

Second day the count is (x+6)

Third day ——— (x+12)

Fourth day ——— (x+18)

Fifth day ——— (x+24)

Total is $5x + 6(1+2+3+4) = 100$

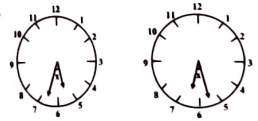
$5x + 60 = 100$

$x = 8$

Day	Idlis
1	8
2	14
3	20
4	26
5	32

So, on fourth day, number of idlis the man ate were 26.

8.



When hour hand moves 1 minute space, then minute hand moves 12 minute spaces.

Hence when hour hand moves x minutes, then minute hand moves 12x minutes.

$\therefore 12x = 60 - x$

$\Rightarrow x = \frac{60}{13}$

Interchangeable positions occur when original interval between the hands is $4\frac{8}{13}$ minute spaces or a multiple of this.

Hour hand moves through $\frac{60}{13}$ minute spaces.

At 5 O'clock, minute hand is 25 minutes behind time.

It has to gain $25\frac{60}{13} = \frac{385}{13}$ minutes.

$\frac{385}{13}$ minutes will be gained in

$\frac{385}{13} \times \frac{60}{55} = \frac{420}{13} = 32\frac{4}{13}$ minutes.

Hence man went out at $32\frac{4}{13}$ minutes past 5 O'clock.

9. When they are in the same straight line, in opposite direction. Minute hand and hour hand are 30 minute spaces apart.

At 7 p.m. they are 25 minute spaces apart.

So, minute hand has to gain only (30 – 25) = 5 minute spaces.

i.e., $\frac{60}{55} \times 5 = \frac{60}{11} = 5.\frac{5}{11}$ minutes.

Hence hands are in the same straight line at $5\frac{5}{11}$ minutes past 7.

NUMERICAL TYPE QUESTIONS

1. $p(x) = x^5 + x^4 + 5x^2 - 3$

$= (-2)^5 + (-2)^4 + 5(-2)^2 - 3$

$= (-32) + 16 + 5 \times 4 - 3 = 1$

2. $p \propto \frac{1}{v}$

$\therefore \quad p = \frac{k}{v}$

where, $k = 84 \times 1000$

$\therefore \quad v = \frac{k}{p} = \frac{84 \times 1000}{28} = 3000 \text{ cm}^3$

3. Middle term $= 2\sqrt{\text{first term} \times \text{last term}}$

$ax = 2 \times x \times 5 = 10x$

$\therefore \quad a = 10$

4. $\alpha + \beta = \frac{7}{2}$

$\alpha\beta = -\frac{4}{2} = -2$

$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

$\left(\frac{7}{2}\right)^3 = \alpha^3 + \beta^3 + 3(-2)\left(\frac{7}{2}\right)$

$\frac{343}{8} = \alpha^3 + \beta^3 - 21$

$\frac{343}{8} + 21 = \alpha^3 + \beta^3$

$\therefore \quad \frac{511}{8} = \alpha^3 + \beta^3$

5. Let Anita's present age be 'x' years and that of Sunita by 'y' years.

$\therefore \quad x + y = 43$ ...(1)

and $(x + 11) = \frac{7}{6}(y + 11)$

$\Rightarrow \quad 6x + 66 = 7y + 77$

$\Rightarrow \quad 6x - 7y = 11$ ...(2)

Multiplying (1) by 6.

$6x + 6y = 258$ ...(3)

Subtracting (3) from (2),

$-13y = -247$

$\therefore \quad y = 19$ years.

6. Let the number of students in room A be x and number of students in room B be y.

$\therefore \quad x - 5 = y + 5$

$x - y = 10$ ...(1)

Also, $x + 5 = 2(y - 5)$

$\Rightarrow \quad x - 2y = -15$ ...(2)

Solving (1) and (2) we get,

$x = 35$ and $y = 25$

$\therefore \quad x + y = 60$

7. Let P be the number of workers, and let C be the number of boxes. Then, by counting the number of boxes handling in 2 different ways yields $11P = 7C$.

Since P and C are integers, it follows that P must be divisible by 7.

Since $50 \le P \le 60$,

P must equal 56

$\therefore \quad C = 11 \times \frac{56}{7} = 88$

8. $ax = b$

$\Rightarrow \quad (cz)\ x = b$ ....[since cz = a]

$\Rightarrow \quad (by)\ xz = b$ ....[since by = c]

$\Rightarrow \quad xyz = 1$

9. $\frac{1}{\left(1 + \frac{xb}{xa} + \frac{xc}{xa}\right)} + \frac{1}{\left(1 + \frac{xa}{xb} + \frac{xc}{xb}\right)} + \frac{1}{\left(1 + \frac{xb}{xc} + \frac{xa}{xc}\right)}$

$= \frac{xa}{(xa + xb + xc)} + \frac{xb}{(xa + xb + xc)} + \frac{xc}{(xa + xb + xc)}$

$= \frac{(xa + xb + xc)}{xa + xb + xc} = 1$

10. Let tree grows x cm each year

Height of the tree at the end of the sixth year $= (90 + 6x)$ cm

Growth in seventh year,

$x = \frac{1}{9}(90 + 6x)$ cm

$x = 10 + \frac{2x}{3} \quad \therefore \quad x = 30$

Hence height of tree at the end of the twelfth year $= (90 + 12 \times 30) = 450$cm.

11. Let B = price of the bottle

C = price of the cork

Given : $B + C = ₹\ 1.10$ ...(1)

and $B - C = ₹\ 1.00$ ...(2)

From equations (1) and (2)

$B = ₹\ 1.05$

$C = ₹\ 0.05$.

12. $(100 - 1)(100 - 2)(100 - 3)\ldots\ldots$

$(100 - 100)(100+1)(100+2)(100+3)$

$\Rightarrow \quad (100 - 1)(100 - 2)(100 - 3)\ldots\ldots$

$(0)(100 + 1)(100 + 2)(100 + 3) = 0.$

13. $\frac{2 + \frac{1}{19}}{2 + \frac{1}{3 + \frac{4}{5}}} = \frac{2 + \frac{5}{19}}{2 + \frac{5}{19}} = 1$

14. $\frac{b}{c} = 3 \Rightarrow b = 3c$, i.e., b is divisible by 3.

Also, $\frac{b}{c}$ cannot be reduced,

i.e., a is not divisible by 3.

EXERCISE – II

MCQ TYPE QUESTIONS

1. $\frac{5}{7} = x^{-1/3} \Rightarrow \frac{7}{5} = x^{1/3} \Rightarrow \left(\frac{7}{5}\right)^3 = x \Rightarrow x = \frac{343}{125}$

2. $f(x) = 2x + 33 = 2(5) + 33 = 43$

3. For whole numbers, greater the value greater will be its log.

Same logic for power of *e*.

4. $\log \tan 1° + \log \tan 89° = \log(\tan 1° \times \tan 89°)$

$= \log(\tan 1° \times \cot 1°)$

$= \log 1 = 0$

Using the same logic total sum is '0'.

6. The given function is

$f(x) = 1 - |x|$ on $-1 \le x \le 1$

In order to find maximum and minimum value of function in interval [–1, 1]

At $x = -1$, $f(-1) = 1 - |-1| = 1 - 1 = 0$

At $x = -0.5$, $f(-0.5) = 1 - |-0.5| = 1 - 0.5 = 0.5$

At $x = 0$, $f(0) = 1 - |0| = 1$

At $x = 0.5$, $f(0.5) = 1 - |0.5| = 0.5$

At $x = 1$, $f(1) = 1 - |1| = 1 - 1 = 0$

$\therefore$ Hence, maximum value occurs at $x = 0$ and its maximum value is 1.

$\therefore$ Option (c) is correct.

9. $y = 5x^2 + 3$, $\frac{dy}{dx} = 10x$

Slope of tangent $= \left(\frac{dy}{dx}\right)_{x=0, y=3} = 10 \times 0 = 0$

Slope = 0 $\Rightarrow$ tangent is parallel to *x*-axis.

10. $P' = 2^n X^n \left(\frac{1}{2}\right)^m y^m = 2^{n-m} X^n Y^m = 2^{n-m} P$

11. For a very large number of throws, the frequency should be same for unbiased throw. As it not same, then the die is baised.

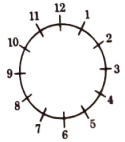
12. $(7526)_8 - (y)_8 = (4364)_8$

$\Rightarrow \quad y_8 = (7526)_8 - (4364)_8$

$$\begin{array}{cccc} & {\scriptstyle 4} & {\scriptstyle (8+2=10)} & \\ 7 & \not{5} & 2 & 6 \\ 4 & 3 & 6 & 4 \\ \hline 3 & 1 & 4 & 2 \end{array}$$

When we have base 8, we borrow 8 instead of 10 as done in normal subtraction.

13.



From option

At 6 a.m.

Both hand – 180° aparts,

At 6 : 20 a.m. – both hand apart

$\Rightarrow \quad 180° + 10° - 120° = 70°$

At 6 : 22 a.m. – both hand apart

$\Rightarrow \quad 180° + 12° - 132° = 60°$ (closest)

14. $ax^2 + bx + c = 0$

Roots are real and positive

$\therefore \quad b^2 - 4ac > 0$

By the above condition we get two positive real roots

Now, $ax^2 + b|x| + c = 0$

This can be written as:

$ax^2 + bx + c$

$\Delta = b^2 - 4ac > 0$

Now for $ax^2 - bx + c$

$(-b)^2 - 4ac \quad \Rightarrow \quad b^2 - 4ac$

Again $b^2 - 4ac > 0$

So again ∴ ...et real roots

Thus we have ... ...l roots of $ax^2 + b|x| + c = 0$

15. $P = 50q - 5q^2$

$\frac{dp}{dq} = 50 - 10q; \frac{d^2p}{dq^2} < 0$

∴ p is maximum at $50 - 10q = 0$ or, $q = 5$

Else check with options

16. P and R always holds true

Else consider a sample set {1, 2, 3, 4} and check accordingly

17. $y = 2x - 0.1x^2$

$\frac{dy}{dx} = 2 - 0.2x$

$\frac{d^2y}{dx^2} < 0$

∴ y maximises at $2 - 0.2x = 0$

$\Rightarrow x = 10$

∴ $y = 20 - 10 = 10$m

18. Let us categorize bags in three groups as

$A_1 A_2 A_3$ $B_1 B_2 B_3$ $C_1 C_2$

1st weighing A vs B

Case-1	**Case-2**
$A_1 A_2 A_3 = B_1 B_2 B_3$	$A_1 A_2 A_3 \neq B_1 B_2 B_3$
Then either C_1 or C_2 is heavier	Either A or B would be heavier (Say A > B)
2nd weighing	
C_1 vs C_2	A_1 vs A_2
If $C_1 > C_2$, then C_1	If $A_1 = A_2$, then A_3
If $C_1 < C_2$, then C_2	If $A_1 > A_2$, then A_1
If $A_1 < A_2$, then A_2	

19. $\log P = \frac{1}{2} \log Q = \frac{1}{3} \log (R) = k$

∴ $P = b^k$, $Q = b^{2k}$, $R = b^{3k}$

Hence, $Q^2 = b^{4k} = b^{3k} b^k = PR$

20. Checking with all options in formula: $(4q+100/q)$ i.e. (V+F). Option (a) gives the minimum cost.

21. Let total number of toffees in bowl be x

R took $\frac{1}{3}$ of toffees and returned 4 to the bowl

∴ Number of toffees with R $= \frac{1}{3}x - 4$

Remaining of toffees in bowl $= \frac{2}{3}x + 4$

Number of toffees with S $= \frac{1}{4}\left(\frac{2}{3}x + 4\right) - 3$

Remaining toffees in bowl $= \frac{3}{4}\left(\frac{2}{3}x + 4\right) + 4$

Number of toffees with T $= \frac{1}{2}\left(\frac{3}{4}\left(\frac{2}{3}x + 4\right) + 4\right) + 2$

Remaining toffees in bowl

$$= \frac{1}{2}\left[\frac{3}{4}\left(\frac{2}{3}x + 4\right) + 4\right] + 2$$

Given : $\frac{1}{2}\left[\frac{3}{4}\left(\frac{2}{3}x + 4\right) + 4\right] + 2 = 17$

$\Rightarrow \frac{3}{4}\left(\frac{2}{3}x + 4\right) = 27$

$\Rightarrow x = 48$

NUMERICAL TYPE QUESTIONS

1. Let area of triangle PQR be 'A'

$$\frac{SQ}{PQ} = \frac{1}{1+3} = \frac{1}{4}$$

$$\frac{QT}{QR} = \frac{2}{2+5} = \frac{2}{7}$$

∴ Area of $\Delta QTS = \frac{1}{2} \times SQ \times QT$

$$= \frac{1}{2} \times \left(\frac{1}{4}PQ\right) \times \left(\frac{2}{7}QR\right)$$

$$= \frac{1}{4} \times \frac{2}{7} \times \left(\frac{1}{2} \times PQ \times QR\right)$$

$$= \frac{1}{14} \times \text{Area of } \Delta PQR$$

Given $20 \text{ cm}^2 = \frac{1}{14} \times A$

∴ $A = 14 \times 20 = 280 \text{ cm}^2$

2. $fg(h(2, 5, 7, 3), 4, 6, 8) = fg(1, 4, 6, 8)$

$= f(1, 4, 6, 8) \times g(1, 4, 6, 8)$

$= 8 \times 1 = 8$

Engineering Mathematics

1 CHAPTER

Linear Algebra

DETERMINANTS

A determinant of order n has n rows and n columns. It has $n \times n$ elements.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{vmatrix}$$

MINOR

If a_{ij} is an element which is in the i^{th} row and j^{th} column of a squre matrix A, then determinant of the matrix obtained by deleting i^{th} row and j^{th} column of A is called minor of a_{ij} and is denoted by M_{ij}

e.g. If $A = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, then

M_{11} = minor of $a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

M_{12} = minor of $a_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

$= a_{21}a_{33} - a_{31}a_{23}$ and so on.

9 minors can be obtained corresponding to 9 elements of Δ.

e.g. Minor of the element

$$a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = M_{11}$$

$$a_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = M_{12}$$ and so on.

Determinant of a Square Matrix

Only square matrix can have determinants because in a determinant, number of rows is equal to number of columns.

Let $A = [a_{ij}]_{max}$ be a square matrix of order n.

Then $$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{vmatrix}$$

is called *determinant of matrix A.*

Properties of Determinants

(*i*) If all the elements of one column or one row of a determinant are multiplied by the same number "C", then value of the new determinant is "C" times the value of the given determinant.

(*ii*) Value of a determinant does not change when rows and columns are interchanged.

(*iii*) If any two columns or two rows of a determinant are interchanged, then value of the determinant is multiplied by – 1.

(*iv*) In a determinant, sum of the products of the element of any column (or row) with the cofactors of the corresponding elements of any other column (or row) is zero.

(*v*) If two columns (or two rows) of a determinant are identical, then value of the determinant is zero.

(*vi*) If in a determinant each element in any column (or row) consists of sum of two terms, then determinant can be expressed as the sum of two determinants of the same order.

(*vii*) If the elements of a column (or row) of a determinant are added K times the corresponding elements of another column (or row), then value of the determinant obtained is equal to value of the original determinant.

CO-FACTOR

If a_{ij} is an element which is in the i^{th} row and j^{th} column of a square matrix A, then product of $(-1)^{i+j}$ and minor of a_{ij} is called cofactor of a_{ij} and is denoted by A_{ij}.

e.g. If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then

A_{11} = cofactor of $a_{11} = (-1)^{1+1} M_{11}$

$= +1 \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$

A_{12} = cofactor of $a_{12} = (-1)^{1+2} M_{11}$

$= -1 \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = (a_{31}a_{23} - a_{21}a_{33})$

and so on.

and if $A=\begin{bmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{bmatrix}$, then

A_1 = cofactor of $a_1 = (-1)^{1+1}\begin{vmatrix}b_2 & c_2\\ b_3 & c_3\end{vmatrix}$

$= b_2c_3 - b_3c_2$

B_1 = cofactor of $b_1 = (-1)^{1+2}\begin{vmatrix}a_2 & c_2\\ a_3 & c_3\end{vmatrix}$

$= -(a_2c_3 - a_3c_2) = a_3c_2 - a_2c_3$

C_1 = cofactor of c_1

$= (-1)^{1+3}\begin{vmatrix}a_2 & b_2\\ a_3 & b_3\end{vmatrix} = a_2b_3 - a_3b_2$ and so on.

Example. If $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$, then find all cofactors of element of A.

Solution. $A_{11} = (-1)^{1+1}\begin{vmatrix}5 & 6\\ 8 & 9\end{vmatrix} = +(45-48) = -3$

$A_{12} = (-1)^{1+2}\begin{vmatrix}4 & 6\\ 7 & 9\end{vmatrix} = -(36-42) = 6$

$A_{13} = (-1)^{1+3}\begin{vmatrix}4 & 5\\ 7 & 8\end{vmatrix} = +(32-35) = -3$

$A_{21} = (-1)^{2+1}\begin{vmatrix}2 & 3\\ 8 & 9\end{vmatrix} = -(18-24) = 6$

$A_{22} = (-1)^{2+2}\begin{vmatrix}1 & 3\\ 7 & 9\end{vmatrix} = +(9-21) = -12$

$A_{23} = (-1)^{2+3}\begin{vmatrix}1 & 2\\ 7 & 8\end{vmatrix} = -(8-14) = 6$

$A_{31} = (-1)^{3+1}\begin{vmatrix}2 & 3\\ 5 & 6\end{vmatrix} = +(12-15) = -3$

$A_{32} = (-1)^{3+2}\begin{vmatrix}1 & 3\\ 4 & 6\end{vmatrix} = -(6-12) = 6$

$A_{33} = (-1)^{3+3}\begin{vmatrix}1 & 2\\ 4 & 5\end{vmatrix} = +(5-8) = -3$

ALGEBRA OF MATRICES

Matrix

A set of m,n numbers (real or complex) arranged in the form of a rectangular may have m rows and n columns is called an $m \times n$ *matrix.*

An $m \times n$ matrix is written as

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13} & \dots & a_{1n}\\ a_{21} & a_{22} & a_{23} & \dots & a_{2n}\\ \dots & \dots & \dots & \dots & \dots\\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn}\end{bmatrix}$$

Element a_{ij} belongs to i^{th} row and j^{th} column and is sometimes called as $(i,j)^{th}$ element of the matrix.

Note: In a matrix, number of rows and columns need not be equal.

Types of matrix

1. Square matrix

In this matrix, the number of rows is equal to the number of columns.

e.g. $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}_{2\times2}$ and $\begin{bmatrix}1 & 1 & 2\\ 0 & 1 & 2\\ 1 & 0 & 3\end{bmatrix}_{3\times3}$

In these square matrices, m = n

Elements a_{ij} of a square matrix $A = [a_{ij}]_{n\times n}$ for which $i = j$ are called *diagonal elements* and line along which they lie is called *principal diagonal* of the matrix.

2. Null matrix or Zero matrix

The $m \times n$ matrix whose all elements are 0 is called *null matrix* (or zero matrix) of the type $m \times n$. It is denoted by 0.

e.g. $\begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}_{3\times4}$ is zero matrix of type 3×4

Above matrix can be written as

$A = [a_{11}\, a_{22}\, a_{33}\, a_{44}]$

3. Diagonal matrix

In this matrix, elements above and below the principal diagonal are all zero.

e.g. $A=\begin{bmatrix}a_{11} & 0 & 0 & 0\\ 0 & a_{22} & 0 & 0\\ 0 & 0 & a_{33} & 0\\ 0 & 0 & 0 & a_{44}\end{bmatrix}$

4. Scalar matrix

In this matrix, all diagonal elements are equal.

e.g. $A=\begin{bmatrix}C & 0 & 0\\ 0 & C & 0\\ 0 & 0 & C\end{bmatrix}$

- If A be a matrix and $A^2 = A$, then A is called *idempotent.*

  If AB = A and BA = B, then A and B are called *idempotent.*
- If A be a matrix and satisfies $A^2 = I$, then A is called *invalutory.*

5. Rectangular matrix

A matrix of order m × n is called a *rectangular matrix* if $m \neq n$

e.g. $\begin{bmatrix}1 & 2 & 3\\ 4 & 0 & 1\end{bmatrix}_{2\times3}$, $\begin{bmatrix}1 & 0\\ 0 & 2\\ 1 & 2\end{bmatrix}_{3\times2}$, $\begin{bmatrix}1 & 2 & 5 & 7\\ 2 & 4 & 6 & 8\end{bmatrix}_{2\times4}$

6. Unit matrix or Identity matrix

A square matrix each of whose diagonal elements is 1 and each of whose non-diagonal elements is equal to zero is called a *unit matrix or an identity matrix*. It is denoted by I_n when n is order of the matrix.

e.g. $I_2=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, $I_3=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$, $I_4=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$

are unit matrices of order 2, 3, 4, respectively.

7. Column matrix

Any $m \times 1$ matrix which has m rows and only one column is a column matrix or a column vector.

Row matrix

Any $1 \times n$ matrix which has only one row and n columns is called a row matrix or as row vector.

e.g. $X_1 = [1\ 2\ 3\ 4\ 5\ 6]_{1\times6}$ is a row matrix of the type 1×6

$X_2=\begin{bmatrix}1\\ 2\\ 3\\ 4\end{bmatrix}_{4\times1}$ is a column matrix of type 4×1

8. Triangular matrix

It is of two types

(i) Lower triangular matrix. In this matrix, all the elements above the principal diagonal are zero.

e.g. $\begin{bmatrix}a_{11} & 0 & 0\\ a_{21} & a_{22} & 0\\ a_{31} & a_{32} & a_{33}\end{bmatrix}_{3\times3}$

The diagonal and lower triangular elements may or may not be zero.

(ii) Upper triangular matrix. In this matrix, all the elements below the principal diagonal are zero.

e.g. $\begin{bmatrix}a_{11} & a_{12} & a_{13} & a_{14}\\ 0 & a_{22} & a_{23} & a_{24}\\ 0 & 0 & a_{33} & a_{34}\\ 0 & 0 & 0 & a_{44}\end{bmatrix}_{4\times4}$

The diagonal and upper triangular elements may or may not be zero.

Types of Square Matrices

(i) Nilpotent matrix

A square matrix A is called a *nilpotent matrix* if there exists a positive integer n such that $A^n = 0$ and $A^{x-1} \neq 0$

If n is least positive integer such that $A^n = 0$, then n is called *index of the nilpotent* matrix A.

(ii) Symmetrical matrix

It is a square matrix in which $a_{ij} = a_{ji}$ for all i, j. A symmetric matrix is necessarily a square one.

If A is symmetric, then $A^T = A$

e.g. $\begin{bmatrix}a & h & g\\ h & b & f\\ g & f & c\end{bmatrix}$

(a) $A\ A^T$ is always symmetric matrix.

(b) $\frac{A+A^T}{2}$ is always symmetric matrix.

(c) If A and B are symmetric then A + B and A – B also symmetric AB, BA may or May not by symmetric.

(iii) Skew-symetrical matrix

It is a square matrix in which $a_{ij} = -a_{ji}$ for all i and j.

In a skew symmetrical matrix, all elements along the diagonal are zero.

e.g. $\begin{bmatrix}0 & h & g\\ h & 0 & f\\ g & f & 0\end{bmatrix}$

For any matrix A, the matrix $\frac{A-A^T}{2}$ is always skew symmetric.

Theorem:

Every square matrix can be uniquely expressed as a sum of a symmetric matrix and a skew symmetric matrix.

(iv) Hermitian matrix

It is a square matrix A in which $(i,j)^{th}$ element is equal to conjugate complex of the $(j,i)^{th}$ element.

i.e. $a_{ij} = \bar{a}_{ji}$ for all i and j

e.g. $\begin{bmatrix}x & 3-4i & 1+2i\\ 3+4i & 0 & 5-4i\\ 1-2i & 5+4i & 1\end{bmatrix}$

A necessary and sufficient condition for a matrix A to be hermitian is that

$A = A^\theta$

where A^θ is transposed conjugates of A.

(v) Skew-Hermitian matrix

It is a square matrix $A = [a_{ij}]$ in which $a_{ij} = -\bar{a}_{ji}$ for all i and j.

The diagonal elements of a skew-Hermitian matrix must be pure imaginary numbers or zero

A necessary and sufficient condition for a matrix A to be skew-Hermitian is that

$A^\theta = -A$

(vi) Orthogonal matrix

A square matrix A is called *orthogonal matrix if*

$$AA^T = A^TA = I$$

e.g. If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and

$$A^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

then $AA^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

$$= \begin{bmatrix} \cos^2\theta+\sin^2\theta & 0 \\ 0 & \sin^2\theta+\cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

(vii) Adjoint matrix

Let A be the square matrix given by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

Let A_{ij} be the cofactor a_{ij} then transpose of the matrix $[A_{ij}]$ is called *adjoint of the matrix A.*

$$\text{Adjoint } A = \begin{bmatrix} A_{11} & A_{21} & \dots & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & \dots & A_{n2} \\ \dots & \dots & \dots & \dots & \dots \\ A_{1n} & A_{2n} & \dots & \dots & A_{nn} \end{bmatrix}$$

If A be an n-rowed square matrix, then $(\text{Adj } A)A = A(\text{Adj } A) = |A|\, I_n$

where I_n is the $n \times n$ identity matrix.

Sub-matrix

Any matrix obtained by omitting some rows and columns from a given $(m \times n)$ matrix B is called *submatrix of B.*

The matrix B itself is a sub-matrix of B as it can be obtained from B by omitting no rows or columns.

Principal Sub-matrix

A square sub-matrix of a square matrix B whose diagonal elements are also diagonal elements of matrix B is called *principal sub-matrix.*

Addition of Matrices

Let A and B be two matrices of the same type $m \times n$. If $A = [a_{ij}]_{m\times n}$, $B = [b_{ij}]_{m\times n}$, then

$$A + B = [a_{ij} + b_{ij}]_{m\times n}$$

e.g. If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$

and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b23 & \dots & b_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ b_{m1} & b_{m2} & b_{m3} & \dots & b_{mn} \end{bmatrix}$

then, $A + B = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} & \dots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} & \dots & a_{2n}+b_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & a_{m3}+b_{m3} & \dots & a_{mn}+b_{mn} \end{bmatrix}$

Properties of Matrix Addition

(i) Matrix addition is Commulative

If A and B be two $m \times n$ matrices, then

$$A + B = B + A$$

(ii) Matrix addition is Associative

If A, B, C be three matrices each of type $m \times n$, then

$$(A + B) + C = A + (B + C)$$

(iii) Existence of Additive identity

If 0 be the $m \times n$ matrix each of whose element is zero, then $A + 0 = A = 0 + A$

where A is a $m \times n$ matrix

(iv) Existence of additive inverse

If $B = [b_{ij}]_{m\times n}$, then negative of matrix B is defined as he matrix $[-b_{ij}]_{m\times n}$ and is denoted by –B.

$$-B + B = 0 = B + (-B)$$

0 is null matrix of order m× n

(v) Cancellation Law

If A, B, C are three main matrices, then

$$B + A = C + A$$

$$\Rightarrow \quad B = C \quad \text{(right cancellation law)}$$

$$A + B = A + C$$

$$\Rightarrow \quad B = C \quad \text{(left cancellation law)}$$

These cancellation laws hold good in the case of addition of matrices.

(vi) The equation $B + X = 0$ has a unique solution in the set of all $m \times n$ matrices.

EQUALITY OF TWO MATRICES

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if

(i) both are the same size, and

(ii) $a_{ij} = b_{ij}$ for each pair of subscripts i and j.

If two matrices A and B are equal, then we write

$$A = B$$

Multiplication of a Matrix

1. By a scalar

Let C any real number called scaler and B be any $m \times n$ matrix, then

$$CB = BC = [cb_{ij}]_{m\times n}$$

where, $[b_{ij}]_{m\times n} = B$

Properties

(i) If p and q are two scalars and B is any $m \times n$ matrix, then

$$p(qB) = (pq)B$$

(ii) $(p + q)B = pB + qB$

(iii) If A and B are two matrices each of the type $m \times n$, then

$$p(A + B) = p(A) + p(B)$$

where p is any scalar.

(iv) If B be any $m \times n$ matrix, then

$$(-p)B = (-pB) = p(-B)$$

(a) $1B = B$

(b) $(-1)B = -B$

2. By a matrix

Let $A = [a_{ij}]_{l\times m}$ and $B = [b_{jk}]_{m\times n}$ be two matrices such that number of columns in A is equal to number of rows in B. Then in $l \times n$ matrix

$$C = [C_{ik}]_{l\times n} \text{ such that } C_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}$$

is called *product of the matrices A and B.*

We can write $C = AB$

Properties

(i) Associative laws

If A, B, C are $l \times m$, $m \times n$ and $n \times p$ matrices respectively, then

$$A(BC) = (AB)C$$

(ii) Distributive laws

$$A(B + C) = AB + AC$$

where A, B, C are any three $l \times m$, $m \times n$ and $n \times p$ matrices respectively.

(iii) Commutative

Multiplication of matrices is not always commutative.

For commutative $AB = BA$

For anti commutative $AB = -BA$

(iv) Product of two matrices can be zero matrix while neither of them is a zero matrix.

(v) If A be an $l \times m$ matrix. and I_m denotes m rowed unit matrix, then

$$AI_m = A = I_lA$$

TRANSPOSE OF A MATRIX

If row and column are interchanged in a matrix A, then matrix obtained is called *transpose matrix A* and is denoted by A^T.

- If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then $A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$
- If $A = [a_{ij}]$, then $A^T = [a_{ji}]$
- If A is a square matrix, then $|A| = |A^T|$

Properties of transpose

Theorem 1

If A is any matrix, then $(A^T)^T = A$.

Theorem 2

If A and B are two matrices of same type, then

$$(A + B)^T = A^T + B^T.$$

Note : If A is any matrix and k is any complex number, then $(kA)^T = kA^T$.

Theorem 3

If A and B are two matrices for which conformability for multiplication is assured, then

$$(AB)^T = B^TA^T.$$

Corollary 1 : If $A_1, A_2, \dots A_n$ are n matrices which are conformable for multiplication, then

$$(A_1, A_2, \dots A_n)^T = A_n^T A_{n-1}^T \dots A_1^T.$$

Note : If A is any matrix, then AA^T and A^TA are defined. They are square matrices but their order may be different.

e.g. If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$ then $A^T = \begin{bmatrix} 1 & -2 \\ 2 & 0 \\ 0 & 3 \end{bmatrix}$

and $AA^T = \begin{bmatrix} 5 & -2 \\ -2 & 13 \end{bmatrix}$

$$A^TA = \begin{bmatrix} 5 & 2 & -6 \\ 2 & 4 & 0 \\ -6 & 0 & 9 \end{bmatrix}$$

Inverse of Matrix

Let A and B are two square matrices of the same order, such that $AB = BA = I$

then B is called *inverse of A* and is denoted by A^{-1}.

$$A^{-1} = \frac{(adj A)}{|A|}$$

Properties

(*i*) A^{-1} exists only when A is non singular, i.e. $|A| \neq 0$

(*ii*) The inverse of a matrix is unique.

(*iii*) $(AB)^{-1} = B^{-1} A^{-1}$

(*iv*) $(A^T)^{-1} = (A^{-1})^T$

(*v*) $(A^{-1})^\theta = (A^\theta)^{-1}$

Inverse of a Square Matrix

A square matrix A is said to be an invertible matrix if there exists a square matrix B such that

$AB = BA = I$, *then matrix B is called inverse of A.*

- A rectangular matrix can not be invertible.
- Every square matrix need not be invertible.

Theorem 1

An invertible matrix has unique inverse.

If A is an invertible matrix, then its inverse is denoted by A^{-1}

Hence $AA^{-1} = A^{-1}A = I$.

Theorem 2

If A is an invertible matrix, and A^{-1} is also invertible then $(A^{-1})^{-1} = A$

Theorem 3

If A and B are two invertible matrices of same type, then AB is also invertible and $(AB)^{-1} = B^{-1} A^{-1}$.

Corollary 1: If $A_1, A_2, \ldots A_n$ are invertible matrices of same type, then $A_1 A_2 \ldots A_n$ is also invertible and

$$(A_1 A_2 \ldots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \ldots A_1^{-1}$$

If A is an invertible matrix and k is a non zero complex number, then kA is invertible and

$$(kA)^{-1} = k^{-1} A^{-1}$$

Theorem 4

If A is an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$

Theorem 5

If A is a non singular matrix, then A is invertible and

$$A^{-1} = \frac{\text{Adj } A}{\det A}$$

If A is a square matrix, then

$$A (\text{Adj } A) = (\det A) I$$

Theorem 6

If A is an invertible matrix, then A is non singular.

If A is an invertible matrix, then

$$\det (A^{-1}) = \frac{1}{\det A}$$

Theorem 7

A square matrix A is non singular if A is invertible.

Corollary 2 : If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad \neq bc$, then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & b \\ c & a \end{bmatrix}$$

ORTHOGONAL MATRIX

An invertible square matrix is called orthogonal, if

$$A^T = A^{-1}$$

$$\therefore \quad AA^T = I$$

$$\Rightarrow \quad |A| = |A^T|$$

$$\Rightarrow \quad |A| = \pm 1$$

Example. If $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, then find A^{-1}.

Solution. $A^{-1} = \frac{1}{14 - 15} \begin{bmatrix} 7 & -3 \\ -5 & 2 \end{bmatrix}$

$$= -1 \begin{bmatrix} 7 & -3 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ -5 & 2 \end{bmatrix}$$

Theorem 8

Let A, B, C be three square matrices of same type and A be a non-singular matrix. Then

$$AB = AC \Rightarrow B = C$$

and $BA = CA \Rightarrow B = C.$

Theorem 9

If A and B are non singular matrices of same type, then

$$\text{Adj } (AB) = (\text{Adj } B)(\text{Adj } A)$$

Theorem 10

If A is a square matrix of type n, then

$\det (\text{Adj } A) = (\det A)^{n-1}$

Example. If A is a square matrix of type 3 and $\det A = 10$, then find (Adj A).

Solution. $\det (\text{Adj } A) = (\det A)^{3-1} = 10^2 = 100$

RANK OF A MATRIX

If A is not a null matrix; then rank of the matrix A is positive integer, if it possesses following two properties

(i) Each minor of order $n + 1$ of the matrix A vanishes (i.e. is zero).

(ii) There exists at least one minor of order n of the matrix A which is not zero.

If A is a null matrix, then rank of A is defined as *zero*.

A need not be square matrix.

Properties

- Rank of A is denoted by $p(A)$ and is UNIQUE.
- Every matrix will have a rank.
- If A is a matrix of order $m \times n$, then $p(A) \leq m$ and n (smaller of the two).
- If $P(A) = n$, then every minor of order $n + 1, n + 2,$ etc. is zero.
- If A is a matrix of order $n \times n$, then A is non-singular $\Leftrightarrow p(A) = n$.
- Rank of $I_n = n$.
- A is a matrix of order $m \times n$. If every kth order minor $(k < m, k < n)$ is zero, then $P(A) < k$
- A is a matrix of order $m \times n$. If there is a minor of order $k (k < m, k < n)$ which is not zero, then $p(A) \geq k$
- Rank of a zero matrix is defined to be zero. The rank of a unit matrix is defined to be one.
- If x is a non-zero column matrix and y is a non-zero row matrix, then $p(xy) = 1$
- Rank of a matrix is greater than or equal to (not less than) rank of every sub-matrix thereof.
- If A is any n-rowed square matrix of rank, $n - 1$, then adj $A \neq 0$.
- Rank of the transpose of a matrix is equal to rank of the original matrix.
- Rank of a matrix does not change by pre-multiplication or post-multiplication with a non-singular matrix.
- If $A \sim B$, then $p(A) = p(B)$.
- Rank of a product of two matrices cannot exceed rank of either matrix.
- Rank of the sum of two matrices cannot exceed sum of their ranks.
- Elementary transformations do not change the rank of a matrix.

Echelon form of a Matrix

Let A be the given matrix of order m × n, then it can be reduced to Echelon form by row tranformation.

If we can make all the entries equal to zero, then

R (A) = Number of none zero rows

Consistency and Solution of Algebraic equations

In argument matrix A*, we reduce echelon form.

(*i*) If Rank of A* = Rank of A, then equation are consistent.

(*ii*) If Rank of A* ≠ Rank of A, then system is inconsistent (i.e. no solution)

Homogenous system of equations AX = 0

(*i*) If rank = number of unknowns, then it has zero solution.

(*ii*) If rank < number of unknowns, then it has infinitely may solution.

Elementary Transformation of a Matrix

Following operations on a matrix are called *elementary transformations*.

(*i*) Interchange of any two rows (or columns).

(*ii*) Multiplication of elements of any row (or column) by any non-zero number.

(*iii*) Addition to elements of any other row (or column) the corresponding elements of any other row (or column) multiplied by any number.

Solution of Simultaneous Linear Equations by Matrix

Simultaneous linear equations of two or three variables are solved by using matrices.

1. Homogeneous linear equations

Let system of m homogeneous equations in n unknown $x_1, x_2, \ldots, x_n$ be

$$\left.\begin{aligned} &a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = 0 \\ &a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = 0 \\ &\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ &a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n = 0 \end{aligned}\right\} \quad \ldots(i)$$

Let $A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}_{m \times n}$

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \ldots \\ \ldots \\ x_n \end{bmatrix}_{n \times 1}$; $B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \ldots \\ \ldots \\ 0 \end{bmatrix}_{m \times 1}$

where A, X and B are $m \times n$, $n \times 1$ and $m \times 1$ matrices respectively.

Equation *(i)* can be written in the form of a single matrix equation

$$AX = B \quad \ldots(ii)$$

The matrix A is called *coefficient matrix* of the system of equation *(i)*

Solutions of equation AX = B

Working Rule

Reduce the coefficient matrix A to Echelon form by applying elementary row transformations only.

From this we know rank of the matrix A.

Let matrix A is of the type $m \times n$ and its rank is r.

- If $r < m$, then $(m - n)$ equations will be eliminated. Thus given system of m equations will be replaced by an equivalent system of r equations. Solving these r equations we can express values of some unknown in terms of remaining $n - r$ unknowns. These $n - r$ unknown can be given any arbitrarily chosen values.
- If $r = n$, then zero solution (trivial solution) will be the only solution.
- If $r < n$, there will be an infinity of solutions.

2. Linear non-homogeneous equations

Let system of m simultaneous equations in n unknowns $x_1, x_2, \ldots, x_n$ be

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = K_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = K_2$$
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n = K_m$$

Let $A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ \ldots \\ x_n \end{bmatrix}_{n \times 1}$,

$K = \begin{bmatrix} K_1 \\ K_2 \\ \ldots \\ \ldots \\ K_m \end{bmatrix}_{m \times 1}$

where A, X, K are $m \times n$, $n \times 1$ and $m \times 1$ matrices respectively.

Above equations can be written in the form of a single matrix equation AX = K

Consistent and Inconsistent equations

If above system has a solution, then equations are *consistent* otherwise equations are *called inconsistent equations*.

- The system of equation AX = K is consistent, i.e. possesses a solution, if and only if coefficient matrix A and augmented matrix [A K] are of the same rank.
- If A be an n-rowed non-singular matrix, X be an $n \times 1$ matrix, K be an $m \times 1$ matrix, then system of equation AX = K has a unique solution.

Solution of equation AX = K

Working rule

Let coefficient matrix A is of the type $m \times n$.

Write augmented matrix [A K] and reduce it to a Echelon form by applying only elementary row transformation on it. This Echelon form will give ranks of the augmented matrix [A K] and coefficient matrix A.

Different cases

Case I : *Rank A < Rank [A K]*

In this case, equation AX = K are inconsistent, i.e. they have no solution.

Case II : *Rank A = Rank [A K] = r (suppose)*

In this case, equation AX = K are consistent.

- If $r < m$, then in the process of reducing the matrix [A K] to Echelon form, $(m - r)$ equations will be eliminated. Then given system of m equations will be replaced by an equivalent system of r equations. From these r equations we shall be able to express the values of some r unknowns in terms of the remaining $n - r$ unknowns which can be given any arbitrary chosen values.
- If $r = n$, there will be a unique solution.
- If $r < n$, then $n-r$ variables can be assigned arbitrary values. So there will be an infinite number of solutions. Only $n - r + 1$ solutions will be linearly independent and rest of the solutions will be linear combinations of them.
- If $m < n$, then $r \le m < n$.

  Thus $n - r > 0$.

Therefore when number of equations is less than the number of unknowns, equations will always have an infinite number of solutions, provided they are consistent.

Matrix Inversion Method

Theorem 1

If A is a non singular matrix, then $A^{-1}B$ is a solution $AX = B$.

Proof : A is non singular

$\Rightarrow$ A^{-1} exists and $AA^{-1} = A^{-1}A = I$

$\Rightarrow$ $A(A^{-1}B) = (AA^{-1})B = I\,B = B$

$\therefore$ $X = A^{-1}B$ is a solution of AX = B.

Theorem 2

If A is a non singular matrix, then AX = B has a unique solution.

Proof : A is non singular

$\Rightarrow$ A^{-1} exists and $AA^{-1} = A^{-1}A = I$

By theorem 1, $X = A^{-1}B$ is a solution of AX = B

Let $X = X_1$, $X = X_2$ be two solutions of AX = B.

$\therefore$ $AX_1 = B$, $AX_2 = B$

$\Rightarrow$ $AX_1 = AX_2$

$\Rightarrow$ $A^{-1}(AX_1) = A^{-1}(AX_2)$

$\Rightarrow$ $(A^{-1}A)X_1 = (A^{-1}A)X_2$

$\Rightarrow$ $IX_1 = IX_2$

$\Rightarrow$ $X_1 = X_2$

Hence AX = B has a *unique solution.*

Theorem 3

If A is a non singular matrix, then $X = A^{-1}B$ is the unique solution of AX = B.

Method of finding solution $X = A^{-1}B$ of AX = B is called **matrix inversion method**

Example. Solve following equations by matrix inversion method

$$x + y + z = 6$$
$$2x - y + z = 3$$
$$x + 2y - z = 2.$$

Solution

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$

Given equations can be written as

$$AX = B$$

By matrix inversion method solution is,

$$X = A^{-1}B$$

$$\det A = 1(1-2) - 1(-2-1) + 1(4+1)$$
$$= -1 + 3 + 5 = 7.$$

Cofactors of elements of A are

$A_{11} = +(1-2) = -1$

$A_{12} = -(-2-1) = 3$

$A_{13} = +(4+1) = 5$

$A_{21} = -(-1-2) = 3$

$A_{22} = +(-1-1) = -2$

$A_{23} = -(2-1) = -1$

$A_{31} = +(1+1) = 2$

$A_{32} = -(1-2) = 1$

$A_{33} = +(-1-2) = -3$

$\therefore$ $\text{Adj } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} = \begin{bmatrix} -1 & 3 & 2 \\ 3 & -2 & 1 \\ 5 & -1 & -3 \end{bmatrix}$

$$A^{-1} = \frac{\text{Adj } A}{\det A} = \frac{1}{7}\begin{bmatrix} -1 & 3 & 2 \\ 3 & -2 & 1 \\ 5 & -1 & -3 \end{bmatrix}$$

$$X = A^{-1}B = \frac{1}{7}\begin{bmatrix} -1 & 3 & 2 \\ 3 & -2 & 1 \\ 5 & -1 & -3 \end{bmatrix}\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$$

$$= \frac{1}{7}\begin{bmatrix} -6 & +9 & +4 \\ 18 & -6 & +2 \\ 30 & -3 & -6 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 7 \\ 14 \\ 21 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore$ $x = 1, y = 2, z = 3$ is the solution.

CRAMER'S RULE

Theorem 4

Let

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2 \quad \ldots(i)$$
$$a_3x + b_3y + c_3z = d_3$$

be a system of linear equations.

If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$, $\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$,

$\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

then solution of equation, (i) is

$x = \frac{\Delta_1}{\Delta}$, $y = \frac{\Delta_2}{\Delta}$, $z = \frac{\Delta_3}{\Delta}$ is

Proof : Let $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$.

Then $\det A = \Delta \neq 0$

$\therefore$ AX = B has unique solution

Hence equation (i) has unique solution.

Let, A_1, A_2, A_3 be the cofactors of a_1, a_2, a_3 respectively.

Thus

$$(a_1x + b_1y + c_1z)\,A_1 + (a_2x + b_2y + c_2z)\,A_2 + (a_3x + b_3y + c_3z)\,A_3$$
$$= d_1A_1 + d_2A_2 + d_3A_3$$

$\Rightarrow$ $x[a_1A_1 + a_2A_2 + a_3A_3] + y[b_1A_1 + b_2A_2 + b_3A_3] + z[c_1A_1 + c_2A_2 + c_3A_3] = \Delta_1$

$\Rightarrow$ $x\Delta + y0 + z0 = x = \Delta_1$

$\Rightarrow$ $x = \frac{\Delta_1}{\Delta}$

Similarly, $y = \frac{\Delta_2}{\Delta}$, and $z = \frac{\Delta_3}{\Delta}$

Hence solution of (i) is $x = \frac{\Delta_1}{\Delta}$, $y = \frac{\Delta_2}{\Delta}$, $z = \frac{\Delta_3}{\Delta}$

- $\Delta_1, \Delta_2, \Delta_3$ are also denoted as $\Delta_x, \Delta_y, \Delta_z$ respectively.
- $\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$

Example. Solve following equations by Cramer's rule.

$$x + y + z = 6$$
$$2x - y + z = 3$$
$$x + 2y - z = 2.$$

Solution $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}$

$= 1(1-2) - 1(-2-1) + 1(4+1)$

$= -1 + 3 + 5 = 7$

$\Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ 3 & -1 & 1 \\ 2 & 2 & -1 \end{vmatrix}$

$= 6(1-2) - 1(-3-2) + 1(6+2)$

$= -6 + 5 + 8 = 7$

$\Delta_2 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & -1 \end{vmatrix}$

$= 1(-3-2) - 6(-2-1) + 1(4-3)$

$= -5 + 18 + 1 = 14$

$\Delta_3 = \begin{vmatrix} 1 & 1 & 6 \\ 2 & -1 & 3 \\ 1 & 2 & 2 \end{vmatrix}$

$= 1(-2-6) - 1(4-3) + 6(4+1)$

$= -8 - 1 + 30 = 21$

By Cramer's rule,

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta}$$

$$\therefore \quad \frac{x}{7} = \frac{y}{14} = \frac{z}{21} = \frac{1}{7}$$

Hence solution is, $x = 1, y = 2, z = 3$

Coefficient and Augmented matrix

Let $a_1x + b_1y + c_1z = d_1$

$a_2x + b_2y + c_2z = d_2$ ... (i)

$a_3x + b_3y + c_3z = d_3$

be a system of linear equations.

Then matrix, $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is called **Coefficient matrix.**

and matrix $K = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$ is called **Augmented matrix.**

EIGEN VALUES AND EIGEN VECTORS

Let $A = [a_{ij}]$ be an $n \times n$ matrix.

- Matrix $A - \lambda I$ is called *characteristic matrix* of A, where I is identity matrix.
- Determinant $|A - \lambda I|$ is called *characteristic polynomial of A.*
- Equation $|A - \lambda I| = 0$ is called *characteristic equation* of A and its roots are called *eigen values.*
- Problem of finding the eigen values of a matrix is called *eigen-value problem.*
- If λ is a characteristic root or eigen values of an $n \times n$ matrix A, then a non-zero vector X such that $AX = \lambda X$ is called *characteristic vector* or *eigen vector of* A corresponding to the characteristic root λ.

  The set of eigen values of A is called the spectrum of A.

Relations between Eigen values and Eigen vectors

- λ is a eigen values of a matrix A if and only if there exists a non-zero vector X such that
  $$AX = \lambda X$$
- If X is a characteristic vector of a matrix A corresponding to the characteristic value λ, then KX is also a characteristic vector of A corresponding to the same characteristic value λ. Here K is any non-zero scalar.
- If X is a eigen vector of a matrix A, then X cannot correspond to more than one eigen value of A.
- Eigen vectors corresponding to distinct characteristic roots of a matrix are linearly independent.

Properties of Eigen values

(*i*) Sum of all the eigen values of a matrix is equal to sum of the elements of the principal diagonal.

(*ii*) Product of the eigen values of a matrix is equal to its determinant.

(*iii*) If λ is an eigen value of a matrix A, then $\frac{1}{\lambda}$ is the eigen value of A^{-1}.

(*iv*) If λ is an eigen value of an orthogonal matrix, then $\frac{1}{\lambda}$ is also its eigen value.

(*v*) If $\lambda_1, \lambda_2 \ldots\ldots\ldots\ldots \lambda_m$ are the eigen values of a matrix A, then A^n has the eigen values $\lambda_1^n, \lambda_2^n \ldots\ldots\ldots\ldots \lambda_m^n$.

(*vi*) Eigen value of A = Eigen value of A^T

(*vii*) Maximum number of distinct eigen values = size of A

(*viii*) In a triangular and diagonal matrix, Eigen values are diagonal elements themselves.

(*ix*) If A and B are two matrices of same order then the matrix AB and BA will have same characteristic roots.

Cayley-Hamilton theorem.

Every square matrix satisfies its characteristic equation, i.e. if for a square matrix A of order n

$|A - \lambda I| = (-1)^n [\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \ldots\ldots + a_n]$

then matrix equation

$X^n + a_1 X^{n-1} + a_2 X^{n-2} + a_3 X^{n-3} + \ldots\ldots + a_n I = 0$

is satisfied by $X = A$

i.e., $A^n + a_1 A^{n-1} + a_2 A^{n-2} + \ldots\ldots + a_n I = 0$

SOLVED PROBLEMS

1. Find inverse of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

Solution

Determinant of given matrix A is

$$\Delta = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

If $A_1, A_2 \ldots$ be the co-factors of $a_1, a_2 \ldots$ in Δ, then

$A_1 = -24, \quad A_2 = -8, \quad A_3 = -12,$

$B_1 = 10, \quad B_2 = 2, \quad B_3 = 6,$

$C_1 = 2, \quad C_2 = 2, \quad C_3 = 2$

Thus, $D = a_1A_1 + a_2A_2 + a_3A_3 = -8$

$$\therefore \text{ Adj. } A = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} -24 & -8 & -12 \\ 10 & 2 & 6 \\ 2 & 2 & 2 \end{bmatrix}$$

Hence inverse of matrix A

$$= \frac{\text{adj.A}}{\Delta} = \frac{1}{-8}\begin{bmatrix} -24 & -8 & -12 \\ 10 & 2 & 6 \\ 2 & 2 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & 1 & \frac{3}{2} \\ -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

2. Determine rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$

Solution

Operate $R_2 - R_1$ and $R_3 - 2R_1$, therefore

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} = A \text{ (say)}$$

Obviously, 3rd oder minor of A vanishes. All its second order minors formed by 2nd and 3rd rows are zero.

But another minor

$$\begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix} = -1 \neq 0$$

Hence rank of A is 2

3. Investigate value of λ and μ, so that the equations

$2x + 3y + 5z = 9$

$7x + 3y - 2z = 8$

$2x + 3y + \lambda z = \mu$

have

(*i*) no solution,

(*ii*) a unique solution, and

(*iii*) an infinite number of solutions

Solution

We have $\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$

The system admits of unique solution if, and only if, coefficent matrix is of rank 3. This requires in

$$\begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{vmatrix} = 15(5 - \lambda) \neq 0$$

Thus for a unique solution $\lambda \neq 5$ and m may have any value.

If $\lambda = 5$, the system wil have no solution for those values of μ for which the matrices

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix}$$

and $K = \begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & 5 & \mu \end{bmatrix}$

are not of the same rank. But A is of rank 2 and K is not of rank 2 unless $\mu = 9$.

Thus if $\lambda = 5$ and $\mu \neq 9$, the system will have no solution.

If $\lambda = 5$ and $\mu = 9$, the system will have an infinite number of solutions.

4. Find eigen values and eigen vectors of the matrix.

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution

Characteristic equation is

$$|A - \lambda I| = \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & 4 \\ 2 & -4 & 3-\lambda \end{bmatrix}$$

$= \lambda^3 + 18\lambda^3 - 45\lambda = 0$

$\Rightarrow \quad \lambda(\lambda - 3)(\lambda - 15) = 0$

$\therefore \quad \lambda = 0, 3, 15$

Thus eigen values of A are 0, 3, 15.

If x, y, z be the component of an eigen vectors corresponding to the eigen value λ, then

$$[A - \lambda I]X = \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Putting $\lambda = 0$, we have

$$8x - 6y + 2z = 0,$$
$$-6x + 7y + 4z = 0,$$
$$2x - 4y + 3z = 0$$

These equations determine a single linearly independent solution which may be takne as (1, 2, 2), so that every non-zero multiple of this vector is an eigen vector corresponding to $\lambda = 0$.

Similarly, eigen vectors corresponding to $\lambda = 3$ and $\lambda = 15$ are arbitrary non-zero multiples of the vectors (2, 1, –2) and (2, –2, 1) which are obtained from above equation.

Hence three vectors may be taken as (1, 2, 2), (2, 1, – 2), (2, –2, 1)

5. If $A\begin{bmatrix} -2 & 1 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 7 \\ -1 & 20 \end{bmatrix}$, find matrix A

Solution

$$A = \begin{bmatrix} -1 & 7 \\ -1 & 20 \end{bmatrix}\begin{bmatrix} -2 & 1 \\ 3 & 5 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} -1 & 7 \\ -1 & 20 \end{bmatrix}\frac{1}{-10-3}\begin{bmatrix} 5 & -1 \\ -3 & -2 \end{bmatrix}$$

$$= \frac{-1}{13}\begin{bmatrix} -1 & 7 \\ -1 & 20 \end{bmatrix}\begin{bmatrix} 5 & -1 \\ -3 & -2 \end{bmatrix}$$

$$= \frac{-1}{13}\begin{bmatrix} -26 & -13 \\ -65 & -39 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

6. If ω is cube root of –1, then find vlaue of

$$\begin{bmatrix} 1 & -\omega & \omega^2 \\ -\omega & \omega^2 & 1 \\ \omega^2 & 1 & -\omega \end{bmatrix}$$

Solution

Since ω is cube root of –1, therefore

$$\omega^3 = -1$$

i.e., $\omega^3 + 1 = 0$

$\Rightarrow (\omega + 1)(\omega^2 - \omega + 1) = 0$

Hence either $\omega = -1$ or $\omega^2 - \omega + 1 = 0$

Let $A = \begin{bmatrix} 1 & -\omega & \omega^2 \\ -\omega & \omega^2 & 1 \\ \omega^2 & 1 & -\omega \end{bmatrix}$

$C_1 \to C_1 + C_2 + C_3$

$$A = \begin{bmatrix} 1-\omega+\omega^2 & -\omega & \omega^2 \\ -\omega+\omega^2+1 & \omega^2 & 1 \\ \omega^2+1-\omega & 1 & -\omega \end{bmatrix} = \begin{bmatrix} 0 & -\omega & \omega^2 \\ 0 & \omega^2 & 1 \\ 0 & 1 & -\omega \end{bmatrix}$$

$= 0$ (since $1 - \omega + \omega^2 = 0$)

7. If $A = \begin{bmatrix} -3 & 5 \\ 2 & 1 \end{bmatrix}$, find $|A^{-1}|$.

Solution

$$A = \begin{bmatrix} -3 & 5 \\ 2 & 1 \end{bmatrix}$$

$\therefore \quad |A| = -3 - 10 = -13 \neq 0$

and $A^{-1} = \dfrac{A/\text{dj}\,A}{|A|} = \dfrac{\begin{bmatrix} 1 & -5 \\ -2 & -3 \end{bmatrix}}{-13}$

$$|A^{-1}| = \frac{-1}{13}(-3 - 10) = 1$$

8. If α, β are roots of the equation $\begin{vmatrix} 2x+3 & 1 \\ x & x+2 \end{vmatrix} = 0$, then find $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$

Solution

Given: $\begin{vmatrix} 2x+3 & 1 \\ x & x+2 \end{vmatrix} = 0$

$\therefore \quad (2x + 3)(x + 2) - x = 0$

$\Rightarrow \quad 2x^2 + 6x + 6 = 0$

$\Rightarrow \quad x^2 + 3x + 3 = 0$

Since α, β are roots of this equation, therefore

$$\alpha + \beta = -3, \quad \alpha\beta = 3$$

$\therefore \quad \dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{\alpha+\beta}{\alpha\beta} = \dfrac{-3}{3} = -1$

EXERCISE 1.13

MCQ TYPE QUESTIONS

1. Eigen values of the matrix $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$ are

(a) 1, 1, 1 (b) 1, 1, 2 (c) 1, 4, 4 (d) 1, 2, 4

2. Rank of the following $(n+1)\times(n+1)$ matrix, where a is a real number, is

$$\begin{bmatrix} 1 & a & a^2 & \cdots & a^n \\ a & a & a^2 & \cdots & a^n \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ 1 & a & a^2 & \cdots & a^n \end{bmatrix}$$

(a) 1
(b) 2
(c) n
(d) dependent upon the value of A

3. Matrices $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication

(a) if $a = b$ or $\theta = n\pi$, where n is an integer
(b) always
(c) never
(d) if $a \cos\theta \neq b \sin\theta$

4. If $A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$, then

(a) $A' = -A$ (b) $A' = A$

(c) $A' = -\dfrac{1}{A}$ (d) none of these

5. Eigen values of the system represented by

$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} X$ are

(a) 0, 0, 0, 0 (b) 1, 1, 1, 1
(c) 0, 0, 0, –1 (d) 1, 0, 0, 0

6. Matrix, $A = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is not

(a) orthogonal (b) non–singular
(c) singular (d) have A^{-1} exists

7. In the matrix equation $Px = q$, which of the following is a necessary condition for the existence of at least one solution for the unknown vector x ?

(a) Augmented matrix $[Pq]$ must have the same rank as matrix P
(b) Vector q must have only non-zero elements
(c) Matrix P must be singular
(d) Matrix P must be square

8. Eigen vectors of a real symmetric matrix corresponding to different eigen values are

(a) orthogonal (b) singular
(c) non-singular (d) none of these

9. An arbitrary vector X is an eigen vector of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{bmatrix}$, if (a, b) =

(a) (0, 0) (b) (1, 1) (c) (0, 1) (d) (1, 2)

10. For which value of k, following system is consistent ?

$$2x - 5ky + 6z = 0$$
$$kx + 2y - 2z = 0$$
$$2x + 2y - kz = 0$$

(a) 1 (b) 2 (c) 3 (d) 5

11. The system of equations

$$a_1x + a_2y = 0$$
$$b_1x + b_2y = 0$$

where a_1, a_2, b_1, b_2, are real numbers, has a non-trivial solution if

(a) $a_1b_1 = a_2b_2$ (b) $a_1b_2 = b_1a_2$
(c) $a_1a_2 = b_1b_2$ (d) none of these

12. Solution of the given matrix equation is

$$\begin{bmatrix} 2 & 2 & 2 \\ 1 & 5 & -3 \\ -1 & -3 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(a) $x_1 = 0, x_2 = 0, x_3 = 0$
(b) $x_1 = 1, x_2 = 1, x_3 = 0$
(c) $x_1 = 0, x_2 = 1, x_3 = 1$
(d) $x_1 = 2, x_2 = -5, x_4 = -1$

13. If product of matrix

$$A = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \text{ and}$$

$$B = \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

is a null matrix, then θ and ϕ differ by an

(a) odd multiple of π (b) even multiple of π
(c) odd multiple of $\pi/2$ (d) even multiple of $\pi/2$

14. Inverse of the matrix $\begin{Bmatrix} ab & 0 \\ 0 & 1 \end{Bmatrix}$ is

(a) $\begin{Bmatrix} a & 0 \\ 0 & a \end{Bmatrix}$ (b) $\begin{Bmatrix} b & 0 \\ a & b \end{Bmatrix}$

(c) $\begin{pmatrix} 1/a & 0 \\ 0 & 1/b \end{pmatrix}$ (d) $\begin{Bmatrix} a & 0 \\ 0 & 1/b \end{Bmatrix}$

15. The system of equations
$$2x + 4y = 10,$$
$$5x + 10y = 25$$
has
(a) no unique solution (b) only one solution
(c) only two solutions (d) infinite solutions

16. Sum of the eigen values of the matrix $\begin{pmatrix} 3 & 4 \\ x & 1 \end{pmatrix}$ for real and negative values of x is
(a) greater than zero
(b) less than zero
(c) zero
(d) dependent on the value of x

17. The system of equations
$$4x + 6y + 8z = 0$$
$$7x + 8y + 9z = 0$$
$$3x + 2y + z = 0$$
has
(a) no solution
(b) only one solution
(c) two solutions
(d) infinite number of solutions

18. Eigen values of $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ are
(a) 0, 0 0 (b) 0, 0 1 (c) 0, 0, 3 (d) 1, 1, 1

19. For following set of simulataneous equations
$$1.5x - 0.5y = 2$$
$$4x + 2y + 3z = 9$$
$$7x + y + 5z = 10$$
(a) solution is unique
(b) infinitely many solutions exist
(c) equations are incompatible
(d) finite number of multiple solutions exist

20. If $R = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 2 & 3 & 2 \end{bmatrix}$, then top row of R^{-1} is
(a) [5 6 4] (b) [5 −3 1]
(c) [2 0 −1] (d) [2 −1 1/2]

21. The system of simultaneous equations
$$x + 2y + z = 6$$
$$2x + y + 2z = 6$$
$$x + y + z = 5$$
has
(a) unique solution
(b) infinite number of solutions
(c) no solution
(d) exactly two solutions

22. For which value of x, will the matrix $\begin{bmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$ become singular?

23. Eigen vectors of the matrix $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ is (are)
(a) (1, 0) (b) (0, 1) (c) (1, 1) (d) (1, −1)

24. Inverse of the matrix $\begin{bmatrix} -3 & 5 \\ 2 & 1 \end{bmatrix}$ is
(a) $\begin{bmatrix} \frac{5}{13} & \frac{-1}{13} \\ \frac{2}{13} & \frac{3}{13} \end{bmatrix}$ (b) $\begin{bmatrix} \frac{2}{13} & \frac{5}{13} \\ \frac{-1}{13} & \frac{3}{13} \end{bmatrix}$
(c) $\begin{bmatrix} \frac{-1}{13} & \frac{5}{13} \\ \frac{2}{13} & \frac{3}{13} \end{bmatrix}$ (d) $\begin{bmatrix} \frac{1}{13} & \frac{-5}{13} \\ \frac{2}{13} & \frac{3}{13} \end{bmatrix}$

25. If A and B are square matrices of size n × n, then which of the following statement is not true ?
(a) det (AB) = det (A) det (B)
(b) det (kA) = k^n det (A)
(c) det (A + B) = det (A) + det (B)
(d) det (A^T) = 1/det (A^{-1})

26. If matrix A is m × n and B is n× p, then number of multiplication operations and addition operations needed to calculate the matrix AB, respectively are
(a) mn^2p, mpn (b) mpn, $(n-1)$
(c) mpn, mpn (d) mn^2p, $(m+p)n$

27. If A be an invertible matrix and inverse of 7A is $\begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$, then matrix A is
(a) $\begin{bmatrix} 1 & 2/7 \\ 4/7 & 1/7 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & 2 \\ 4 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & -4/7 \\ -2/7 & 1/7 \end{bmatrix}$ (d) $\begin{bmatrix} 7 & 4 \\ 2 & 1 \end{bmatrix}$

28. Solution(s) to the equations
$$2x + 3y = 1$$
$$x - y = 4$$
$$4x - y = a$$
will exist if a is equal to
(a) −33 (b) 0 (c) 9 (d) 12

29. Matrix B = A^T, where A is any matrix is
(a) skew symmetric
(b) symmetric about the secondary diagonal
(c) always symmetric
(d) another general matrix

30. If $A^T = A^{-1}$, where A is a real matrix, then A is
(a) normal (b) symmetric
(c) Hermitian (d) orthogonal

31. If A and B are non-zero squarematrices, then AB = I implies
(a) A and B are orthogonal
(b) A and B are singular
(c) B is singular
(d) A is singular

32. Eigen values of a square symmetric matrix are always
(a) positive (b) real and imaginary
(c) negative (d) real

33. If A = $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then state transition matrix e^{At} is
(a) $\begin{bmatrix} 0 & e^{-t} \\ e^{-t} & 0 \end{bmatrix}$ (b) $\begin{bmatrix} e^t & 0 \\ 0 & e^t \end{bmatrix}$
(c) $\begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-t} \end{bmatrix}$ (d) $\begin{bmatrix} 0 & e^t \\ e^t & 0 \end{bmatrix}$

34. For matrix $p = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, if one of the eigen values is equal to −2, then which of the following is an eigen vector ?
(a) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ (d) $\begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$

35. Eigen values and corresponding eigen vectors of a 2 × 2 matrix are given by

Eigenvalue	Eigenvector
$\lambda_1 = 4$	$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
$\lambda_2 = 8$	$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is
(a) $\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$

36. A second order system starts with an initial condition of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ without any external input. The state transition matrix for the system is given by $\begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix}$. The state of the system at the end of 1 second is given by
(a) $\begin{bmatrix} 0.271 \\ 1.100 \end{bmatrix}$ (b) $\begin{bmatrix} 0.135 \\ 0.368 \end{bmatrix}$
(c) $\begin{bmatrix} 0.271 \\ 0.736 \end{bmatrix}$ (d) $\begin{bmatrix} 0.135 \\ 1.100 \end{bmatrix}$

NUMERICAL TYPE QUESTIONS

1. Rank of the matrix A = $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 0 \\ 1 & 0 & 0 & 0 \\ 4 & 0 & 3 & 0 \end{bmatrix}$ is _____

2. Rank of the diagonal matrix is_____

3. Rank of the matrix A = $\begin{bmatrix} 1 & 3 & 5 & 1 \\ 2 & 4 & 8 & 0 \\ 3 & 1 & 7 & 5 \end{bmatrix}$ is_____

4. Rank of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is_____

5. The value of λ for which the equations
$$2x + y + 2z = 0$$
$$x + y + 3z = 0$$
$$4x + y + \lambda z = 0$$
have a non-zero solution, is_____

6. The system of linear equations
$$(4d - 1)x + y + z = 0$$
$$-y + z = 0$$
$$(4d - 1)z = 0$$
has a non-trivial solution, if d equals _____

7. Rank of a 3 × 3 matrix C (= AB), found by multiplying a non-zero column matrix A of size 3 × 1 and a non-zero row matrix B of size 1 × 3, is_____

8. Sum of eigen values of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ is_____

9. Rank of the matrix, A = $\begin{bmatrix} 1 & -2 & -1 \\ -3 & 3 & 0 \\ 2 & 2 & 4 \end{bmatrix}$ is_____

10. If $A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix}$, then $(a + b)$ equals _____

11. For matrix $\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, eigen value corresponding to the eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$ is_____

ANSWERS

EXERCISE

MCQ Type Questions

1. (c)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (d)	**6.** (b,d)	**7.** (a)	**8.** (a)	**9.** (b)	**10.** (b)
11. (b)	**12.** (a)	**13.** (c)	**14.** (c)	**15.** (d)	**16.** (a)	**17.** (b)	**18.** (c)	**19.** (c)	**20.** (b)
21. (c)	**22.** (a)	**23.** (c,d)	**24.** (c)	**25.** (c)	**26.** (b)	**27.** (a)	**28.** (d)	**29.** (c)	**30.** (d)
31. (a)	**32.** (d)	**33.** (b)	**34.** (d)	**35.** (a)	**36.** (a)				

Numerical Type Questions

1. 3	**2.** 4	**3.** 3	**4.** 1	**5.** 8	**6.** $\frac{1}{4}$	**7.** 1	**8.** 7	**9.** 2

10. $\frac{7}{20}$ **11.** 6

EXPLANATIONS

MCQ TYPE QUESTIONS

1. Characteristic equation is

$$[A - \lambda I] = 0$$

$$\Rightarrow \begin{bmatrix} 3-\lambda & -1 & -1 \\ -1 & 3-\lambda & -1 \\ -1 & -1 & 3-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (3-\lambda)\{(3-\lambda)^2 - 1\} - 1\{(3-\lambda) + 1\} - 1\{(3-\lambda) + 1\} = 0$$

$$\Rightarrow \{(3-\lambda) + 1\}\{(3-\lambda)(3-\lambda-1)\} - 2\} = 0$$

$\lambda = 1, 4, 4$

$\therefore$ $\lambda = 1, 4, 4$ are eigen values.

4. $A' = \begin{bmatrix} 0 & -2 & -3 \\ 2 & 0 & -5 \\ 3 & 5 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix} = -A.$

This is a skew-symmetric matrix.

6. Determinant $A = 1(\cos^2\theta - \sin^2\theta)$

Hence A is non-singular and A^{-1} exists.

7. According to *Rouche's theorem,* the system is consistent if and only if coefficient matrix and augmented matrix K are of the same rank, otherwise the system is inconsistent.

8. Let A be a real symmetric matrix, therefore

$$A^T = A$$

Let α_1 and α_2 be different eigen values of the matrix A, and X_1 and X_2 be the corresponding vectors, then

$$AX_1 = \alpha_1 X_1 \quad \text{and} \quad AX_2 = \alpha_2 X_2$$

Taking transpose of the second equation

$$(AX_2)^T = (\alpha_2 X_2)^T$$

$$\Rightarrow \quad X_2^T A^T = \alpha_2 . X_2^T$$

But $A^T = A$,

$$\therefore \quad X_2^T A = \alpha_2 . X_2^T$$

Post multiply by X_1, we get

$$X^T_2 AX_1 = \alpha_2 X_2^T X_1$$

But $AX_1 = \alpha_1 X_1$

$$\therefore \quad X^T_2 \alpha_1 X_1 = \alpha_2 X_2^T X_1$$

$$\Rightarrow \quad (\alpha_1 - \alpha_2) X_2^T X_1 = 0$$

Since $\alpha_1 \neq \alpha_2$, $\alpha_1 - \alpha_2 \neq 0$

$$\therefore \quad X_2^T X_1 = 0$$

i.e. X_2 and X_1 are orthogonal.

9. Since matrix is triangular, hence eigen values are a, a, b.

If (X_1, X_2, X_3) is an arbitrary eigen vector, say corresponding to 1, then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = 1 \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$\therefore$ X_2, X_3 being not zero, we have

$$X_1 = X_1 ; aX_2 = X_2$$

which gives $a = 1$

and $bX = X_3$

$\Rightarrow$ $b = 1$

$\therefore$ $(a, b) = (1, 1).$

10. $A = \begin{bmatrix} 2 & -5k & 6 \\ k & 2 & -2 \\ 2 & 2 & -k \end{bmatrix}$

Equations are consistent, if rank of A and that of k are equal. But in this case it is always true. Hence equations will have a trivial solution if $|A| \neq 0$ Therefore only non-trivial solution will exist if $|A| = 0$

i.e. $\begin{bmatrix} 2 & -5k & 6 \\ k & 2 & -2 \\ 2 & 2 & -k \end{bmatrix} = 0$

$$\Rightarrow 2(-2k + 4) + 5k(-k^2 + 4) + 6(2k - 4)$$

$$\Rightarrow -5k^3 + 20k - 4k + 8 + 12k - 24 = 0$$

$$\Rightarrow 5k^3 - 28k + 16 = 0$$

$$\Rightarrow 5k^3 - 10k^2 + 10k^2 - 20k - 8k + 16 = 0$$

$$\Rightarrow (5k^2 + 10k - 8)(k - 2) = 0$$

$$k = \frac{-10 \pm \sqrt{100 + 160}}{2} \text{ or } 2$$

$$= -1 \pm \sqrt{13/5} \text{ or } 2$$

11. $\frac{x}{y} = -\frac{a_2}{a_1}$ and $\frac{x}{y} = -\frac{b_2}{b_1}$

Equations are consistent only if

$$-\frac{a_2}{a_1} = -\frac{b_2}{b_1} \Rightarrow a_1 b_2 = b_1 a_2$$

12. Let matrics, $A = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 5 & -3 \\ -1 & -3 & 1 \end{bmatrix}$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad B_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore$ $AX = B$

Multiplying both sides by A^{-1},

$$X = A^{-1} B$$

But as $B = 0$; therefore $X = 0$

Hence $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\therefore$ $x_1 = 0, \quad x_2 = 0, \quad x_3 = 0.$

13. AB

$$= \begin{bmatrix} \cos\theta.\cos\phi\cos(\theta - \phi) & \cos\theta.\sin\phi.\cos(\theta - \phi) \\ \cos\phi\sin\theta.\cos(\theta - \phi) & \sin\theta.\sin\phi\cos(\theta - \phi) \end{bmatrix}$$

= A null matrix, when $\cos(\theta - \phi) = 0$

i.e. if $(\theta - \phi)$ is an odd multiple of $\left(\frac{\pi}{2}\right)$.

16. Eigen values are given by the solution of equation,

$$\begin{vmatrix} 3-\lambda & 4 \\ x & 1-\lambda \end{vmatrix} = 0$$

Since x is real and negative, put $x = -k$, where k is positive constant

$$\therefore \quad (3-\lambda)(1-\lambda) + 4k = 0$$

$$\Rightarrow \quad \lambda^2 - 4\lambda + 3 + 4k = 0$$

If λ_1 and λ_2 be the solutions of above equation, then λ_1 and λ_2 are eigen values.

Now, Sum of eigen values

= Sum of roots of above equation

i.e. $\lambda_1 + \lambda_2 = \frac{-(-4)}{1} = 4\ (> 0)$

17. For given system of equations,

$$\Delta = \begin{vmatrix} 4 & 6 & 8 \\ 7 & 8 & 9 \\ 3 & 2 & 1 \end{vmatrix} = 4(8 - 18) - 6(7 - 27) + 8(14 - 24)$$

$$= -40 + 120 - 80 = 0$$

Since $\Delta = 0$, hence given system of equations has unique solution, *i.e.* only one solution.

18. Given matrix is $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Now $\begin{bmatrix} (1-\lambda) & 1 & 1 \\ 1 & (1-\lambda) & 1 \\ 1 & 1 & (1-\lambda) \end{bmatrix} = 0$

$R_1 \to R_1 + R_2 + R_3$

$$\Rightarrow \begin{bmatrix} (3-\lambda) & (3-\lambda) & (3-\lambda) \\ 1 & (1-\lambda) & 1 \\ 1 & 1 & (1-\lambda) \end{bmatrix} = 0$$

$$\Rightarrow (3-\lambda)\begin{vmatrix} 1 & 1 & 1 \\ 1 & (1-\lambda) & 1 \\ 1 & 1 & (1-\lambda) \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1$

$C_3 \to C_3 - C_1$

$$\Rightarrow (3-\lambda)\begin{vmatrix} 1 & 0 & 0 \\ 1 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{vmatrix} = 0$$

Hence eigen values are 0, 0, 3

20. Adj $R = \begin{bmatrix} 5 & -6 & 4 \\ -3 & 4 & -3 \\ 1 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 5 & -3 & 1 \\ -6 & 4 & -1 \\ 4 & -3 & 1 \end{bmatrix}$

Since, $|R| = 1$, therefore top row $\rightarrow [5\ -3\ 1]$

21. Given equations are :

$x + 2y + z = 6$

$2x + y + 2z = 6$

$x + y + z = 5$

Given system can be written as

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 5 \end{bmatrix}$$

Applying row operation

$R_2 = R_2 - 2R_1$, $R_3 = R_3 - R_1$, we get

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \\ -1 \end{bmatrix}$$

Applying $R_3 = 3R_3 - R_2$, we get

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -6 \\ 3 \end{bmatrix}$$

Since rank of co-efficient matrix is 2 and rank of argument matrix is 3, which is not equal. Hence system has no solution.

22. For singular matrix,

$$\begin{vmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{vmatrix} = 0$$

$\Rightarrow \quad 8[0 - 12] - x[0 - 2 \times 12] = 0$

$\Rightarrow \quad -96 + 24x = 0$

$\Rightarrow \quad x = 4$

23. Eigen value are roots of the equation

$$\begin{bmatrix} 1-\alpha & -1 \\ -1 & 1-\alpha \end{bmatrix} = 0$$

i.e. $\quad (1-\alpha)^2 - 1 = 0$

$\Rightarrow \quad \alpha^2 - 2\alpha = 0$

$\Rightarrow \quad \alpha = 0 \text{ or } \alpha = 2$

When $\alpha = 0$, $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

This gives, $\quad x = y$

When $\alpha = 2$, $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2$

$\Rightarrow \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$.

This gives, $\quad x = -y$

Hence, eigen vectors corresponding to 0 and 2 are $\begin{bmatrix} x \\ x \end{bmatrix}$ and $\begin{bmatrix} x \\ -x \end{bmatrix}$, or $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

24. For matrix $A = \begin{bmatrix} -3 & 5 \\ 2 & 1 \end{bmatrix}$

cofactors are :

$a_{11} = 1, a_{21} = -5, a_{12} = -2, a_{22} = -3$

$\therefore$ Matrix by cofactors $= \begin{bmatrix} 1 & -2 \\ -5 & -3 \end{bmatrix}$

Adjoint $A = \begin{bmatrix} 1 & -5 \\ -2 & -3 \end{bmatrix}$

$|A| = -3(1) - 5(2)$

$= -3 - 10 = -13$

$\therefore \quad A^{-1} = \dfrac{\text{Adjoint } A}{|A|} \begin{bmatrix} -\frac{1}{13} & \frac{5}{13} \\ \frac{2}{13} & \frac{3}{13} \end{bmatrix}$

27. Inverse of $7A = \begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$

$\det |\Delta| = \begin{vmatrix} -1 & 2 \\ 4 & -7 \end{vmatrix} = 7 - 8 = -1$

$\therefore$ Matrices of cofactor $= \begin{bmatrix} -7 & -4 \\ -2 & -1 \end{bmatrix}$,

and it's transpose, $A^T = \begin{bmatrix} -7 & -2 \\ -4 & -1 \end{bmatrix}$

$\therefore$ Inverse $= \dfrac{A^T}{|\Delta|} = \begin{bmatrix} 7 & 2 \\ 4 & 1 \end{bmatrix}$

$\therefore \quad A = \frac{1}{7}\left[\dfrac{A^T}{|\Delta|}\right] = \begin{bmatrix} 1 & \frac{2}{7} \\ \frac{4}{7} & \frac{1}{7} \end{bmatrix}$

28. Equations are consistent when

$$\begin{vmatrix} 2 & 3 & -1 \\ 1 & -1 & -4 \\ 4 & -1 & -a \end{vmatrix} = 0$$

$\Rightarrow \quad 2(a-4) - 3(-a+16) - 1(-1+4) = 0$

$5a - 8 - 48 - 3 = 0$

$5a = 59$

$a = \dfrac{59}{5} \approx 12$

33. Here $\quad [sI - A]^{-1} = \begin{bmatrix} s-1 & 0 \\ 0 & s-1 \end{bmatrix}^{-1}$

Then, $\quad 2^{-1}[sI - A]^{-1} = 2^{-1}\left\{\begin{bmatrix} \frac{1}{s-1} & 0 \\ 0 & \frac{1}{s-1} \end{bmatrix}\right\}$

$= \begin{bmatrix} e^t & 0 \\ 0 & e^t \end{bmatrix}$

34. $|P - \lambda I| = \begin{vmatrix} 3-\lambda & -2 & 2 \\ 0 & -2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$

$\therefore \quad \lambda = -2, 3, 1$

Putting $\lambda = -2$, we get

$5x - 2y + 2z = 0$

$z = 0$

So. $\quad 5x = 2y$ eigen vector $= \begin{bmatrix} 5 \\ 2 \\ 0 \end{bmatrix}$

35. Given matrix : $\quad A = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$

We know $\quad |\lambda I - A| = 0$

$\Rightarrow \quad \begin{vmatrix} \lambda - 6 & -2 \\ -2 & \lambda - 6 \end{vmatrix} = 0$

$\Rightarrow \quad \lambda^2 - 12\lambda + 32 = 0$

$\Rightarrow \quad \lambda = 4, 8$ (eigen values)

For $\lambda_1 = 4$,

$(\lambda_1 I - A) = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}$, $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda_1 = 8$,

$(\lambda_2 I - A) = \begin{bmatrix} 2 & -6 \\ -6 & 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

36. State of the system at time t is,

$X(t) = [sI - A]^{-1}X(0) = \phi(t)X(0)$

$= \begin{bmatrix} e^{-2t} & 0 \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2e^{-2t} \\ 3e^{-t} \end{bmatrix}$

At $\quad t = 1, \quad X(1) = \begin{bmatrix} 2e^{-2} \\ 3e^{-1} \end{bmatrix} = \begin{bmatrix} 0.271 \\ 1.100 \end{bmatrix}$

NUMERICAL TYPE QUESTIONS

1. Determinant $A = 0$

$\Rightarrow \quad \begin{vmatrix} 4 & 2 & 3 \\ 1 & 0 & 0 \\ 4 & 0 & 3 \end{vmatrix} = (-1) \times 6 = -6 \neq 0$

Hence rank of $A = 3$

2. Since the number of non-zero elements on this diagonal matrix is four, hence the rank is four.

3. Given matrix A possesses a minor of order 3

viz. $\begin{vmatrix} 1 & 3 & 1 \\ 2 & 4 & 0 \\ 3 & 1 & 5 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ 2 & 4 & 0 \\ -2 & -14 & 5 \end{vmatrix}$

Replacing C_1 and C_2 by $C_1 - C_3$ and $C_2 - 3C_3$.

$= \begin{vmatrix} 2 & 4 \\ -2 & -14 \end{vmatrix}$ expanding with respect to R_1

$= 2(-14) - (4)(-2) = -28 + 8 \neq 0$

$\therefore \quad p(A) \geq 3 \qquad \ldots (i)$

Also A does not possess any minor of order 4, *i.e.* 3 + 1

$\therefore \quad p(A) \leq 3 \qquad \ldots (ii)$

From equations (i) and (ii), we get

$p(A) = 3$ *i.e.* rank of A is 3.

5. Equivalent matrix equation is

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

In order that given system of equations may have non-zero solution, rank of A should be less than 3. This requires that

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & \lambda \end{bmatrix} = 0$$

Interchanging R_1 and R_2,

$$\begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 2 \\ 4 & 3 & \lambda \end{bmatrix} = 0$$

By $(R_2 - 2R_1)$ and $(R_3 - 4R_1)$,

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & -1 & -4 \\ 0 & -1 & \lambda - 12 \end{bmatrix} = 0$$

$\Rightarrow \begin{bmatrix} -1 & -4 \\ -1 & \lambda - 12 \end{bmatrix} \Rightarrow -1(\lambda - 12) - (-1)(4) = 0$

$\Rightarrow \quad \lambda = 8$

6. The system of homogeneous linear equations has a non-trivial solution if

$$\begin{bmatrix} 4d-1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 4d-1 \end{bmatrix} = 0$$

$\Rightarrow -(4d-1)^2 = 0 \quad \Rightarrow d = \dfrac{1}{4}$

7. Let $A = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$; $B = [a_2\ b_2\ c_2]$

$$\therefore \quad C = \begin{bmatrix} a_1a_2 & a_1b_2 & a_1c_2 \\ b_1a_2 & b_1b_2 & b_1c_2 \\ c_1a_2 & c_2b_2 & c_1c_2 \end{bmatrix}$$

$$= a_1a_2\,[b_1\,b_2\,c_1\,c_2 - b_1b_2\,c_1c_2] = 0$$

Cofactor $= \begin{vmatrix} a_1a_2 & a_1b_2 \\ b_1a_2 & b_1a_2 \end{vmatrix} = 0$

So rank = 1

8. Sum of eigen values of given matrix
= Sum of element of diagonal of given matrix
= 1 + 5 + 1 = 7

9. Since, $det\ A = \begin{vmatrix} 1 & -2 & -1 \\ -3 & 3 & 0 \\ 2 & 2 & 4 \end{vmatrix} = 12 - 24 + 12 = 0$

and $M_{11} = 12 \neq 0$. Hence rank is 2.

11. Given : Matrix $M = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$

$$\therefore \quad M - \lambda I = \begin{bmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix}$$

For the eigen vector $\begin{bmatrix} 101 \\ 101 \end{bmatrix}$

$$(4-\lambda)(101) + 2(101) = 0$$

$$\Rightarrow \quad 4 - \lambda + 2 = 0 \quad \Rightarrow \quad \lambda = 6 \text{ (eigen value)}$$

■■

2 CHAPTER

Calculus

FUNCTIONS OF SINGLE VARIABLE

LIMITS

By limit we are able to find out behaviour of a function $f(x)$ near a point c, when distance between x and c is very small i.e. $|x-c|$ is small.

Specifically, we would like to know whether there is some real number k such that $f(x)$ approaches k as x approaches c.

We write as $\lim\limits_{x \to c} f(x) = k$

Right hand and Left hand limits

Left-hand limit: $\lim\limits_{x \to c^-} f(x) = k$

Right-hand limit: $\lim\limits_{x \to c^+} f(x) = k$

If $\lim\limits_{x \to c^-} f(x) = \lim\limits_{x \to c^+} f(x)$, then limit of $f(x)$ as $x \to c$ exists and is written as $\lim\limits_{x \to c} f(x)$.

The limit of a function f(x) may not exist in any of the following situations :

(*i*) $\lim\limits_{x \to c^-} f(x)$ does not exist

(*ii*) $\lim\limits_{x \to c^+} f(x)$ does not exist

(*iii*) Both $\lim\limits_{x \to c^-} f(x)$, and $\lim\limits_{x \to c^+} f(x)$ exist but are unequal

Theorems on Limits

If $\lim\limits_{x \to c} f(x)$ and $\lim\limits_{x \to c} g(x)$ exist, then

(1) $\lim\limits_{x \to c} (f(x) \pm g(x)) = \lim\limits_{x \to c} f(x) \pm \lim\limits_{x \to c} g(x)$

(2) $\lim\limits_{x \to c} (f(x).g(x)) = \lim\limits_{x \to c} f(x).\lim\limits_{x \to c} g(x)$

(3) $\lim\limits_{x \to c} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x \to c} f(x)}{\lim\limits_{x \to c} g(x)}$ where $\left(\lim\limits_{x \to c} g(x) \neq 0\right)$

Some useful Limits

(1) $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = 1 = \lim\limits_{x \to 0} \cos x = \lim\limits_{x \to 0} \dfrac{\tan x}{x}$

(2) $\lim\limits_{x \to 0} (1+x)^{\frac{1}{x}} = e = \lim\limits_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^x$

(3) $\lim\limits_{x \to 0} \dfrac{\log_a(1+x)}{x} = \log_a e \quad (a > 0, a \neq 0)$

(4) $\lim\limits_{x \to 0} \dfrac{\log(1+x)}{x} = 1$

(5) $\lim\limits_{x \to 0} \dfrac{a^x - 1}{x} = \log a \quad (a > 0)$

(6) $\lim\limits_{x \to 0} \dfrac{(1+x)^m - 1}{x} = m$

(7) $\lim\limits_{x \to x} \dfrac{\log x}{x^m} = 0 \quad (m > 0)$

(8) $\lim\limits_{x \to 0} \dfrac{x^n - a^n}{x - a} = na^{n-1}$

L' HOSPITAL'S RULE

It states that for functions f and g if

$\lim\limits_{x \to c} f(x) = \lim\limits_{x \to c} g(x) = 0$

or $\lim\limits_{x \to c} g(x) = \pm\infty$

(condition that $\lim\limits_{x \to c} f(x) = \pm\infty$ is not necessary)

and if $\lim\limits_{x \to c} \dfrac{f'(x)}{g'(x)}$ exists then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$$

Note : If f' and g' again satisfy above conditions of f and g, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f''(x)}{g''(x)} \text{ and so on}$$

e.g. $\lim\limits_{x \to 0} \dfrac{\sin x}{x} = \dfrac{0}{0}$

By L' Hospital Rule

$$\lim_{x \to 0} \frac{\sin x}{x} = \lim_{x \to 0} \frac{\cos x}{x} = \frac{1}{1} = 1$$

e.g. $\lim\limits_{x \to 0} \dfrac{x^2 + x - \sin x}{x^2} = \dfrac{0}{0}$

By L' Hospital Rule

$$\Rightarrow \lim_{x \to 0} \frac{x^2 + x - \sin x}{x^2} = \lim_{x \to 0} \frac{2x + 1 - \cos x}{2x} = \frac{0}{0}$$

$$\Rightarrow \lim_{x\to 0}\frac{x^2+x-\sin x}{x^2} = \lim_{x\to 0}\frac{2+\sin x}{2} = 1$$

e.g. $\lim_{x\to 0}\frac{x^n}{x^m} = \frac{0}{0}$

By L' Hospital Rule

$$\lim_{x\to 0}\frac{x^n}{x^m} = \frac{nx^{n-1}}{mx^{m-1}} = \frac{0}{0}$$

$$\Rightarrow \lim_{x\to 0}\frac{x^n}{x^m} = \frac{n(n-1)x^{n-2}}{m(m-1)x^{m-2}} = \frac{0}{0}$$

$$\lim_{x\to 0}\frac{x^n}{x^m} = \frac{n(n-1).....1}{m(m-1).....1} = \frac{n!}{m!}$$

so on

CONTINUITY AND DISCONTINUITY

Continuity

- The function f is said to be *continuous* at $x = c$ if

  $\lim_{x\to c} f(x) = f(c)$

- The function f is said to be continuous on an open interval (a, b) if it is continuous at each point of (a,b)
- The function is said to be continuous on a closed interval $[a, b]$ if

  (*i*) f is continuous at each point of (a, b)

  (*ii*) $\lim_{x\to a} f(x) = f(a)$

  (*iii*) $\lim_{x\to b} f(x) = f(b)$

Note : Graphically, $f(x)$ is continuous if its graph can be drawn on paper without raising the pencil.

Discontinuity

A function f is said to be discontinuous at c if it is not continuous at that point. A function f is said to be discontinuous on an interval if it is discontinuous at least one point of the interval.

A function is discontinuous at c under any of the following circumstances:

(*i*) f is not defined at c

(*ii*) $\lim_{x\to c} f(x)$ does not exist

(*iii*) f is defined at c and $\lim_{x\to c} f(x)$ exists but $\lim_{x\to c} f(x) \neq f(c)$

Note : If f and g are continuous at $x = c$, then $f + g$, $f.g$, αf, and $\frac{f}{g}$ is also continuous at $x = c$ $(g(c) \neq 0)$

DIFFERENTIABILITY

A function $f(x)$ is said to be differentiable at $x = a$ if

both $\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$, $h > 0$

and $\lim_{h\to 0}\frac{f(a-h)-f(a)}{-h}$, $h > 0$

exist and have common value (finite). This common value is called *derivative of* $f(x)$ at the point $x = a$ and is denoted by $f'(a)$.

FUNCTIONS OF TWO VARIABLES

Let $f(x,y)$ be any function of two independent varibles x and y.

Limit

A function f tends to a limit l, when (x, y) tends to (a, b)

$$\lim_{(x,y)\to(a,b)} f(x,y) = l$$

Note : If $\lim_{(x,y)\to(a,b)} f(x,y) = l$ and if $y = \phi(x)$ is any function such that $\phi(x) \to b$ when $x \to a$, then $\lim_{(x,y)\to(a,b)} f(x,\phi(x))$ must exist and be equal to l.

Repeated Limits

If a function f is defined in some neigbourhood of (a, b), then limit $\lim_{y\to b} f(x,y)$ if exists, is a function of x, say $\phi(x)$.

If limit $\lim_{x\to a}\phi(x)$ exists and is equal to K, we write

$$\lim_{x\to a}\lim_{y\to b} f(x,y) = K$$

Here K is repeated limits of f as $y \to b, x \to a$

If we change order of taking the limit, we get other repeated limit.

$$\lim_{y\to b}\lim_{x\to a} f(x,y) = K'$$

when first $x \to a$ and then $y - b$

These two limits may or may not be equal.

Note :

(*i*) If repeated limits are not equal, then simultaneous limits cannot exist.

(*ii*) In case simultaneous limit exists, these two repeated limits if they exist are necessarily equal but the converse is not true.

Continuity

A function f is said to be *continuous* at a point (a,b) of its domain to definition if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

Computing the Derivative

Rules of Differentiation

$(f+g)' = f' + g'$ → Sum Rule

$(f-g)' = f' - g'$ → Difference Rule

$(fg)' = fg' + gf'$ → Product Rule

$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$ → Quotient Rule

$\frac{1}{dx}(f(g(x))) = \frac{df}{dg}\cdot\frac{dg}{dx}$ → Chain Rule

Some Standard Derivative

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin h\, x$	$\cos h\, x$
$\ln x$	$1/x$	$\cos h\, x$	$\sin h\, x$
$\log_a x$	$\log_a e.\frac{1}{x}$	$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$
e^x	e^x	$\cos^{-1}x$	$\frac{-1}{\sqrt{1-x^2}}$
a^x	$a^x \log_a e$	$\tan^{-1}x$	$\frac{1}{1+x^2}$
$\sin x$	$\cos x$	$\operatorname{cosec}^{-1}x$	$\frac{-1}{x^2\sqrt{x^2-1}}$
$\cos x$	$\sin x$		
$\tan x$	$\sec^2 x$	$\sec^{-1}x$	$\frac{1}{x\sqrt{x^2-1}}$
$\sec x$	$\sec x \tan x$	$\cot^{-1}x$	$\frac{1}{1+x^2}$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$	$\lvert x\rvert\frac{x}{\lvert x\rvert}$	$(x \neq 0)$
$\cot x = -\operatorname{cosec}^2 x$			

DIFFERENTIATION BY SUBSTITUTION

If a function contain as expression of the form

(1) $a^2 - x^2$; put $x = a \sin t$ or $x = a \cos t$

(2) $a^2 + x^2$; put $x = a \tan t$ or $x = a \cot t$

(3) $x^2 - a^2$; put $x = a \sec t$ or $x = a \operatorname{cosec} t$

(4) $\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$; put $x = a \cos t$

(5) $a \cos x \pm b \sin x$; put $a = r \cos\theta$, and $b = r \sin\theta$, $r > 0$

Partial Derivatives

The partial differential coefficient of $z = f(x, y)$ with respect to x is defined as

$$\lim_{\delta x\to 0}\frac{f(x+\delta x, y) - f(x,y)}{\delta x}$$

provided this limit exists and is denoted by

$$\frac{\partial f}{\partial x} \text{ or } \frac{\partial z}{\partial x} \text{ or } f_x \text{ or } z_x$$

Similarly, partial derivatives of $f(x, y)$ with respect to y is defined as

$$\lim_{\delta y\to 0}\frac{f(x, y+\delta y) - f(x, y)}{\delta y}$$

provided this limit exists and is denoted by

$$\frac{\partial f}{\partial y} \text{ or } \frac{\partial z}{\partial y} \text{ or } f_y \text{ or } z_y$$

$\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are called *first order partial derivatives of z.*

- *The partial derivation of $f(x, y)$ with respect to y is the ordinary derivative of $f(x, y)$ when x is regarded as constant.*

Example. If $z = \log(x^2 + y^2)$, find $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y}$

Solution

Given : $z = \log(x^2 + y^2)$

$$\therefore \quad \frac{\partial z}{\partial x} = \frac{2x}{x^2+y^2}$$

$$\Rightarrow \quad x\frac{\partial z}{\partial x} = \frac{2x^2}{x^2+y^2}$$

$$\frac{\partial z}{\partial y} = \frac{2y}{x^2+y^2}$$

$$\Rightarrow \quad y\frac{\partial z}{\partial y} = \frac{2y^2}{x^2+y^2}$$

$$\therefore \quad x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{2(x^2+y^2)}{(x^2+y^2)} = 2$$

Partial Derivatives of Higher Order

Since f_x and f_y are also functions of x and y and so we can further differentiate f_x and f_y partially with respect to x or y. The further partial derivatives of f_x or f_y are called *partial derivatives f higher order of u.*

- Partial derivatives of $\frac{\partial u}{\partial x}$ with respect to x and y are $\frac{\partial^2 u}{\partial^2 x}$ or $\frac{\partial^2 u}{\partial y\,\partial x}$ respectively.
- Partial derivatives $\frac{\partial^2 u}{\partial^2 x^2}$, $\frac{\partial^2 u}{\partial y\,\partial x}$, $\frac{\partial^2 u}{\partial x\partial y}$, $\frac{\partial^2 u}{\partial y^2}$ are also denoted by f_{xx}, f_{xy}, f_{yx}, f_{yy} respectively.

Example. If $u = e^{xyz}$, then find $\frac{\partial^3 u}{\partial x\partial y\partial z}$.

Solution

Given : $u = e^{xyz}$

$\therefore \quad \frac{\partial u}{\partial z} = e^{xyz}\, xy = uxy, \quad \frac{\partial u}{\partial y} = uxz$

and $\quad \frac{\partial u}{\partial x} = uyz$

Now, $\quad \frac{\partial^2 u}{\partial y \partial z} = ux + xy$

$$\frac{\partial u}{\partial y} = ux + (xy)\,(uxz) = ux + ux^2\, yz$$

$$\therefore \quad \frac{\partial^3 u}{\partial x \partial y \partial z} = u + x\frac{\partial u}{\partial x} + 2u\, xyz + x^2 yz\, \frac{\partial u}{\partial x}$$

$$= u + x(uyz) + 2uxyz + x^2 yz\,(uyz)$$

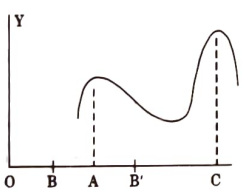
$$= e^{xyz}(1 + 3xyz + x^2\, y^2 z^2)$$

MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

A function $f(x, y)$ is said to have a maximum or minimum at $x = a, y = b$, according as $f(a + h, b + k)$ is less or greater than $f(a, b)$ for all positive or negative small values of h and k.

In other words, if $\Delta = f(a + h, b + k) - f(a, b)$ is of the same sign for all small values of h, k and if this sign is negative, then $f(a, b)$ is a maximum. If this sign is positive, then $f(a, b)$ is a minimum.

A maximum or minimum value of a function is called its *extreme value.*



Above figure shows a graph of the function $f(x)$ and $OA = a$, *i.e.*, $f(x)$ has a maximum value for $x = a$ because $f(a)$ has a value more than the values of $f(x)$ for every value of x between B and B'. $f(x)$ is said to be a *maximum* at $x = a$, even though value of $f(x)$ at $x = a$ should be greater than all other values of $f(x)$ in some small neighbourhood. Thus a maximum value of $f(x)$ is not necessarily the greatest value of $f(x)$. In fact, a curve might have several maxima (and minima).

Condition for Maxima and Minima

Necessary conditions for existence of a maximum or a minimum of $f(x)$ at $x = a, y = b$ is

$$\left(\frac{\partial f}{\partial x}\right)_{\substack{x=a\\y=b}} = 0; \text{ and } \left(\frac{\partial f}{\partial y}\right)_{\substack{x=a\\y=b}} = 0$$

Sufficient condition for Maxima and Minima

Let $\quad r = \left(\frac{\partial^2 f}{\partial x^2}\right)_{\substack{x=a\\y=b}}, \; s = \left(\frac{\partial^2 f}{\partial x \partial y}\right)_{\substack{x=a\\y=b}}$

and $\quad t = \left(\frac{\partial^2 f}{\partial y^2}\right)_{\substack{x=a\\y=b}}$

Case I: $rt - s^2 > 0$

$f(x, y)$ will have maximum or a minimum of $x = a, y = b$. Again, $f(x, y)$ is a maximum or a minimum according as r is negative or positive

Case II: $rt - s^2 < 0$

$f(x,y)$ will have neither a maximum nor a minimum at $x = a, y = b$.

Case III: $rt - s^2 = 0$

This case is doubtful and further investigation is needed to determine whether

(x, y) is maximum or a minimum at $x = a, y = b$ or not.

Finding Absolute Maxima and Minima Values

Working rules

If f is a differentiable function in $[a\ b]$ except at finitely many points, then to find absolute maximum and absolute minimum values, adopt following procedure :

(*i*) Evaluate $f(x)$ at the points, where $f'(x) = 0$.

(*ii*) Evaluated $f(a)$ and $f(b)$.

Then maximum of these values is the **absolute maxima** and minimum of these values is called **absolute minima.**

Finding Local Maxima and Local Minima

Working rules

Condition for maxima

- Function $f(x)$ has a maximum value $f(a)$ if $f'(a) = 0$ and $f'(x)$ changes sign from +ve to –ve as x passes through 'a' from left to right.
- In general, for any even number n, $f'(a) = f''(a) = f'''(a) \ \ldots = f^{n-1}\ (a) = 0$ and $f^n(a) < 0$, then $f(a)$ is a maximum value of $f(x)$.
- In particular, if $n = 2$, then $f'(a) = 0$ and $f''(a) < (0)$, and the function is maximum at $x = a$.

Condition for minima :

- Function $f(x)$ has a minimum value $f(a)$ if $f'(a) = 0$ and $f'(x)$ changes sign from –ve to +ve as x passes through $x = a$ from left to right.
- In general, for any even number n, if $f'(a) = f''(a) = f'''(a) = \ \ldots f^{n-1}\ (a) = 0$, and $f^n(a) > 0$, then $f(a)$ is minimum value of $f(x)$
- In particular, if $n = 2$, then $f'(a) = 0$ and $f''(a) > 0$, the function is minimum at $x = a$.

Note : If n is a even number such that $f'(a) = f''(a) = \ldots f^n(a) = 0, f^{n+1}(a) \neq 0$, then $f(a)$ is neither maximum nor minimum of the function $f(x)$.

In particular if $n = 2$ and $f'(a) = f''(a) = 0$, $f'''(a) \neq 0$, then $f(a)$ is neither maximum nor minimum values of $f(x)$. Then the point $x = a$ is called **point of inflexion.**

Extreme value

Either a maximum value or minimum value $f(a)$ of the function $f(x)$ is called *extreme value.*

Stationary value

If $f'(a) = 0$, then $f(a)$ is called *stationary value.*

Every extreme value is stationary but every stationary value of the function need not to be extreme value

Example. Find the extreme value of

$$f(x) = x^5 - 5x^4 + 5x^3 - 1$$

Solution

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

and $\quad f'(0) = 0$

Hence, $f(0)$ is stationary value but $f(0)$ is not an extreme value because $f''(0) = 0$ but $f'''(0) \neq 0$.

Greatest Value

The greatest value of the function in an interval (a, b) is either a maximum value in an interval or end value (i.e. at $x = a$ or $x = b$) of $f(x)$ whichever is greater.

Least Values

The least value of $f(x)$ in an interval (a, b) is either a minimum value of $f(x)$ at a point inside the interval or an end value (i.e. at $x = a$ or $x = b$) of $f(x)$ whichever is less.

Example. Find greatest and least value of the function $2x^2 - 9x - 1$, in the internal [2, 5]

Solution

Let $\quad f(x) = 2x^2 - 9x - 1$

$\Rightarrow \quad f'(x) = 4x - 9$

At $x = \frac{9}{4} \quad f'(x) = 0$

$\therefore \quad f''(x) = 4 > 0$

$\therefore \quad f\left(\frac{9}{4}\right) = -\frac{89}{8}$ is a minimum value.

Also, $\quad f(2) = -11, f(5) = 4$

$\therefore$ Greatest value = 4, and least value = $-\frac{89}{8}$

SOME STANDARD INTEGRATIONS

1. $\int x^n = \frac{x^{n+1}}{n+1}, \ (n \neq -1)$
2. $\int \frac{1}{x} dx = \log x$
3. $\int e^x dx = e^x$
4. $\int a^x dx = \frac{a^x}{\log a}$

5. $\int \sin x\, dx = -\cos x$
6. $\int \cos x\, dx = \sin x$
7. $\int \sec^2 x\, dx = \tan x$
8. $\int \text{cosec}^2 dx = -\cot x$
9. $\int \sec x \tan x\, dx = \sec x$
10. $\int \text{cosec}\, x \cot x\, dx = -\text{cosec}\, x$
11. $\int \tan x\, dx = \log \sec x = -\log \cos x$
12. $\int \cot x\, dx = \log \sin x$
13. $\int \sec x\, dx = \log(\sec x + \tan x)$
14. $\int \text{cosec}\, x\, dx = -\log(\text{cosec}\, x + \cot x) = \log \tan \frac{x}{2}$
15. $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x \text{ or } -\cos^{-1} x$
16. $\int \frac{1}{\sqrt{1+x^2}} dx = \sin h^{-1} x$
17. $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$
18. $\int \frac{1}{x\sqrt{x^2 - 1}} dx = \sec^{-1} x = \cos^{-1} \frac{1}{x}$
19. $\int \sin hx\, dx = \cos hx$
20. $\int \cos hx\, dx = \sin hx$
21. $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log\left(x + \sqrt{x^2 - a^2}\right) = \cos h^{-1} \frac{x}{a}$
22. $\int \frac{1}{\sqrt{a^2 + x^2}} dx = \log\left(x + \sqrt{a^2 + x^2}\right)$
23. $\int \frac{1}{\sqrt{a^2 + x^2}} dx = \sin h^{-1}\left(\frac{x}{a}\right)$
24. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a} = \frac{1}{a} \cos$
25. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$
26. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log\left(\frac{x-a}{x+a}\right)$ if $x > a$

27. $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log\left(\frac{a+x}{a-x}\right)$ if $x < a$

28. $\int u \cdot v\, dx = u \int v\, dx - \int\left[\left(\frac{du}{dx}\right)\left(\int v dx\right)\right] dx$

where u and v are the function of x. u is a first function and v as a second function.

29. $\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + c$

30. $\int \frac{f'(x)}{f(x)} dx = \log f(x) + C$

Note : I L A T E
- I → Inverse
- L → Logarithmic function
- A → Algebraic
- T → Trigonometric
- E → Exponential

Example. Calculate $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$

Solution

Let $\tan^{-1} x = y \Rightarrow \frac{1}{1+x^2} dx = dy$

then $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx = \int e^y dy = e^y + C = e^{\tan^{-1}\frac{1}{x}} + C$

Example. Calculate $\int 1 \cdot \sin^{-1} x\, dx$

Solution

Take $u = \sin^{-1} x$ and $v = 1$

Then $\int \sin^{-1} x dx = \sin^{-1} x \int dx - \int\left[\frac{d(\sin^{-1} x)}{dx} \int dx\right] dx$

$= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}\, x\, dx$

$= x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$

$= x \sin^{-1} x + \sqrt{1-x^2} + C$

Example. Calculate $\int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx$

Solution

Let $y = e^x - e^{-x} \Rightarrow dy = (e^x + e^{-x})\, dx$

$\therefore \int \frac{(e^x + e^{-x})}{e^x - e^{-x}} = \int \frac{dy}{y} = \log y + C = \log (e^x + e^{-x}) + C$

DEFINITE INTEGRAL

If a function is defined in the interval $[a, b]$, then definite integral of $f(x)$ is

$$\int_a^b f(x)\, dx = [F(x)]_a^b = F(b) - F(a)$$

where $F(x)$ is an integral of $f(x)$

$f(b)$ is the value of $f(x)$ at $x = b$, and

$f(a)$ is the value of $f(x)$ at $x = a$.

a and b are called *lower upper limit respectively.*

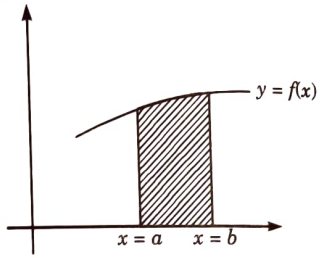
Geometrically, definite integral represents the area bounded by curve $y = f(x)$, x-axis and the lines $x = a$ and $x = b$.

Properties of Definite Integrals

Let $\int f(x)\, dx = F(x)$. Then

1. $\int_a^b f(x)\, dx = \int_a^b f(y)\, dy$

i.e. in definite integral the variable of integration is dummy.



2. $\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx$

3. $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx$

where c is a point lies between the interval $[a, b]$. and $f(x)$ integrable over $[a, c]$ and $[c, b]$.

4. $\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$

5. $\int_{-a}^a f(x)\, dx = \int_0^a f(x)\, dx + \int_0^a f(-x)\, dx$

If function is even, i.e. $f(-x) = f(x)$, then

$$\int_{-a}^a f(x)\, dx = 2\int_0^a f(x)\, dx$$

If function is odd, i.e. $f(-x) = -f(x)$, then

$$\int_{-a}^a f(x)\, dx = 0$$

6. $\int_0^{2a} f(x)\, dx = \int_0^a f(x)\, dx + \int_0^a f(2a-x)\, dx$

If $f(x) = f(2a - x)$, then

$$\int_0^{2a} f(x)\, dx = 2\int_0^a f(x)\, dx$$

and if $f(x) = -f(2a - x)$, then

$$\int_0^{2a} f(x)\, dx = 0$$

Definite Integral as Limit of sum

If a function $f(x)$ is either increasing or decreasing on $[a, b]$, then

$$\int_a^b f(x)\, dx = \underset{h\to 0}{\text{Lt}}\, h[f(a) + f(a+h) + \ldots + f(a+(n-1)h]$$

$$= \underset{h\to 0}{\text{Lt}}\, h\,[f(a+h) + f(a+2h) + \ldots + f(a+nh)]$$

where, $h = \frac{b-a}{n}$

Example. From definition of a definite integral as the limit of a sum, evaluate $\int_1^2 (x + x^2)\, dx$.

Solution

$$\int_1^2 (x + x^2)\, dx$$

$$= \underset{h\to 0}{\text{Lt}} \sum_{i=1}^n h\,[(1 + ih) + (1 + ih)^2 \ldots\ldots] \text{ where } \left(h = \frac{1}{n}\right)$$

$$= \underset{h\to 0}{\text{Lt}}\, h \sum_{i=1}^n [1 + ih + 1 + 2ih + i^2 h^2]$$

$$= \underset{h\to 0}{\text{Lt}}\, h \sum_{i=1}^n (2 + 3hi) + i^2 h^2$$

$$= \underset{h\to 0}{\text{Lt}} \left[2h \sum_{i=1}^n 1 + 3h^2 \sum_{i=1}^n i + h^3 \sum_{i=1}^n i^2\right]$$

$$= \underset{h\to 0}{\text{Lt}} \left[2nh + \frac{3}{2}h^2 n(n+1) + \frac{1}{6}h^3 n(n+1)(2n+1)\right]$$

$$= \underset{h\to 0}{\text{Lt}} \left[2 + \frac{3}{2}(1+h) + \frac{1}{6}(1+h)(2+h)\right] \quad \ldots\text{since } nh = 1$$

$$= 2 + \frac{3}{2} + \frac{1}{3} = \mathbf{\frac{23}{6}}$$

Improper Integral

Definite intgral $\int_0^a f(x) dx$ is *called improper integral* if

(i) range of integration is infinite and integrand is bounded.

(ii) range of integration is definite and integrand is unbounded.

(iii) neither range of integration is finite nor integrand is bounded over it.

MULTIPLE INTEGRALS

Let a single-valued and bounded function $f(x,y)$ of two independent variables x,y be defined in a closed region R of the xy-plane. Divide the region R into sub-regions by drawing lines parallel to co-ordinate axes. Number the rectangles which lie entirely inside the region R from 1 to n.

Let (x_r, y_r) be any point inside the rth rectangle whose area is δA_r

$$f(x_1,y_1)\,\delta A_1 + f(x_2,y_2)\,\delta A_2 + \ldots\ldots f(x_n,y_n)\,\delta A_n$$

$$= \sum_{r=1}^n f(x_r,y_r)\delta A_r \quad \ldots\ldots (i)$$

Let number of these sub-regions increases indefinitely such that the largest linear dimension (i.e.diagonal) of δA_r approaches zero. The limit of the sum(i), if it exists, irrespective of the mode of sub-division, is called *double integral of* $f(x,y)$ over the region R and is denoted by $\iint_R f(x,y)\, dA$.

In other words, $\underset{\delta A_r \to 0}{\overset{Lt}{n\to\infty}} \sum_{r=1}^n f(x_r, y_r)\,\delta A_r = \iint_R f(x,y)\, dA$

which is also expressed as $\iint_R f(x,y)\, dxdy$

or $\iint_R f(x,y)\, dy\, dx$

Let the region R be bounded by the curves

$x = x_1$ and $y = y_2$

$x = x_2$ and $y = y_2$

Properties of a Double Integral

(i) When x_1, x_2 are functions of y and y_1, y_2 are constants.

Let AB and CD be the curves $x_1 = \phi_1(y)$ and $x_2 = \phi_2(y)$. Then

$$\iint_R f(x,y)\, dxdy = \int_{y_1}^{y_2} \int_{x_1=\phi_1(y)}^{x_2=\phi_2(y)} f(x,y)\, dx\, dy$$

(ii) When y_1, y_2 are functions of x and x_1, x_2 are constants.

Let AB and CD be the curves

$y_1 = \phi_1(x)$ and $y_2 = \phi_2(x)$. Then

$$\iint_R f(x,y)\, dx\, dy = \int_{x_1}^{x_2} \int_{y_1=\phi_1(x)}^{y_2=\phi_2(x)} f(x,y)\, dy\, dx$$

(iii) When x_1, x_2, y_1, y_2 are constants.
Here the region of integration R is the rectangle. Therefore

$$\iint_R f(x,y)\,dx\,dy = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dx\,dy$$

$$= \int_{x_1}^{x_2}\int_{y_1}^{y_2} f(x,y)\,dy\,dx$$

(i) If region R is partitioned into two parts, say R_1 and R_2, then

$$\iint_R f(x,y)\,dx\,dy = \iint_{R1} f(x,y)\,dx\,dy + \iint_{R2} f(x,y)\,dx\,dy$$

(ii) Double integral of the algebraic sum of a fixed number of functions is equal to algebraic sum of the double integrals taken far each term. Thus

$$\iint_R [f_1(x,y) + f_2(x,y) + f_3(x,y) + \ldots]\,dx\,dy = \iint_R f_1(x,y)\,dx\,dy + \iint_R f_2(x,y)\,dx\,dy + \iint_R f_3(x,y)\,dx\,dy + \ldots$$

(iii) A constant factor may be taken outside the integral sign. Thus

$$\iint_R K\, f(x,y)\,dx\,dy = K \iint_R f(x,y)\,dx\,dy$$

where K is constant

Evaluation of Double Integrals

During evaluating double intergrals, first integrate with respect to the variable limits, treating other varibles as constant, and then integrate with respect to the variable with constant limit.

CHANGE OF ORDER OF INTEGRATION

In a double integral, if limits of integration are constant, then order of integration is immaterial, provided limits of integration are changed accordingly. Thus

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx$$

But if limits of integration are variable, a change in the order of integration changes in the limits of integration.

TRIPLE INTEGRALS

Consider a function $f(x,y,z)$ which is continuous at every point of the 3-dimensional finite region V.
Divide V into n elementary volumes $\delta V_1, \delta V_2, \ldots \delta V_n$.
Let (x_r, y_r, z_r) be any point within rth sub-division δV_r.

Consider the sum $\sum_{r=1}^{\infty} f(x_r, y_r, z_r)\,\delta V_r$

The limit of this sum, if it exists, as $n \to \infty$ and $\delta V_r \to 0$ is called *triple integral* of $f(x,y,z)$ over the region V and is denoted by $\iiint f(x,y,z)\,dV$

It can also be expressed as the repeated integral

$$\int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} f(x,y,z)\,dx\,dy\,dz.$$

FOURIER SERIES

If numbers $a_0, a_1, \ldots, a_n, \ldots, b_1, b_2, \ldots, b_n, \ldots$ are derived from a function f by means of Euler-Fourier formulas

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \qquad (n = 0,1,2,\ldots)$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \qquad (n = 1,2,\ldots)$$

then series $a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$ is called *Fourier series.*

Dirichlet's Condition

Assumptions for expansion in a Fourier's series :

(i) Given function $f(x)$ is assumed to be defined and single valued in the given range $(-l, l)$.

(ii) $f(x)$ is periodic outside $(-l, l)$ with period $2l$.

(iii) Series $a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$ is uniformly convergent, so that term by term integration of the series is possible.

Even and Odd function of x (sine series and cosine series)

Case I: *When $f(x)$ is an even function*

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

where, $$a_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$$b_n = 0$$

Case II: *When $f(x)$ is an odd function*

$$f(x) = \sum_{n=1}^{\infty}(b_n \sin nx)$$

where, $$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

$$a_0 = 0$$

$$a_n = 0$$

Half Range Expansions

A series which contains only sine terms or cosine terms is called *half range Fourier sine or cosine series* respectively.
In this case, function $f(x)$ is generally defined in the interval $(0, l)$.

Now if $f(x)$ is an even function, then Fourier cosine series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{nx\pi}{l}$$

where $$a_0 = \frac{1}{l}\int_0^l f(x)\,dx;$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$

If $f(x)$ is an odd function of x, then Fourier sine series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{nx\pi}{l}$$

where, $a_0 = o$ $$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

Fouriers series in the interval (a, b)

If function $f(x)$ satisfies Dirichlet's conditions in the interval (a, b), then

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{2n\pi x}{b-a} + b_n \sin\frac{2n\pi x}{b-a}\right)$$

where, $$a_0 = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$a_n = \frac{2}{b-a}\int_a^b f(x)\cos\frac{2n\pi x}{b-a}\,dx$$

$$b_n = \frac{2}{b-a}\int_a^b f(x)\sin\frac{2n\pi x}{b-a}\,dx$$

CONVERGENCE SERIES

SEQUENCE

A set of numbers $u_1, u_2, \ldots u_n, \ldots$ formed according to some definite rule is called a *sequence*. A sequence is generally denoted by $\{u_n\}$.

Finite Sequence

A sequence is called a *finite sequence* if it has a finite number of terms.

Infinite Sequence

If sequene has infinite number of terms, then it is called an *infinite sequence.*

Positive Term Sequence

A series $u_1 + u_2 + u_3 \ldots$ is called *positive term series*, if $u_n > 0\ \forall\ n$.

SERIES

Sum of the corresponding terms of the sequence $u_1, u_2, u_3, \ldots$, *i.e.*, $u_1 + u_2 + u_3 \ldots$ is called *series.*
A series is called a *finite* or *infinite* according as the corresponding sequence is *finite* or *infinite.*
Infinites series $u_1 + u_2 + u_3 + \ldots + u_n + \ldots$ is denoted by $\sum u_n$ and sum of the first n terms series is denoted by S_n.

$$\therefore \quad S_n = u_1 + u_2 + u_3 + \ldots + u_n$$

Alternating series

A series $u_1 - u_2 + u_3 \ldots$ is called *alternating series.* if $u_n > 0\ \forall\ n$.

Nature of Series

(1) Convergent
Infinite series $u_1 + u_2 + u_3 + \ldots$ is said to be convergent, if
$\lim_{n\to\infty} S_n = \lim_{n\to\infty}(u_1 + u_2 + \ldots + u_n)$ is finite.

(2) Divergent
Infinite series $u_1 + u_2 + u_3 + \ldots$ is said to be divergent, if
$\lim_{n\to\infty} S_n = \lim_{n\to\infty}(u_1 + u_2 + \ldots u_n) = +\infty$ or $-\infty$

(3) Oscillatory
Infinite series $u_1 + u_2 + u_3 + \ldots$ is said to be an oscillatory series, if
$\lim_{n\to\infty} S_n = \lim_{n\to\infty}(u_1 + u_2 + \ldots u_n)$
is not defined or oscillates between two limites.

Properties of Convergent and Divergent Series

(i) Nature of an infinite series remains unaltered by adding or removing a finite number of terms.

(ii) Nature of an inifinite series remains unaltered by multiplying or dividing the terms by some fixed non zero constant.

(iii) If $\sum u_n$ and $\sum v_n$ are two series such that $0 \le u_n \le v_n$ for all n and $\sum v_n$ is convergent, then $\sum u_n$ is also convergent.

(iv) If $\sum u_n$ and $\sum v_n$ are two series such that $u_n \ge v_n \ge 0$ for all n and $\sum v_n$ is divergent, then $\sum u_n$ is also divergent.

(v) If infinite series $\sum u_n$ and $\sum v_n$ both are convergent, then $\sum(u_n + v_n)$, $\sum(u_n - v_n)$ and $\sum(\lambda u_n + \lambda v_n)$ are also convergent.

(vi) If series $\sum u_n$ of positive terms is convergent, then $\lim_{n\to\infty} u_n = 0$ (only necessary not sufficient).

(vii) The series $\sum u_n$ is not convergent, $\lim_{n\to\infty} u \ne 0$

Absolutely Convergent Series

Series $\sum u_n$ is called *absolutely convergent* if series $\sum|u_n|$ is convergent.

Conditionally (or semi) Convergent series

A convergent series $\sum u_n$ is called conditionlly convergent series, if $\sum|u_n|$ is divergent.

Series Tests

1. Leibnitz's test (Alternating series test)

Infinite series $u_1 - u_2 + u_3 - u_4 + \ldots$ is convergent, if

(*i*) $u_1 > u_2 > u_3 > u_4 + \ldots\ldots$ and

(*ii*) $u_n \to 0$ as $n \to \infty$

2. Geomteric Series test

Geometric series $a + ar + ar^2 + \ldots$ is

(*i*) convergent, if $|r| < 1$

(*ii*) divergent, if $r \geq 1$

(*iii*) osillatory, when $r \leq -1$

3. Hyperharmonic or *p*-Series test

Inifinite series $\sum \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \ldots\ldots$ is

(*i*) convergent if $p > 1$

(*ii*) divergent if $p \leq 1$

4. Comparison test

If $\sum u_n$ and $\sum v_n$ are two series of positive terms such that $\lim_{n\to\infty} \frac{u_n}{v_n}$ is finite ($\neq 0$), then $\sum u_n$ and $\sum v_n$ behave alike.

Note : Test series $\sum v_n$ of the comparison test is called *auxiliary series* and

$$v_n = \frac{\text{highest power term in numerator of } u_n}{\text{highest power term in denominator in } u_n}$$

5. D' Alembert's ratio test

If $\sum u_n$ is a series of positive terms and, if

$$\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = L,$$

then the series is

(*i*) convergent, if $L > 1$

(*ii*) divergent if, $L < 1$

(*iii*) the test fails, if $L = 1$

Note:

I. If $\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = 1$ and $\frac{u_n}{u_{n+1}}$ contains e, then Logarithmic test is used.

II. If $\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = 1$ and $\frac{u_n}{u_{n+1}}$ does not contains e, then either Raabe's test of Gauss's test is used.

6. Raabe's test

$\sum u_n$ is a series of positive terms and, if $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = L$, then the series is

(*i*) convergent, if $L > 1$

(*ii*) divergent, if $L < 1$

(*iii*) The test fails, if $L = 1$

7. Gauss's test

If $\sum u_n$ is a series of positive terms and if ratio $\frac{u_n}{u_{n+1}}$ is expressed in the form

$$\frac{u_n}{u_{n+1}} = 1 + \frac{\lambda}{n} + \frac{\mu}{n^2} + \frac{\alpha}{n^3} + \ldots\ldots$$

then the series is

(*i*) convergent, if $\lambda > 1$

(*ii*) divergent if $\lambda \leq 1$

8. Logrithmic test

If $\sum u_n$ is series of positive terms and if $\lim_{n\to\infty}\left(n \log \frac{u_n}{u_{n+1}}\right) = L$, then the series is

(*i*) convergent, if $L > 1$

(*ii*) divergent, if $L < 1$

(*iii*) the test fails, if $L = 1$

9. De Morgan and Bertrand's test

If $\sum u_n$ is series of positive terms and if

$$\lim_{n\to\infty}\left[\left\{n\left(\log \frac{u_n}{u_{n+1}} - 1\right) - 1\right\}\log n\right] = L,$$

then the series is

(*i*) convergent, if $L > 1$

(*ii*) divergent if, $L < 1$

(*iii*) the test fails, if $L = 1$

10. Higher Logarithmic test

If $\sum u_n$ is series of positive terms and if

$$\lim_{n\to\infty}\left\{\left(n \log \frac{u_n}{u_{n+1}} - 1\right)\log n\right\} = L,$$

then the series is

(*i*) convergent, if $L > 1$

(*ii*) divergent, if $L < 1$

(*iii*) the test fails, if $L = 1$

11. Cauchy's Root test

If $\sum u_n$ is a series of positive terms and if $\lim_{n\to\infty} (u_n)^{1/n} = L$, then the series is

(*i*) convergent, if $L < 1$

(*ii*) divergent, if $L > 1$

(*iii*) the test fails, if $L = 1$

12. Cauchy's Condensation test

If $\sum u_n$ is series of positive terms and if $u_{n+1} < u_n$ then series $\sum u_n$ and $\sum a^n u_n$ ($a > 1$ any integer) behave alike.

13. Cauchy's Integral test

If $\sum u_n$ is series of positive terms and $f(x)$ be a continuous positive decreasing function of ($x > 0$), then $\sum u_n$ and $\int_1^\infty f(x)dx$ behave alike.

14. Kummer's test

Let a divergent series of positive terms $\sum (d_n)^{-1}$ is given, and let a terms of a series of positive terms $\sum u_n$ be such that

$$\lim_{n\to\infty}\left(d_n \frac{u_n}{u_{n+1}} - d_n + 1\right) = L,$$

then the series is

(*i*) convergent, if $L > 0$

(*ii*) divergent, if $L < 0$

(*iii*) the test fails, if $L = 1$

VECTOR

Vector is completely specified by its magnitude and direction.

Vector is represented by a directed line segment. Thus $\overline{PQ}$ represents a vector where magnitude is the length PQ and direction is form P (starting point) to Q (end point)

Unit Vector

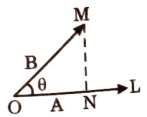
A vector of unit magnitude is called *unit vector.* Unit vector corresponding to vector A is written as $\hat{A}$.

Null Vector

A vector of zero magnitude, which can have no direction associated with it is called *zero (or null) vector* and is denoted by O-a thick zero.

PRODUCT OF TWO VECTORS

1. Scalar or Dot Product



Scalar or dot product of two vectors A and B is defined as scalar ab cos θ, where θ is angle between A and B.

Mathematically, $A.B = ab \cos\theta$

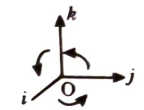
Properties

(*i*) For two vectors to be perpendicular, their scalar product should be zero, i.e.

$$A.B = ab\cos 90° \quad \ldots[\theta = 90°]$$

$$\Rightarrow \quad A.B = 0$$

(*ii*) Mutually perpendicular unit vectors i, j, k, are have relation



$$i.j = j.k = k.i = 0 \quad \ldots(i = 0)$$

and $\quad i^2 = j^2 = k^2 = 1$

2. Vector or Cross Product

Vector or cross product of two vectors A and B is defined as a vector ab sin θ, where 9 is angle between A and B.

Mathematically, $A \times B = ab \sin\theta N$

N is unit vector normal to the plane A and B.

Properties

(*i*) For two non zero vectors to be parallel, their vector product should be zero.

$$\theta = 0° \text{ or } 180°$$

$$\therefore \quad A \times B = \sin 0° \text{ or } \sin 180°$$

$$\Rightarrow \quad A \times B = 0$$

(*ii*) For the orthonormal vectors i, j and k, we have the relation

$$i \times i = j \times j = k \times k = 0,$$

$$i \times j = k, j \times i = -k$$

$$j \times k = i, \; k \times j = -i$$

$$k \times i = j, i \times k = -j$$

Differentiation of Vector

If a vector R varies continuously as a scalar variable t changes, then R is said to be function of t and is written as R = f(t)

In scalar calculus, we define derivative of vector function R = f(t) as

$$\lim_{\delta t \to 0} = F\frac{(t + dt) - F(t)}{dt}$$

and it is written as $\frac{dR}{dt}$ or $\frac{df}{dt}$ or F'(t)

- If F(t) has constant magnitude, then $F.\frac{df}{dt} = 0$

  For $\quad F(t).F(t) = [F(t)]^2 = \text{constant}$

  $$F.\frac{dF}{dt} = 0, \text{ i.e. } \frac{dF}{dt} \text{ is perpendicular to F}$$

- If F(t) has constant or fixed direction, then

  $$F \times \frac{dF}{dt} = 0$$

VECTOR CALCULUS

Consider a vector operator ∇ (del) defined by

$$\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$$

Gradient

Del (∇) applied to vector function ∇f is defined as the gradient of the scalar point function f and written as grad f. Thus gradient f defined as

$$\text{Grad } f = \nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

Grad f is a vector normal to surface f = constant, and has a magnitude equal to the rate of change of f along this normal.

DIRECTIONAL DERIVATIVE

If δr denotes the length PP' and N' is a unit vector in the direction PP', then limiting value of $\frac{\delta f}{\delta r}$ as $\delta r \to 0$ is called *directional derivative of f* at P along the direction PP'.

Since $\delta r = \frac{\delta N}{\cos\alpha} = \frac{\delta N}{N} \cdot N'$

$$\therefore \quad \frac{\partial f}{\partial r} = \lim_{\delta r \to 0}\left[N.N\frac{\partial f}{\partial n}\right] = \vec{N}\frac{\partial f}{\partial n}\vec{N} = N'\nabla f$$

Thus directional derivative of f in the direction of N' is the resolved part of ∇f in the direction of N'

Since $\nabla f N' = |\nabla f| \cos\alpha \le |\nabla f|$.

therefore ∇f gives the maximum rate of change of f.

Vector Identities

1. grad (U + V) = grad U + grad V,
   i.e. $\nabla(U+V) = \nabla U + \nabla V$
2. $\text{div}\left(\vec{A}+\vec{B}\right) = \text{div A} + \text{div B}$,
   i.e. $\nabla . A + \nabla . B$
3. $\text{Curl}\left(\vec{A}+\vec{B}\right) = \text{Curl A} + \text{Curl B}$,
   i.e. $\nabla \times (A+B) = \nabla \times A + \nabla \times B$
4. $\nabla . \left(U\vec{A}\right) = (\nabla U) . A + U(\nabla . A)$
5. $\nabla \times \left(U\vec{A}\right) = (\nabla U) \times A + U(\nabla \times A)$
6. $\nabla . \left(\vec{A} \times \vec{B}\right) = B(\nabla \times A) - A . (\nabla \times B)$
7. $\nabla \times \left(\vec{A} \times \vec{B}\right) = (B.\nabla)A - B(\nabla.A) - (A.\nabla)B + A(\nabla.B)$
8. $\nabla(A.B) = (B.\nabla)A + (A.\nabla)B + B \times (\nabla \times A) + A \times (\nabla \times B)$
9. $\nabla . (\nabla U) = \nabla^2 U$
10. $\nabla \times (\nabla U) = 0$, i.e. curl of grad of U = 0
11. $\nabla (\nabla \times A) = 0$, i.e. div of the curl A = 0
12. $\nabla \times (\nabla \times A) = \nabla(\nabla . A) - \nabla^2 A$

Note : (i) $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$; $\nabla(f.g) = f\nabla g - g\nabla f$

(ii) $\text{div}\left(\vec{r} \times \vec{a}\right) = 0$, if $\vec{a}$ is a constant vector

(iii) $\text{curl}\left(\vec{r} \times \vec{a}\right) = -2\vec{a}$, if $\vec{a}$ is a constant vector

(iv) If div $f = 0$, then f is called **solenoidal.**

Example. Find the unit vector normal to the surface $xy^3z^2 = 4$ at the point (–1, –1, 2).

Solution

A vector normal to the surface is given by

$$\nabla(xy^3z^2) = i\frac{\partial}{\partial x}(xy^3z^2) + j\frac{\partial}{\partial y}(xy^3z^2) + \hat{k}\frac{\partial}{\partial z}(xy^3z^2)$$

$$= i(y^3z^2) + j(3xy^2z^2) + \hat{k}(2xy^3z)$$

$$= -4i - 12j + 4\hat{k} \text{ at } (-1,-1,2)$$

Hence desired unit normal to the surface

$$= \frac{-4i - 12j + 4k}{\sqrt{(-4)^2 + (-12)^3 + 4^2}} = -\frac{1}{\sqrt{11}}(i + 3j - k)$$

Example. Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point (2, –1, 1) in the direction of vector $i + 2j + 2k$.

Solution $\nabla f = y^2 i + (2xy + z^3)j + (3yz^2)\hat{k}$

$= i - 3j - 3k$ at (2, –1, 1)

Hence, directional derivative of f in the direction $i + 2j + 2k$

$$= (i - 3j - 3k) \cdot \frac{(i + 2j + 2k)}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{(1 - 6 - 6)}{3} = -\frac{11}{3}$$

Example. Prove that curl $r = 0$

Solution

$$\text{Curl } \vec{r} = \nabla \times \vec{r}$$

$$= \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right) \times \left(xi + yj + z\hat{k}\right)$$

$$= i \times i + j \times j + k \times k = 0 + 0 + 0 = 0$$

DIVERGENCE

The divergence of a continuously differentiable vector point function F is denoted by divergence F and is define by the equation

$$\text{div } F = \nabla F = i.\frac{\partial F}{\partial x} + j.\frac{\partial F}{\partial y} + k.\frac{\partial F}{\partial z}$$

If F = fi + ϕj + ψk, then

$$\text{div } F = \nabla.F = \left(i.\frac{\partial}{\partial x} + j.\frac{\partial}{\partial y} + k.\frac{\partial}{\partial z}\right)(fi + \phi j + \psi k)$$

$$\text{div } F = \left(i.\frac{\partial f}{\partial x} + j.\frac{\partial \phi}{\partial y} + k.\frac{\partial \psi}{\partial z}\right)$$

CURL

The curl of a continuously differentiable vector point function F is defined by the equation

$$\text{Curl } F = \nabla \times F = i \times \frac{\partial F}{\partial x} + j \times \frac{\partial F}{\partial y} + k \times \frac{\partial F}{\partial x}$$

If F = f i + ϕ j + ψ k, then $F = \nabla \times F$

$$= \left(i.\frac{\partial}{\partial x} + j.\frac{\partial}{\partial y} + k.\frac{\partial}{\partial z}\right) \times (fi + \phi j + \psi k)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & \phi & \psi \end{vmatrix}$$

$$= i\left(\frac{\partial \psi}{\partial y} - \frac{\partial \phi}{\partial z}\right) + j\left(\frac{\partial \psi}{\partial y} - \frac{\partial f}{\partial z}\right) + k\left(\frac{\partial \phi}{\partial x} - \frac{\partial f}{\partial y}\right)$$

Del (∇) Applied Twice to Point Function

- Div grad $f = \nabla . \nabla f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$
- Curl grad $f = \nabla \times \nabla f = 0$
- Div curl $F = \nabla . \nabla \times F = 0$
- Curl Curl F = grad div F – ∇^2 f
  i.e. $\nabla \times (\nabla \times F) = \nabla(\nabla . f) - \nabla^2 f$
- Grad div F = curl Curl F + ∇^2 F
  i.e. $\nabla(\nabla.F) = \nabla \times (\nabla \times F) + \nabla^2 F$

Del(∇) Applied to Product of Point Functions

- $\nabla(fg) = f\nabla g + g\nabla f$
- $\nabla.(fG) = \nabla f.G + f\nabla.G$
- $\nabla \times (fG) = \nabla f \times G + f\nabla \times G$
- $\nabla(F.G) = (F.\nabla)G + (G.\nabla)F + F \times (\nabla \times G) + G \times (\nabla \times F)$
- $\nabla.(F \times G) = G.(\nabla \times F) - F.(\nabla \times F)$
- $\nabla \times (F \times G) = F(\nabla.G) - G(\nabla.F) + G.\nabla F - F.\nabla G$

MCQ TYPE QUESTIONS

1. $\lim_{x\to\infty} x^n e^{-x}$ is equal to

(a) ∞ (b) 1 (c) $\lfloor n$ (d) 0

2. $\lim_{n\to 0} e^{\left(\frac{1}{n \log n}\right)}$ is

(a) 1 (b) 0
(c) ∞ (d) does not exist

3. If $|x|$ denotes the greatest integer not greater than 2, then $\lim_{x\to 2}[x]$ is

(a) 0 (b) 1
(c) 2 (d) does not exist

4. If two functions $f(x) = \begin{cases} x \sin\frac{1}{x} & , x \neq 0 \\ 0 & , x \neq 0 \end{cases}$ $g(x) = |x|$ are given, then at $x = 0$

(a) both f and g continuous
(b) f is continuous and g is discontinuous
(c) f is discontinuous and g is continuous
(d) both f and g are discontinuous

5. If $f(x) = |x|$, then $f(x)$ is

(a) discontinuous at $x = 0$
(b) continuous only at $x = 0$
(c) continuous at all values of x
(d) discontinuous at $x = 1$

6. The function $f(x) = \begin{cases} ax^2 + b & \text{if } x > 2 \\ 2 & \text{if } x = 2 \\ 2ax - b & \text{if } x < 2 \end{cases}$ is continuous at $x = 2$, if

(a) $a = 0, b = \frac{1}{2}$ (b) $a = \frac{1}{2}, b = 0$

(c) $a = -\frac{1}{2}, b = 0$ (d) $a = 0, b = -\frac{1}{2}$

7. If $f(x) = x - [x]$, then $f(x)$ is discontinuous

(a) $\forall x \in Z$
(b) $\forall x \in Q$
(c) $\forall x \in R$
(d) at very real number except integers

8. Maxima and Minima occur

(a) simultaneously (b) once
(c) alternately (d) rarely

9. If $x + y = k, x > 0, y > 0$, then xy is the maximum when

(a) $x = ky$ (b) $kx = y$
(c) $x = y$ (d) none of these

10. The triangle of maximum area inscribed in a circle of radius r is

(a) a right angled triangle with hypotenuse measuring $2r$
(b) an equilateral triangle
(c) an isosceles triangle of height r
(d) does not exist

11. Sum of the perimeters of a circle and a square is l. If the sum of area is least, then

(a) side of the square is double the radius of the circle
(b) side of the square is half of the radius of the circle
(c) side of the square is equal to radius of the circle
(d) none of these

12. Which of the following is false?

(a) $f(a)$ is an extreme value of $f(x)$ if $f'(a) = 0$
(b) If $f(a)$ is an extreme value of $f(x)$, then $f'(a) = 0$
(c) If $f'(a) = 0$, then f(a) is an extreme value of $f(x)$
(d) None of these

13. Let $f(x) = |x|$, then

(a) $f'(0) = 0$
(b) $f(x)$ is maximum at $x = 0$
(c) $f(x)$ is minimum at $x = 0$
(d) none of these

14. The greatest and the least value of $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 1$ in $[0, 2]$ are

(a) 0, 8 (b) 0, −8 (c) 1, 8 (d) 1, −8

15. The function $f(x) = x^5 - 5x^4 + 5x^3 - 1$ has

(a) one minima and two maxima
(b) two minima and one maxima
(c) two minima and two maxima
(d) one minima and one maxima

16. The maximum value of $\frac{\log x}{x}$ in $(0, \infty)$ is

(a) e (b) $\frac{1}{e}$
(c) 1 (d) none of these

17. Maximum volume of a cylinder in a cone with semi vertical angle 'α' and height 'b' is

(a) $\frac{4}{27}\pi b^3 \tan^2\alpha$ (b) $\frac{4}{27}\pi b^2 \tan\alpha$

(c) $\frac{4}{27}\pi b \tan^3\alpha$ (d) none of these

18. The maximum point on the curve $x = e^x y$ is

(a) $(1, e)$ (b) $(1, e^{-1})$
(c) $(e, 1)$ (d) $(e^{-1}, 1)$

19. The value of function $f(x) = x + \frac{1}{x}$ at the points of minimum and maximum are respectively

(a) −2 and 2 (b) 2 and −2
(c) −1 and 1 (d) 1 and −1

20. The series $1 - \frac{1}{2} + 1 - \frac{3}{4} + 1 - \frac{7}{8} + \ldots$ is

(a) convergent
(b) conditionally convergent
(c) absolutely convergent
(d) oscillatory

21. The series $1 + 2x + 3x^2 + 4x^3 + \ldots$ where $0 < x < 1$, is

(a) convergent
(b) divergent
(c) oscillatory
(d) semi-convergent

22. For geometric series $1 - k + k^2 - k^3 + \ldots$ which of the following is false?

(a) The series is convergent if $k = -\frac{1}{3}$
(b) The series is divergent if $k = -3$
(c) The series is oscillatory if $k = -1$
(d) The series is divergent if $k = -2$

23. The series $\frac{1}{1.2.3} + \frac{3}{2.3.4} + \frac{5}{3.4.5} + \ldots$ is

(a) convergent (b) divergent
(c) oscillatory (d) none of these

24. The series $\frac{2}{1^p} + \frac{3}{2^p} + \frac{4}{3^p} + \ldots$ is

(a) convergent if $p \geq 2$ and divergent if $p < 2$
(b) convergent if $p > 2$ and divergent if $p \leq 2$
(c) convergent if $p \leq 2$ and divergent if $p > 2$
(d) convergent if $p < 2$ and divergent if $p \geq 2$

25. The series $\sum \frac{\sqrt{n}}{\sqrt{n^2+1}} x^n$ is

(a) convergent if $x < 1$ and divergent if $x \geq 1$
(b) convergent if $x \leq 1$ and divergent if $x > 1$
(c) convergent if $x > 1$ and divergent if $x \leq 1$
(d) convergent if $x \geq 1$ and divergent if $x < 1$

26. The series $\frac{x}{1} + \frac{1}{2}\cdot\frac{x^3}{3} + \frac{1.3}{2.4}\cdot\frac{x^5}{5} + \frac{1.3.5}{2.4.6}\cdot\frac{x^7}{7} + \ldots$ is

(a) convergent if $x^2 > 1$ and divergent is $x^2 \leq 1$
(b) convergent if $x^2 \geq 1$ and divergent if $x^2 < 1$
(c) convergent if $x^2 < 1$ and divergent if $x^2 \geq 1$
(d) convergent if $x^2 \leq 1$ and divergent is $x^2 > 1$

27. Sum of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$ is

(a) zero
(b) infinite
(c) log 2
(d) not defined as the series is not convergent

28. The series $x + \frac{2^2x^2}{\lfloor 2} + \frac{3^3x^3}{\lfloor 3} + \frac{4^4x^4}{\lfloor 4} + \ldots$ is convergent if

(a) $0 < x < \frac{1}{e}$ (b) $x > \frac{1}{e}$

(c) $\frac{2}{e} < x < \frac{3}{e}$ (d) $\frac{3}{e} < x < \frac{4}{e}$

29. The alternating series $u_1 - u_2 + u_3 - u_4 + \ldots (0 < u_n < n)$ is convergent if

(a) $u_n < u_{n+1}$ and $u_n \to 0$ as $n \to \infty$
(b) $u_n < u_{n+1}$ and $u_n \to \infty$ as $n \to \infty$
(c) $u_n > u_{n+1}$ and $u_n \to 0$ as $n \to \infty$
(d) $u_n > u_{n+1}$ or $u_n \to \infty$ as $n \to \infty$

30. The series $\frac{a}{b} + \frac{a(a+1)}{b(b+1)} + \frac{a(a+1)(a+2)}{b(b+1)(b+2)} + \ldots$ is

(a) convergent if $b - a < 1$ and divergent if $b - a \geq 1$
(b) convergent if $b \leq 1 + a$ and divergent if $b > 1 + a$
(c) convergent if $b > 1 + a$ and divergent if $b \leq 1 + a$
(d) convergent if $b \geq 1 + a$ and divergent if $b < 1 + a$

31. The series $x + x^{1+1/2} + x^{1+1/2+1/3} + x^{1+1/2+1/3+1/4} + \ldots$ is

(a) convergent if $x \geq \frac{1}{e}$
(b) divergent if $x \geq \frac{1}{e}$
(c) convergent if $x > \frac{1}{e}$
(d) divergent if $x \leq \frac{1}{e}$

32. The series $\frac{1}{n(\log n)^p}$ is divergent if

(a) $p > 1$ (b) $p \geq 1$ (c) $p < 1$ (d) $p \leq 1$

33. Let Σu_n be a series of positive terms and let $\Sigma \frac{1}{d^n}$ be a divergent series of positive terms such that $\lim_{n\to\infty}\left(d_n \frac{u_n}{u_{n+1}} - d_{n+1}\right) = k$, then the series is

(a) convergent if $k > 0$
(b) divergent if $k \geq 0$
(c) divergent if $k < 0$
(d) both (a) and (c)

34. The series $\sum_{n=3}^{\infty} \frac{1}{n \log n (\log\log n)^p}, p > 0$ is

(a) convergent if $p < 1$ and divergent if $p \geq 1$
(b) convergent if $p \leq 1$ and divergent if $p > 1$
(c) convergent if $p > 1$ and divergent if $p \leq 1$
(d) convergent if $p \geq 1$ and divergent if $p < 1$

35. The series $x + \frac{x(x+1)}{\underline{|2}} + \frac{x(x+1)(x+2)}{\underline{|3}} + \ldots$ is

(a) convergent if $x > 0$ (b) divergent if $x > 0$
(c) convergent if $x > 1$ (d) divergent if $-1 < x < 0$

36. Which of the following series is convergent?

(a) $\sum \sin\frac{1}{n}$

(b) $\sum \cos\frac{1}{n}$

(c) $\sum \frac{1}{n}\sin\frac{1}{n}$

(d) $\sum \sqrt{n^2+1} - \sqrt{n^2}$

37. The series $\sum \frac{\underline{|n}.2^n}{n^n}$ is

(a) convergent
(b) divergent
(c) conditionally convergent
(d) none of these

38. The series $\sum \frac{4.7\ldots(3n+1)}{1.2\ldots n} x^n$ is convergent, if

(a) $|x| < 1$ (b) $|x| < \frac{1}{3}$

(c) $|x| < \frac{1}{4}$ (d) $|x| < \frac{1}{2}$

39. The series $\frac{2}{1^2} - \frac{3}{2^2} + \frac{4}{3^2} - \frac{5}{4^2} + \ldots$ is

(a) conditionally convergent
(b) absolutely convergent
(c) divergent
(d) none of these

40. The sequence $\left(\frac{n}{e^n}\right)_{n=1}^{\infty}$ is

(a) divergent
(b) convergent
(c) monotonic increasing
(d) oscillating

41. Which of the following series is divergent?

(a) $1 + \frac{2.1}{3.4} + \frac{2.4.1}{3.5.6} + \frac{2.4.6.1}{3.5.7.8} + \ldots$

(b) $\sum \frac{n^3 + a}{2^n + a}$

(c) $\sum \frac{n^2}{\underline{|n}}$

(d) $\sum \frac{\sqrt{n}}{n + \sqrt{n}}$

42. The series $\sum_{n=1}^{\infty} \sin\frac{\pi}{n^p}$ is

(a) convergent for all value of p
(b) convergent for $p \leq 1$ and divergent for $p > 1$
(c) convergent for $p > 1$ and divergent for $p \leq 1$
(d) divergent for all values of p

43. The series $\sum \cos\left(\frac{1}{n}\right)$ is

(a) convergent (b) divergent
(c) oscillatory (d) none of these

44. The series $x \log x + x^2 \log 2x + \ldots + x^n \log nx + \ldots$ is convergent if

(a) $x < 1$
(b) $x \leq 1$
(c) $x \geq 1$
(d) none of these

45. If p and q are positive real numbers, then the series $\frac{2^p}{1^q} + \frac{3^p}{2^q} + \frac{4^p}{3^q} + \ldots$ *ad inf*, is converges for

(a) $p < q - 1$ (b) $p < q + 1$
(c) $p \geq q - 1$ (d) $p \geq q + 1$

NUMERICAL TYPE QUESTIONS

1. $\lim_{x\to 0} \frac{e^x - 1}{x}$ is equal to _____

2. If $\lim_{x\to 1} \frac{f(x) - 2}{f(x) + 2} = 0$, then $\lim_{x\to 1} f(x)$ is equal to _____

3. $\lim_{x\to 0} \frac{1 - \cos x}{2}$ is equal to _____

4. $\lim_{x\to 1} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$ is given by _____

5. $\lim_{x\to\infty} x \sin\frac{1}{x}$ is _____

6. $\lim_{x\to\infty}\left[\frac{1}{n^2} + \frac{2}{n^2} + \frac{3}{n^2} + \ldots + \frac{1}{n}\right]$ is _____

7. The minimum value of $|x^2 - 5x + 2|$ is _____

8. The maximum value of $\frac{1}{\sqrt{2}}(\sin x - \cos x)$ is _____

9. Maximum slope of the curve $-x^3 + 6x^2 + 2x + 1$ is _____

10. A rectangular box with square base in open at the top. The maximum volume of the box made from 300 m^3 wood is _____

11. A series $\sum_1^{\infty} u_n$ of positive terms is convergent, if $\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = l$ and l is greater than _____

ANSWERS

MCQ Type Questions

1. (d)	2. (b)	3. (d)	4. (a)	5. (c)	6. (b)	7. (d)	8. (c)	9. (d)	10. (b)
11. (a)	12. (d)	13. (c)	14. (d)	15. (d)	16. (b)	17. (a)	18. (b)	19. (d)	20. (a)
21. (a)	22. (c)	23. (a)	24. (b)	25. (a)	26. (c)	27. (c)	28. (a)	29. (a)	30. (c)
31. (b)	32. (d)	33. (d)	34. (c)	35. (b)	36. (c)	37. (a)	38. (b)	39. (a)	40. (b)
41. (d)	42. (c)	43. (b)	44. (a)	45. (a)					

Numerical Type Questions

1. 1	2. 2	3. 0	4. 1	5. 0	6. $\frac{1}{2}$	7. 0	8. 1
9. 14	10. 500	11. 1					

MCQ TYPE QUESTIONS

1. $\lim_{x\to\infty} x^n e^{-x} = \lim_{x\to\infty} \frac{x^n}{e^x}$ $\left[\text{form } \frac{\infty}{\infty}\right]$

By L' Hospital Rule Differentiating denominator and numerator n times, we get

$$\lim_{x\to\infty} \frac{x^n}{e^x} = \lim_{x\to\infty} \frac{\lfloor n}{e^x} = 0$$

2. Let $A = \lim_{n\to 0} e^{\frac{1}{n \log n}}$

$\therefore \log_e A = \lim_{n\to 0} \frac{1}{n \log n}$

$\Rightarrow \log_e A = \lim_{n\to 0} \frac{1}{\log n} \frac{n}{1}$ $\left[\text{form } \frac{\infty}{\infty}\right]$

$$= \lim_{n\to 0} \frac{-1/n^2}{1\ n} = \lim_{n\to 0} -\frac{1}{n}$$

$\log_e A = -\infty$

$A = e^{-\infty} = 0$

4. L. $\lim_{x\to 0} f(x) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} -h\sin\left(-\frac{1}{h}\right)$

$$= \lim_{h\to 0} + h\sin\frac{1}{h} = 0$$

R. $\lim_{x\to 0} f(x) = \lim_{h\to 0} f(0+h)$

$$= \lim_{h\to 0} -h\sin\left(\frac{1}{h}\right) = 0$$

Also $f(0) = 0$

Since L. $\lim_{x\to 0} f(x)$ = R. $\lim_{x\to 0} f(x)$ = f(0) = 0

Hence $f(x)$ is continuous at $x = 0$

Now, L. $\lim_{x\to 0} g(x) = \lim_{h\to 0} g(0-h)$

$$= \lim_{h\to 0} |-h| = \lim_{h\to 0} h = 0$$

R. $\lim_{x\to 0} g(x) = \lim_{h\to 0} g(0+h)$

$$= \lim_{h\to 0} |h| = \lim_{h\to 0} h = 0$$

$g(0) = |0| = 0$

$\therefore$ L. $\lim_{x\to 0} g(x)$ = R. $\lim_{x\to 0} g(x) = g(0) = 0$

Hence $g(x)$ is also continuous at $x = 0$

5. Let a∈ R (real numbers)

L. $\lim_{x\to a} f(x) = \lim_{h\to 0} |a-h| = \lim_{h\to 0} (a-h) = a$

R. $\lim_{x\to a} f(x) = \lim_{h\to 0} |a+h| = \lim_{h\to 0} a+h = a$

and $f(a) = |a| = a$

$\therefore \lim_{x\to a} |x| = a = f(a)$

Hence $\lim_{x\to a} |x|$ continuous for all values of x.

6. L. $\lim_{x\to 2} f(x) = \lim_{h\to 0} 2a(2-h) - b = 4a - b$

R. $\lim_{x\to 2} f(x) = \lim_{h\to 0} a(2+h)^2 + b = 4a + b$

The function will be continuous if

L. $\lim_{x\to 2} f(x)$ = R. $\lim_{x\to 2} f(x) = f(2)$

$\therefore \quad 4a - b = 2$

$4a + b = 2$

Solving we get, $a = \frac{1}{2}$, $b = 0$

7. Given function is, $f(x) = x - [x]$

Let a is a real number except integer and h be a very small real number, then

$[a-h] = b$ (an integer less than to a)

$[a+h] = b$ (an integer greater than to a)

Also $[a] = b$

$\therefore$ L. $\lim_{x\to 0} f(x) = \lim_{h\to 0} (a-h) - [a-h]$

$$= \lim_{h\to 0} (a-h) - b = a - b$$

Also $f(a) = (a) - [a] = a - b$

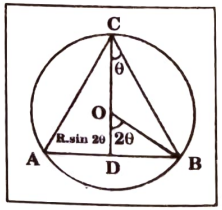
$\therefore$ L. $\lim_{x\to a} f(x)$ = R. $\lim_{x\to a} f(x) = f(a) = a - b$

$\therefore \quad f(x) = x - [x]$

is continuous all real except integers.

10. Let ABC be a triangle inscribed in the circle with centre O and radius r.

If area of this triangle is maximum, then vertex C should be at a maximum distance from the base AB i.e., CD must be perpendicular to AB. Hence ABC is an isosceles triangle.



If $\angle BCD = \theta$,

where D is the mid-point of AB, then

$\angle BOD = 2\theta$

$\therefore \quad AB = 2BD = 2r \sin 2\theta$

$CD = CO + OD = r + r\cos 2\theta$

If S be the area of the triangle ABC, then

$$S = \frac{1}{2} AB \times CD = \frac{1}{2} \times 2r \sin 2\theta\,(r + r\cos 2\theta)$$

$$\therefore \frac{ds}{d\theta} = r^2[\sin 2\theta\,(-2\sin 2\theta) + (1+\cos 2\theta)(2\cos 2\theta)]$$

For maximum and minimum,

$$\frac{dS}{d\theta} = 0$$

$\therefore \quad \cos 4\theta + \cos 2\theta = 0$

$\Rightarrow \quad 2\cos 3\theta \cos\theta = 0$

Hence either $\cos 3\theta = 0$, or $\cos\theta = 0$

If $\cos\theta = 0$, then $\theta = \frac{\pi}{2}$

If $\cos 3\theta = 0$, then $3\theta = \frac{\pi}{2}$

$\Rightarrow \quad \theta = \frac{\pi}{6}$

$\left(\frac{d^2S}{d\theta^2}\right)_{\theta=\pi/6}$ is negative.

Hence S is maximum for $\theta = \frac{1}{6}\pi$

$$\angle ACB = 2\theta = 2\left(\frac{\pi}{6}\right) = \frac{\pi}{3}$$

$= \angle ABC = \angle BAC$

Hence ΔABC is an equalateral triangle.

11. Let radius of circle is r and side of a sqaure is x.

$\therefore \quad$ Perimeter $= 2\pi r + 4x = l$ ...(i)

Let area be A

$\therefore \quad A = \pi r^2 + x^2$

$$= \pi r^2 + \left(\frac{l - 2\pi r}{4}\right)^2 \quad \text{[from equation (i)]}$$

$$\frac{dA}{dr} = 2\pi r + \frac{2}{16}(1 - 2\pi r).(-2\pi)$$

and $\frac{d^2A}{dr^2} = 2\pi + \frac{\pi^2}{2}$

For maximum and minimum,

$$\frac{dA}{dr} = 0$$

$\Rightarrow \quad 2\pi r - \frac{\pi}{4}(l - 2\pi r) = 0$ ...(ii)

$\Rightarrow \quad l = 8r + 2\pi r$

$\frac{d^2A}{dr^2} > 0$, hence area is least

From equations (i) and (ii),

$2\pi r + 4x = 8r + 2\pi r$

$\Rightarrow 4x = 8r \quad \Rightarrow \quad x = 2r$

Hence side of a square is double the radius of circle.

12. Since statements (a), (b), (c) are all correct, hence the statement (d) is false.

14. $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 1$

$f(0) = 1$

$f(2) = 2^4 - 8.2^3 + 22.2^2 - 24.2 + 1 = -7$

Now $f'(x) = 4x^3 - 24x^2 + 44x - 24$

For maximum and minimum,

$f'(x) = 0$

$\Rightarrow \quad 4x^3 - 24x^2 + 44x - 24 = 0$

$\Rightarrow \quad 4(x-1)(x-2)(x-3) = 0$

$\therefore \quad x = 1, 2, 3$

Since $x = 3$ does not lie in [0, 2]

Hence, consider only $x = 1$ and $x = 2$.

We have

$f(1) = 1^4 - 8.1^3 + 22.1^2 - 24.1 + 1 = -8$

Greatest of $f(x)$ = largest of $\{1, -7, -8\} = 1$

Least of $f(x)$ = smallest of $\{1, -7, -8\} = -8$

15. $f(x) = x^5 - 5x^4 + 5x^3 - 1$

$f'(x) = 5x^4 - 20x^3 + 15x^2$

$f''(x) = 20x^3 - 60x^2 + 30x$

$f'''(x) = 60x^2 - 120x + 30$

For maximum and minimum

$f'(x) = 0$

$\therefore \quad x^4 - 4x^3 + 3x^2 = 0$

$\Rightarrow \quad x^2(x-1)(x-3) = 0$

$\therefore \quad x = 0, 1, 3$

$f''(0) = 0$

$f'''(0) = 30 \neq 0$

$\therefore f(x)$ is neither maximum nor minimum at $x = 0$

$f''(1) = 20 - 60 + 30 = -10 < 0$

$\therefore f(x)$ is maximum at $x = 1$

$f''(3) = 540 - 540 + 90 = 90 > 0$

$\therefore f(x)$ is maximum at $x = 3$

Hence there is one maximum and one minimum for $f(x)$.

16. Given function is

$$f(x) = \frac{\log x}{x}$$

$$\therefore \quad f'(x) = \frac{x.\frac{1}{x} - \log x.1}{x^2} = \frac{1 - \log x}{x^2}$$

and $f''(x) = \dfrac{x^2.\left(-\frac{1}{x}\right) - (1 - \log x).2x}{x^4}$

$$= \frac{2\log x - 3}{x^3}$$

For maximum and minimum of $f(x)$,

$f'(x) = 0 \Rightarrow 1 - \log x = 0 \quad \Rightarrow \quad x = e$

$\therefore \quad f''(e) = \dfrac{2-3}{e^3} = -\dfrac{1}{e^3} < 0$

Hence $f(x)$ is maximum at $x = e$

$\therefore$ Maximum value, $f(e) = \dfrac{\log e}{e} = \dfrac{1}{e}$

NUMERICAL TYPE QUESTIONS

1. $\displaystyle\lim_{x\to 0}\frac{e^x-1}{x} = \lim_{x\to 0}\frac{\left(1+x+\frac{x^2}{\lfloor 2}+\frac{x^3}{\lfloor 3}+\ldots\right)-1}{x}$

$= \displaystyle\lim_{x\to 0}\left(1+\frac{x}{\lfloor 2}+\frac{x^2}{\lfloor 3}+\ldots\right) = 1$

By L' Hospital's Rule

$\displaystyle\lim_{x\to 0}\frac{e^x-1}{x}\left[\text{form}\ \frac{0}{0}\right] = \lim_{x\to 0}\frac{e^x}{1} = 1$

4. $\displaystyle\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}$

$= \displaystyle\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x} \times \left\{\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right\}$

$= \displaystyle\lim_{x\to 0}\frac{1+x-1+x}{x(\sqrt{1+x}+\sqrt{1+x})}$

$= \displaystyle\lim_{x\to 0}\frac{2x}{x(\sqrt{1+x}+\sqrt{1-x})}$

$= \dfrac{2}{\sqrt{1+0}+\sqrt{1-0}} = \dfrac{2}{1+1} = 1$

5. $\displaystyle\lim_{x\to\infty} x\sin\frac{1}{x}$ = Let $\dfrac{1}{x} = n$

So, $\quad x \to \infty,\ n \to 0$

$\displaystyle\lim_{n\to 0}\frac{\sin n}{n} = 1$

6. $\displaystyle\lim_{n\to\infty}\left[\frac{1}{n^2}+\frac{2}{n^2}+\frac{3}{n^2}+\ldots+\frac{1}{n}\right]$

$= \displaystyle\lim_{n\to\infty}\left[\frac{1+2+3+\ldots+n}{n^2}\right] = \lim_{n\to\infty}\frac{n(n+1)/2}{n^2}$

$= \displaystyle\lim_{n\to\infty}\frac{1}{2}\left(1+\frac{1}{n}\right) = \frac{1}{2}(1+0) = \frac{1}{2}$

7. Value of a mod function cannot be less than zero, therefore

$f(x) = |x^2 - 5x + 2|$ is zero.

8. Let $\quad f(x) = \dfrac{1}{\sqrt{2}}(\sin x - \cos x)$

$= \sin x \cos\dfrac{\pi}{4} - \cos x \sin\dfrac{\pi}{4} = \sin\left(x-\dfrac{\pi}{4}\right)$

Since maximum value of sin θ is 1, hence maximum value of $f(x) = 1$

9. Let $\quad f(x) = -x^3 + 6x^2 + 2x + 1$

Slope of the function is,

$f'(x) = -3x^2 + 12x + 2 = F(x)$ (say)

Now to find minimum or maximum of $F(x)$.

$\therefore \quad F'(x) = -6x + 12$

and $\quad F''(x) = -6$

For maxima and minima

$F'(x) = 0$

$\Rightarrow \quad x = 2$

$\therefore \quad F''(2) = -6 < 0$

Hence $F(x)$ (slope) is maximum at $x = 2$

$\therefore$ Maximum slope $= F(2) = -3(2)^2 + 12(2) + 2 = 14$

10. Volume is maximum at $x = 10$

At $x = 10$, $\quad y = \dfrac{300-100}{40} = 5$

$\therefore$ Maximum volume $= x^2y = (10)^2 \times 5 = 500$

CHAPTER **3**

Differential Equations

An equation which contains independent variable, dependent variables and their derivatives with respect to independent variables, is called *differential equation.*

I. ORDINARY DIFFERENTIAL EQUATION

A differential equation which involves only one independent variables is called *ordinary differential equation.*

e.g. $\dfrac{dy}{dx} + xy = 0$

and $x\left(\dfrac{dy}{dx}\right)^2 - y\dfrac{dy}{dx} + 1 = 0$

Order of Differential equation

It is the order of the highest derivative involving in the equation.

Degree of Differential equation

It is the degree of the highest order derivative involving in the equation, when the equation is free from radicals and fractional powers.

e.g. $\dfrac{dy}{dx} + xy = 0$

is equation of first order and first degree.

$\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} = 2$

is equation of second order and first degree.

$\dfrac{d^2y}{dx^2} = \left[1+\left(\dfrac{dy}{dx}\right)\right]^{1/3}$

is equation of second order and third degree.

Solution of a Differential equation

It is any relation between variables involved which satisfies the differential equation.

General or Complete Solution of a Differential equation

Solution of a differential equation in which number of arbitrary constant is equal to order of the differential equation is called *general solution.*

Variable Separables

$f_1(x)\,dx = f_2(y)\,dy$

It's solution is

$\int f_1(x) = \int f_2(y)\,dy + c$

Homogeneous equation

$\dfrac{dy}{dx} = \dfrac{\phi_1(x,y)}{\phi_2(x,y)}$

where, ϕ_1, ϕ_2 are of same degree

$\dfrac{dy}{dx} = f\left(\dfrac{y}{x}\right)$

Put $\dfrac{y}{x} = v$; $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$\therefore \dfrac{dv}{f(v)-v} = \dfrac{dx}{x}, \Rightarrow \int\dfrac{dv}{f(v)-v} = \log x + c$

Equation Reducible to Homogeneous Equation

$\dfrac{dy}{dx} = \dfrac{ax+by+c}{a'x+b'y+c'}$

$a : b \neq a' : b'$

Putting $x = x + h$, $y = y + k$

$\dfrac{dy}{dx} = \dfrac{ax+by+ah+bk+c}{a'x+b'y+a'h+b'k+c'}$

Choosing h, k such that

$ah + bk + c = a'h + b'k + c'$

$\dfrac{dy}{dx} = \dfrac{ax+by}{a'x+b'y} = \dfrac{y}{x} = v$

Special Case

When $\quad a : b = a' : b'$

then $\quad ax + by = v$

Linear Differential Equation

$\dfrac{dy}{dx} + py = Q$

where P and Q are functions of x only.

Integrating factor $\quad$ I.F. $= e^{\int pdx}$

$y.e^{\int pdx} = \int Q.e^{\int pdx}dx + C$

Equations Reducible to Linear Form

$$\frac{dy}{dx} + Py = Qy^n$$

Extended form of linear equation are Bernoulli's equation

$$\frac{1}{y^n}\frac{dy}{dx} + \frac{P}{y^{n-1}} = Q$$

Put $y^{1-n} = t$

$$\Rightarrow \quad dt = (1-n)\,y^{-n}dy \quad \Rightarrow \quad \frac{dy}{y^n} = \frac{1}{(1-n)}dt$$

$$\therefore \quad \frac{dt}{dx} + P(1-n)t = Q(1-n)$$

$$\frac{dy}{dx} + y\frac{d\phi}{dx} = \phi(x)\frac{d\phi}{dx}$$

$$\Rightarrow \quad \frac{\frac{dy}{dx} + y\frac{d\phi}{dx} = \phi(x)\frac{d\phi}{dx}}{\left(\frac{d\phi}{dx}\right)}$$

$$\Rightarrow \quad \frac{dy}{d\phi} + y = \phi(x)$$

$$ye^x = \int \phi(x)e^x dx + C$$

Exact Differential Equation

$$Mdx + Ndy = 0$$

$$\frac{\partial M}{\partial Y} = \frac{\partial N}{\partial X}$$

(*a*) Integrate M with respect to x regarding Y as a constant.

(*b*) Integrate N with respect to Y and then retain only that term which have not been already obtained by integrating M, *i.e.* in (*a*) above.

(*c*) Add the two expression obtained in (*a*) and (*b*) above and equate the result to an arbitrary constant which gives the required solution.

Differential Equations of First Order and First Degree

Type I : Variable Separation Form

The equation of this type can be put in the form,

$$f(x)\,dx + g(y)dy = 0$$

Now, $\int f(x)\,dx + \int g(y)\,dy = C$

Example. Solve $(e^y + 1)\,dx + xe^y\,dy = 0$

Solution

Given equation can be written as

$$e^y\,dx + xe^y\,dy + dx = 0 \quad \Rightarrow \quad d\left(xe^y\right) + dx = 0$$

Integrating $\int d\left(xe^y\right) + \int dx = \int 0 \quad \Rightarrow \quad xe^y + x = C$

Type II : Linear Differential Equations

A linear equations is called linear if the dependent variable and its derivative occuring in it are of the first degree and are not multiplied together.

Linear differential equation is of the form

$$\frac{dy}{dx} + Py = Q,$$

where P and Q are functions of x only.

Solution of the linear differential equations is

$$y \cdot e^{\int pdx} = \int\left(Q \cdot e^{\int pdx}\right)dx + C$$

where, $e^{\int pdx}$ is called I.F

Example. Solve $(x + 2y^3)\,dy = ydx$

Solution

Given equation can be written as

$$y\frac{dx}{dy} = x + 2y^3 \quad \Rightarrow \quad \frac{dx}{dy} - \frac{1}{y}x = 2y^2$$

Here, $P = -\frac{1}{y}$ and $Q = 2y^2$

$$\therefore \quad \text{I.F.} = e^{\int -\frac{1}{y}dy} = e^{-\log y} = e^{\log(1/y)} = \frac{1}{y}$$

Solution of the equation is,

$$x(\text{IF}) = \int(Q \cdot \text{IF} \cdot dy)$$

$$\therefore \quad x\left(\frac{1}{y}\right) = \int 2y^2\frac{1}{y}dy$$

$$\Rightarrow \frac{x}{y} = y^2 + C \quad \Rightarrow \quad x = y\,(C + y^2)$$

Type III : Homogeneous Differential Equations

This equation is of the form, $\frac{dy}{dx} = \frac{f(x,y)}{\phi(x,y)}$

where $f(x, y)$ and $\phi(x, y)$ are homogeneous functions of same degree.

This can be solved by putting $y = vx$

$$\therefore \quad \frac{dy}{dx} = v + x\frac{dy}{dx}$$

Substitute the value of y and then solve by variable-separable method.

Example. Solve $x^2dy + y\,(x + y)\,dx = 0$

Solution

Here $\frac{dy}{dx} = -\frac{y(x+y)}{x^2}$

Let $y = vx,$

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Now $v + x\frac{dv}{dx} = -\frac{x(vx) + v^2x^2}{x^2} = -(v^2 + v)$

$$\Rightarrow \quad x\frac{dv}{dx} = -(v^2 + 2v)$$

Integrating $\int\frac{dv}{v^2+2v} = -\int\frac{dx}{x}$

$$\Rightarrow \quad \int\frac{dv}{v(v+2)} = -\int\frac{dx}{x}$$

$$\Rightarrow \quad \int\frac{1}{2}\left(\frac{1}{v} - \frac{1}{v+2}\right)dv = -\int\frac{dx}{x}$$

$$\Rightarrow \quad \frac{1}{2}\ln v - \frac{1}{2}\ln(v+2) = -\ln x + C$$

$$\Rightarrow \quad \frac{1}{2}\log\frac{v}{v+2} = -\log x + C$$

$$\Rightarrow \quad \frac{1}{2}\log\frac{(y/x)}{(y/x)+2} = -\log x + C$$

$$\Rightarrow \quad \log\frac{y}{2x+y} = \ln\frac{1}{x^2} + C$$

$$\Rightarrow \quad \frac{y}{2x+y} = \frac{C'}{x^2}$$

$$\Rightarrow \quad x^2y = K'(2x + y)$$

Type IV : Exact Differential Equations

A differential equation is said to be exact if it can be derived from its primitive by direct differentiation without any further transformation such as elimination, etc.

e.g. $(x^2 - ay)\,dx + (y^2 - ax)\,dy = 0$

is exact in as much as it can be derived from its primitive

$x^3 - 3ayx + y^3 = C$ by direct differentiation.

Theorem : The necessary and sufficient condition for the differential equation,

$$M + N\frac{dy}{dx} = 0$$

$$\Rightarrow \quad Mdx + Ndy = 0$$

to be exact is that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

If equation $Mdx + Ndy = 0$ is exact, then it can be integrated as follows :

(*i*) Firstly integrate M with respect to x regarding y as a constant.

(*ii*) Then integrate with respect to y those of the terms in N which do not involve x.

(*iii*) Sum of the two expressions thus obtained equated to a constant is the required solution.

i.e. $\int Mdx + \int Ndy = C$

$\therefore$ y = constant terms contain $x = 0$

Example. Show that

$(x^4 - 2xy^2 + y^4)\,dx - (2x^2y - 4xy^2 + \sin y)\,dy = 0$

is exact and also solve it.

Solution

Here, $M = x^4 - 2xy^2 + y^4$

and $N = -(2x^2y - 4xy^3 + \sin y)$

$$\therefore \quad \frac{\partial M}{\partial y} = -4xy + 4y^3$$

and $\frac{\partial N}{\partial x} = -4xy + 4y^3 \quad \Rightarrow \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$

Thus the given equation is exact.

Now, $\int\left(x^4 - 2xy^2 + y^4\right)dx$

$$-\int\left(2x^2y - 4xy^2 + \sin y\right)dy = C$$

(since terms contains $x = 0$)

$$\Rightarrow \quad \frac{1}{5}x^5 - x^2y^2 + xy^4 + \cos y = C$$

Integrating Factors (I.F.)

Sometimes an equation which is not exact may become so on multiplication by some suitable function known as an *integrating factor.*

Consider an equation $xdy - ydx = 0$, which is not exact

It will become exact on multiplication by $\frac{1}{y^2}$. Then

$$\frac{x}{y^2}dy - \frac{1}{y}dx = 0$$

Hence solution is, $-\int\frac{1}{y}dx + \int\frac{x}{y^2}dy = C$

$$\Rightarrow \quad -\frac{x}{y} + 0 = C \quad \Rightarrow \quad \frac{x}{y} = C$$

Rule 1 : If equation $Mdx + Ndy = 0$ is homogeneous and $Mx + Ny \neq 0$, then

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

Rule 2 : If equation is of the form $f_1(x, y)\,ydx + f_2(x, y)\,xdy = 0$ and $Mx - Ny \neq 0$, then

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

Rule 3 : If equation $Mdx + Ndy = 0$

and $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x)$, is a function of x, then

$$\text{I.F.} = e^{\int f(x)dx}$$

Rule 4 : If equation $Mdx + Ndy = 0$ and

$$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = f(y), \text{ then}$$

$$\text{I.F.} = e^{\int f(y)dy}$$

Integrating Factors

- $d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$
- $d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$
- $d(xy) = xdy + ydx$
- $d\left(\frac{x^2}{y}\right) = \frac{2yxdx - x^2dy}{y^2}$
- $d\left(\frac{y^2}{x^2}\right) = \frac{2.y.x^2dy - 2xy^2dx}{x^4}$
- $d\left(\tan^{-1}\frac{x}{y}\right) = \frac{ydx - xdy}{x^2 + y^2}$
- $d\left(\tan^{-1}\left(\frac{y}{x}\right)\right) = \frac{xdy - ydx}{x^2 + y^2}$
- $d\left(\frac{1}{2}\log(x^2 + y^2)\right) = \frac{xdx + ydy}{x^2 + y^2}$
- $d\left(-\frac{1}{xy}\right) = \frac{xdy + ydx}{x^2y^2}$
- $d\log\left(\frac{x}{y}\right) = \frac{ydx - xdy}{xy}$
- $d\log\left(\frac{y}{x}\right) = \frac{xdy - ydx}{xy}$
- $d\left(\frac{e^x}{y}\right) = \frac{ye^xdx - e^xdy}{y^2}$

Example. Solve $(x^2 + y^2 + 2x)\,dx + 2ydy = 0$.

Solution

Here, $M = x^2 + y^2 + 2x$

$\Rightarrow \quad \frac{\partial M}{\partial y} = 2y$ and, $N = 2y$; $\frac{\partial N}{\partial x} = 0$

$\therefore \frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = 1$ which is function of x

Now $\text{I.F.} = e^{\int 1.\,dx} = e^x$

Multiplying by I.F, we get

$$e^x(x^2 + y^2 + 2x)\,dx + 2ye^x\,dy = 0$$

which is exact equation.

Hence the solution is,

$$\int_{y=\text{const}} e^x\left(x^2 + y^2 + 2x\right)dx + \int_{f(x)} 2ye^x dy = C$$

$\Rightarrow \quad e^x(x^2 + y^2) = C$

Example. Solve $\cos^2 x\left(\frac{dy}{dx}\right) + y = \tan x$ ...(i)

Solution

Given equation can be written as

$$\frac{dy}{dx} + \sec^2 xy = \sec^2 x \tan x$$

It is linear equation in y.

Here $P = \sec^2 x$

$\therefore \quad \text{I.F.} = e^{\int pdx} = e^{\int \sec^2 x dx} = e^{\tan x}$

Multiplying equation (ii) by I.F., we get

$$\left(\frac{dy}{dx} + y\sec^2 x\right)e^{\tan x} = \tan x \sec^2 x\, e^{\tan x}$$

Integrating, we get

$$y.\, e^{\tan x} = C + \int \tan x \sec^2 x.\, e^{\tan x}\, dx$$

where C is an arbitrary constant

Putting $t = \tan x$

$$dt = \sec^2 x dt$$

$\therefore \quad y\, e^{\tan x} = C + \int t\, e^t\, dt$

Integration by parts

$y\, e^{\tan x} = C + (te^t - e^t) \quad \Rightarrow \quad y\, e^{\tan x} = C + e^{\tan x}(\tan x - 1)$

Clairaut's Equation

Equation of the form $y = Px + f(P)$ is called *Clairaut's equation.*

Here $P = \frac{dy}{dx}$

Solution of the equation is,

$$y = cx + f(c)$$

Example. Solve $P = \tan(Px - y)$

Solution

$Px - y = \tan^{-1} P \quad \Rightarrow \quad y = Px - \tan^{-1} P$

Hence general solution of the equation is

$$y = cx - \tan^{-1} C$$

Singular Solution

Let $f(x, y, p) = 0$ be a given differential equation and its general solution is $\phi(x, y, c) = 0$.

Envelope of the family of curves $\phi(x, y, c) = 0$ is contained in the locus obtained by eliminating c between $\phi(x, y, c) = 0$ and $\frac{\partial}{\partial c}\phi(x, y, c) = 0$.

Example. Find singular solution of the differential equation,

$$P^3 + Px - y = 0$$

Solution

Given equation is, $y = Px + P^3$

Hence, general solution is,

$$y = cx + c^3$$

Let $\phi(x, y, c) = cx + c^3 - y = 0$ ...(i)

$\therefore \quad \frac{\partial}{\partial c}\phi(x, y, c) = x + 3c^2 = 0$ ...(ii)

Eliminating c between equations (i) and (ii), we get

$$y^2 = c^2(x + c^2)^2$$

$\Rightarrow \quad y^2 = -\frac{x}{3}\left(x - \frac{x}{3}\right)^2 \quad \Rightarrow \quad 27y^2 + 4x^3 = 0$

Linear Differential Equation with of n^{th} Order (Higher Order)

Linear equations are the equations in which the dependent variable and its derivatives appear only in the first degree and are not multiplied together.

$$\frac{d^n y}{dx^n} + P_1\frac{d^{n-1}}{dx^{n-1}} + P_2\frac{d^{n-2}}{dx^{n-2}} + \ldots + P_n y = X$$

where $P_1, P_2, \ldots P_n$ are functions of x only.

The operator $\frac{d}{dx}$ is denoted by D.

$\therefore \quad D^n y + P_1 D^{n-1} y + \ldots + P_n y = X$

$\Rightarrow \quad f(D)y = X$

where, $f(D) = D^n + P_1D^{n-1} + \ldots + P_n = 0$

Case I : *When A.E.has Distinct and real roots, i.e.*

$$m_1 \neq m_2 \neq m_3 \ldots$$

$$y = c_1e^{m_1x} + c_2e^{m_2x} + c_3e^{m_3x} + \ldots$$

Case II : *When A.E. has real and some equal roots.*

If $m_1 = m_2 \neq m_3 \neq m_4 \ldots$

then $y = (c_1 + c_2x)\,e^{m_1x} + c_3e^{m_3x} + \ldots$

If $m_1 = m_2 = m_3 \neq m_4 \neq m_5 \ldots$

then $y = (c_1 + c_2x + c_3x^2)\,e^{m_1x} + c_4e^{m_4x} + \ldots$

Case III : *When A. E. has complex roots,*

i.e. $m_1, m_2 = \alpha \pm i\beta$, $m_3, m_4 \ldots$

$y = (A\cos\beta x + B\sin\beta x)\,e^{\alpha x} + C_3e^{m_3x} + C_4e^{m_4x} + \ldots$

If complex roots are equal, then

$$y = [(A_1 + A_2x)\cos\beta x + (B_1 + B_2x)\sin\beta x]\,e^{\alpha x} + c_5e^{m_5x} + \ldots$$

Example. Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

Solution

A.E. is, $D^2 - 5D + 6 = 0$

$\therefore \quad D = 3, 2$

Hence solution is, $y = c_1e^{3x} + c_2e^{2x}$

PARTICULAR INTEGRAL

When the equation is, $f(D)\,y = X$

then general solution is, $y = (C.F.) + (P.I.)$

A particular integral is given by $\frac{1}{f(D)}X$

I. When $X = e^{ax}$, then

$$\text{P.I.} = \frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}, \text{ if } f(a) \neq 0$$

II. When $f(a) = 0$, then

$$\text{P.I.} = \frac{1}{f(D)}e^{ax}$$

$$= \frac{x}{f'(a)}e^{ax}, \text{ if } f'(a) \neq 0$$

III. If $f'(a) = 0$, then

$$\text{P.I.} = \frac{1}{f(D)}e^{ax} = \frac{x^2}{f''(a)}e^{ax} \text{ if } f''(a) \neq 0$$

Example. Find P.I. when $(D^2 + 4)\,y = \sin 2x + \cos 3x$

Solution

$$\text{P. I.} = \frac{1}{D^2 + 4}\sin 2x + \frac{1}{D^2 + 4}\cos 3x$$

$$= \frac{x}{2D}\sin 2x + \frac{1}{-3^2 + 4}\cos 3x = \frac{x}{2D^2}D\sin 2x - \frac{1}{5}\cos 3x$$

$$= \frac{2x}{-8}\cos 2x - \frac{1}{5}\cos 3x = -\frac{x}{4}\cos 2x - \frac{1}{5}\cos 3x$$

Example. Find general solution of the equation

$$(D^2 + D - 2)\,y = e^x$$

Solution

A.E. is, $D^2 + D - 2 = 0$

$D = -1, 2$

$\Rightarrow \quad C.F. = C_1e^{-x} + C_2e^{2x}$

Now $\text{P. I.} = \frac{1}{D^2 + D - 2}e^x = x\frac{1}{2D + 1}e^x$

Hence general solution is,

$$y = c_1 e^{-x} + c_2 e^{2x} + \frac{x}{3} e^{x}$$

IV. $\text{P.I.} = \dfrac{1}{F(D^2)} \sin ax$

$$= \frac{1}{F(-a^2)} \sin ax, \quad \text{if } F(-a^2) \neq 0$$

V. $\text{P.I.} = \dfrac{1}{F(D^2)} \cos ax$

$$= \frac{1}{F(-a^2)} \cos ax \quad \text{if } F(-a^2) \neq 0$$

VI. $\text{P.I.} = \dfrac{1}{F(D)} x^m$

Expansion is to be carried upto the term D^m, since $D^{m+1} x^m = 0$, $D^{m+2} x^m = 0$.

VII. When $X = e^{ax} v(x)$, then

$$\text{P.I.} = \frac{1}{f(D)} e^{ax} v(x) = e^{ax} \frac{1}{f(D+a)} v(x)$$

Example. Solve $(D^2 - 4D + 1) y = e^{2x} \sin 2x$

Solution

A.E. is $m^2 - 4m + 1 = 0$

$$m = 2 \pm \sqrt{3}$$

$$\text{C.F.} = c_1 e^{(2+\sqrt{3})x} + c_2 e^{(2-\sqrt{3})x}$$

$$= e^{2x} \left[c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x} \right]$$

Now, $\text{P.I.} = \dfrac{1}{D^2 - 4D + 1} e^{2x} \sin 2x$

$$= e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 1} \sin 2x$$

$$= e^{2x} \frac{1}{D^2 + 4D + 4 - 4D - 8 + 1} \sin 2x$$

$$= e^{2x} \frac{1}{D^2 - 3} \sin 2x = e^{2x} \frac{1}{-4 - 3} \sin 2x$$

$$= \frac{-1}{7} e^{2x} \sin 2x$$

Hence general solution is

$$y = \left(c_1 e^{\sqrt{3}x} + c_2 e^{-\sqrt{3}x} \right) e^{2x} - \frac{1}{7} e^{2x} \sin 2x$$

VIII. When $X = x\, v(x)$, then

$$\text{P.I.} = \frac{1}{f(D)} x v(x) = \left[x - \frac{f'(D)}{f(D)} \right] \frac{1}{f(D)} v(x)$$

Example. Find PI of $(D^2 - 1) y = x \sin x$

Solution

$$\text{P.I.} = \frac{1}{D^2 - 1} x \sin x$$

$$= \left[x - \frac{2D}{D^2 - 1} \right] \frac{1}{D^2 - 1} \sin x$$

$$= \left(x - \frac{2D}{D^2 - 1} \right) \frac{\sin x}{-2}$$

$$= -\frac{1}{2} \left[x \sin x - \frac{2D \sin x}{-2} \right]$$

$$= -\frac{1}{2} [x \sin x + \cos x]$$

IX. $Q = e^{ax}$ and $F(a) = 0$

$$\text{P.I.} = \frac{1}{\phi(a)} e^{ax} \cdot \frac{x^n}{n} = \frac{e^{ax}}{(D+a)}$$

X. To find P.I., when Q is of the form xV, where V is any function of x.

By Leibnitz's theorem,

$$\frac{1}{F(D)} (x.V) = x \frac{1}{F(D)} V - \frac{F'(D)}{\{F(D)\}^2}$$

XI. To find P.I. when Q is of the form x^m. V where V is any function of x.

(*i*) If $V = x^n$, then $x^m . x^n = x^{m+n}$

(*ii*) If $V = e^{ax}$, then $x^m v = x^m e^{ax}$ [evaluate as (X)]

(*iii*) If $V = \cos ax$, then $x^m v = x^m \cos ax$

$$\text{P.I.} = \frac{1}{F(D)} x^m \cos ax = \frac{1}{F(D)} (\text{Real part of } x^m e^{iax})$$

$$= \text{Real part of } \frac{1}{F(D)} x^m e^{iax}$$

If $V = \sin ax$, then

$$\text{P.I.} = \text{I.P. of } \frac{1}{F(D)} x^m e^{iax}$$

and calculate as above.

XII. $\dfrac{1}{D-a} Q = e^{ax} \int Q e^{-ax} dx$

XIII. $\dfrac{1}{(D^2 - a^2)} = \text{cosec } ax - \dfrac{1}{2ia} \left[\left(\dfrac{1}{D - ia} \right) - \left(\dfrac{1}{D + ia} \right) \right]$

Method of Variation of Parameters

Finding Particular Integral

Consider linear differential equation

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = X$$

Complementary function $= C_1 y_1 + C_2 y_2$

and Particular integral $= u y_1 + v y_2$

where $u = \int \dfrac{-y_2 X}{y_1 y_2' - y_1' y_2} dx$

and $v = \int \dfrac{y_1 X}{y_1 y_2' - y_1' y_2} dx$

Complete solution = Complementary function + Particular integral.

Method of Undetermined Coefficients to Find Particular Integral

Consider a linear differential equation

$$D^n y + p_1 D^{n-1} y + \ldots\ldots + p_n y = X$$

For particular integral, assume a trial solution containing unknown constants which are determined by substitution in the given equation. The trial solution to be assumed in each case, depends on the form of X. Thus

(1) When $X = 2e^{3x}$

Trial solution $= ae^{3x}$

(2) When $X = 3 \sin 2x$

Trial solution $= a_1 \sin 2x + a_2 \cos 2x$

(3) When $X = 2x^3$

Trial solution $= a_1 x^3 + a_2 x^2 + a_3 x + a_4$

However when $X = \tan x$ or $\sec x$, this method fails, since number of terms obtained by differentiating $X = \tan x$ or $\sec x$ is infinite.

This method holds as long as no term in the trial solution appears in the C.F.

If any term of the trial solution appears in the C.F., multiply this trial solution by the lowest positive integral power of x which is large enough so that none of the terms which are then present, appear in the C.F.

Example : Solve $\left(\dfrac{d^4 y}{dx^4} + m^4 \right) y = 0$

Solution

Given equation can be written as

$(D^4 + m^4) y = 0$

here, $D = \dfrac{d}{dx}$

Its auxiliary equation is

$M^4 + m^4 = 0$

$\Rightarrow (M^2 + m^2)^2 - 2M^2 m^2 = 0$

$\Rightarrow (M^2 + m^2)^2 - (\sqrt{2}Mm)^2 = 0$

$\Rightarrow (M^2 + m^2 + \sqrt{2}Mm)(M^2 + m^2 - \sqrt{2}Mm) = 0$

$$M = \frac{-\sqrt{2}m \pm \sqrt{2m^2 - 4m^2}}{2}, \frac{\sqrt{2}m \pm \sqrt{2m^2 - 4m^2}}{2}$$

$$= \frac{-\sqrt{2}m \pm \sqrt{2}mi}{2}, \frac{\sqrt{2}m \pm \sqrt{2}mi}{2}$$

$$= \frac{-m}{\sqrt{2}} \pm \frac{m}{\sqrt{2}} i, \frac{m}{\sqrt{2}} \pm \frac{m}{\sqrt{2}} i$$

Hence required solution,

$$e^{mx/\sqrt{2}} \{C_1 \cos mx/\sqrt{2}) + C_2 \sin (mx/\sqrt{2})\}$$

$$+ e^{-mx/\sqrt{2}} \{C_3 \cos mx/\sqrt{2}) + C_4 \sin (mx/\sqrt{2})\}$$

where C_1, C_2, C_3, C_4 are arbitrary constants.

Example : Solve $(D^2 - 2D + 5) y = e^{-x}$

Solution

Given equation is $(D^2 - 2D + 5) y = e^{-x}$

Auxiliary equation is $m^2 - 2m + 5 = 0$,

where roots are $m = -1 \pm 2i$

Complementary Function,

$$\text{C.F.} = e^{-x} (C_1 \cos 2x + C_2 \sin 2x)$$

where C_1 and C_2 are arbitrary constants.

and $\text{P.I.} = \dfrac{1}{D^2 - 2D + 5} e^{-x} = \dfrac{e^{-x}}{(-1)^2 - 2(-1) + 5}$

$$= \frac{e^{-x}}{1 + 2 + 5} = \frac{e^{-x}}{8}$$

Hence required solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$= e^{-x} (C_1 \cos 2x + C_2 \sin 2x) + \frac{e^{-x}}{8}$$

First Order Equations (Linear and Non-Linear)

Order of a differential equation means order of the highest differential coefficient which appears in it.

Linear Differential Equations

$$\frac{dy}{dx} + Py = Q$$

where P and Q are constant or functions of x alone.

This equation is called *linear differential* of the first order in y.

It's integrating factor is, I.F. $= e^{\int P dx}$

Multiplying both sides by I.F., we get

$$\left(\frac{dy}{dx} + Py\right)e^{\int P dx} = Q\,e^{\int P dx} \qquad ...(i)$$

Now, $$\frac{d}{dx}(ye^{\int P dx}) = y.\frac{d}{dx}\left[e^{\int P dx}\right] + e^{\int P dx}.\frac{dy}{dx}$$

$$= y.e^{\int P dx}.P + e^{\int P dx}.\frac{dy}{dx}$$

$$\frac{d}{dx}(ye^{\int P dx}) = \left(Py + \frac{dy}{dx}\right)e^{\int P dx}$$

From equation (i), $$\frac{d}{dx}\left[ye^{\int P dx}\right] = Q\,e^{\int P dx}$$

Integrating both sides with respect to x, we get

$$y.e^{\int P dx} = C + \int Q.e^{\int P dx} dx$$

where, C is a integration constant.

This equation is complete solution.

Cauchy's Homogeneous Linear Equation

$$x^n\frac{d^n y}{dx^n} + a_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + a_2x^{n-2}\frac{d^{n-2}y}{dx^{n-2}} + ... + a_n y = X(x)$$

where $a_1, a_2, ..., a_n$ are constants and X is a function of x.

This can be reduced to linear differential equation with constant co-efficients by the substitution

$$x = e^t \qquad \Rightarrow \qquad \log x = t$$

which can be solved by above methods.

Legendre's Homogeneous Linear Equation

$$(ax+b)^n\frac{d^n y}{dx^n} + a_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}}$$

$$+ a_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}} + ... + a_n y = X$$

where $a_1, a_2, ... a_n$ are constants and X is a function of x.

If we substitute $ax+b=t$, then above equation will reduce to the Cauchy's form and then a further substitution $t = e^z$ reduces it to the linear differential equation with constant co-efficients. A single substitution $ax+b=e^z$ will reduce the above equation to the linear differential equation with constant coefficients.

Example. Solve by the method of variation of parameters : $$\frac{d^2y}{dx^2} + y = \text{cosec } x.$$

Solution

Its A.E. is $D^2 + 1 = 0 \Rightarrow D = \pm i$

$\therefore$ C.F. $= C_1\cos x + C_2\sin x = C_1F_1 + C_2F_2$ (say)

where, $F_1 = \cos x$ and $F_2 = \sin x$

Now P.I. $= PF_1 + QF_2$

where, $$P = -\int\frac{F_2X}{F_1F_2' - F_2F_1'}dx$$

and, $$Q = \int\frac{F_1X}{F_1F_2' - F_2F_1'}dx$$

$F_1 = \cos x \Rightarrow F_1' = -\sin x$

and $F_2 = \sin x \Rightarrow F_2' = \cos x$

$\therefore$ $F_1F_2' - F_2F_1' = \cos^2 x - (\sin x)(-\sin x)$

$= \cos^2 x + \sin^2 x = 1$

$$\therefore \quad P = -\int\frac{\sin x\,\text{cosec } x}{1}dx = -x$$

Similarly, $$Q = \int\frac{F_1X}{F_1F_1' - F_2F_1'}dx$$

$$= -\int\frac{\cos x}{1}\text{cosec } x\,dx = \int\frac{\cos x}{\sin x}dx$$

$$= \int\cot x\,dx = \log\sin x.$$

$\therefore$ P.I. $= PF_1 + QF_2$

$= -x\cos x + \log(\sin x).\sin x$

$= -x\cos x + \sin x\log(\sin x)$

Hence complete solution is,

$y = C_1\cos x + C_2\sin x - x\cos x + \sin x.\log(\sin x)$.

Euler-Cauchy Equations

Homogeneous

If the ODE is of the form

$$ax^2y'' + bxy' + cy = 0, \qquad ...(1)$$

where a, b and c are constants; then (1) is called homogeneous Euler-Cauchy equation. Two linearly independent solutions (i.e. basis) depend on the quadratic equation

$$am^2 + (b-a)m + c = 0. \qquad ...(2)$$

Equation (2) is called *characteristic equation* for (1). The ODE (1) is singular at $x = 0$. Hence, we solve (1) for $x \neq 0$. We consider the case when $x > 0$.

Theorem

(i) If the roots of (2) are real and distinct, say m_1 and m_2, then two linearly independent (LI) solutions of (1) are x^{m1} and x^{m2}. Thus, the general solution to (1) is

$$y = C_1x^{m1} + C_2x^{m2}.$$

(ii) If the roots of (2) are real and equal, say $m_1 = m_2 = m$, then two LI solutions of (1) are x^m and $x^m\ln x$. Thus, the general solution to (1) is

$$y = (C_1 + C_2\ln x)x^m.$$

(iii) If the roots of (2) are complex conjugate, say $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, then two real LI solutions of (1) are $x^\alpha\cos(\beta\ln x)$ and $x^\alpha\sin(\beta\ln x)$. Thus, the general solution to (1) is

$$y = x^\alpha(C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x)).$$

Example. *Solve* $x^2y'' - xy' - 3y = 0$

Solution

The characteristic equation is $m^2 - 2m - 3 = 0$

$\Rightarrow m = -1, 3$. The general solution is $y = C_1/x + C_2x^3$

(ii) If $p(m) = 0$ has real equal roots $m_1 = m_2 = m$, then x^m is a solution of (1). To find the other solution, note that if m is repeated root, then $p(m) = p'(m) = 0$. This suggests differentiating (3) w.r.t. m. Since L consists of differentiation w.r.t. x only,

$$\frac{\partial}{\partial m}\left(L(x^m)\right) = L\left(\frac{\partial}{\partial m}x^m\right)$$

$$= L(x^m\ln x).$$

Now $L(x^m\ln x) = (p'(m) + p(m)\ln x)x^m$

where $'$ represents the derivative. Since, m is a repeated root, the RHS is zero. Thus, $x^m\ln x$ is also a solution to (1) and it is independent of x^m. Hence, the general solution to (1) is

$$y = (C_1 + C_2\ln x)x^m.$$

Example. *Solve* $x^2y'' - 3xy' + 4y = 0$

Solution

The characteristic equation is $m^2 - 4m + 4 = 0$

$\Rightarrow m = 2, 2$.

The general solution is $y = (C_1 + C_2\ln x)x^2$.

(iii) If $p(m) = 0$ has complex conjugate roots, say $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, then two LI solutions are

$$Y_1 = x^{(\alpha+i\beta)} = x^\alpha e^{i\beta\ln x},$$

and $$Y_2 = x^\alpha e^{-i\beta\ln x}.$$

But these are complex valued. Note that if Y_1, Y_2 are LI, then so are $y_1 = (Y_1 + Y_2)/2$ and $y_2 = (Y_1 - Y_2)/2i$. Hence, two real LI solutions of (1) are $y_1 = x^\alpha\cos(\beta\ln x)$ and $y_2 = x^\alpha\sin(\beta\ln x)$. Thus, the general solution to (1) is

$$y = x^\alpha(C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x).$$

Nonhomogeneous

If the ODE is of the form

$$ax^2y'' + bxy' + cy = r(x), \qquad ...(4)$$

where a, b and c are constants; then (4) is called nonhomogeneous Euler-Cauchy equation. We can use the method of variation of parameters as follows. First divide (4) by ax^2 so that the coefficient of y'' becomes unity:

$$y'' + \frac{b}{ax}y' + \frac{c}{ax^2}y = r(x) \qquad ...(5)$$

where $r(x) = \tilde{r}(x)/ax^2$. Now we already know two LI solutions y_1, y_2 of the homogeneous part. Hence, the particular solution to (4) is given by

$$y_p(x) = -y_1(x)\int\frac{y_2(x)r(x)}{W(y_1,y_2)}dx + y_2(x)\int\frac{y_1(x)r(x)}{W(y_1,y_2)}dx.$$

Thus, the general solution to (4) is

$$y(x) = C_1y_1(x) + C_2y_2(x) + y_p(x).$$

Example. *Consider*

$$x^2y'' - xy' - 3y = \frac{\ln x}{x},\ x > 0.$$

The characteristic equation is $m^2 - 2m - 3 = 0$

$\Rightarrow m = -1, 3$.

Hence $y_1 = 1/x$ and $y_2 = x^3$. Hence,

$$y_p(x) = y_1(x)u(x) + y_2(x)v(x)$$

where $$u(x) = -\int\frac{y_2(x)r(x)}{W(y_1,y_2)}dx$$

$$v(x) = \int\frac{y_1(x)r(x)}{W(y_1,y_2)}dx.$$

Now $W(y_1,y_2) = 4x$ and $\boxed{r(x) = \ln x/x^2}$. Hence,

$$u(x) = -\int\frac{\ln x}{4x}dx$$

$$= -\frac{(\ln x)^2}{8}$$

$$v(x) = \int\frac{\ln x}{4x^5}dx$$

$$= -\frac{\ln x}{16x^4} - \frac{1}{16x^4}$$

Hence, $$y_p(x) = -\frac{(\ln x)^2}{8x} - \frac{\ln x}{16x} - \frac{1}{64x}$$

Hence the general solution is $y = c_1y_1 + c_2y_2 + y_p$, i.e.

$$y(x) = \frac{A}{x} + Bx^3 - \frac{(\ln x)^2}{8x} - \frac{\ln x}{16x}.$$

Note that last term of y_p is absorbed with y_1.

Aliter: Let us make the transformation $t = \ln x$. Then the given transformed to

$$\ddot{y} - 2\dot{y} - 3y = te^t,$$

where $\dot{} = d/dt$. This is the same problem we have solved in lecture 9 using method of undetermined coefficients.

The solution is (see lecture 9)

$$y(t) = C_1 e^{-t} + C_2 e^{2t} - \frac{te^{-t}}{16}(2t+1).$$

which in terms of original x variable becomes

$$y(x) = \frac{C_1}{x} + C_2 x^2 - \frac{\ln x}{16x}(2\ln x + 1).$$

Solutions to Differential Equations Using Laplace Transformation

As promised, we can now solve differential equations using Laplace Transforms. What we do is transform the differential equation into an algebraic relation using Lapalce Transforms, solve for the Laplace Transform of the solution, then use inverse Laplace Transforms to bring the solution back into the time domain. The most important property of the Laplace transform in this case is the differentiation property:

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0^-),$$

$$\mathcal{L}\left\{\frac{d^2}{dt^2}f(t)\right\} = s\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} - \frac{df}{dt}(0^-)$$

$$= s^2F(s) - sf(0^-) - \frac{dt}{dt}(0^-)$$

$$\mathcal{L}\left\{\frac{d^n}{dt^n}f(t)\right\} = s^nF(s) - s^{n-1}f(0^-) -$$

$$\ldots - s^nf^{(n-2)}(0^-) - f^{n-1}(0^-).$$

A good starting point to remember this is: if the initial conditions are zero, we can replace n^{th} order differentiation with s^n. Note that because we have defined the Laplace Transform with the lower limit of integration zero, we must always take the starting time as 0, which is no problem for linear constant coefficient differential equations.

Example. Here is a simple differential equation:

$$\frac{dx}{dt} = 1 \quad t \geq 0.$$

Since the functions are equal on the left and right, the Laplace Transforms will be equal:

$$sX(s) - x(0^-) = \frac{1}{s}.$$

Solve for $X(s)$:

$$X(s) = \frac{1}{s^2} + \frac{x(0^-)}{s}$$

We recognize the Laplace Transforms on the right hand side, resulting in the solution

$$x(t) = t + x(0^-) \quad t \geq 0.$$

A. Initial Value Problems

- These are the types of problems we have been solving with RK methods

$$\frac{dy_1}{dt} = f_1(t, y_1, y_2)$$

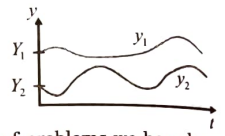
$$\frac{dy_2}{dt} = f_2(t, y_1, y_2)$$

Subject to :

$$t = 0$$

$$y_1(t=0) = Y_1$$

$$y_2(t=0) = Y_2$$



- These are the types of problems we have been solving with RK methods

$$\frac{dy_1}{dt} = f_1(t, y_1, y_2)$$

$$\frac{dy_2}{dt} = f_2(t, y_1, y_2)$$

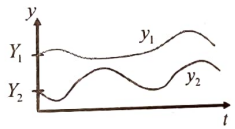
All conditions are specified at the same value of the independent variable!

Subject to :

$$t = 0$$

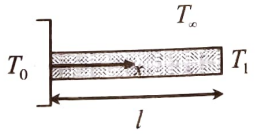
$$y_1(t=0) = Y_1$$

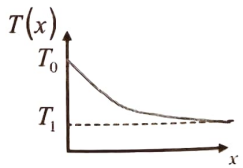
$$y_2(t=0) = Y_2$$



B. Boundary Value Problems

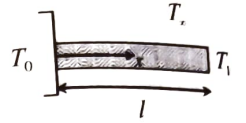
- Auxiliary conditions are specified at the boundaries (not just a one point like in initial value problems)

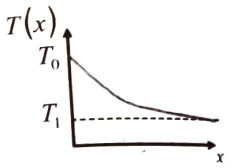




Two Methods: Shooting Method Finite Difference Method

- Auxiliary conditions are specified at the boundaries (not just a one point like in initial value problems)





Conditions are specified at different values of the independent variable!

Two Methods: Shooting Method Finite Difference Method

C. Shooting Method

- Applicable to both linear & non-linear Boundary Value (BV) problems.
- Easy to implement
- No guarantee of convergence

D. Approach

- Convert a BV problem into an initial value problem
- Solve the resulting problem iteratively (trial & error)
- Linear ODEs allow a quick linear interpolation
- Non-linear ODEs will require an iterative approach similar to our root finding techniques.

II. PARTIAL DIFFERENTIAL EQUATIONS

If a dependent variable is a function of two or more independent variables, an equation involving partial differential coefficients is called *partial differential equation*. The order of a partial differential equation is the same as that of the highest order differential coefficient in it.

If $z = f(x, y)$ where x and y are independent variables, then partial differntial coefficients $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ will be denoted by p and q respectively.

Formation of Partial Differential Equation

The partial differential equations can be formed either by elimination of arbitrary constants from a relation between x, y, z or by the elimination of arbitrary functions of these variables.

Solutions of Partial Differential Equations

- The solution $f(x, y, z, a, b) = 0$ of a first order partial differential equation, which contains two orbitrary constants is called *complete solution* or *complete integral*.
- If in this solution, we put $b = \phi(a)$ and find evnelope of the family of surfaces $f(x, y, z, a, \phi(a)) = 0$, we get a solution involving an arbitrary function ϕ. This is called *general solution* or *general integral*.
- A solution obtained from the complete integral by giving particular values to the arbitrary constant is called *particular solution* or *particular integral*.

Equations Solved by Direct Integration

Those equations which contain **only one** partial derivative can be solved by direct integration. However, in place of the constants of integration, we must use arbitrary functions of the variable kept constant.

Linear Partial Differential Equations of First Order

A differential equation involving first order partial derivatives p and q only is called *partial differential equation* of the first order. If p and q both occur in the first degree only and are not multiplied together, then it is called a *linear partial differential equation of the first order*.

Lagrange's Linear Equation

Partial differential equation of the form $Pp + Qq = R$, where P, Q and R are functions of x, y, z is the standard form of a linear partial differential equation of the first order and is called *Lagrange's linear equation*.

WORKING PROCEDURE TO SOLVE $P_p + Q_q = R$

(i) Form Auxiliary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

(ii) Find two independent solutions of the auxiliary equations.

Let $u = c_1$ and $v = c_2$ be two solutions of these equations. Then $f(u, v) = 0$ or $u = f(v)$ is solution of the given equation.

Example. The equation of any sphere of radius r having its centre in the XOY plane is

$$(x-a)^2 + (y-b)^2 + z^2 = r^2$$

Form a partial differential equation by eliminating the constants a, b.

Solution

Given : $(x-a)^2 + (y-b)^2 + z^2 = r^2$

Differentiating with respect to x and y, we get

$$2(x-a) + 2z\frac{\partial z}{\partial x} = 0 \quad \Rightarrow \quad x - a = -z.\frac{\partial z}{\partial x}$$

and $$2(y-b) + 2z\frac{\partial z}{\partial y} = 0 \quad \Rightarrow \quad y - b = -z.\frac{\partial z}{\partial y}$$

Substituting values, we get

$$z^2\left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1\right] = r^2$$

Applications of Partial Differential Equations

Method of Separation of Variables (Product Method)

This method describes a solution which breaks up into a product of functions each of which depends only on one variable.

i.e. if u is a dependent variable and depends on two independent variables x and t, *i.e.* $u\ (x, t)$, then we assume solution of the equation in the form

$$u = X(x)\,T(t)$$

where X is a function of x alone and T is a function of t alone.

Substitution for u from above equation in the given differential equation reduces to ordinary differential equations which can be conveniently studied.

Wave Equations

(*i*) One dimensional wave equation or equation of the vibration of a string.

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$$

(*ii*) Two dimensional wave equation or equation of the vibration of a rectangular membrance.

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

(*iii*) Three dimensional wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

Solution of Wave Equation by Separation of Variables

Wave equation is, $$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \qquad ...(i)$$

Let its solution be, $y(x, t) = X(x)\,T(t)$ ...(*ii*)

Substituting from equations (*ii*) in equation (*i*), we get

$$X\frac{d^2T}{dt^2} = c^2T\frac{d^2X}{dx^2}$$

$$\Rightarrow \quad \frac{1}{T}\frac{d^2T}{dt^2} = \frac{c^2}{X}\frac{d^2X}{dx^2} \qquad ...(iii)$$

Since x and t are independent variables and left side of equation (*iii*) is a function of t alone while right side is a function of x; equation (*iii*) will be true only when each side is equal to a constant quantity.

Following three possibilities arise:

(*i*) *Constant is negative* ($-k^2$).

General solution is,

$$y(x, t) = (A\cos kt + B\sin k)\left(C\cos\frac{k}{c}x + D\sin\frac{k}{c}x\right)$$

(*ii*) *Constant is zero.*

General solution is,

$$y(x, t) = (At + B)(Cx + D)$$

(*iii*) *Constant is positive* (k^2).

General solution is

$$y(x, t) = (Ae^{kt} + Be^{-kt})(Ce^{(k/c)x} + De^{-(k/c)x})$$

Example. Solve by method of separation of variables the equation

$$\frac{\partial^2 u}{\partial t^2} - 2\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0.$$

Solution

Let $u = XT$ ...(*i*)

$$\therefore \quad X\frac{d^2T}{dt^2} - 2X\frac{dT}{dt} + T\frac{dX}{dx} = 0$$

$$\Rightarrow \quad \frac{1}{T}\left(\frac{d^2T}{dt^2} - 2\frac{dT}{dt}\right) = -\frac{1}{X}\frac{dX}{dx} = k \text{ (say)}$$

[since x and t are independent of each other]

$$\frac{d^2T}{dt^2} - 2\frac{dT}{dt} - kT = 0 \qquad ...(ii)$$

and $$\frac{dX}{dx} + kX = 0 \qquad ...(iii)$$

Solutions of equations (*ii*) and (*iii*) are

$T = A\,e^{\{1+\sqrt{(1+k)}\}t} + B\,e^{\{1-\sqrt{(1+k)}\}t}$ and $X = C\,e^{-kx}$

Substituting for X and T in equation (*i*), we get

$$u = \left[A'\,e^{\{1+\sqrt{(1+k)}\}t} + B'\,e^{\{1-\sqrt{(1+k)}\}t}\right]e^{-kx}$$

where $A' = AC$ and $B' = BC$.

EXERCISE

MCQ TYPE QUESTIONS

1. The solution for the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0 \text{ is}$$

(*a*) $C_1e^{-2t} + C_2e^{3t}$ (*b*) $C_1\sin 2t + C_2\cos 2t$

(*c*) $C_1e^{2t} + C_2e^{-2t}$ (*d*) $C_1e^{-2t} + C_2e^{-3}$

2. Integrating factor for the differential equation

$$\left(\cos^2 x\right)\frac{dy}{dx} + y = \tan x, \text{ is}$$

(*a*) $e^{\tan x}$ (*b*) $\cos 2x$

(*c*) $e^{-\tan x}$ (*d*) $\sin 2x$

3. Find range of values for a constant K to yield a stable system in the following set of time dependent differential equations

$$\frac{dy_1}{dt} = -5\,y_2 + (4-K)\,y_2;\quad \frac{dy_2}{dt} = y_1 - 2y_2$$

(*a*) $0 < K < 7$ (*b*) $6.25 < K < 10$

(*c*) $-6 < K \le 6.25$ (*d*) $0 \le K \le 7$

4. Differential equation,

$$\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$$

will have a solution of the form

(*a*) $(C_1 + C_2\,t)\,e^{-5t}$ (*b*) C_1e^{-2t}

(*c*) $C_1e^{-5t} + C_2e^{5t}$ (*d*) $C_1e^{-5t} + C_2e^{2t}$

where C_1 and C_2 are constants.

5. Differential equation for the variation of the amount of salt x in a tank with time t is given by

$$\frac{dx}{dt} + \frac{x}{20} = 10$$

where x is in kg and t is in minutes.

Assuming that there is no salt in the tank initially, the time (in min.) at which the amount of salt increases to 100 kg is

(*a*) $10\ln 2$ (*b*) $20\ln 2$

(*c*) $50\ln 2$ (*d*) $100\ln 2$

6. Solve the differential equation

$$xdx + ydy + 2(x^2 + y^2)\,dx = 0$$

(*a*) $x^2 - y^2 = ce^{-4x}$

(*b*) $x^2 + y^2 = ce^{+4x}$

(*c*) $x^2 + y^2 = ce^{-4x}$

(*d*) $x^2 - y^2 = ce^{4x}$

7. The differential equation

$$\frac{d^2y}{dx^2} + \sin x\,\frac{dy}{dx} + ye^x = \sinh x$$

is

(*a*) first order and linear

(*b*) first order and non-linear

(*c*) second order and linear

(*d*) second order and non-linear

8. For the differential equation $\frac{dy}{dt} + 5y = 0$ with $y(0) = 1$, the general solution is

(*a*) e^{5t} (*b*) e^{-5t}

(*c*) $5e^{-5t}$ (*d*) $e^{\sqrt{-5}t}$

9. The necessary and sufficient for the differential equation of the from

$$M(x, y)\,dx + N(x, y)\,dy = 0$$

to be exact is

(*a*) $M = N$ (*b*) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$

(*c*) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ (*d*) $\frac{\partial^2 M}{\partial x^2} = \frac{\partial^2 N}{\partial y^2}$

10. The differential equation

$$y'' + (x^3\sin x)^5\,y' + y = \cos x^3 \text{ is}$$

(*a*) homogeneous

(*b*) nonlinear

(*c*) second order linear

(*d*) non-homogeneous with constant coefficients

11. For the differential equation,

$$f(x, y)\frac{dy}{dx} + g(x, y) = 0$$

to be exact

(*a*) $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$ (*b*) $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}$

(*c*) $f = g$ (*d*) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 g}{\partial y^2}$

12. If $f(D)\,y = 0$ be a linear differential equation with constant coefficients, then its auxiliary equation is

(*a*) $f(m^3) = 0$

(*b*) $f(m) = 0$

(*c*) $f(D - m) = 0$

(*d*) $f(x) = 0$

13. The general solution of the linear differential equation $(D^4 - 81)y = 0$ is given by

(a) $(C_1 + C_2x)e^{3x} + (C_3 + C_4x)\sin 3x$

(b) $(C_1 + C_2x)e^{3x} + (C_3 + C_4x)e^{-3x} + (C_5 + C_6x)\cos 3x + (C_7 + C_8x)\sin 3x$

(c) $C_1e^{3x} + C_2e^{-3x} + C_3\cos 3x + C_4\sin 3x$

(d) $C_1e^{3x} + C_2e^{-3x} + e^{3x}(C_3\cos x + C_4\sin x)$

14. $\frac{1}{D-a}Q(x)$ is equal to

(a) $e^{ax}\int Q(x)dx$ (b) $e^{-ax}\int e^{ax}Q(x)dx$

(c) $e^{-ax}\int Q(x)dx$ (d) $e^{ax}\int e^{-ax}Q(x)dx$

15. P.I. of $(D^2 - 2D)y = e^x \sin x$ is

(a) $-\frac{1}{2}e^x \sin x$ (b) $e^x \cos x$

(c) $-\frac{1}{2}e^x \cos x$ (d) none of these

16. General solution of the differential equation $(D^2 - m^2)y = 0$ is

(a) $y = (C_1 + C_2x)e^{mx}$

(b) $y = C_1 \sin x + C_2 \cos x$

(c) $y = C_1 \cosh x + C_2 \sinh x$

(d) none of these

17. $\frac{1}{\phi(D)(D-a)^n}e^{ax}$ is equal to

(a) $\frac{x^n e^{ax}}{\phi(a)}$ (b) $\frac{x^n e^{ax}}{n!\phi(a)}$

(c) $\frac{x^n e^{ax}}{\phi(a)}, \phi(a) \neq 0$ (d) $\frac{x^n e^{ax}}{n!\phi(a)}, \phi(a) \neq 0$

18. If V be any function of x, then $\frac{1}{f(D)}xV$ is equal to

(a) $\left[x + \frac{f(D)}{f'(D)}\right]f'(D).V$

(b) $\left[x - \frac{f'(D)}{f(D)}\right]f(D).V$

(c) $\left[x - \frac{f'(D)}{f(D)}\right]\frac{1}{f'(D)}.V$

(d) $\left[x - \frac{f'(D)}{f(D)}\right]\frac{1}{f(D)}.V$

19. If $y = 1 + \cos x$ and $y = 1 + \sin x$ solutions of the differential equation $\frac{d^2y}{dx^2} + y = 1$, then its solution will be also

(a) $y = 2(1 + \cos x)$

(b) $y = 2 + \cos x + \sin x$

(c) $y = \cos x - \sin x$

(d) $y = 1 + \cos x + \sin x$

20. Solution of the differential equation

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3e^{2x},$$

where $y(0) = 0$ and $y'(0) = -2$ is

(a) $y = e^{-x} - e^{2x} + xe^{2x}$ (b) $y = e^x - e^{-2x} - xe^{2x}$

(c) $y = e^{-x} + e^{2x} + \frac{x}{2}e^{2x}$ (d) $y = e^x - e^{-2x} - \frac{x}{2}e^{2x}$

21. The general solution of the differential equation $\frac{d^2y}{dx^2} + n^2y = 0$ is

(a) $c_1\sqrt{\cos nx} + c_2\sqrt{\sin nx}$

(b) $c_1 \cos nx + c_2 \sin nx$

(c) $c_1 \cos^2 nx + c_2 \sin^2 nx$

(d) $c_1 \cos^3 nx + c_2 \sin^3 nx$

22. The particular integral for the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{3x}$ is

(a) xe^{-x} (b) x^2e^{-x}

(c) $(x-1)e^{3x}$ (d) $(x-2)e^{4x}$

23. Solution of the differential equation $\frac{d^2y}{dx^2} = x$ is

(a) $y = \frac{x^3}{6} + c_1x^2 + c_2$

(b) $y = \frac{x^3}{6} + c_1x + c_2$

(c) $y = \frac{x^2}{6} + c_1x + c_2$

(d) $y = -\frac{x^2}{6} + c_1x + c_2$

24. Particular integral of $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$ is

(a) $\frac{x^2}{3} + 4x$ (b) $\frac{x^3}{3} + 4x$

(c) $\frac{x^3}{3} + 4$ (d) $\frac{x^3}{3} + 4x^2$

NUMERICAL TYPE QUESTIONS

1. The value of y as $t \to \infty$ for an initial value of $y(1) = 0$, for the differential equation

$$\left(4t^2 + 1\right)\frac{dy}{dt} + 8yt - t = 0 \text{ is } ____$$

2. The number of boundary conditions required to solve the following differential equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0 \text{ is/are } ____$$

ANSWERS

MCQ Type Questions

1. (d)	2. (c)	3. (c)	4. (a)	5. (b)	6. (c)	7. (d)	8. (b)	9. (c)	10. (c)
11. (b)	12. (b)	13. (c)	14. (d)	15. (a)	16. (c)	17. (d)	18. (d)	19. (d)	20. (a)
21. (b)	22. (c)	23. (b)	24. (b)						

Numerical Type Questions

1. (1/8) 2. (2)

EXPLANATIONS

MCQ TYPE QUESTIONS

3. Given : $\frac{dy_1}{dt} = -5y_2 + (4-k)y_2$ ...(i)

and $\frac{dy_2}{dt} = y_1 - 2y_2$ ...(ii)

Putting $\frac{d}{dt} = D$, we have from equation (i),

$$Dy_1 + 5y_2 - (4-K)y_2 = 0$$

$$\Rightarrow (D+5)y_1 - (4-K)y_2 = 0 \quad ...(iii)$$

From equation (ii)

$$Dy_2 = y_1 - 2y_2$$

$$\Rightarrow (D+2)y_2 - y_1 = 0 \quad ...(iv)$$

$$\Rightarrow y_1 = (D+2)y_2$$

Putting in equation (iii), we get

$$(D+5)(D+2)y_2 - (4-K)y_2 = 0$$

$$\Rightarrow [D^2 + 7D + (6+K)]y_2 = 0$$

$$\Rightarrow D^2 + 7D + 6K = 0$$

$$\Rightarrow D = \frac{-7 \pm \sqrt{49 - 4(6+K)}}{2 \times 2}$$

For real value of D,

$$49 - 4(6+K) \geq 0$$

$$\Rightarrow 49 - 24 - 4K \geq 0$$

$$\Rightarrow 4K \leq 25$$

$$\Rightarrow K \leq \frac{25}{4} \Rightarrow K \leq 6.25$$

4. $\frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 25x = 0$

$\Rightarrow (D^2 + 10D + 25)x = 0 \Rightarrow (D+5)^2 = 0$

$\Rightarrow D = -5, -5$

Hence solution is $(C_1 + C_2t)e^{-5t}$

5. Given differential equation is,

$$\frac{dx}{dt} + \frac{x}{20} = 10$$

$$\Rightarrow \frac{dx}{dt} = 10 - \frac{x}{20} = \frac{200 - x}{20}$$

Integrating,

$$20\int_0^{100}\frac{dx}{200-x} = \int_0^t dt$$

$$\therefore t = 20\left[\ln\frac{1}{200-x}\right]_0^{100} = 20\ln$$

7. Given: $\frac{d^2y}{dx^2} + \sin x\frac{dy}{dx} + ye^x = \sinh x$

The above equation is second order and non linear.

8. Given : $\frac{dy}{dt} + 5y = 0$

$$\Rightarrow \frac{dy}{y} = -5\,dt$$

Integrating, we get $\log_e y = c - 5t$

When $t = 0, y = 1$.

$$\therefore \log_e 1 = c - 5 \times 0$$

$$\Rightarrow c = 0$$

$$\therefore \log_e y = -5t$$

$$\Rightarrow y = e^{-5t}$$

10. y'' indicates that it is of second order. Degree of y'', y' and y is one. Hence given differential equation is second order linear equation.

11. For the differential equation $Mdx + Ndy = 0$, to be exact, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Hence for the differential equation

$f(x, y)\frac{dy}{dx} + g(x, y) = 0$ to be exact $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial y}$

13. $(D^4 - 81)y = 0$

Auxiliary equation is

$m^4 - 81 = 0 \Rightarrow (m + 3)(m - 3)(m + 3i)(m - 3i) = 0$

Hence Solution is

$y = C_1e^{3x} + C_2e^{-3x} + C_3 \cos 3x + C_4 \sin 3x$

15. P.I. $= \frac{1}{D^2 - 2D}e^x \sin x = e^x \frac{1}{(D+1)^2 - 2(D+1)} \sin x$

$= e^x \frac{1}{D^2 - 1}\sin x = e^x \frac{1}{(-1^2) - 1}\sin x$

$= -\frac{1}{2}e^x \sin x$

16. Auxiliary equation of the given equation is,

$D^2 - m^2 = 0 \quad \Rightarrow D = \pm m$

Hence C.F. is $y = C_1e^x + C_2e^{-x}$

$= C_1 \cosh x + C_2 \sinh x$

17. $\frac{1}{\phi(D)(D-a)^n}e^{ax} = \frac{1}{\phi(a)} \cdot \frac{1}{(D-a)^n}e^{ax}$

$= \frac{1}{\phi(a)}x.\frac{1}{\frac{d}{dD}(D-a)^n}e^{ax} (\because f(a) = 0)$

$= \frac{1}{\phi(a)}x.\frac{1}{n(D-a)^{n-1}}e^{ax}$

$= \frac{1}{\phi(a)}x^2.\frac{1}{n\frac{d}{dD}(D-a)^{n-1}}e^{ax} \quad (\because f'(a) = 0)$

$= \frac{1}{\phi(a)}x^2.\frac{1}{n(n-1)(D-a)^{n-2}}e^{ax}$

$= \frac{1}{\phi(a)}x^n.\frac{1}{n(n-1)(n-2)...2.1}e^{ax}$

$= \frac{1}{\phi(a)} \cdot \frac{x^n e^{ax}}{n!}$

19. $y = 1 + \cos x + \sin x$

$\therefore \frac{dy}{dx} = -\sin x + \cos x$ and $\frac{d^2y}{dx^2} = -\cos x - \sin x$

$\therefore \quad y + \frac{d^2y}{dx^2} = 1$

Hence $y = 1 + \cos x + \sin x$ is solution of, $\frac{d^2y}{dx^2} + y = 1$

20. Auxiliary equation is,

$m^2 - m - 2 = 0$

$\therefore \quad m = -1, 2$

$CF = c_1e^{-x} + c_2e^{2x}$

P.I. $= \frac{1}{D^2 - D - 2}.3.e^{2x} = 3.\frac{1}{D^2 - D - 2}e^{2x}$

$= 3x\frac{1}{2D - 1}e^{2x} = 3x\frac{1}{3}.e^{2x} = xe^{2x}$

$\therefore \quad y = c_1e^{-x} + c_2e^{2x} + xe^{2x}$

$\frac{dy}{dx} = -c_1e^{-x} + c_2e^{2x} + 2xe^{2x} + e^{2x}$

$y(0) = c_1 + c_2 = 0$ (given)

$y'(0) = -c_2 + 2c_2 + 1 = -2$ (given)

$\therefore \quad c_1 + c_2 = 0$ ...(i)

$c_1 - 2c_2 = 3$ ...(ii)

Solving equations (i) and (ii), we get

$c_1 = 1, c_2 = -1$

Hence required solution is

$y = e^{-x} - e^{2x} + xe^{2x}$

22. Given equation is,

$(D^2 - 5D + 6) = e^{3x}$

$\therefore \quad$ P.I. $= \frac{1}{D^2 - 5D + 6}e^{3x} = \frac{1}{(D-3)(D-2)}e^{3x}$

$= \frac{1}{D-3}e^{3x} - \frac{1}{D-2}e^{3x}$

$= x.e^{3x} - \frac{1}{3-2}e^{3x} = (x-1)e^{3x}$

24. Given differential equation is, $D^2 + D = x^2 + 2x + 4$

$\therefore \quad$ P.I. $= \frac{1}{D(D+1)}(x^2 + 2x + 4)$

$= \frac{1}{D}(D+1)^{-1}(x^2 + 2x + 4)$

$= \frac{1}{D}(1 - D + D^2)(x^2 + 2x + 4)$

$= \frac{1}{D}(x^2 + 2x + 4 - 2x - 2 + 2) = \frac{x^3}{3} + 4x$

NUMERICAL TYPE QUESTION

1. Given: $(4t^2 + 1)\frac{dy}{dt} + 8yt - t = 0$

$\Rightarrow \quad \frac{dy}{dt} + \frac{8t}{4t^2 + 1}y = \frac{t}{4t^2 + 1}$ ...(i)

$\therefore \quad$ I.F. $= e^{\int .8t/(4t^2+1)} = (4t^2 + 1)$

Hence solution of equation (i) is

$y(4t^2 + 1) = \int \frac{t}{(4t^2 + 1)}.(4t^2 + 1)\,dt + C = \frac{t^2}{2} + C$

$\Rightarrow \quad y = \frac{t^2}{2(4t^2 + 1)} + \frac{C}{(4t^2 + 1)}$ ...(ii)

But $y(1) = 0$,

$\therefore \quad 0 = \frac{1}{10} + \frac{C}{5} \quad \Rightarrow \quad C = -\frac{1}{2}$

From equation (ii), we have

$y = \frac{t^2}{2(4t^2 + 1)} - \frac{1}{2(4t^2 + 1)}$

$\therefore \quad y|_{t\to\infty} = \underset{t\to\infty}{\text{Lt}} \frac{1}{2(4 + 1/t^2)} - \frac{1}{2(4t^2 + 1)}$

$= \frac{1}{2(4 + 0)} - 0 = \frac{1}{8}$

■■